

Supplementary material document for:
*Proportional-limit asymptotics for Diaconis–Ylvisaker-penalised
logistic regression with fitted intercept*

Philipp Sterzinger 

September 9, 2026

A Simulations

A.1 Simulation setup

We use one common data-generating model and five contrasting prior specifications. The aspect ratio, true signal strength, true event probability and prior configurations are held fixed, while the slope and intercept information supplied by the prior slope varies across settings. Throughout we set

$$n \in \{125, 250, 500, 1000, \dots, 4000\}, \quad p = \kappa n, \quad \kappa = 0.2, \quad \gamma = \delta = 3/2.$$

We construct the signal β_0 as follows: Let $I_0 = \{1, \dots, 20\}$ denote a fixed block of true-null coordinates. We generate a standard Gaussian vector $\mathbf{g}_1 \in \mathbb{R}^{p-20}$ and define $\mathbf{u}_1$ by normalising $\mathbf{g}_1$ and prepending 20 zero coordinates. Independently, we generate $\mathbf{g}_2 \sim \mathcal{N}(\mathbf{0}_p, \mathbf{I}_p)$ and set

$$\tilde{\mathbf{u}}_2 = \mathbf{g}_2 - \langle \mathbf{g}_2, \mathbf{u}_1 \rangle \mathbf{u}_1, \quad \mathbf{u}_2 = \frac{\tilde{\mathbf{u}}_2}{\|\tilde{\mathbf{u}}_2\|_2},$$

so that $\mathbf{u}_1$ and $\mathbf{u}_2$ are orthonormal, $u_{1,j} = 0$, $j \in I_0$, while $u_{2,j} \neq 0$ for every $j \in I_0$ almost surely. At each value of p , the two directions are generated once, and are then held fixed over all replications, prior settings and all three tuning streams at that sample size.

We then set

$$\beta_0 = \sqrt{p} \gamma \mathbf{u}_1, \quad \beta_P(r) = \sqrt{p} \delta \left\{ r \mathbf{u}_1 + \sqrt{1 - r^2} \mathbf{u}_2 \right\},$$

where $\delta = \gamma = 1.5$ for every non-null prior. Hence,

$$\frac{\|\beta_0\|_2^2}{p} = \gamma^2, \quad \frac{\|\beta_P(r)\|_2^2}{p} = \delta^2, \quad \frac{\langle \beta_0, \beta_P(r) \rangle}{p} = r \gamma \delta.$$

All non-null prior slopes therefore have the same strength and differ only in their alignment with the true slope. Since the set of correlations used in the simulations is finite and satisfies $|r| < 1$, every non-null $\beta_P(r)$ is dense and on the block I_0 ,

$$\beta_{0,j} = 0, \quad \beta_{P,j}(r) = \sqrt{p} \delta \sqrt{1 - r^2} u_{2,j} \neq 0, \quad j \in I_0,$$

almost surely. Thus I_0 is a null block for the true slope, but not for the non-null prior slopes. Setting ‘N’ remains the sole exception, with an identically zero prior slope.

We choose the intercepts through marginal event probabilities. For $s \geq 0$ and $\pi \in (0, 1)$, let $a(s, \pi)$ be the unique solution of

$$\mathbb{E}[\rho'(a(s, \pi) + sZ)] = \pi, \quad Z \sim \mathcal{N}(0, 1).$$

The true marginal event probability is fixed at $\pi_0 = 0.2$, giving $\theta_0 = a(1.5, 0.2) \approx -1.9163$ and $a(1.5, 0.8) \approx 1.9163$. For the non-null prior settings, $\theta_P = a(1.5, \pi_P)$. The five settings are given in Table 3.

Settings ‘A’ and ‘O’ isolate the effect of prior-slope alignment, settings ‘A’ and ‘I’ isolate prior-intercept misspecification, and settings ‘I’ and ‘M’ compare aligned and anti-aligned prior slopes under the same incorrect intercept. Setting ‘N’ provides a neutral baseline and a rank-deficient prior geometry.

For each setting $s \in \{\text{‘A’}, \text{‘O’}, \text{‘I’}, \text{‘M’}, \text{‘N’}\}$, we use three setting-specific numerically selected tuning parameters for the limiting adjusted slope MSE, unadjusted slope MSE, and unadjusted classification error. For this, let $(\mathbf{v}_{\alpha,s}^*, \sigma_{\alpha,s}^*)$ denote the relevant solution components of the population equations in (8), let $\mathbf{e}_1 = [1, 0]^\top$, and define the numerical search interval $I_\alpha = \{0.01, 0.05, 0.10, \dots, 0.95, 0.975, 0.99, 0.995\}$. We define the numerical-search criteria

$$\begin{aligned} \mathcal{Q}_{s,\text{adj}}(\alpha) &= \frac{(\sigma_{\alpha,s}^*)^2}{(v_{1,\alpha,s}^*)^2} \\ \mathcal{Q}_{s,\text{unadj}}(\alpha) &= (\sigma_{\alpha,s}^*)^2 + (\mathbf{v}_{\alpha,s}^* - \mathbf{e}_1)^\top \mathbf{\Gamma}_s (\mathbf{v}_{\alpha,s}^* - \mathbf{e}_1) \\ \mathcal{Q}_{s,\text{pred}}(\alpha) &= \frac{(\mathbf{e}_1^\top \mathbf{\Gamma}_s \mathbf{e}_1) \{(\mathbf{v}_{\alpha,s}^*)^\top \mathbf{\Gamma}_s \mathbf{v}_{\alpha,s}^* + (\sigma_{\alpha,s}^*)^2\}}{(\mathbf{e}_1^\top \mathbf{\Gamma}_s \mathbf{v}_{\alpha,s}^*)^2}. \end{aligned}$$

Table 3: Simulation settings. The common true model has signal strength $\gamma = 1.5$ and marginal event probability $\pi_0 = 0.2$. For the non-null prior settings, $\delta = 1.5$, $r = \text{corr}(\mathbf{x}^\top \boldsymbol{\beta}_0, \mathbf{x}^\top \boldsymbol{\beta}_P)$, and $\pi_P = \mathbb{E}[\rho'(\theta_P + \mathbf{x}^\top \boldsymbol{\beta}_P)]$.

Setting	δ	r	π_P	θ_P	Prior information
‘A’: aligned and calibrated	1.5	0.8	0.2	-1.9163	Aligned slope and correctly calibrated intercept
‘O’: orthogonal and calibrated	1.5	0	0.2	-1.9163	Orthogonal slope and correctly calibrated intercept
‘I’: aligned, incorrect intercept	1.5	0.8	0.8	1.9163	Aligned slope and oppositely calibrated intercept
‘M’: misleading	1.5	-0.8	0.8	1.9163	Anti-aligned slope and oppositely calibrated intercept
‘N’: null	0	—	0.5	0	Identically zero prior slope

We denote by $\alpha_{s,\text{slope}}^{\text{adj}}$, $\alpha_{s,\text{slope}}^{\text{DY}}$, and $\alpha_{s,\text{pred}}^{\text{DY}}$ the values returned by the numerical search below for the adjusted slope-MSE, unadjusted slope-MSE, and unadjusted classification-error criteria, respectively. We evaluate each criterion over I_α .

We use $R = 10,000$ independent Monte Carlo replications at each value of n . Unless a display explicitly shows the sample-size grid, the main-text illustrations use $n = 4000$ and $p = 800$. For replication $b = 1, \dots, R$, we generate one training sample from the common true model and reuse it across all five prior settings and all three tuning streams. For out-of-sample prediction, we evaluate the finite-sample conditional test error in (15). The adjusted rule uses the fit at $\alpha_{s,\text{slope}}^{\text{adj}}$, whereas the unadjusted rule uses the fit at $\alpha_{s,\text{pred}}^{\text{DY}}$. In each case we use the corresponding population cutoff and orientation in (26). For testing, the common true-null block I_0 is used for the fixed-coordinate Gaussian experiment. In the adjusted-tuning stream, we evaluate the statistic in (160) for $I = \{1\}$ and $b_I^0 = 0$ in every replication. For the penalised likelihood-ratio experiment, we also use the fit at $\alpha_{s,\text{slope}}^{\text{adj}}$ and take $I = \{1, \dots, 10\}$. The oracle restriction centre is

$$\mathbf{m}_{I,s}^{0,\text{adj}} = v_{1,s,\text{adj}}^* \boldsymbol{\beta}_{0,I} + v_{2,s,\text{adj}}^* \boldsymbol{\beta}_{P,s,I} = v_{2,s,\text{adj}}^* \boldsymbol{\beta}_{P,s,I},$$

and record

$$\frac{\lambda_{s,\text{adj}}^*}{(\sigma_{s,\text{adj}}^*)^2} 2\Lambda_{I,s}^{(b)}.$$

Limiting parameter estimation for Section 7 is performed on replications $b \in \{10, 20, \dots, R\}$. For each fit, we retain the SLOE variance estimate $\hat{v}_{\text{SLOE},s}^2$ described in Section A.1.1 of the supplementary material, which removes the fitted-intercept contribution from the estimated variance of the fitted score. The response-moment estimator, estimates γ using $\hat{\zeta}^2$ in (47), and estimates the remaining response geometry using the covariance-centred aligned moment $\hat{b}$ and the signed-truncated ratio $\hat{\varphi}$ in (50). For each successful estimate $(\hat{\theta}_0, \hat{\gamma}^2, \hat{\varphi})$ produced by either procedure, we form

$$\hat{\mathbf{\Gamma}} = \begin{bmatrix} \hat{\gamma}^2 & \hat{\varphi} \\ \hat{\varphi} & \hat{\delta}^2 \end{bmatrix},$$

and solve the population first-order equations at the common value $\alpha_{s,\text{slope}}^{\text{DY}}$ used for the underlying unadjusted MDYPL fit, recording the resulting estimates of $(\theta^*, \mathbf{u}^*, \mathbf{v}^*, \boldsymbol{\chi}^*, \sigma^*, \lambda^*)$.

A.1.1 SLOE-based estimation of the limiting parameters

The original SLOE idea (Yadlowsky *et al.*, 2021) is to reparameterise the ML analogue of the limiting system in (8) in terms of moments of the fitted limiting score. Applied to the present setting, Theorem 1 gives

$$\frac{1}{p} \left\| \hat{\boldsymbol{\beta}}^{\text{DY}} \right\|_2^2 \xrightarrow{\text{a.s.}} \nu^2 = (\sigma^*)^2 + \mathbf{v}^{*\top} \mathbf{\Gamma} \mathbf{v}^*,$$

and

$$\frac{1}{p} \langle \hat{\boldsymbol{\beta}}^{\text{DY}}, \boldsymbol{\beta}_P \rangle \xrightarrow{\text{a.s.}} \chi_P = \varphi v_1^* + \delta^2 v_2^*.$$

Since

$$\mathbf{v}^{*\top} \mathbf{\Gamma} \mathbf{v}^* = \gamma^2 (v_1^*)^2 + 2\varphi v_1^* v_2^* + \delta^2 (v_2^*)^2,$$

the two limiting identities can, when $v_1^* \neq 0$, be inverted as

$$\varphi = \frac{\chi_P - \delta^2 v_2^*}{v_1^*},$$

and

$$\gamma^2 = \frac{\nu^2 - (\sigma^*)^2 - 2\chi_P v_2^* + \delta^2 (v_2^*)^2}{(v_1^*)^2}.$$

Substituting these expressions into (8) gives a system for $(\theta_0^*, v_1^*, v_2^*, \sigma^*, \lambda^*)$ parameterised by $(\kappa, \theta_P^*, \theta^*, \delta^2, \nu^2, \chi_P)$, for which the direct plug-in estimates are $p/n, \theta_P, \hat{\theta}^{\text{DY}}, \delta_P^2 = \|\beta_P\|_2^2/p, \hat{\nu}^2 = \|\hat{\beta}^{\text{DY}}\|_2^2/p, \hat{\chi}_P = \langle \hat{\beta}^{\text{DY}}, \beta_P \rangle/p$. For numerical stability, the implementation instead replaces $(\mathbf{v}^*, \sigma^*)$ by (χ, ν^2) , where $\chi = (\text{cov}(S^{\text{DY}}, Q_1), \chi_P)^\top$, treats $(\hat{\chi}_P, \hat{\nu}^2)$ as plug-ins, and solves for $(\theta_0^*, \gamma^2, \varphi, \lambda^*, \chi)$. On the full-rank region with $v_1^* \neq 0$, this is algebraically equivalent to the direct parameterisation above.

Yadlowsky *et al.* (2021) further provide a leave-one-out alternative to the direct estimation of $\hat{\nu}^2$. Let $\hat{\eta}_i = \hat{\eta}^{\text{DY}}(\mathbf{x}_i)$, set $\tilde{\mathbf{W}} = \text{diag}\{\rho''(\hat{\eta}_i)\}_{i=1}^n$, and let $\hat{h}_i$ be the i th diagonal element of the corresponding weighted hat matrix for the augmented design. Adapting Yadlowsky *et al.* (2021) to the MDYPL pseudo-responses, as in Sterzinger and Kosmidis (2026) equation (11), define

$$\hat{\nu}_{\text{SLOE}}^2 = \frac{1}{n} \sum_{i=1}^n (\hat{\eta}_i^{\text{SLOE}} - \bar{\eta}^{\text{SLOE}})^2, \quad \hat{\eta}_i^{\text{SLOE}} = \hat{\eta}_i - \frac{\hat{h}_i}{1 - \hat{h}_i} \frac{\tilde{Y}_i - \rho'(\hat{\eta}_i)}{\rho''(\hat{\eta}_i)}, \quad \bar{\eta}^{\text{SLOE}} = \frac{1}{n} \sum_{i=1}^n \hat{\eta}_i^{\text{SLOE}}.$$

The centring removes the fitted-intercept contribution, while Theorem 1 supplies the required full-sample intercept and slope-norm limits. If the corresponding leave-one- and leave-two-out stability results hold for the augmented MDYPL problem, the argument of Yadlowsky *et al.* (2021) Proposition 2) yields $\hat{\nu}_{\text{SLOE}}^2 \xrightarrow{p} \nu^2$.

While the SLOE reparameterisation is algebraically valid, it does not by itself establish invertibility of the nuisance map. In particular, even with a fitted intercept, a fixed-intercept slice of the map from the true signal strength γ to the corrupted fitted-score strength ν can fold. Distinct values of γ may therefore produce the same fitted-score moments, and the inverse SLOE system may have more than one admissible root. Moreover, consistency of a selected root also requires local identification, uniform convergence of the plug-in equations, existence of a nearby sample root and a rule selecting that root when several solutions are present.

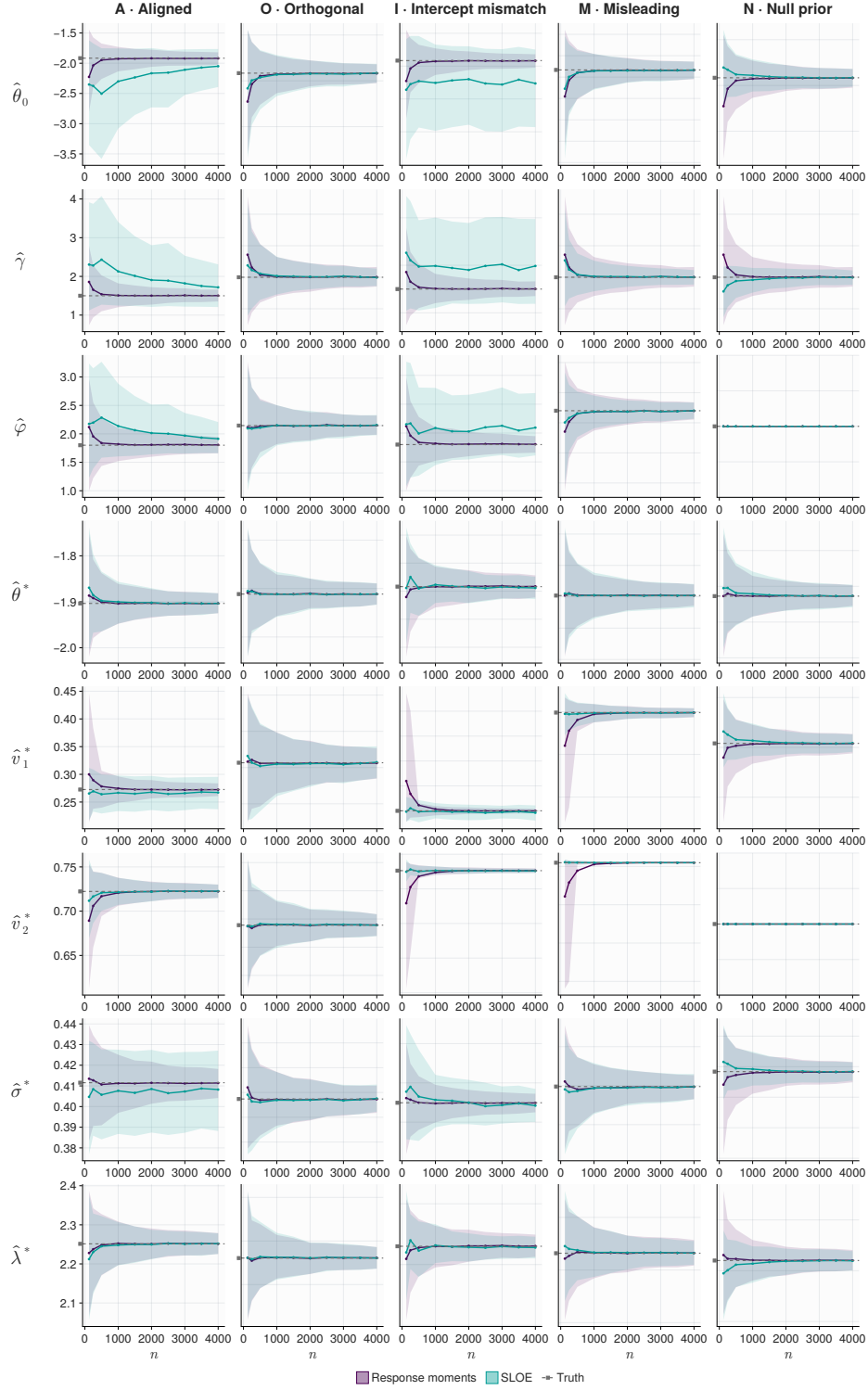


Figure 8: Finite-sample behaviour of the response-moment and fitted-score SLOE estimates of the unknown model parameters and induced limiting state parameters for varying n . Solid curves and shaded bands show means and pointwise 10–90th percentiles over successful unknown-parameter estimates. Grey dashed lines mark the corresponding population targets.

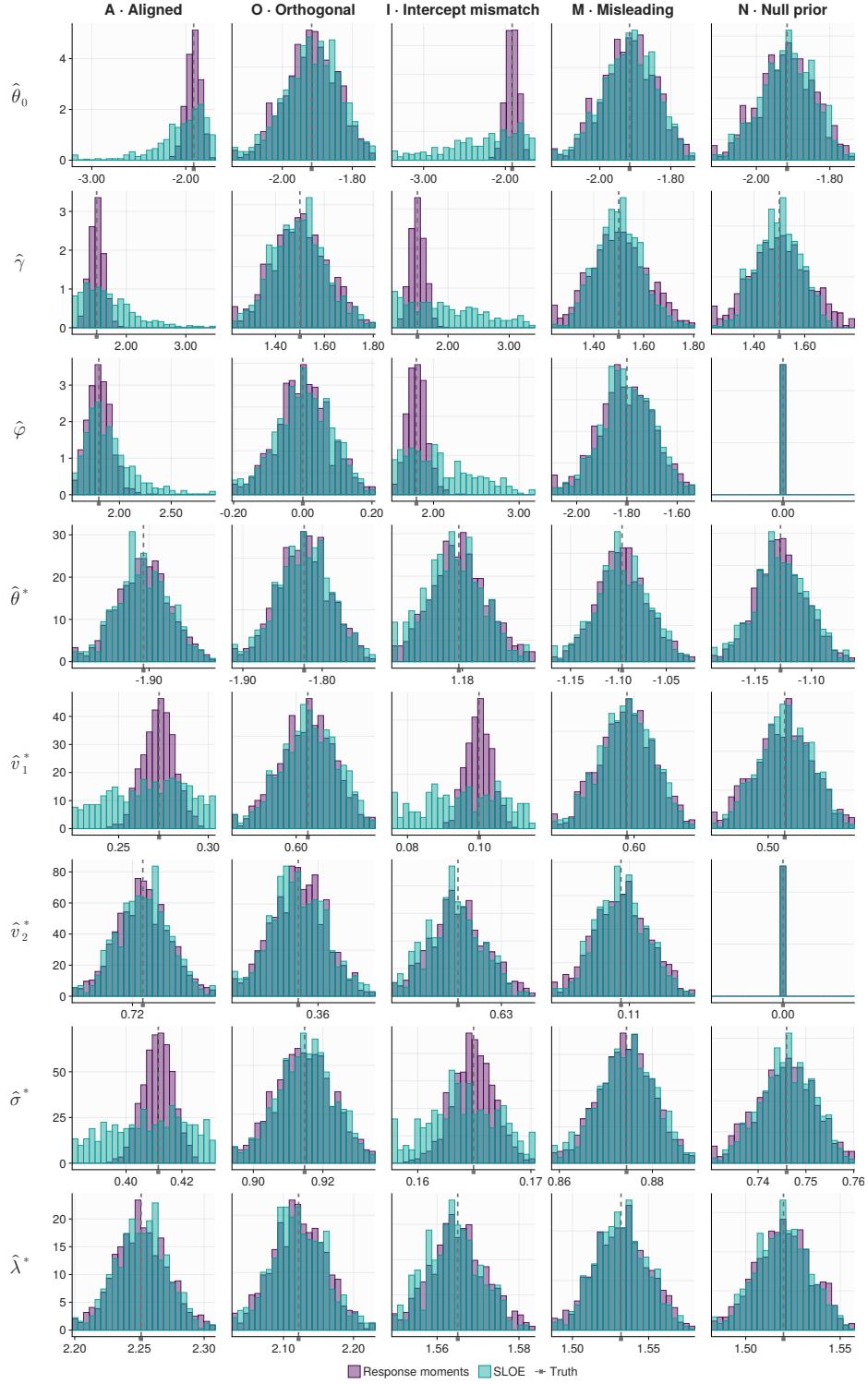


Figure 9: Monte Carlo distributions of the response-moment and fitted-score SLOE estimates of the unknown model parameters and induced limiting state parameters at $n = 4000$ and $p = 800$, which are obtained by re-solving the oracle equations at $\alpha_{s,\text{slope}}^{\text{DY}}$. Grey dashed vertical lines mark the corresponding population targets.

A.2 Additional simulation output

A.2.1 Additional output for Section 3.4

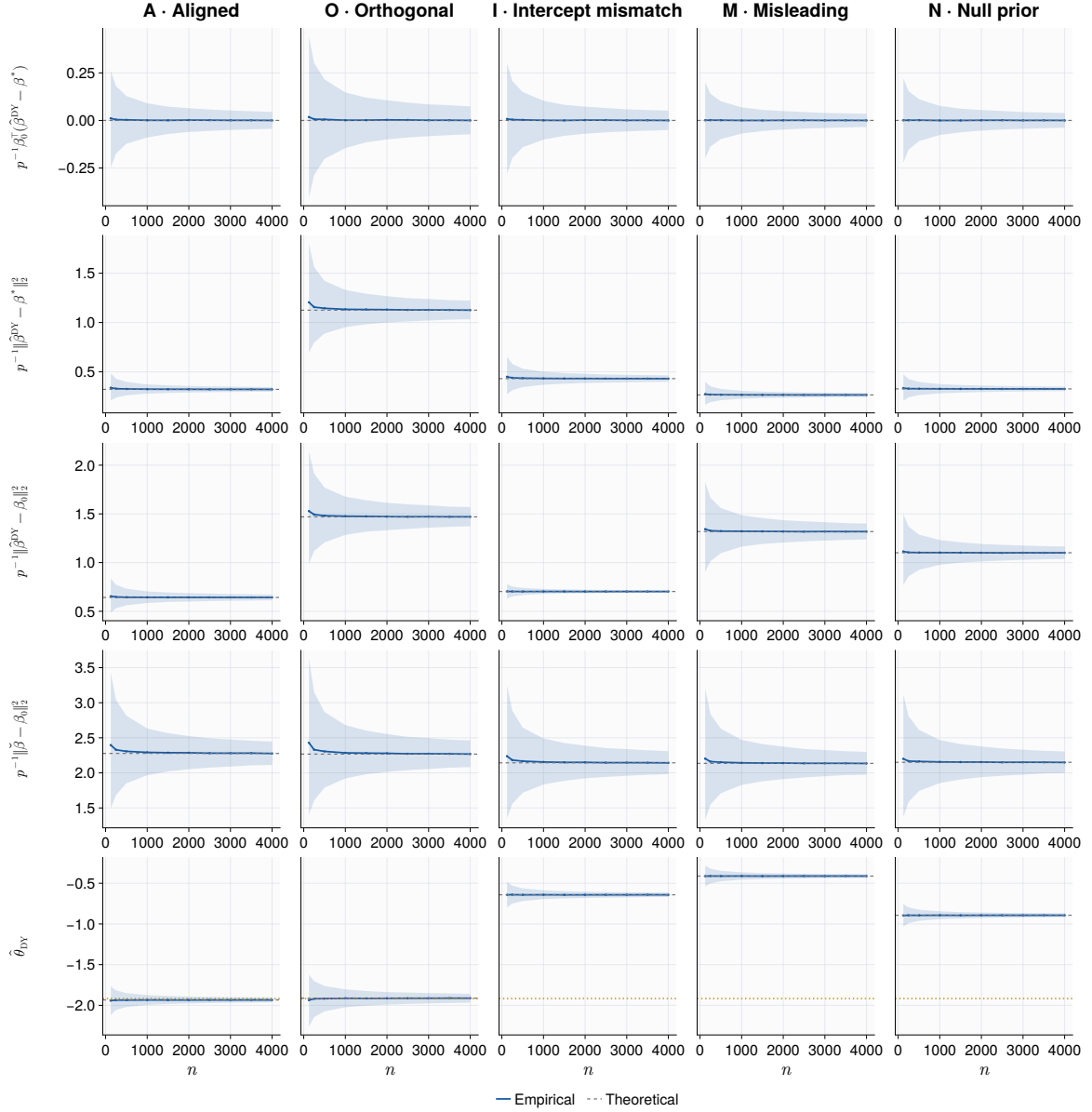


Figure 10: Aggregate finite-sample convergence toward the limits in Theorem 1 for the consequences summarised in Table 1. The third row uses $\alpha_{s,\text{slope}}^{DY}$, while the remaining rows use $\alpha_{s,\text{slope}}^{\text{adj}}$. Solid blue curves show Monte Carlo means, shaded bands span the empirical 10–90th percentiles, and grey dashed lines mark the corresponding state-equation limits. In the fitted-intercept panels, gold dotted lines mark the data-generating value θ_0 .

A.2.2 Additional output for Section 5

Setting	1%	5%	10%	25%	50%	75%	90%	95%	99%
N(0, 1) reference	-2.326	-1.645	-1.282	-0.674	0.000	0.674	1.282	1.645	2.326
A · Aligned	-2.397	-1.617	-1.256	-0.648	0.001	0.670	1.260	1.609	2.320
O · Orthogonal	-2.381	-1.623	-1.244	-0.651	0.000	0.673	1.258	1.608	2.290
I · Wrong intercept	-2.405	-1.631	-1.253	-0.644	0.001	0.681	1.240	1.621	2.309
M · Misleading	-2.381	-1.620	-1.247	-0.643	0.006	0.678	1.258	1.629	2.303
N · Null prior	-2.397	-1.639	-1.247	-0.656	-0.002	0.676	1.259	1.627	2.338

Table 4: Empirical quantiles of the oracle-adjusted statistic $\mathbf{Z}_1^{\text{adj}}(0)$ for the first true-null coordinate, compared with the corresponding N(0, 1) quantiles. Results are based on 10,000 replications with $n = 4000$ and $p = 800$.

Nominal coverage	Setting				
	‘A’	‘O’	‘I’	‘M’	‘N’
90%	0.907 (0.003)	0.907 (0.003)	0.905 (0.003)	0.904 (0.003)	0.902 (0.003)
95%	0.952 (0.002)	0.952 (0.002)	0.951 (0.002)	0.951 (0.002)	0.951 (0.002)
99%	0.990 (0.001)	0.990 (0.001)	0.989 (0.001)	0.990 (0.001)	0.990 (0.001)

Table 5: Empirical coverage of nominal 90%, 95%, and 99% two-sided confidence intervals for the null coordinate $\beta_{0,1} = 0$, based on the oracle-adjusted statistic $\mathbf{Z}_1^{\text{adj}}(0)$, along with Monte Carlo standard errors. Results are based on 10,000 replications with $n = 4000$ and $p = 800$.

Setting	1%	5%	10%	25%	50%	75%	90%	95%	99%
χ_{10}^2 reference	2.558	3.940	4.865	6.737	9.342	12.549	15.987	18.307	23.209
A · Aligned	2.644	3.966	4.870	6.788	9.317	12.607	16.034	18.507	23.353
O · Orthogonal	2.557	3.912	4.812	6.812	9.359	12.592	16.076	18.400	23.509
I · Wrong intercept	2.598	3.949	4.838	6.784	9.360	12.548	16.113	18.211	23.560
M · Misleading	2.597	3.990	4.895	6.781	9.336	12.525	16.097	18.293	23.459
N · Null prior	2.657	3.956	4.908	6.761	9.368	12.553	16.015	18.267	23.628

Table 6: Empirical quantiles of the candidate oracle-rescaled penalised likelihood-ratio statistic $2\lambda^* \Lambda_I / (\sigma^*)^2$, for $I = \{1, \dots, 10\}$, compared with the corresponding χ_{10}^2 quantiles. Results are based on 10,000 replications with $n = 4000$ and $p = 800$.

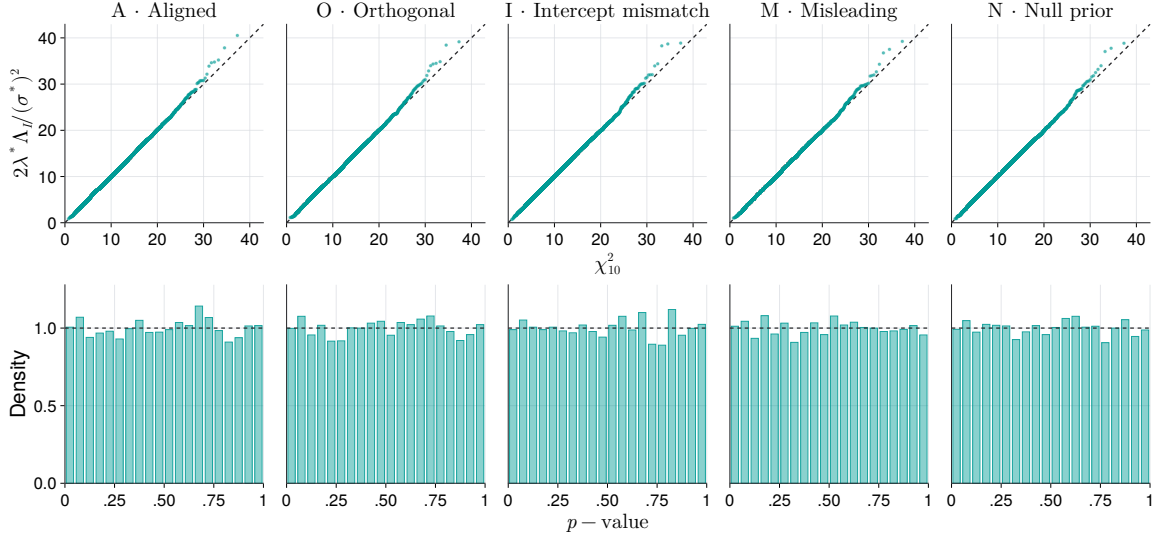


Figure 11: Finite-sample calibration of the oracle-adjusted penalised likelihood-ratio statistic for the true-null block $I = \{1, \dots, 10\}$ in the settings of Section [3.4](#) based on 10,000 replications with $n = 4000$ and $p = 800$. Top: Q-Q plots of $2\lambda^* \Lambda_I / (\sigma^*)^2$ against the χ^2_{10} distribution. Bottom: densities of the corresponding one-sided p -values. Black dashed lines mark the identity line and the $\text{Unif}([0, 1])$ reference density, respectively.

B Proof of Theorem 1

Theorem 1. Assume the conditions of Section 2, and let θ^* , σ^* and β^* be as in (7) and (9). Then the MDYPL estimator in (6) satisfies the following.

(i)

$$\hat{\theta}^{\text{DY}} \xrightarrow{\text{a.s.}} \theta^*.$$

(ii) For every pseudo-Lipschitz function $\psi : \mathbb{R}^3 \rightarrow \mathbb{R}$ of order two ,

$$\frac{1}{p} \sum_{j=1}^p \psi \left(\hat{\beta}_j^{\text{DY}} - \beta_j^*, \beta_{0,j}, \beta_{P,j} \right) \xrightarrow{\text{a.s.}} \mathbb{E} \left[\psi(\sigma^* G, \bar{\beta}_0, \bar{\beta}_P) \right], \quad (51)$$

where $(\bar{\beta}_0, \bar{\beta}_P) \sim \pi_{\{\bar{\beta}_0, \bar{\beta}_P\}}$ and $G \sim \mathcal{N}(0, 1)$ is independent of $(\bar{\beta}_0, \bar{\beta}_P)$.

B.1 Proof roadmap

By Lemma 9, on $\mathcal{E}_n$ the compact PO has a unique primal minimiser $(\hat{\mathbf{v}}, \hat{\mathbf{w}}, \hat{\theta}, \hat{\boldsymbol{\eta}})$. Set

$$\hat{\boldsymbol{\beta}} = \mathbf{B}\hat{\mathbf{v}} + \mathbf{E}\hat{\mathbf{w}}, \quad \hat{\sigma} = \frac{\|\hat{\mathbf{w}}\|_2}{\sqrt{p}}.$$

The direction $\mathbf{d}$ of (121) is defined also when $\hat{\sigma} = 0$ and satisfies

$$\hat{\boldsymbol{\beta}} = \underbrace{\mathbf{B}\hat{\mathbf{v}}}_{\text{signal-aligned component}} + \underbrace{\sqrt{p}\hat{\sigma}\mathbf{d}}_{\text{signal-orthogonal component}}. \quad (52)$$

On the compactification event of Proposition B.3, $(\hat{\theta}, \hat{\boldsymbol{\beta}}) = (\hat{\theta}^{\text{DY}}, \hat{\boldsymbol{\beta}}^{\text{DY}})$. The localisation argument identifies the limits of $\hat{\mathbf{u}} = \mathbf{\Gamma}_p^{1/2}\hat{\mathbf{v}}$, $\hat{\sigma}$ and $\hat{\theta}$, while Lemma 1 converts (52) into a standard Gaussian coupling for the actual MDYPL slope.

The route from the estimator (6) to these limits comes from a CGMT comparison with two branches. The left branch embeds MDYPL into a PO, whose optimiser is the object we want to localise. The right branch studies the corresponding AO, where the actual scalar and limiting calculations are performed. The two branches are then brought back together through restricted value comparisons, as summarised in Figure 12

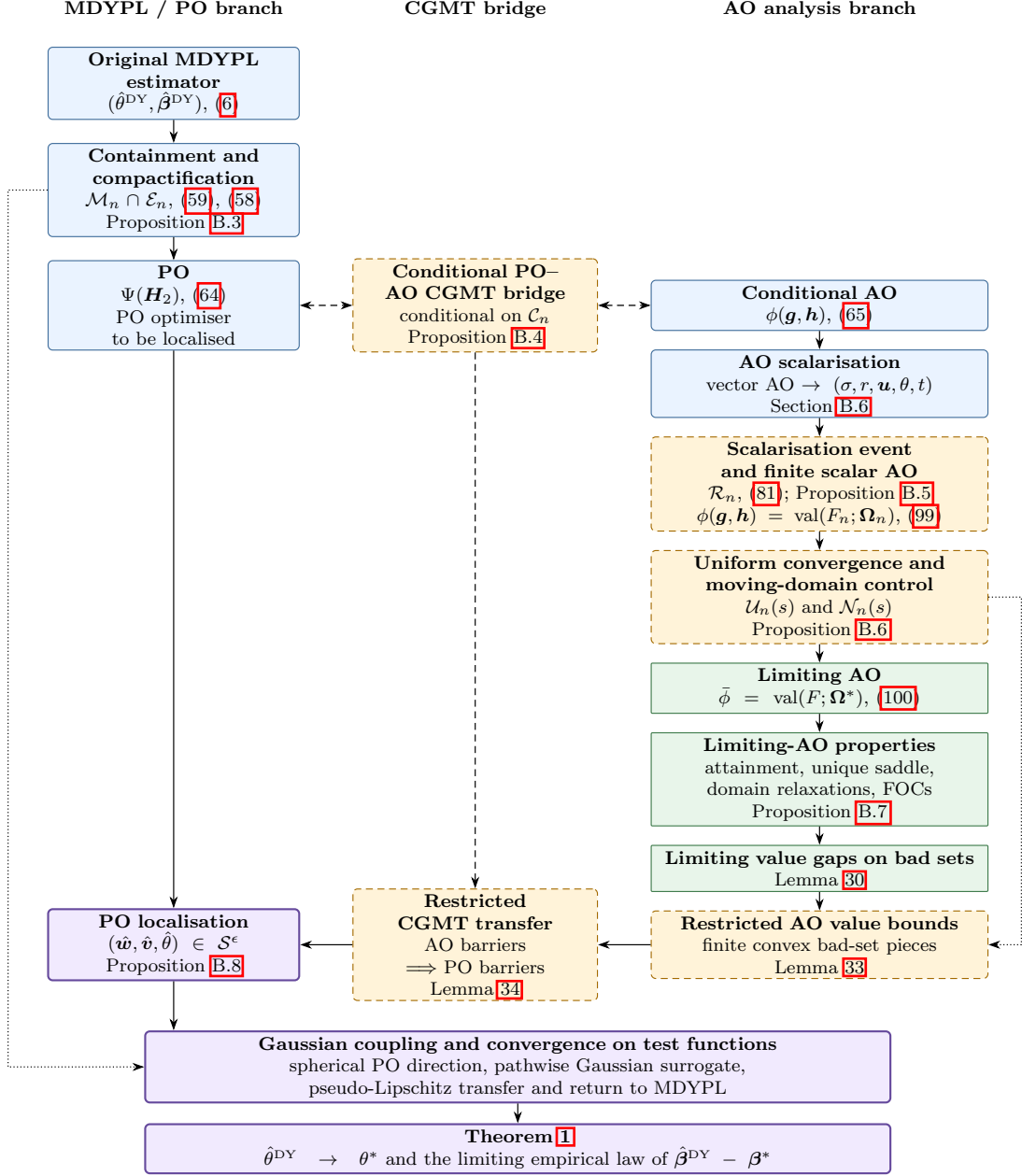


Figure 12: Proof roadmap for Theorem 1. The left branch embeds MDYPL, given a containment event, into a compact primary optimisation (PO). The middle branch applies the conditional CGMT to unrestricted and restricted PO-AO values. The right branch scalarises the auxiliary optimisation (AO), controls the finite and moving domains, identifies the limiting AO and establishes attainment, its unique saddle, the required domain relaxations, and the associated first-order conditions, and obtains bad-set gaps and restricted AO bounds. These bounds localise the unique PO minimiser. Its exact spherical representation and Gaussian surrogate feed the test-function assembly, which combines them with the containment event into the theorem. Blue nodes denote constructions and reformulations, amber dashed nodes denote probabilistic finite- n comparisons and controls, green nodes denote deterministic limiting-AO arguments, and violet nodes denote conclusions feeding into Theorem 1.

In words, the hierarchy in Figure 12 is as follows. First, Lemma 3 and Proposition B.3 put the MDYPL estimator on the event $\mathcal{M}_n \cap \mathcal{E}_n$ inside a compact convex–concave primary optimisation. This is the PO branch of the argument: it produces the PO optimiser that we ultimately want to show belongs to $\mathcal{S}^\epsilon$. Second, after conditioning on $\mathcal{C}_n = \sigma(\mathbf{B}, \theta_0, \theta_P, \mathbf{H}_1, \varepsilon)$, the pair (64)–(65) is a valid pathwise CGMT instance, with $\mathbf{H}_2$ random in the PO and $(\mathbf{g}, \mathbf{h})$ random in the AO. Third, the AO branch is analysed on its own: Proposition B.5 reduces it to the scalar problem in $\omega = (\sigma, r, \mathbf{u}, \theta, t)$. Proposition B.6 connects the scalar problem to $\phi = \text{val}(F; \Omega^*)$, and Proposition B.7 establishes attainment, the unique limiting saddle, the required domain relaxations, and the population equations (8). Finally, the deterministic gaps of Lemma 30 and the restricted AO bounds of Lemma 33 are inserted back into the CGMT comparison. This rules out PO minimisers outside $\mathcal{S}^\epsilon$ and gives localisation by Proposition B.8. The spherical representation in Lemma 35 and the Gaussian surrogate in (123), with law and approximation bound (124) and (125), supply the Gaussian replacement of the localised PO slope, and Proposition B.9 assembles it, conditionally on $\mathcal{G}_n$, with the containment and full-rank event $\mathcal{M}_n \cap \mathcal{E}_n$, the conditional triangular-array law, and the W_2 signal replacement into Theorem 1. Lemma 1 records the resulting vector coupling for the MDYPL slope.

B.2 CGMT

For completeness, we restate the form of the CGMT used in the conditional PO–AO comparison. Its application to the present problem is recorded in Proposition B.4

Theorem B.2 (CGMT, Thrampoulidis *et al.* (2018) Theorem VI.1, (i)–(ii)). *Consider the primary optimisation (PO) and auxiliary optimisation (AO)*

$$\Phi(\mathbf{G}) = \min_{\mathbf{w} \in \mathcal{S}_{\mathbf{w}}} \max_{\mathbf{u} \in \mathcal{S}_{\mathbf{u}}} \{ \mathbf{u}^\top \mathbf{G} \mathbf{w} + \psi(\mathbf{w}, \mathbf{u}) \}, \quad (53)$$

and

$$\phi(\mathbf{g}, \mathbf{h}) = \min_{\mathbf{w} \in \mathcal{S}_{\mathbf{w}}} \max_{\mathbf{u} \in \mathcal{S}_{\mathbf{u}}} \{ \|\mathbf{w}\|_2 \mathbf{g}^\top \mathbf{u} + \|\mathbf{u}\|_2 \mathbf{h}^\top \mathbf{w} + \psi(\mathbf{w}, \mathbf{u}) \}, \quad (54)$$

where $\mathbf{G} \in \mathbb{R}^{m \times n}$, $\mathbf{g} \in \mathbb{R}^m$, $\mathbf{h} \in \mathbb{R}^n$, $\mathcal{S}_{\mathbf{w}} \subset \mathbb{R}^m$, $\mathcal{S}_{\mathbf{u}} \subset \mathbb{R}^n$, and $\psi : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}$. Let $\mathbf{w}_\Phi = \mathbf{w}_\Phi(\mathbf{G})$ and $\mathbf{w}_\phi = \mathbf{w}_\phi(\mathbf{g}, \mathbf{h})$ denote any optimal minimisers of (53) and (54), respectively.

Assume that $\mathcal{S}_{\mathbf{w}}$ and $\mathcal{S}_{\mathbf{u}}$ are compact, that ψ is continuous on $\mathcal{S}_{\mathbf{w}} \times \mathcal{S}_{\mathbf{u}}$, and that $\mathbf{G}, \mathbf{g}, \mathbf{h}$ have i.i.d. standard normal entries.

(i) For every $c \in \mathbb{R}$,

$$\Pr(\Phi(\mathbf{G}) < c) \leq 2 \Pr(\phi(\mathbf{g}, \mathbf{h}) \leq c).$$

(ii) If, in addition, $\mathcal{S}_{\mathbf{w}}$ and $\mathcal{S}_{\mathbf{u}}$ are convex and ψ is convex–concave on $\mathcal{S}_{\mathbf{w}} \times \mathcal{S}_{\mathbf{u}}$, then for every $c \in \mathbb{R}$,

$$\Pr(\Phi(\mathbf{G}) > c) \leq 2 \Pr(\phi(\mathbf{g}, \mathbf{h}) \geq c).$$

In particular, for every $\mu \in \mathbb{R}$ and $t > 0$,

$$\Pr(|\Phi(\mathbf{G}) - \mu| > t) \leq 2 \Pr(|\phi(\mathbf{g}, \mathbf{h}) - \mu| \geq t).$$

B.3 Conditioning architecture

To employ the CGMT machinery of Theorem B.2 we work with a collection of good events. The proof is organised in two stages. First, we derive the PO–AO comparison and the scalar AO reduction on these events. Second, after the limiting AO has been defined, we prove that these events hold with sufficiently large probability.

The guiding convention is the following. Let

$$\mathcal{G}_n = \sigma(\mathbf{B}, \theta_0, \theta_P), \quad \mathcal{C}_n = \sigma(\mathbf{B}, \theta_0, \theta_P, \mathbf{H}_1, \varepsilon). \quad (55)$$

The field $\mathcal{C}_n$ is used for the pathwise CGMT comparison, because conditional on $\mathcal{C}_n$ the function ψ in the PO is fixed and $\mathbf{H}_2$ remains a fresh Gaussian matrix. The field $\mathcal{G}_n$ is used for the probability bounds on the AO events, because conditional on $\mathcal{G}_n$ the randomness in $(\mathbf{H}_1, \varepsilon, \mathbf{g}, \mathbf{h})$ is still available.

B.4 Setting up the PO

The purpose of this section is to replace the original MDYPL problem (6), on a high-probability containment event, by a compact convex–concave primary optimisation suitable for CGMT. The output of the section is Proposition B.3

To bring the optimisation problem of (6) into a CGMT-friendly min–max form, let us first rewrite it as a constrained optimisation problem

$$\min_{\substack{\beta \in \mathbb{R}^p \\ \theta \in \mathbb{R} \\ \eta \in \mathbb{R}^n}} \frac{1}{n} \{ \mathbf{1}^\top \rho(\eta) - \tilde{\mathbf{y}}^\top \eta \}, \quad \text{such that } \eta = \theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H} \beta, \quad \mathbf{H} = \sqrt{p} \mathbf{X}. \quad (56)$$

The Lagrangian of (56) is then

$$\min_{\substack{\beta \in \mathbb{R}^p \\ \theta \in \mathbb{R} \\ \eta \in \mathbb{R}^n}} \max_{\lambda \in \mathbb{R}^n} \frac{1}{n} \left\{ \mathbf{1}^\top \rho(\eta) - \tilde{\mathbf{y}}^\top \eta + \lambda^\top \left(\eta - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H} \beta \right) \right\}. \quad (57)$$

Next, let us introduce a containment event on the MDYPL estimates that allows us to constrain the min- and max-variables of (57) to CGMT-conformal, convex sets. To start, by Rignon and Aliverti (2023) Theorem 1), the MDYPL estimates exist and are unique, whenever the matrix $\tilde{\mathbf{H}} = [\mathbf{1} \quad \mathbf{H}]$ has full column rank. Thus define the event

$$\mathcal{E}_n = \{ \text{rank}([\mathbf{1} \quad \mathbf{H}]) = p + 1 \}. \quad (58)$$

Since $p/n \rightarrow \kappa \in (0, 1)$, one has $n \geq p + 1$ for all sufficiently large n . For every such n , $\Pr(\mathcal{E}_n) = 1$, since, conditionally on the preceding columns, each Gaussian column of $\mathbf{H}$ has a density and the span of $\mathbf{1}$ and those columns is a proper subspace of $\mathbb{R}^n$. Further, define $\hat{\eta}^{\text{DY}} = \hat{\theta}^{\text{DY}} \mathbf{1} + \mathbf{H} \hat{\beta}^{\text{DY}} / \sqrt{p}$ and $\hat{\lambda}^{\text{DY}} = \tilde{\mathbf{y}} - \rho'(\hat{\eta}^{\text{DY}})$. Define the compact containment event

$$\mathcal{M}_n = \left\{ \frac{\|\hat{\beta}^{\text{DY}}\|_2}{\sqrt{p}} \leq C_\beta, \quad |\hat{\theta}^{\text{DY}}| \leq C_\theta, \quad \frac{\|\hat{\eta}^{\text{DY}}\|_2}{\sqrt{n}} \leq C_\eta, \quad \frac{\|\hat{\lambda}^{\text{DY}}\|_2}{\sqrt{n}} \leq 1 \right\}, \quad (59)$$

where the constants $C_\beta, C_\theta, C_\eta$ are chosen as in Lemma 3 and may be enlarged later without invalidating the containment bound. We henceforth enlarge C_β and C_θ , if necessary, so that $C_\beta, C_\theta > B_a$ where B_a is the deterministic constant from Lemma 24.

Now, denoting the d -dimensional Euclidean ball centred at zero, with radius $C\sqrt{d}$, as $\mathcal{B}_C^d = \{v \in \mathbb{R}^d : \|v\|_2 / \sqrt{d} \leq C\}$, on $\mathcal{E}_n \cap \mathcal{M}_n$, the optimiser of (57) coincides with the optimiser of

$$\min_{\substack{\beta \in \mathcal{B}_{C_\beta}^p \\ |\theta| \leq C_\theta \\ \eta \in \mathcal{B}_{C_\eta}^n}} \max_{\lambda \in \mathcal{B}_1^n} \frac{1}{n} \left\{ \mathbf{1}^\top \rho(\eta) - \tilde{\mathbf{y}}^\top \eta + \lambda^\top \left(\eta - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H} \beta \right) \right\}. \quad (60)$$

Now note that the bilinear form $\lambda^\top \mathbf{H} \beta$ is not independent of the vector $\tilde{\mathbf{y}}$ whose elements (5), depend on $\mathbf{X} = \mathbf{H} / \sqrt{p}$. Hence, following the idea of Salehi et al. (2019), we decompose β , and consequently $\mathbf{H} \beta$, into a part that depends on $\tilde{\mathbf{y}}$ and one that is independent. Towards this, for $\mathbf{B} = [\beta_0, \beta_P]$, define the orthogonal projector on the column space of $\mathbf{B}$:

$$\mathbf{P} = \mathbf{B} (\mathbf{B}^\top \mathbf{B})^+ \mathbf{B}^\top,$$

where $\mathbf{A}^+$ is the Moore–Penrose pseudoinverse of $\mathbf{A}$ (cf. Magnus and Neudecker (2019) Chapter 2). Let $\mathbf{P}^\perp = \mathbf{I}_p - \mathbf{P}$ and for $s = \text{rank}(\mathbf{B})$, fix once and for all, separately on each rank stratum $\{\text{rank}(\mathbf{B}) = s\}$, a Borel-measurable map

$$\mathbf{B} \mapsto \mathbf{E}(\mathbf{B}) \in \mathbb{R}^{p \times (p-s)} : \quad \mathbf{E} \mathbf{E}^\top = \mathbf{P}^\perp, \quad \mathbf{E}^\top \mathbf{E} = \mathbf{I}_{p-s}.$$

For example, a Borel-measurable choice is obtained by applying Gram–Schmidt to $\mathbf{P}^\perp \mathbf{e}_1, \dots, \mathbf{P}^\perp \mathbf{e}_p$, selecting the smallest admissible index at each step and retaining the first $p-s$ nonzero residuals. Then, for any $\boldsymbol{\beta} \in \mathbb{R}^p$, we can uniquely decompose the linear predictor

$$\mathbf{H}\boldsymbol{\beta} = \mathbf{H}\mathbf{P}\boldsymbol{\beta} + \mathbf{H}\mathbf{P}^\perp\boldsymbol{\beta} = \mathbf{H}\mathbf{B}(\mathbf{B}^\top \mathbf{B})^+ \mathbf{B}^\top \boldsymbol{\beta} + \mathbf{H}\mathbf{E}\mathbf{E}^\top \boldsymbol{\beta} = \mathbf{H}_1 \mathbf{v} + \mathbf{H}_2 \mathbf{w},$$

for $\mathbf{H}_1 = \mathbf{H}\mathbf{B}$, $\mathbf{H}_2 = \mathbf{H}\mathbf{E}$, and $\mathbf{v} = (\mathbf{B}^\top \mathbf{B})^+ \mathbf{B}^\top \boldsymbol{\beta} \in \text{range}(\mathbf{B}^\top) \subseteq \mathbb{R}^2$, $\mathbf{w} = \mathbf{E}^\top \boldsymbol{\beta} \in \mathbb{R}^{p-s}$.

By construction of $\mathbf{E}$, the rows of $\mathbf{H}_2$ are of the form $\mathbf{h}_i^\top \mathbf{E}$ and are thus i.i.d. $\mathcal{N}(\mathbf{0}_{p-s}, \mathbf{I}_{p-s})$. Similarly, the rows of $\mathbf{H}_1$ are of the form $\mathbf{h}_i^\top \mathbf{B}$, which are i.i.d. $\mathcal{N}(\mathbf{0}_2, \mathbf{B}^\top \mathbf{B})$. Finally, note that conditional on $\mathbf{B}$,

$$\text{cov}(\mathbf{B}^\top \mathbf{h}_i, \mathbf{E}^\top \mathbf{h}_j) = \mathbf{B}^\top \mathbf{E} = \mathbf{0}_{2 \times (p-s)},$$

by Lemma 7 and therefore $\mathbf{H}_1, \mathbf{H}_2$ are independent. Furthermore, since $\mathbf{H}\boldsymbol{\beta}_0$ and $\mathbf{H}\boldsymbol{\beta}_P$ are the two columns of $\mathbf{H}_1 = \mathbf{H}\mathbf{B}$, the response vector $\tilde{\mathbf{y}}$ is measurable with respect to $\sigma(\mathbf{B}, \theta_0, \theta_P, \mathbf{H}_1, \boldsymbol{\varepsilon}) = \mathcal{C}_n$, and conditional on $\mathcal{C}_n$, $\mathbf{H}_2$ is independent of $\tilde{\mathbf{y}}$.

Since, by Lemma 7 the decomposition $\boldsymbol{\beta} = \mathbf{B}\mathbf{v} + \mathbf{E}\mathbf{w}$ is componentwise unique for $\mathbf{v} \in \text{range}(\mathbf{B}^\top)$, we can rewrite (60) equivalently as

$$\min_{\substack{(\mathbf{v}, \mathbf{w}) \in \mathcal{K}_{C_\beta}^p \\ |\theta| \leq C_\theta \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n}} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \frac{1}{n} \left\{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} + \boldsymbol{\lambda}^\top \left(\boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \frac{1}{\sqrt{p}} \mathbf{H}_2 \mathbf{w} \right) \right\}, \quad (61)$$

where $\mathcal{K}_{C_\beta}^p = \{\mathbf{v} \in \text{range}(\mathbf{B}^\top) \subseteq \mathbb{R}^2, \mathbf{w} \in \mathbb{R}^{p-s} : \|\mathbf{B}\mathbf{v} + \mathbf{E}\mathbf{w}\|_2^2 \leq C_\beta^2 p\}$. To obtain a product domain, choose $C_\sigma \geq C_\beta$, enlarging it if necessary so that $C_\sigma > B_a$, and define

$$V_p = \{\mathbf{v} \in \text{range}(\mathbf{B}^\top) : \|\mathbf{B}\mathbf{v}\|_2 \leq C_\beta \sqrt{p}\}, \quad W_p = \{\mathbf{w} \in \mathbb{R}^{p-s} : \|\mathbf{w}\|_2 \leq C_\sigma \sqrt{p}\}.$$

Since $\mathbf{B}^\top \mathbf{E} = \mathbf{0}$ and $\mathbf{E}^\top \mathbf{E} = \mathbf{I}_{p-s}$, every $(\mathbf{v}, \mathbf{w}) \in \mathcal{K}_{C_\beta}^p$ satisfies

$$\|\mathbf{B}\mathbf{v}\|_2^2 + \|\mathbf{w}\|_2^2 = \|\mathbf{B}\mathbf{v} + \mathbf{E}\mathbf{w}\|_2^2 \leq C_\beta^2 p,$$

and hence $\mathcal{K}_{C_\beta}^p \subseteq V_p \times W_p$. Both V_p and W_p are nonempty, compact, and convex: W_p is a Euclidean ball, while V_p is the closed ball of radius $C_\beta \sqrt{p}$ for the norm $\mathbf{v} \mapsto \|\mathbf{B}\mathbf{v}\|_2$ on $\text{range}(\mathbf{B}^\top)$, since $\text{range}(\mathbf{B}^\top) \cap \ker(\mathbf{B}) = \{\mathbf{0}_2\}$. The particular form of V_p is required to ensure compactness and to allow for a uniform treatment of all constellations of prior and signal alignment. By Lemma 8 replacing the coupled coordinate set $\mathcal{K}_{C_\beta}^p$ in (61) by the Cartesian product $V_p \times W_p$ does not change the optimiser in $(\theta, \boldsymbol{\beta}, \boldsymbol{\eta})$ on $\mathcal{M}_n \cap \mathcal{E}_n$. Hence, on $\mathcal{M}_n \cap \mathcal{E}_n$, (61) is equivalent to

$$\min_{\substack{\mathbf{v} \in V_p \\ \mathbf{w} \in W_p \\ |\theta| \leq C_\theta \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n}} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \frac{1}{n} \left\{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} + \boldsymbol{\lambda}^\top \left(\boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \frac{1}{\sqrt{p}} \mathbf{H}_2 \mathbf{w} \right) \right\}. \quad (62)$$

For each fixed $\mathbf{w} \in W_p$, the objective in (62) is continuous on the compact product domain, convex in $(\mathbf{v}, \theta, \boldsymbol{\eta})$, and affine, hence concave, in $\boldsymbol{\lambda}$, because $\boldsymbol{\rho}$ is convex and all remaining terms are linear. Hence Sion's min–max theorem (see, for example, [Simons1995, Theorem 3]) applies to the compact convex sets $V_p \times [-C_\theta, C_\theta] \times \mathcal{B}_{C_\eta}^n$ and $\mathcal{B}_1^n$. Thus, we get that

$$\begin{aligned} & \min_{\substack{\mathbf{v} \in V_p \\ \mathbf{w} \in W_p \\ |\theta| \leq C_\theta \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n}} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \frac{1}{n} \left\{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} + \boldsymbol{\lambda}^\top \left(\boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \frac{1}{\sqrt{p}} \mathbf{H}_2 \mathbf{w} \right) \right\} \\ &= \min_{\mathbf{w} \in W_p} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} - \frac{1}{n\sqrt{p}} \boldsymbol{\lambda}^\top \mathbf{H}_2 \mathbf{w} + \min_{\substack{\mathbf{v} \in V_p \\ |\theta| \leq C_\theta \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n}} \frac{1}{n} \left\{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} + \boldsymbol{\lambda}^\top \boldsymbol{\eta} - \theta \mathbf{1}^\top \boldsymbol{\lambda} - \frac{1}{\sqrt{p}} \boldsymbol{\lambda}^\top \mathbf{H}_1 \mathbf{v} \right\} \\ &= \min_{\mathbf{w} \in W_p} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} - \frac{1}{n\sqrt{p}} \boldsymbol{\lambda}^\top \mathbf{H}_2 \mathbf{w} + \psi(\boldsymbol{\lambda}), \end{aligned} \quad (63)$$

for

$$\psi(\boldsymbol{\lambda}) = \min_{\substack{\mathbf{v} \in V_p \\ |\theta| \leq C_\theta \\ \boldsymbol{\eta} \in \mathcal{B}_{\mathcal{C}_\eta}^n}} \frac{1}{n} \left\{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} + \boldsymbol{\lambda}^\top \boldsymbol{\eta} - \theta \mathbf{1}^\top \boldsymbol{\lambda} - \frac{1}{\sqrt{p}} \boldsymbol{\lambda}^\top \mathbf{H}_1 \mathbf{v} \right\}.$$

By Lemma 10(i), applied with $Z = Z_p$, $S_\lambda = \mathcal{B}_1^n$, and $S_w = W_p$, ψ is concave in $\boldsymbol{\lambda}$ and continuous. Hence, given our PO-candidate in (63), we are finally in a place to formulate an AO in line with the CGMT of Theorem B.2.

We thus conclude this section with the following result about the equivalence of MDYPL from (6) and the PO of (63):

Proposition B.3 (MDYPL compactification and PO link). *There exist constants $C_\sigma, C_\beta, C_\theta, C_\eta, C, c > 0$ and $N \in \mathbb{N}$ such that, for all $n > N$,*

$$\Pr(\mathcal{M}_n \cap \mathcal{E}_n) \geq 1 - C \exp\{-cn\},$$

where $\mathcal{M}_n, \mathcal{E}_n$ are defined in (59) and (58), respectively, and $C_\sigma, C_\beta, C_\theta, C_\eta, C, c$ depend only on

$$\kappa, \alpha, \boldsymbol{\Gamma}, \theta_0^*, \theta_P^*,$$

and on the constants in the probability bounds of Section 2.

Furthermore, on $\mathcal{E}_n$, let $\hat{\mathbf{w}}$ be the unique minimiser of the profiled PO in the final line of (63), and let $(\hat{\mathbf{v}}, \hat{\theta}, \hat{\boldsymbol{\eta}})$ be the inner minimising variables at any saddle point of the corresponding fixed- $\hat{\mathbf{w}}$ problem. By Lemma 9, these variables are unique. Extend $(\hat{\mathbf{v}}, \hat{\mathbf{w}}, \hat{\theta}, \hat{\boldsymbol{\eta}})$ by zero on $\neg \mathcal{E}_n$, and define

$$\hat{\boldsymbol{\beta}} = \mathbf{B}\hat{\mathbf{v}} + \mathbf{E}\hat{\mathbf{w}}.$$

Then, on $\mathcal{M}_n \cap \mathcal{E}_n$,

$$\hat{\theta} = \hat{\theta}^{\text{DY}}, \quad \hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^{\text{DY}}, \quad \hat{\boldsymbol{\eta}} = \hat{\boldsymbol{\eta}}^{\text{DY}}.$$

Consequently, for all $n > N$,

$$\Pr\left(\{\hat{\theta} = \hat{\theta}^{\text{DY}}\} \cap \{\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^{\text{DY}}\} \cap \{\hat{\boldsymbol{\eta}} = \hat{\boldsymbol{\eta}}^{\text{DY}}\}\right) \geq 1 - C \exp\{-cn\}.$$

Proof. By Lemma 3,

$$\Pr(\neg \mathcal{M}_n) \leq C \exp\{-cn\}.$$

Since $p/n \rightarrow \kappa < 1$, the Gaussian augmented design has full column rank almost surely for all sufficiently large n , and hence $\Pr(\mathcal{E}_n) = 1$. After increasing N if necessary,

$$\Pr(\mathcal{M}_n \cap \mathcal{E}_n) = \Pr(\mathcal{M}_n) \geq 1 - C \exp\{-cn\}.$$

Work now on $\mathcal{M}_n \cap \mathcal{E}_n$. By Lemma 9, $(\hat{\mathbf{v}}, \hat{\mathbf{w}}, \hat{\theta}, \hat{\boldsymbol{\eta}})$ is the unique tuple of minimising variables of the compact saddle problem and hence is a product-domain optimiser. Lemma 8 therefore gives

$$\hat{\theta} = \hat{\theta}^{\text{DY}}, \quad \hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^{\text{DY}}, \quad \hat{\boldsymbol{\eta}} = \hat{\boldsymbol{\eta}}^{\text{DY}}$$

on $\mathcal{M}_n \cap \mathcal{E}_n$. The final probability bound follows from the first part. $\square$

B.5 Conditional PO–AO pair

To apply the CGMT PO–AO value-link to (63), we require that the function $\psi(\boldsymbol{\lambda})$ is non-random. Hence, we condition on the sigma field $\mathcal{C}_n = \sigma(\mathbf{B}, \theta_0, \theta_P, \mathbf{H}_1, \boldsymbol{\varepsilon})$ from (55). Conditional on $\mathcal{C}_n$, the function ψ , the feasible sets, and the response vector $\tilde{\mathbf{y}}$ are deterministic, while $\mathbf{H}_2$ is a random matrix with standard normal entries. Our conditional PO is given by

$$\Psi(\mathbf{H}_2) = \min_{\mathbf{w} \in W_p} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} - \frac{1}{n\sqrt{p}} \boldsymbol{\lambda}^\top \mathbf{H}_2 \mathbf{w} + \psi(\boldsymbol{\lambda}), \quad (64)$$

and the corresponding conditional AO is given by

$$\phi(\mathbf{g}, \mathbf{h}) = \min_{\mathbf{w} \in W_p} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} - \frac{1}{n\sqrt{p}} \{ \|\mathbf{w}\|_2 \mathbf{g}^\top \boldsymbol{\lambda} + \|\boldsymbol{\lambda}\|_2 \mathbf{w}^\top \mathbf{h} \} + \psi(\boldsymbol{\lambda}), \quad (65)$$

where $\mathbf{g} \in \mathbb{R}^n$, $\mathbf{h} \in \mathbb{R}^{p-s}$, and conditional on $\mathcal{C}_n$, the only remaining PO randomness is $\mathbf{H}_2$, and the only remaining AO randomness is $(\mathbf{g}, \mathbf{h})$. Thus, conditional on $\mathcal{C}_n$, the PO/AO pair in (64)–(65) is a valid pathwise CGMT instance. The following proposition records the corresponding conditional value comparison, in a form that can later be applied to the unrestricted PO and to the restricted values used in Section B.9.

Proposition B.4 (Conditional PO–AO CGMT bridge). *Let W_0 and D_0 be $\mathcal{C}_n$ -measurable compact-valued random sets such that almost surely*

$$W_0 \subseteq W_p, \quad D_0 \subseteq V_p \times [-C_\theta, C_\theta],$$

both sets are nonempty, and D_0 is convex. Set

$$Z_0 = Z_p(D_0) = D_0 \times \mathcal{B}_{\mathcal{C}_n}^n.$$

For $\boldsymbol{\zeta} = (\mathbf{v}, \theta, \boldsymbol{\eta}) \in Z_0$, write

$$L(\boldsymbol{\zeta}; \boldsymbol{\lambda}) = \frac{1}{n} \left\{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} + \boldsymbol{\lambda}^\top \boldsymbol{\eta} - \theta \mathbf{1}^\top \boldsymbol{\lambda} - \frac{1}{\sqrt{p}} \boldsymbol{\lambda}^\top \mathbf{H}_1 \mathbf{v} \right\},$$

and define

$$\psi_{Z_0}(\boldsymbol{\lambda}) = \min_{\boldsymbol{\zeta} \in Z_0} L(\boldsymbol{\zeta}; \boldsymbol{\lambda}).$$

Let

$$\Psi_{W_0, Z_0}(\mathbf{H}_2) = \min_{\mathbf{w} \in W_0} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \left\{ -\frac{1}{n\sqrt{p}} \boldsymbol{\lambda}^\top \mathbf{H}_2 \mathbf{w} + \psi_{Z_0}(\boldsymbol{\lambda}) \right\},$$

and

$$\phi_{W_0, Z_0}(\mathbf{g}, \mathbf{h}) = \min_{\mathbf{w} \in W_0} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \left\{ -\frac{1}{n\sqrt{p}} (\|\mathbf{w}\|_2 \mathbf{g}^\top \boldsymbol{\lambda} + \|\boldsymbol{\lambda}\|_2 \mathbf{w}^\top \mathbf{h}) + \psi_{Z_0}(\boldsymbol{\lambda}) \right\}, \quad (66)$$

where $\mathbf{g} \in \mathbb{R}^n$ and $\mathbf{h} \in \mathbb{R}^{p-s}$ have i.i.d. standard normal entries and are independent of $\mathcal{C}_n$. Then, almost surely for every $a \in \mathbb{R}$,

$$\Pr(\Psi_{W_0, Z_0}(\mathbf{H}_2) < a \mid \mathcal{C}_n) \leq 2 \Pr(\phi_{W_0, Z_0}(\mathbf{g}, \mathbf{h}) \leq a \mid \mathcal{C}_n).$$

If, in addition, W_0 is convex, then, almost surely for every $a \in \mathbb{R}$,

$$\Pr(\Psi_{W_0, Z_0}(\mathbf{H}_2) > a \mid \mathcal{C}_n) \leq 2 \Pr(\phi_{W_0, Z_0}(\mathbf{g}, \mathbf{h}) \geq a \mid \mathcal{C}_n).$$

Proof. Work conditionally on $\mathcal{C}_n$ from (55). Outside a $\mathcal{C}_n$ -measurable null set, W_0 , D_0 , Z_0 , $\tilde{\mathbf{y}}$, and $\mathbf{H}_1$ are deterministic, while $\mathbf{H}_2$ is an independent Gaussian matrix with i.i.d. standard normal entries.

By Lemma 10(i) applied with $Z = Z_0$, $S_\lambda = \mathcal{B}_1^n$, and $S_w = W_0$, the function ψ_{Z_0} is continuous and concave on $\mathcal{B}_1^n$. Set

$$\tilde{W}_0 = \left\{ \frac{\mathbf{w}}{n\sqrt{p}} : \mathbf{w} \in W_0 \right\}.$$

Then $\tilde{W}_0$ is compact, and

$$\Psi_{W_0, Z_0}(\mathbf{H}_2) = \min_{\tilde{\mathbf{w}} \in \tilde{W}_0} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \left\{ \boldsymbol{\lambda}^\top (-\mathbf{H}_2) \tilde{\mathbf{w}} + \psi_{Z_0}(\boldsymbol{\lambda}) \right\}.$$

Since $-\mathbf{H}_2$ has the same law as $\mathbf{H}_2$, Theorem B.2(i) gives

$$\Pr(\Psi_{W_0, Z_0}(\mathbf{H}_2) < a \mid \mathcal{C}_n) \leq 2 \Pr(\tilde{\phi}_{W_0, Z_0}(\mathbf{g}, \mathbf{h}) \leq a \mid \mathcal{C}_n),$$

where

$$\tilde{\phi}_{W_0, Z_0}(\mathbf{g}, \mathbf{h}) = \min_{\tilde{\mathbf{w}} \in \tilde{W}_0} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \left\{ \|\tilde{\mathbf{w}}\|_2 \mathbf{g}^\top \boldsymbol{\lambda} + \|\boldsymbol{\lambda}\|_2 \tilde{\mathbf{w}}^\top \mathbf{h} + \psi_{Z_0}(\boldsymbol{\lambda}) \right\}.$$

Changing back to $\mathbf{w} = n\sqrt{p}\tilde{\mathbf{w}}$ and using the symmetry $(\mathbf{g}, \mathbf{h}) \stackrel{d}{=} (-\mathbf{g}, -\mathbf{h})$ shows that $\tilde{\phi}_{W_0, Z_0}(\mathbf{g}, \mathbf{h})$ has the same law as $\phi_{W_0, Z_0}(\mathbf{g}, \mathbf{h})$. This proves the lower-tail comparison.

For the upper tail, note that if W_0 is convex, then $\tilde{W}_0$ is convex. Since $\mathcal{B}_1^n$ is convex and Lemma 10(i) gives the required convex-concave structure, the upper-tail part of Theorem B.2 applies and yields

$$\Pr(\Psi_{W_0, Z_0}(\mathbf{H}_2) > a \mid \mathcal{C}_n) \leq 2 \Pr(\tilde{\phi}_{W_0, Z_0}(\mathbf{g}, \mathbf{h}) \geq a \mid \mathcal{C}_n).$$

Again using the equality in law between $\tilde{\phi}_{W_0, Z_0}$ and ϕ_{W_0, Z_0} gives the stated upper-tail comparison. $\square$

Note that for $W_0 = W_p$ and $D_0 = V_p \times [-C_\theta, C_\theta]$, the definition of ψ_{Z_0} coincides with the reduced function ψ in (63). The corresponding values are therefore precisely those in (64) and (65).

Now that we have the desired PO-AO link, we analyse the limiting behaviour of the AO. The restricted comparisons needed to localise the PO are carried out in Section B.9, after the scalar limiting value and the limiting-AO properties have been established in Sections B.7 and B.8

B.6 AO scalarisation

The next step is to reduce the optimisations over vector valued arguments to an equivalent min-max problem with scalar optimisers. Let $Z_p = V_p \times [-C_\theta, C_\theta] \times \mathcal{B}_{C_\eta}^n$, and, for $\boldsymbol{\zeta} = (\mathbf{v}, \theta, \boldsymbol{\eta}) \in Z_p$, define

$$\psi(\boldsymbol{\lambda}) = \min_{\boldsymbol{\zeta} \in Z_p} \left\{ G_n(\boldsymbol{\eta}) + \frac{1}{n} \boldsymbol{\lambda}^\top \mathbf{a}(\boldsymbol{\zeta}) \right\}, \quad G_n(\boldsymbol{\eta}) = \frac{1}{n} \{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} \}, \quad \mathbf{a}(\boldsymbol{\zeta}) = \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v}. \quad (67)$$

Denote by $\phi(\mathbf{g}, \mathbf{h})$ the AO of (66) with $W_0 = W_p$ and $D_0 = V_p \times [-C_\theta, C_\theta]$. Moreover, since $s \leq 2$ and $p \rightarrow \infty$, we may increase N if necessary and assume throughout this scalarisation that $p - s \geq 1$.

Now, as a first scalarisation step, we decompose the noise-aligned component $\mathbf{w}$ into a length and a direction. Since

$$W_p = \{ \mathbf{w} \in \mathbb{R}^{p-s} : \|\mathbf{w}\|_2 \leq C_\sigma \sqrt{p} \},$$

every nonzero $\mathbf{w} \in W_p$ can be written as

$$\mathbf{w} = \sqrt{p} \sigma \mathbf{q}, \quad \sigma = \frac{\|\mathbf{w}\|_2}{\sqrt{p}} \in [0, C_\sigma], \quad \mathbf{q} \in \mathbb{S}^{p-s-1}. \quad (68)$$

The case $\mathbf{w} = \mathbf{0}$ is represented by $\sigma = 0$, with arbitrary $\mathbf{q} \in \mathbb{S}^{p-s-1}$. Substituting (68) into (66) gives

$$\phi(\mathbf{g}, \mathbf{h}) = \min_{\sigma \in [0, C_\sigma]} \min_{\mathbf{q} \in \mathbb{S}^{p-s-1}} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \left\{ \psi(\boldsymbol{\lambda}) - \frac{\sigma}{n} \mathbf{g}^\top \boldsymbol{\lambda} - \frac{\sigma}{n} \|\boldsymbol{\lambda}\|_2 \mathbf{q}^\top \mathbf{h} \right\}.$$

For $p > s$ and every $\mathbf{q} \in \mathbb{S}^{p-s-1}$, $\mathbf{q}^\top \mathbf{h} \leq \|\mathbf{h}\|_2$ by Cauchy-Schwarz (Rudin, 1976 Theorem 1.37(d)). Since $\sigma \geq 0$ and $\|\boldsymbol{\lambda}\|_2 \geq 0$, it follows pointwise in $\boldsymbol{\lambda}$ that

$$\psi(\boldsymbol{\lambda}) - \frac{\sigma}{n} \mathbf{g}^\top \boldsymbol{\lambda} - \frac{\sigma}{n} \|\boldsymbol{\lambda}\|_2 \mathbf{q}^\top \mathbf{h} \geq \psi(\boldsymbol{\lambda}) - \frac{\sigma}{n} \mathbf{g}^\top \boldsymbol{\lambda} - \frac{\sigma}{n} \|\boldsymbol{\lambda}\|_2 \|\mathbf{h}\|_2. \quad (69)$$

Equality in (69) is attained by taking $\mathbf{q} = \mathbf{h}/\|\mathbf{h}\|_2$ if $\mathbf{h} \neq \mathbf{0}$, and is immediate for every $\mathbf{q}$ if $\mathbf{h} = \mathbf{0}$. Hence, taking maxima over $\boldsymbol{\lambda}$ on both sides yields

$$\max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \psi(\boldsymbol{\lambda}) - \frac{\sigma}{n} \mathbf{g}^\top \boldsymbol{\lambda} - \frac{\sigma}{n} \|\boldsymbol{\lambda}\|_2 \mathbf{q}^\top \mathbf{h} \geq \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \psi(\boldsymbol{\lambda}) - \frac{\sigma}{n} \mathbf{g}^\top \boldsymbol{\lambda} - \frac{\sigma}{n} \|\boldsymbol{\lambda}\|_2 \|\mathbf{h}\|_2,$$

and thus

$$\min_{\sigma \in \mathbb{S}^{p-s-1}} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \psi(\boldsymbol{\lambda}) - \frac{\sigma}{n} \mathbf{g}^\top \boldsymbol{\lambda} - \frac{\sigma}{n} \|\boldsymbol{\lambda}\|_2 \mathbf{q}^\top \mathbf{h} \geq \min_{\mathbf{q} \in \mathbb{S}^{p-s-1}} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \psi(\boldsymbol{\lambda}) - \frac{\sigma}{n} \mathbf{g}^\top \boldsymbol{\lambda} - \frac{\sigma}{n} \|\boldsymbol{\lambda}\|_2 \|\mathbf{h}\|_2. \quad (70)$$

Since the expression on the right-hand side of (70) does not depend on $\mathbf{q}$ and the left-hand side can attain this value, it follows that

$$\phi(\mathbf{g}, \mathbf{h}) = \min_{\sigma \in [0, C_\sigma]} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \left\{ \psi(\boldsymbol{\lambda}) - \frac{\sigma}{n} \mathbf{g}^\top \boldsymbol{\lambda} - \frac{\sigma}{n} \|\boldsymbol{\lambda}\|_2 \|\mathbf{h}\|_2 \right\}. \quad (71)$$

Now, we seek to decompose $\boldsymbol{\lambda}$ into a radius $\tilde{r} \in [0, \sqrt{n}]$ and a direction $\mathbf{d} \in \mathbb{S}^{n-1}$, the latter of which we will profile out in a similar fashion to (70). For this, fix $\sigma \in [0, C_\sigma]$ and expand ψ using (67). Define

$$\mathbf{a}_\sigma(\boldsymbol{\zeta}) = \mathbf{a}(\boldsymbol{\zeta}) - \sigma \mathbf{g} = \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g}. \quad (72)$$

Then the inner maximisation in (71) is

$$\max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \left\{ \psi(\boldsymbol{\lambda}) - \frac{\sigma}{n} \mathbf{g}^\top \boldsymbol{\lambda} - \frac{\sigma}{n} \|\boldsymbol{\lambda}\|_2 \|\mathbf{h}\|_2 \right\} = \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \min_{\boldsymbol{\zeta} \in \mathcal{Z}_p} \left\{ G_n(\boldsymbol{\eta}) + \frac{1}{n} \boldsymbol{\lambda}^\top \mathbf{a}_\sigma(\boldsymbol{\zeta}) - \frac{\sigma \|\mathbf{h}\|_2}{n} \|\boldsymbol{\lambda}\|_2 \right\}.$$

Now, we apply Lemma 10(ii) (with $R = \sqrt{n}$, $\mathbf{b} = \sigma \mathbf{g}$, $c = \sigma \|\mathbf{h}\|_2$) pathwise for each σ . In particular, we apply two min-max swaps and profile out the direction $\mathbf{d}$ using the identity $\max_{\mathbf{u} \in \mathbb{S}^{n-1}} \mathbf{u}^\top \mathbf{a}_\sigma(\boldsymbol{\zeta}) = \|\mathbf{a}_\sigma(\boldsymbol{\zeta})\|_2$, that is just the Cauchy-Schwarz inequality (Rudin 1976, Theorem 1.37(d)), to get

$$\max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \min_{\boldsymbol{\zeta} \in \mathcal{Z}_p} \left\{ G_n(\boldsymbol{\eta}) + \frac{1}{n} \boldsymbol{\lambda}^\top \mathbf{a}_\sigma(\boldsymbol{\zeta}) - \frac{\sigma \|\mathbf{h}\|_2}{n} \|\boldsymbol{\lambda}\|_2 \right\} = \max_{\tilde{r} \in [0, \sqrt{n}]} \min_{\boldsymbol{\zeta} \in \mathcal{Z}_p} \left\{ G_n(\boldsymbol{\eta}) + \frac{\tilde{r}}{n} \|\mathbf{a}_\sigma(\boldsymbol{\zeta})\|_2 - \frac{\sigma \tilde{r}}{n} \|\mathbf{h}\|_2 \right\}. \quad (73)$$

Since (73) holds for every fixed $\sigma \in [0, C_\sigma]$, combining it with (71) and then taking the outer minimum over σ gives

$$\phi(\mathbf{g}, \mathbf{h}) = \min_{\sigma \in [0, C_\sigma]} \max_{\tilde{r} \in [0, \sqrt{n}]} \left\{ -\frac{\sigma \tilde{r}}{n} \|\mathbf{h}\|_2 + \min_{\boldsymbol{\zeta} \in \mathcal{Z}_p} \left[\frac{\tilde{r}}{n} \|\mathbf{a}_\sigma(\boldsymbol{\zeta})\|_2 + G_n(\boldsymbol{\eta}) \right] \right\}. \quad (74)$$

Finally, reparameterise

$$r = \frac{\tilde{r}}{\sqrt{n}} \in [0, 1]. \quad (75)$$

Using (72) and (75) in (74) yields the scalar AO

$$\phi(\mathbf{g}, \mathbf{h}) = \min_{\sigma \in [0, C_\sigma]} \max_{r \in [0, 1]} -\sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \min_{\substack{\mathbf{v} \in V_p \\ |\theta| \leq C_\theta \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^p}} \frac{r}{\sqrt{n}} \left\| \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} \right\|_2 + \frac{1}{n} \{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} \}. \quad (76)$$

For the analysis of the radial variable r it is convenient to profile out $(\mathbf{v}, \theta)$. Recall the empirical loss $G_n(\boldsymbol{\eta})$ from (67) and set

$$\boldsymbol{\eta}_{\tilde{\mathbf{y}}} = \text{logit}(\tilde{\mathbf{y}})$$

coordinatewise, so that $\rho'(\boldsymbol{\eta}_{\tilde{\mathbf{y}}, i}) = \tilde{Y}_i$ and $\boldsymbol{\eta}_{\tilde{\mathbf{y}}}$ is the unconstrained minimiser of G_n . Define the compact convex predictor set

$$\mathcal{R}_0 = \left\{ \theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} : |\theta| \leq C_\theta, \mathbf{v} \in V_p \right\},$$

and, for $\sigma \in [0, C_\sigma]$ and $\boldsymbol{\eta} \in \mathbb{R}^n$,

$$M_n(\sigma, \boldsymbol{\eta}) = \frac{1}{\sqrt{n}} \text{dist}_2(\boldsymbol{\eta} - \sigma \mathbf{g}, \mathcal{R}_0) = \inf_{\substack{|\theta| \leq C_\theta \\ \mathbf{v} \in V_p}} \frac{1}{\sqrt{n}} \left\| \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} \right\|_2.$$

For $r \in [0, 1]$, define the radial profile

$$A_n(\sigma, r) = -\sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \min_{\boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^p} \{ G_n(\boldsymbol{\eta}) + r M_n(\sigma, \boldsymbol{\eta}) \}. \quad (77)$$

Since the minimum over $(\mathbf{v}, \theta)$ in (76) is exactly $rM_n(\sigma, \boldsymbol{\eta})$, the scalar AO is

$$\phi(\mathbf{g}, \mathbf{h}) = \min_{\sigma \in [0, C_\sigma]} \max_{r \in [0, 1]} A_n(\sigma, r).$$

For $a \in [0, 1]$, write

$$\phi^r(a; \mathbf{g}, \mathbf{h}) = \min_{\sigma \in [0, C_\sigma]} \max_{r \in [a, 1]} A_n(\sigma, r),$$

and, given a constant $c_r > 0$, define the event

$$\mathcal{E}_n^r(c_r) = \{\phi^r(c_r; \mathbf{g}, \mathbf{h}) = \phi^r(0; \mathbf{g}, \mathbf{h}) = \phi(\mathbf{g}, \mathbf{h})\}.$$

A sufficient, stronger property is that, for every $\sigma \in [0, C_\sigma]$,

$$\max_{r \in [0, 1]} A_n(\sigma, r) = \max_{r \in [c_r, 1]} A_n(\sigma, r),$$

which holds as soon as $r \mapsto A_n(\sigma, r)$ is nondecreasing on $[0, c_r]$. This pointwise property is what Proposition F.1 establishes, on the good event introduced before Proposition B.5 below. Given $\mathcal{E}_n^r(c_r)$, the AO can equivalently be written as

$$\phi(\mathbf{g}, \mathbf{h}) = \min_{\sigma \in [0, C_\sigma]} \max_{r \in [c_r, 1]} -\sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \min_{\substack{\mathbf{v} \in V_p \\ |\theta| \leq C_\theta \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n}} \frac{r}{\sqrt{n}} \left\| \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} \right\|_2 + \frac{1}{n} \{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} \}.$$

Next, we use the following identity

$$\|\mathbf{a}\|_2 = \inf_{t > 0} \frac{\|\mathbf{a}\|_2^2}{2t} + \frac{t}{2}, \quad (78)$$

to re-write

$$\begin{aligned} \frac{r}{\sqrt{n}} \left\| \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} \right\|_2 &= r \left\| \frac{1}{\sqrt{n}} \left(\boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} \right) \right\|_2 \\ &= r \inf_{t > 0} \frac{1}{2t} \left\| \frac{1}{\sqrt{n}} \left(\boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} \right) \right\|_2^2 + \frac{t}{2} \\ &= \inf_{t > 0} \frac{r}{2t} \frac{1}{n} \left\| \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} \right\|_2^2 + \frac{rt}{2}. \end{aligned}$$

Now, define

$$\begin{aligned} \phi^{r,t}(a, b; \mathbf{g}, \mathbf{h}) &= \min_{\sigma \in [0, C_\sigma]} \max_{r \in [a, 1]} -\sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \min_{\substack{\mathbf{v} \in V_p \\ |\theta| \leq C_\theta \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n}} \inf_{t \in (0, b]} \frac{r}{2t} \frac{1}{n} \left\| \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} \right\|_2^2 + \frac{rt}{2} \\ &\quad + \frac{1}{n} \{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} \}. \end{aligned} \quad (79)$$

For $b = \infty$ in (79), the same definition is understood with $t \in (0, \infty)$. Given constants c_r, C_t , introduce the event

$$\mathcal{E}_n^t(c_r, C_t) = \{\phi^{r,t}(c_r, C_t; \mathbf{g}, \mathbf{h}) = \phi^{r,t}(c_r, \infty; \mathbf{g}, \mathbf{h}) = \phi^r(c_r; \mathbf{g}, \mathbf{h})\}.$$

Upon $\mathcal{E}_n^r(c_r) \cap \mathcal{E}_n^t(c_r, C_t)$, and since the t - and $(\mathbf{v}, \theta, \boldsymbol{\eta})$ -optimisations are all minimising operations over independent variables, we can re-express the AO as

$$\begin{aligned} \phi(\mathbf{g}, \mathbf{h}) &= \min_{\sigma \in [0, C_\sigma]} \max_{r \in [c_r, 1]} -\sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \inf_{t \in (0, C_t]} \min_{\substack{\mathbf{v} \in V_p \\ |\theta| \leq C_\theta \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n}} \frac{r}{2t} \left\| \frac{1}{\sqrt{n}} \left(\boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} \right) \right\|_2^2 + \frac{rt}{2} \\ &\quad + \frac{1}{n} \{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} \}. \end{aligned}$$

A simple completion of squares yields that

$$\frac{r}{2t} \left\| \mathbf{a} - \frac{t}{r} \tilde{\mathbf{y}} \right\|_2^2 = \frac{r}{2t} \|\mathbf{a}\|_2^2 - \mathbf{a}^\top \tilde{\mathbf{y}} + \frac{t}{2r} \|\tilde{\mathbf{y}}\|_2^2.$$

Hence, for $\mathbf{a} = \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g}$, we get, upon rearrangement, that

$$\begin{aligned} \frac{r}{2t} \left\| \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} \right\|_2^2 - \boldsymbol{\eta}^\top \tilde{\mathbf{y}} &= \frac{r}{2t} \left\| \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} - \frac{t}{r} \tilde{\mathbf{y}} \right\|_2^2 \\ &\quad - \tilde{\mathbf{y}}^\top \left(\theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \sigma \mathbf{g} \right) \\ &\quad - \frac{t}{2r} \|\tilde{\mathbf{y}}\|_2^2. \end{aligned} \quad (80)$$

Hence, the full AO becomes

$$\phi(\mathbf{g}, \mathbf{h}) = \min_{\sigma \in [0, C_\sigma]} \max_{r \in [c_r, 1]} \min_{\substack{\mathbf{v} \in V_p \\ |\theta| \leq C_\theta \\ \boldsymbol{\eta} \in \mathcal{B}_{\mathcal{C}_\eta}^n}} \inf_{t \in (0, C_t]} -\sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \frac{rt}{2} + \frac{1}{n} \left\{ \frac{r}{2t} \left\| \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} - \frac{t}{r} \tilde{\mathbf{y}} \right\|_2^2 - \tilde{\mathbf{y}}^\top \left(\theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \sigma \mathbf{g} \right) \right\}$$

Lastly, we wish to scalarise the n -vector minimisation over $\boldsymbol{\eta}$. For fixed $(\sigma, r, \mathbf{v}, \theta, t)$, the $\boldsymbol{\eta}$ -dependent part is the constrained minimisation

$$\min_{\boldsymbol{\eta} \in \mathcal{B}_{\mathcal{C}_\eta}^n} \frac{1}{n} \left(\mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) + \frac{r}{2t} \left\| \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} - \frac{t}{r} \tilde{\mathbf{y}} \right\|_2^2 \right),$$

which we want to express as an unconstrained, separable, Moreau envelope. For this, define the constrained Moreau envelope

$$M_\rho^{\text{const}}(\mathbf{b}, \lambda) = \min_{\boldsymbol{\eta} \in \mathcal{B}_{\mathcal{C}_\eta}^n} \left\{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) + \frac{1}{2\lambda} \|\boldsymbol{\eta} - \mathbf{b}\|_2^2 \right\},$$

and the unconstrained coordinatewise Moreau envelope

$$M_\rho(\mathbf{b}, \lambda) = \sum_{i=1}^n m_\rho(b_i, \lambda), \quad m_\rho(b, \lambda) = \min_{\eta \in \mathbb{R}} \left\{ \rho(\eta) + \frac{1}{2\lambda} (\eta - b)^2 \right\},$$

where the case $\lambda = 0$, is understood through the continuous extension $M_\rho(\mathbf{b}, 0) = \mathbf{1}^\top \boldsymbol{\rho}(\mathbf{b})$. Then define the AO value with the constrained Moreau envelope by

$$\begin{aligned} \phi^{M, \text{const}}(c_r, C_t; \mathbf{g}, \mathbf{h}) &= \min_{\sigma \in [0, C_\sigma]} \max_{r \in [c_r, 1]} \min_{\substack{\mathbf{v} \in V_p \\ |\theta| \leq C_\theta}} \inf_{t \in (0, C_t]} \frac{1}{n} M_\rho^{\text{const}} \left(\theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \sigma \mathbf{g} + \frac{t}{r} \tilde{\mathbf{y}}, \frac{t}{r} \right) - \sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \frac{rt}{2} \\ &\quad - \frac{1}{n} \left\{ \tilde{\mathbf{y}}^\top \left(\theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \sigma \mathbf{g} \right) + \frac{t}{2r} \|\tilde{\mathbf{y}}\|_2^2 \right\}. \end{aligned}$$

By the completion of squares in (80), $\phi^{M, \text{const}}(c_r, C_t; \mathbf{g}, \mathbf{h}) = \phi^{r, t}(c_r, C_t; \mathbf{g}, \mathbf{h})$. Similarly, define the AO value with the unconstrained Moreau envelope by

$$\begin{aligned} \phi^M(c_r, C_t; \mathbf{g}, \mathbf{h}) &= \min_{\sigma \in [0, C_\sigma]} \max_{r \in [c_r, 1]} \min_{\substack{\mathbf{v} \in V_p \\ |\theta| \leq C_\theta}} \inf_{t \in (0, C_t]} \frac{1}{n} M_\rho \left(\theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \sigma \mathbf{g} + \frac{t}{r} \tilde{\mathbf{y}}, \frac{t}{r} \right) - \sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \frac{rt}{2} \\ &\quad - \frac{1}{n} \left\{ \tilde{\mathbf{y}}^\top \left(\theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \sigma \mathbf{g} \right) + \frac{t}{2r} \|\tilde{\mathbf{y}}\|_2^2 \right\}. \end{aligned}$$

Now, define the Moreau feasibility event as

$$\mathcal{E}_n^M(c_r, C_t, C_\eta) = \{\phi^M(c_r, C_t; \mathbf{g}, \mathbf{h}) = \phi^{M, \text{const}}(c_r, C_t; \mathbf{g}, \mathbf{h}) = \phi^{r, t}(c_r, C_t; \mathbf{g}, \mathbf{h})\}.$$

For constants $c_r > 0$, $C_t < \infty$, and $C_\eta < \infty$, define the scalarisation-valid event

$$\mathcal{R}_n(c_r, C_t, C_\eta) = \mathcal{E}_n^r(c_r) \cap \mathcal{E}_n^t(c_r, C_t) \cap \mathcal{E}_n^M(c_r, C_t, C_\eta), \quad (81)$$

where the quantities ϕ^r , $\phi^{r, t}$, and $\phi^{M, \text{const}}$, and hence the events $\mathcal{E}_n^r$ and $\mathcal{E}_n^t$, also depend on the fixed truncation radius C_η . When the constants have been fixed, we write simply $\mathcal{R}_n$ for $\mathcal{R}_n(c_r, C_t, C_\eta)$.

Next, upon $\mathcal{R}_n$, we wish extend the infimum over $t \in (0, C_t]$ to the compact interval $[0, C_t]$ by continuously extending $M_\rho(\mathbf{x} + \lambda \mathbf{a}, \lambda)/n$ to $\lambda = 0$, with boundary value $\mathbf{1}^\top \boldsymbol{\rho}(\mathbf{x})/n$ for $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{a} \in (0, 1)^n$. To that end, note that since $0 < \rho' < 1$, the function ρ is 1-Lipschitz. Note further then that for every $b \in \mathbb{R}$ and $\lambda > 0$,

$$0 \leq \rho(b) - m_\rho(b, \lambda) \leq \frac{\lambda}{2}.$$

Indeed, the first inequality follows by testing $\eta = b$, while $\rho(\eta) \geq \rho(b) - |\eta - b|$ gives

$$m_\rho(b, \lambda) \geq \rho(b) + \inf_{u \geq 0} \left\{ -u + \frac{u^2}{2\lambda} \right\} = \rho(b) - \frac{\lambda}{2}.$$

Consequently, for every $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{a} \in (0, 1)^n$,

$$\frac{1}{n} |M_\rho(\mathbf{x} + \lambda \mathbf{a}, \lambda) - \mathbf{1}^\top \boldsymbol{\rho}(\mathbf{x})| \leq \frac{3\lambda}{2}.$$

Since $\lambda = t/r \leq t/c_r$, the Moreau term, and hence the full objective defining $\phi^M(c_r, C_t; \mathbf{g}, \mathbf{h})$, extends continuously to $t = 0$. Therefore the infimum over $t \in (0, C_t]$ equals the minimum over $t \in [0, C_t]$. With that, upon $\mathcal{R}_n(c_r, C_t, C_\eta)$, our scalarised AO becomes

$$\begin{aligned} \phi(\mathbf{g}, \mathbf{h}) = & \min_{\sigma \in [0, C_\sigma]} \max_{r \in [c_r, 1]} \min_{\substack{\mathbf{v} \in V_p \\ |\theta| \leq C_\theta \\ t \in [0, C_t]}} \frac{1}{n} M_\rho \left(\theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \sigma \mathbf{g} + \frac{t}{r} \tilde{\mathbf{y}}, \frac{t}{r} \right) - \sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \frac{rt}{2} \\ & - \frac{1}{n} \left(\tilde{\mathbf{y}}^\top \left(\theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \sigma \mathbf{g} \right) + \frac{t}{2r} \|\tilde{\mathbf{y}}\|_2^2 \right). \end{aligned} \quad (82)$$

Before stating the concluding result of this section, it will be useful to introduce a structural event about the signal parameters $\beta_0, \beta_P, \theta_0, \theta_P$. For fixed $t_0 > 0$, define

$$\mathcal{S}_n(t_0) = \left\{ \|\Gamma_P - \Gamma\|_2 \leq t_0, \left\| \begin{bmatrix} \theta_0 \\ \theta_P \end{bmatrix} - \begin{bmatrix} \theta_0^* \\ \theta_P^* \end{bmatrix} \right\|_\infty \leq t_0 \right\}. \quad (83)$$

By Assumptions (i) and (iii), for every fixed $t_0 > 0$, after adjusting constants $C, c > 0$,

$$\Pr(\neg \mathcal{S}_n(t_0)) \leq C \exp\{-cn\}.$$

All constants in the $\mathcal{G}_n$ -conditional AO bounds below are deterministic on $\mathcal{S}_n(t_0)$.

We also need the good event on which the radial boundary $r = 0$ is excluded. Recall G_n , $\boldsymbol{\eta}_{\tilde{Y}}$, $\mathcal{R}_0$, M_n , and the radial profile A_n from (77). For $i = 1, \dots, n$, define the envelope

$$L_i = \left| \eta_{\tilde{Y}, i} \right| + C_\theta + C_\sigma |g_i| + \sup_{\mathbf{v} \in V_p} \left| \frac{1}{\sqrt{p}} (\mathbf{H}_1 \mathbf{v})_i \right|,$$

and, for $a \in (0, 1/2)$ and $K > 0$, the interior index set

$$\mathcal{I}_n(a, K) = \left\{ i \in \{1, \dots, n\} : \tilde{Y}_i \in [a, 1 - a], L_i \leq K \right\}.$$

For $d > 0$, $a \in (0, 1/2)$, and $K, C_L > 0$, define the small- r environment

$$\mathcal{E}_n^{\text{sr}}(d, a, K, C_L) = \left\{ \inf_{\sigma \in [0, C_\sigma]} \left[M_n(\sigma, \boldsymbol{\eta}_{\tilde{Y}}) - \sigma \frac{\|\mathbf{h}\|_2}{\sqrt{n}} \right] \geq d, \right. \\ \left. \frac{1}{n} \sum_{i=1}^n L_i^2 \leq C_L, \quad \frac{1}{n} \sum_{i \notin \mathcal{I}_n(a, K)} L_i^2 \leq \frac{d^2}{64} \right\}.$$

Once $C_\eta \geq \sqrt{C_L}$, as ensured below, the first component is a uniform lower bound on the initial slope $M_n(\sigma, \boldsymbol{\eta}_{\tilde{Y}}) - \sigma \|\mathbf{h}\|_2 / \sqrt{n}$ of $r \mapsto A_n(\sigma, r)$ at $r = 0$; the two envelope components control how far the minimiser of $G_n + r M_n(\sigma, \cdot)$ can move from $\boldsymbol{\eta}_{\tilde{Y}}$ for small $r > 0$. Lemma 11 shows that this event has exponentially high conditional probability, and Proposition F.1 shows that, on it, $A_n(\sigma, \cdot)$ is nondecreasing on a deterministic interval $[0, r_0]$, uniformly in σ .

Proposition B.5 (Scalarisation event bounds). *Fix $t_0 > 0$. After possibly increasing C_η , there exist deterministic constants $d_r > 0$, $a \in (0, 1/2)$, $K, C_L, C_t, C_M \in (0, \infty)$, $c_r \in (0, 1/2]$, such that $C_M \geq \sqrt{C_L}$, $C_\eta \geq C_M + C_t/c_r$, and*

$$C_t > \max \left\{ 1, \frac{(C_\theta + 3C_\beta + 2C_\sigma)^2 + 1 + 4 \log 2}{c_r} \right\}.$$

There are also constants $C, c > 0$ and $N \in \mathbb{N}$ such that, for all $n > N$,

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\neg \mathcal{E}_n^{\text{sr}}(d_r, a, K, C_L) \mid \mathcal{G}_n) \leq C \exp\{-cn\}, \quad (84)$$

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\neg \mathcal{E}_n^r(c_r) \mid \mathcal{G}_n) \leq C \exp\{-cn\}, \quad (85)$$

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\neg \mathcal{E}_n^t(c_r, C_t) \mid \mathcal{G}_n) \leq C \exp\{-cn\}, \quad (86)$$

and

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\neg \mathcal{E}_n^M(c_r, C_t, C_\eta) \mid \mathcal{G}_n) \leq C \exp\{-cn\}. \quad (87)$$

For these constants, abbreviate

$$\mathcal{R}_n = \mathcal{R}_n(c_r, C_t, C_\eta),$$

where $\mathcal{R}_n(c_r, C_t, C_\eta)$ is defined in (81). Then

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\neg \mathcal{R}_n \mid \mathcal{G}_n) \leq C \exp\{-cn\}. \quad (88)$$

Moreover,

$$\Pr(\neg \mathcal{R}_n) \leq C \exp\{-cn\} + \Pr(\neg \mathcal{S}_n(t_0)) \leq C \exp\{-cn\}. \quad (89)$$

The constants depend only on

$$\kappa, \alpha, \boldsymbol{\Gamma}, \theta_0^*, \theta_P^*, t_0, C_\beta, C_\theta, C_\sigma,$$

and on the constants in the probability bounds of Section 2

Proof. Lemma 11 gives deterministic constants d_r, a, K, C_L , constants $C, c > 0$, and $N \in \mathbb{N}$ such that (84) holds with these constants. Set

$$c_{a,K} = \rho''(\log(1-a) + K), \quad r_0 = \min \left\{ \frac{1}{2}, \frac{c_{a,K} d_r}{16} \right\}.$$

Let $\bar{r}$ and c_0 be the deterministic constants from Lemma 24 and Lemma 25 respectively, and set

$$c_r = \min \{r_0, \bar{r}, c_0\}.$$

Let K_t be the constant of Lemma 12 and choose C_t so that

$$C_t > \max \left\{ B_t, 1, \frac{2(K_t + \log 2)}{c_r} \right\},$$

where B_t is the constant from Lemma 24 this is the threshold required by Lemma 12 and by the restricted scalarisation below. Lemma 12 gives constants $C, c > 0$ and $N \in \mathbb{N}$ such that (86) holds.

By Lemma 14 there exist $C_M < \infty$, constants $C, c > 0$, and $N \in \mathbb{N}$ such that, for all $n > N$,

$$\Pr(-\mathcal{E}_n^{\text{ctr}}(C_M) \mid \mathcal{G}_n) \leq C \exp\{-cn\};$$

since $\mathcal{E}_n^{\text{ctr}}(C_M)$ increases with C_M , we may and do take $C_M \geq \sqrt{C_L}$. Increase C_η , if necessary, once and for all so that

$$C_\eta \geq C_M + \frac{C_t}{c_r}. \quad (90)$$

In particular, $C_\eta \geq \sqrt{C_L}$.

On $\mathcal{E}_n^{\text{sr}}(d_r, a, K, C_L)$, Proposition F.1 applied with $\mathcal{R} = \mathcal{R}_0$, gives the pointwise profile equality in (207) for the final cutoff c_r . Taking the outer minimum in σ yields $\mathcal{E}_n^r(c_r)$. Hence

$$\mathcal{E}_n^{\text{sr}}(d_r, a, K, C_L) \subseteq \mathcal{E}_n^r(c_r),$$

and (85) follows from (84).

The bound in Lemma 12 is independent of C_η : its proof uses the feasible point $\boldsymbol{\eta} = \mathbf{0}$ and a lower bound over all $\boldsymbol{\eta} \in \mathbb{R}^n$. Thus (86) remains valid for the final value of C_η . By (90) and Lemma 13

$$\mathcal{E}_n^{\text{ctr}}(C_M) \subseteq \mathcal{E}_n^{\text{M}}(c_r, C_t, C_\eta),$$

which proves (87).

Taking the largest of the thresholds N , the smallest of the exponents c and the largest of the prefactors C obtained above, and increasing N if necessary so that $p - s \geq 1$ for every $n > N$, the four preceding estimates hold with common constants. Since

$$-\mathcal{R}_n \subseteq -\mathcal{E}_n^r(c_r) \cup -\mathcal{E}_n^t(c_r, C_t) \cup -\mathcal{E}_n^{\text{M}}(c_r, C_t, C_\eta),$$

a union bound proves (88).

Finally, $\mathcal{S}_n(t_0) \in \mathcal{G}_n$, so the tower property gives

$$\begin{aligned} \Pr(-\mathcal{R}_n) &\leq \mathbb{E}[\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(-\mathcal{R}_n \mid \mathcal{G}_n)] + \Pr(-\mathcal{S}_n(t_0)) \\ &\leq C \exp\{-cn\} + \Pr(-\mathcal{S}_n(t_0)). \end{aligned}$$

Assumptions (i) and (iii) bound the final probability by $C \exp\{-cn\}$ after adjusting constants. This proves (89). $\square$

B.7 Limiting AO

The scalarisation in Section B.6 reduces the AO to a min-max problem over finite-dimensional variables. Although $\mathbf{v}$ is two-dimensional, its feasible set V_p need not be uniformly scaled in n : when a positive eigenvalue of $\mathbf{\Gamma}_p$ is small, the corresponding coefficient in $\mathbf{v}$ may be large even though the induced signal-aligned predictor remains bounded. We therefore reparameterise the AO at the intrinsic predictor coordinate

$$\mathbf{u} = \mathbf{\Gamma}_p^{1/2} \mathbf{v}.$$

With that, let

$$\mathbf{Q}_n = \frac{1}{\sqrt{p}} \mathbf{H}_1.$$

Conditional on $\mathcal{G}_n$, the rows $\mathbf{q}_i^\top$ of $\mathbf{Q}_n$ are independent with

$$\mathbf{q}_i \sim \mathcal{N}(\mathbf{0}_2, \mathbf{\Gamma}_p),$$

and $\mathbf{Q}_n$ is independent of $(\boldsymbol{\varepsilon}, \mathbf{g}, \mathbf{h})$ conditional on $\mathcal{G}_n$. Hence, for every i ,

$$\text{var}([\mathbf{Q}_n \mathbf{v}]_i \mid \mathcal{G}_n) = \mathbf{v}^\top \mathbf{\Gamma}_p \mathbf{v} = \|\mathbf{u}\|_2^2.$$

Thus $\|\mathbf{u}\|_2$ is the natural intrinsic scale of the signal-aligned predictor, rather than the potentially ill-conditioned Euclidean scale of $\mathbf{v}$. Moreover,

$$\|\mathbf{u}\|_2^2 = \mathbf{v}^\top \mathbf{\Gamma}_p \mathbf{v} = \frac{1}{p} \|\mathbf{B}\mathbf{v}\|_2^2 \leq C_\beta^2,$$

and $\mathbf{u} \in \text{range}(\mathbf{\Gamma}_p)$. Hence

$$\mathbf{u} \in \{\mathbf{u} \in \mathbb{R}^2 : \|\mathbf{u}\|_2 \leq C_\beta\} \cap \text{range}(\mathbf{\Gamma}_p).$$

Conversely, if

$$\mathbf{u} \in \{\mathbf{u} \in \mathbb{R}^2 : \|\mathbf{u}\|_2 \leq C_\beta\} \cap \text{range}(\mathbf{\Gamma}_p),$$

set

$$\mathbf{v} = \mathbf{\Gamma}_p^{+1/2} \mathbf{u}.$$

Then

$$\mathbf{v} \in \text{range}(\mathbf{\Gamma}_p) = \text{range}(\mathbf{B}^\top), \quad \mathbf{\Gamma}_p^{1/2} \mathbf{v} = \mathbf{u},$$

and

$$\frac{1}{p} \|\mathbf{B}\mathbf{v}\|_2^2 = \mathbf{v}^\top \mathbf{\Gamma}_p \mathbf{v} = \|\mathbf{u}\|_2^2 \leq C_\beta^2.$$

Thus $\mathbf{v} \in V_p$. Consequently, the map $\mathbf{v} \mapsto \mathbf{\Gamma}_p^{1/2} \mathbf{v}$ sends V_p onto

$$U_n = U \cap \text{range}(\mathbf{\Gamma}_p), \quad U = \{\mathbf{u} \in \mathbb{R}^2 : \|\mathbf{u}\|_2 \leq C_\beta\}. \quad (91)$$

The change of variables therefore replaces the potentially ill-conditioned set V_p by a moving section U_n of the fixed ambient ball U . Now, let $\mathbf{P}_{\mathbf{\Gamma}_p}$ denote the orthogonal projector onto $\text{range}(\mathbf{\Gamma}_p)$ and define the direct pseudoinverse coordinates

$$\mathbf{Z}_n^0 = \mathbf{Q}_n \mathbf{\Gamma}_p^{+1/2}.$$

Conditional on $\mathcal{G}_n$, the rows of $\mathbf{Z}_n^0$ are centred Gaussian with covariance

$$\mathbf{\Gamma}_p^{+1/2} \mathbf{\Gamma}_p \mathbf{\Gamma}_p^{+1/2} = \mathbf{P}_{\mathbf{\Gamma}_p}.$$

Moreover, for every $\mathbf{v} \in V_p$, with $\mathbf{u} = \mathbf{\Gamma}_p^{1/2} \mathbf{v}$,

$$\mathbf{Z}_n^0 \mathbf{u} = \mathbf{Q}_n \mathbf{\Gamma}_p^{+1/2} \mathbf{\Gamma}_p^{1/2} \mathbf{v} = \mathbf{Q}_n \mathbf{P}_{\mathbf{\Gamma}_p} \mathbf{v} = \mathbf{Q}_n \mathbf{v}.$$

Thus $\mathbf{Z}_n^0$ already represents the finite AO on its actual signal domain U_n . Its row covariance is, however, $\mathbf{P}_{\mathbf{\Gamma}_p}$, which is discontinuous in $\mathbf{\Gamma}_p$ across rank strata: $\mathbf{\Gamma}_p \rightarrow \mathbf{\Gamma}$ does not imply $\mathbf{P}_{\mathbf{\Gamma}_p} \rightarrow \mathbf{P}_{\mathbf{\Gamma}}$. An ambient extension built on $\mathbf{Z}_n^0$ would inherit that discontinuity. Since the finite and limiting objectives will be compared on U , we complete the Gaussian coordinates in the missing directions. Conditional on $\mathcal{G}_n$, let $\tilde{\mathbf{Z}} \in \mathbb{R}^{n \times 2}$ have i.i.d. standard Gaussian entries, independently of $(\mathbf{Q}_n, \boldsymbol{\varepsilon}, \mathbf{g}, \mathbf{h})$, and define

$$\mathbf{Z} = \mathbf{Z}_n^0 + \tilde{\mathbf{Z}} (\mathbf{I}_2 - \mathbf{P}_{\mathbf{\Gamma}_p}) = \mathbf{Q}_n \mathbf{\Gamma}_p^{+1/2} + \tilde{\mathbf{Z}} (\mathbf{I}_2 - \mathbf{P}_{\mathbf{\Gamma}_p}). \quad (92)$$

Conditional on $\mathcal{G}_n$, the two summands are independent, and the rows $\mathbf{z}_i^\top$ of $\mathbf{Z}$ are independent centred Gaussian vectors with covariance

$$\text{cov}(\mathbf{z}_i \mid \mathcal{G}_n) = \mathbf{\Gamma}_p^{+1/2} \mathbf{\Gamma}_p \mathbf{\Gamma}_p^{+1/2} + (\mathbf{I}_2 - \mathbf{P}_{\mathbf{\Gamma}_p})^2 = \mathbf{P}_{\mathbf{\Gamma}_p} + \mathbf{I}_2 - \mathbf{P}_{\mathbf{\Gamma}_p} = \mathbf{I}_2.$$

Thus $\mathbf{Z}$ has i.i.d. standard Gaussian entries conditional on $\mathcal{G}_n$. Since the rows of $\mathbf{Q}_n$ belong to $\text{range}(\mathbf{\Gamma}_p)$ almost surely,

$$\mathbf{Q}_n \mathbf{P}_{\mathbf{\Gamma}_p} = \mathbf{Q}_n,$$

whereas

$$(\mathbf{I}_2 - \mathbf{P}_{\mathbf{\Gamma}_p}) \mathbf{\Gamma}_p^{1/2} = \mathbf{0}_{2 \times 2}.$$

It follows that

$$\frac{1}{\sqrt{p}} \mathbf{H}_1 = \mathbf{Q}_n = \mathbf{Z} \mathbf{\Gamma}_p^{1/2}, \quad (93)$$

almost surely. Moreover, $\mathbf{Z}$ is independent of $(\varepsilon, \mathbf{g}, \mathbf{h})$ conditional on $\mathcal{G}_n$. Writing

$$\mathbf{q}_i = [q_{i,1}, q_{i,2}]^\top = \mathbf{\Gamma}_p^{1/2} \mathbf{z}_i,$$

the pseudo-responses (5) satisfy

$$\tilde{Y}_i = \alpha \mathbf{1} \{ \varepsilon_i < \rho'(\theta_0 + q_{i,1}) \} + (1 - \alpha) \rho'(\theta_P + q_{i,2}).$$

For $\mathbf{u} \in U_n \subseteq \text{range}(\mathbf{\Gamma}_p)$,

$$(\mathbf{I}_2 - \mathbf{P}_{\mathbf{\Gamma}_p}) \mathbf{u} = \mathbf{0}_2,$$

and hence

$$\mathbf{Z} \mathbf{u} = \mathbf{Z}_n^0 \mathbf{u}.$$

Thus the completion leaves $(\mathbf{H}_1, \tilde{\mathbf{y}}, \mathbf{g}, \mathbf{h})$ and the finite AO unchanged on its actual signal domain. The added coordinates define the signal-coordinate part of the objective on the full ambient ball U in a base law that is standard on every rank stratum, so that, by (93), its dependence on $\mathbf{\Gamma}_p$ enters through $\mathbf{\Gamma}_p^{1/2}$.

Combining (91) and (93), for every $\mathbf{v} \in V_p$,

$$\mathbf{Q}_n \mathbf{v} = \mathbf{Z} \mathbf{\Gamma}_p^{1/2} \mathbf{v} = \mathbf{Z} \mathbf{u}, \quad \mathbf{u} = \mathbf{\Gamma}_p^{1/2} \mathbf{v} \in U_n.$$

Accordingly, (82) becomes

$$\begin{aligned} \phi(\mathbf{g}, \mathbf{h}) = & \min_{\sigma \in [0, C_\sigma]} \max_{r \in [c_r, 1]} \min_{\substack{\mathbf{u} \in U_n \\ |\theta| \leq C_\theta \\ t \in [0, C_t]}} \frac{1}{n} M_\rho \left(\theta \mathbf{1} + \mathbf{Z} \mathbf{u} + \sigma \mathbf{g} + \frac{t}{r} \tilde{\mathbf{y}}, \frac{t}{r} \right) - \sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \frac{rt}{2} \\ & - \frac{1}{n} \left\{ \tilde{\mathbf{y}}^\top (\theta \mathbf{1} + \mathbf{Z} \mathbf{u} + \sigma \mathbf{g}) + \frac{t}{2r} \|\tilde{\mathbf{y}}\|_2^2 \right\}. \end{aligned} \quad (94)$$

Now, fix once and for all a baseline structural tolerance $t_\star \in (0, 1]$, and let $d_r, a, K, C_L, c_r, C_t, C_M, C_\eta$ and $\mathcal{R}_n$ be the resulting constants and event from Proposition B.5 with structural tolerance $t_\star$. Throughout this subsection and its auxiliary proofs, every subsequent structural tolerance denoted by t_0 is chosen in $(0, t_\star]$. Since for $t_0 \leq t_\star$, $\mathcal{S}_n(t_0) \subseteq \mathcal{S}_n(t_\star)$, the bounds of Proposition B.5 remain valid on each such event $\mathcal{S}_n(t_0)$.

Recall from (94) that, on the scalarisation-valid event $\mathcal{R}_n$, the AO can be written in terms of the variables $\omega = (\sigma, r, \mathbf{u}, \theta, t)$. Since value convergence will be obtained from uniform convergence of the objective, which compares optimisation values only over a common fixed domain, and since the finite and limiting signal domains U_n and U^* need not converge to one another, we embed both problems into a fixed ambient domain. We use the following domains:

$$\begin{aligned} \Omega &= [0, C_\sigma] \times [c_r, 1] \times U \times [-C_\theta, C_\theta] \times [0, C_t], \\ \Omega_n &= [0, C_\sigma] \times [c_r, 1] \times (U \cap \text{range}(\mathbf{\Gamma}_p)) \times [-C_\theta, C_\theta] \times [0, C_t], \\ \Omega^* &= [0, C_\sigma] \times [c_r, 1] \times (U \cap \text{range}(\mathbf{\Gamma})) \times [-C_\theta, C_\theta] \times [0, C_t]. \end{aligned}$$

The domain Ω_n is the actual finite- n domain of the scalar AO after the change of variables $\mathbf{u} = \mathbf{\Gamma}_p^{1/2} \mathbf{v}$. The ambient domain Ω is fixed and will be used for uniform convergence. The domain Ω^* is the limiting domain.

For $\omega = (\sigma, r, \mathbf{u}, \theta, t) \in \Omega$, write

$$\lambda = \frac{t}{r}, \quad \xi_i = \theta + \mathbf{z}_i^\top \mathbf{u} + \sigma g_i,$$

where $\mathbf{z}_i^\top$ is the i th row of the matrix $\mathbf{Z}$ defined in (92). In the finite- n model, set

$$\mathbf{q}_i = [q_{i,1}, q_{i,2}]^\top = \mathbf{\Gamma}_p^{1/2} \mathbf{z}_i,$$

and

$$\tilde{Y}_i = \alpha \mathbb{1} \{ \varepsilon_i < \rho'(\theta_0 + q_{i,1}) \} + (1 - \alpha) \rho'(\theta_P + q_{i,2}) .$$

Define the objective function

$$F_n(\omega) = \frac{1}{n} \sum_{i=1}^n \left\{ m_\rho \left(\xi_i + \lambda \tilde{Y}_i, \lambda \right) - \tilde{Y}_i \xi_i - \frac{\lambda}{2} \tilde{Y}_i^2 \right\} - \sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \frac{rt}{2} . \quad (95)$$

The continuous extension at $t = 0$ is understood through $m_\rho(b, 0) = \rho(b)$. The conditional mean of the objective in (95) given $\mathcal{G}_n$ is

$$\bar{F}_n(\omega) = \mathbb{E}[F_n(\omega) \mid \mathcal{G}_n] . \quad (96)$$

The limiting objective is defined in the same standard-Gaussian base coordinate $\mathbf{z}$ as follows. Let

$$\mathbf{Q} = [Q_1, Q_2]^\top = \mathbf{\Gamma}^{1/2} \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}_2, \mathbf{I}_2), \quad G \sim \mathcal{N}(0, 1), \quad \varepsilon \sim \text{Unif}([0, 1]),$$

with all variables mutually independent, and let

$$\tilde{Y} = \alpha \mathbb{1} \{ \varepsilon < \rho'(\theta_0^* + Q_1) \} + (1 - \alpha) \rho'(\theta_P^* + Q_2) .$$

For $\xi = \theta + \mathbf{z}^\top \mathbf{u} + \sigma G$, define

$$F(\omega) = \mathbb{E} \left[m_\rho \left(\xi + \lambda \tilde{Y}, \lambda \right) - \tilde{Y} \xi - \frac{\lambda}{2} \tilde{Y}^2 \right] - \sigma r \sqrt{\kappa} + \frac{rt}{2}, \quad (97)$$

where all expectations in (97) are taken with respect to $(\mathbf{z}, G, \varepsilon)$.

The next step is to compare the finite- n objective F_n with the limiting objective F . We will establish uniform convergence on the fixed ambient domain Ω and separately control the replacement of the finite signal domain in Ω_n by the limiting signal domain in Ω^* .

To state the resulting value comparisons compactly, let

$$\mathbf{O} = \Sigma \times R \times \mathbf{K} \subseteq \Omega,$$

where Σ is a domain for σ , R is a domain for r , and $\mathbf{K}$ is a joint domain for $(\mathbf{u}, \theta, t)$. For $f : \mathbf{O} \rightarrow \Re$, define

$$\text{val}(f; \mathbf{O}) = \min_{\sigma \in \Sigma} \max_{r \in R} \min_{(\mathbf{u}, \theta, t) \in \mathbf{K}} f(\sigma, r, \mathbf{u}, \theta, t) . \quad (98)$$

All value operators below use this coordinate order. With this notation, on the scalarisation-valid event $\mathcal{R}_n$,

$$\phi(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \Omega_n) . \quad (99)$$

The limiting AO value is

$$\bar{\phi} = \text{val}(F; \Omega^*) . \quad (100)$$

Now, consider the uniform convergence event for $F_n(\omega)$ to $F(\omega)$ on Ω . For $s > 0$, define

$$\mathcal{U}_n(s) = \left\{ \sup_{\omega \in \Omega} |F_n(\omega) - F(\omega)| \leq s \right\} .$$

The value operator in (98) is monotone and translation equivariant. Hence, for any bounded $f, g : \mathbf{O} \rightarrow \Re$ for which the nested extrema are attained,

$$|\text{val}(f; \mathbf{O}) - \text{val}(g; \mathbf{O})| \leq \sup_{\omega \in \mathbf{O}} |f(\omega) - g(\omega)| . \quad (101)$$

Indeed, with $\Delta = \sup_{\omega \in \mathbf{O}} |f(\omega) - g(\omega)|$, the bounds $g - \Delta \leq f \leq g + \Delta$, together with monotonicity and translation equivariance, give (101). Thus, on $\mathcal{U}_n(s)$, $|\text{val}(F_n; \mathbf{O}) - \text{val}(F; \mathbf{O})| \leq s$ for every scalar domain

$\mathbf{O} \subseteq \mathbf{\Omega}$ equipped with the inherited min–max–min order. It remains to relate the actual finite- n domain $\mathbf{\Omega}_n$ and the limiting domain $\mathbf{\Omega}^*$ to the ambient domain $\mathbf{\Omega}$. For this, define the domain/nullspace event

$$\mathcal{N}_n(s) = \{|\text{val}(F_n; \mathbf{\Omega}_n) - \text{val}(F_n; \mathbf{\Omega})| \leq s\} \cap \{|\text{val}(F; \mathbf{\Omega}) - \text{val}(F; \mathbf{\Omega}^*)| \leq s\}. \quad (102)$$

The event $\mathcal{N}_n(s)$ is controlled jointly with $\mathcal{U}_n(s)$ in Proposition B.6 using the value-Lipschitz property, the exact finite and limiting nullspace identities, and the uniform-convergence and structural-bias bounds below. Combining (99), (100), (101), and (102), we obtain the following implication:

$$\mathcal{R}_n \cap \mathcal{U}_n(s) \cap \mathcal{N}_n(s) \implies |\phi(\mathbf{g}, \mathbf{h}) - \bar{\phi}| \leq 3s.$$

Indeed, on this event,

$$\begin{aligned} |\phi(\mathbf{g}, \mathbf{h}) - \bar{\phi}| &= |\text{val}(F_n; \mathbf{\Omega}_n) - \text{val}(F; \mathbf{\Omega}^*)| \\ &\leq |\text{val}(F_n; \mathbf{\Omega}_n) - \text{val}(F_n; \mathbf{\Omega})| \\ &\quad + |\text{val}(F_n; \mathbf{\Omega}) - \text{val}(F; \mathbf{\Omega})| \\ &\quad + |\text{val}(F; \mathbf{\Omega}) - \text{val}(F; \mathbf{\Omega}^*)| \\ &\leq 3s. \end{aligned}$$

The main probabilistic statement of this section is thus the following.

Proposition B.6 (Uniform convergence and moving-domain control of the scalar AO objective). *For every $s > 0$, there exist $t_0 = t_0(s) \in (0, t_\star]$, constants $C_s, c_s > 0$, and $N = N(s) \in \mathbb{N}$ such that, for all $n > N$, the following hold.*

$$(i) \quad \mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(-\mathcal{U}_n(s) \mid \mathcal{G}_n) \leq C_s \exp\{-c_s n\}. \quad (103)$$

$$(ii) \quad \mathcal{S}_n(t_0) \cap \mathcal{U}_n\left(\frac{s}{4}\right) \subseteq \mathcal{U}_n(s) \cap \mathcal{N}_n(s). \quad (104)$$

Consequently,

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\{-\mathcal{U}_n(s)\} \cup \{-\mathcal{N}_n(s)\} \mid \mathcal{G}_n) \leq C_s \exp\{-c_s n\}. \quad (105)$$

The constants depend only on $s, \kappa, \alpha, \mathbf{\Gamma}, \theta_0^*, \theta_P^*$ and on the constants in Section 2

Proof. Fix $s > 0$ and set

$$q = \frac{s}{4}.$$

(i) By Lemma 18 applied with input tolerance q , there exist $t_B = t_B(q) > 0$ and $N_0 \in \mathbb{N}$ such that, for all $n > N_0$,

$$\mathbb{1}_{\mathcal{S}_n(t_B)} \sup_{\omega \in \mathbf{\Omega}} |\bar{F}_n(\omega) - F(\omega)| \leq \frac{q}{3}.$$

Set $t_0 = \min\{t_\star, t_B\}$. Since $\mathcal{S}_n(t_0) \subseteq \mathcal{S}_n(t_B)$, the same structural-bias bound holds on $\mathcal{S}_n(t_0)$.

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \sup_{\omega \in \mathbf{\Omega}} |\bar{F}_n(\omega) - F(\omega)| \leq \frac{q}{3}. \quad (106)$$

Let L_n be the maximum of the five coordinatewise Lipschitz coefficients in Lemma 20. By that lemma, there exist deterministic constants $\bar{C} < \infty$ and $C, c > 0$ such that, for all sufficiently large n ,

$$\Pr(L_n > \bar{C} \mid \mathcal{G}_n) \leq C \exp\{-cn\}. \quad (107)$$

On the event $\{L_n \leq \bar{C}\}$, Lemma 19 and the Cauchy–Schwarz inequality (Rudin 1976, Theorem 1.37(d)) give

$$|F_n(\omega) - F_n(\omega')| \leq \bar{C} (|\sigma - \sigma'| + |r - r'| + \|\mathbf{u} - \mathbf{u}'\|_2 + |\theta - \theta'| + |t - t'|) \leq \sqrt{5} \bar{C} \|\omega - \omega'\|_2,$$

for every $\boldsymbol{\omega}, \boldsymbol{\omega}' \in \boldsymbol{\Omega}$.

Let $\bar{C}_F$ be the deterministic Lipschitz constant from Lemma 21 and define

$$\bar{C}_0 = \sqrt{5} \bar{C} + \bar{C}_F, \quad \varepsilon = \min \left\{ 1, \frac{q}{3\bar{C}_0} \right\}.$$

Identify $\boldsymbol{\Omega}$ with its natural embedding in $\mathfrak{R}^6$. Since $\boldsymbol{\Omega}$ is a bounded subset of $\mathfrak{R}^6$, the standard volumetric covering bound (see, for example, Vershynin 2018, Section 4.2) provides a deterministic ε -net $\boldsymbol{\Omega}_\varepsilon \subseteq \boldsymbol{\Omega}$ satisfying

$$|\boldsymbol{\Omega}_\varepsilon| \leq \left(1 + \frac{K_\Omega}{\varepsilon} \right)^6, \quad (108)$$

for a constant $K_\Omega < \infty$ depending only on $C_\sigma, C_\beta, C_\theta, C_t, c_r$.

Suppose that $\mathcal{S}_n(t_0)$ and $\{L_n \leq \bar{C}\}$ hold. For any $\boldsymbol{\omega} \in \boldsymbol{\Omega}$, choose $\boldsymbol{\omega}_j \in \boldsymbol{\Omega}_\varepsilon$ such that

$$\|\boldsymbol{\omega} - \boldsymbol{\omega}_j\|_2 \leq \varepsilon.$$

Using (106) and the Lipschitz bounds for F_n and F , we obtain

$$\begin{aligned} |F_n(\boldsymbol{\omega}) - F(\boldsymbol{\omega})| &\leq |F_n(\boldsymbol{\omega}_j) - \bar{F}_n(\boldsymbol{\omega}_j)| + |\bar{F}_n(\boldsymbol{\omega}_j) - F(\boldsymbol{\omega}_j)| \\ &\quad + |F_n(\boldsymbol{\omega}) - F_n(\boldsymbol{\omega}_j)| + |F(\boldsymbol{\omega}) - F(\boldsymbol{\omega}_j)| \\ &\leq \max_{\boldsymbol{\omega}_j \in \boldsymbol{\Omega}_\varepsilon} |F_n(\boldsymbol{\omega}_j) - \bar{F}_n(\boldsymbol{\omega}_j)| + \frac{q}{3} + \bar{C}_0 \varepsilon \\ &\leq \max_{\boldsymbol{\omega}_j \in \boldsymbol{\Omega}_\varepsilon} |F_n(\boldsymbol{\omega}_j) - \bar{F}_n(\boldsymbol{\omega}_j)| + \frac{2q}{3}. \end{aligned}$$

Consequently,

$$\mathcal{S}_n(t_0) \cap \{L_n \leq \bar{C}\} \cap \{-\mathcal{U}_n(q)\} \subseteq \left\{ \max_{\boldsymbol{\omega}_j \in \boldsymbol{\Omega}_\varepsilon} |F_n(\boldsymbol{\omega}_j) - \bar{F}_n(\boldsymbol{\omega}_j)| > \frac{q}{3} \right\}. \quad (109)$$

Since $\mathcal{S}_n(t_0) \in \mathcal{G}_n$, conditioning (109) on $\mathcal{G}_n$ and using (107) gives

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(-\mathcal{U}_n(q) \mid \mathcal{G}_n) \leq \mathbb{1}_{\mathcal{S}_n(t_0)} \Pr \left(\max_{\boldsymbol{\omega}_j \in \boldsymbol{\Omega}_\varepsilon} |F_n(\boldsymbol{\omega}_j) - \bar{F}_n(\boldsymbol{\omega}_j)| > \frac{q}{3} \mid \mathcal{G}_n \right) + C \exp\{-cn\}.$$

The net is deterministic, and the constants in Lemma 17 are uniform over $\boldsymbol{\omega} \in \boldsymbol{\Omega}$. Therefore, a union bound and (108) yield

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr \left(\max_{\boldsymbol{\omega}_j \in \boldsymbol{\Omega}_\varepsilon} |F_n(\boldsymbol{\omega}_j) - \bar{F}_n(\boldsymbol{\omega}_j)| > \frac{q}{3} \mid \mathcal{G}_n \right) \leq C \left(1 + \frac{K_\Omega}{\varepsilon} \right)^6 \exp \left\{ -\frac{cnq^2}{9} \right\}.$$

For fixed $s > 0$, ε and the covering-number factor are deterministic constants independent of n and of the realised value of $\mathcal{G}_n$. After adjusting constants, we conclude that

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(-\mathcal{U}_n(q) \mid \mathcal{G}_n) \leq C_s \exp\{-c_s n\}.$$

Since $\mathcal{U}_n(q) \subseteq \mathcal{U}_n(s)$, it follows that

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(-\mathcal{U}_n(s) \mid \mathcal{G}_n) \leq C_s \exp\{-c_s n\}.$$

This proves part (i), namely (103).

(ii) Let

$$\Delta_n = \sup_{\boldsymbol{\omega} \in \boldsymbol{\Omega}} |F_n(\boldsymbol{\omega}) - \bar{F}_n(\boldsymbol{\omega})|,$$

and

$$\Delta'_n = \sup_{\omega \in \Omega} |\bar{F}_n(\omega) - F(\omega)|.$$

On $\mathcal{S}_n(t_0) \cap \mathcal{U}_n(q)$, the structural-bias bound obtained in part (i) gives $\Delta'_n \leq q/3$. Therefore,

$$\Delta_n \leq \sup_{\omega \in \Omega} |F_n(\omega) - F(\omega)| + \Delta'_n \leq q + \frac{q}{3} = \frac{s}{3}.$$

By (101) and Lemma 15, the triangle inequality gives

$$|\text{val}(F_n; \Omega_n) - \text{val}(F_n; \Omega)| \leq \Delta_n + 0 + \Delta_n = 2\Delta_n \leq \frac{2s}{3} \leq s.$$

Moreover, Lemma 16 gives

$$\text{val}(F; \Omega) = \text{val}(F; \Omega^*).$$

Hence $\mathcal{N}_n(s)$ holds. Since $q = s/4 \leq s$, $\mathcal{U}_n(q) \subseteq \mathcal{U}_n(s)$, and therefore

$$\mathcal{S}_n(t_0) \cap \mathcal{U}_n\left(\frac{s}{4}\right) \subseteq \mathcal{U}_n(s) \cap \mathcal{N}_n(s).$$

Taking complements in (104) gives

$$\mathcal{S}_n(t_0) \cap (\{\neg \mathcal{U}_n(s)\} \cup \{\neg \mathcal{N}_n(s)\}) \subseteq \mathcal{S}_n(t_0) \cap \{\neg \mathcal{U}_n(q)\}.$$

Since $\mathcal{S}_n(t_0) \in \mathcal{G}_n$, the stronger quarter-tolerance bound from part (i) yields

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\{\neg \mathcal{U}_n(s)\} \cup \{\neg \mathcal{N}_n(s)\} \mid \mathcal{G}_n) \leq \mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\neg \mathcal{U}_n(q) \mid \mathcal{G}_n) \leq C_s \exp\{-c_s n\}.$$

This proves (105). Taking $N(s)$ to be the maximum of the finite thresholds required above completes the proof. $\square$

B.8 Properties of the limiting AO

Before deriving the value gaps used in the comparison argument, we establish that the limiting AO is well posed and that its compact, natural-domain, and ambient formulations are equivalent. For the range-restricted and ambient inner domains, write

$$U^* = U \cap \text{range}(\mathbf{\Gamma}), \quad \mathbf{K}^* = U^* \times [-C_\theta, C_\theta] \times [0, C_t], \quad \mathbf{K} = U \times [-C_\theta, C_\theta] \times [0, C_t]. \quad (110)$$

The compact and ambient inner problems are profiled as

$$\begin{aligned} H(\sigma, r) &= \min_{(\mathbf{u}, \theta, t) \in \mathbf{K}^*} F(\sigma, r, \mathbf{u}, \theta, t), & V(\sigma) &= \max_{r \in [c_r, 1]} H(\sigma, r), \\ H_0(\sigma, r) &= \min_{(\mathbf{u}, \theta, t) \in \mathbf{K}} F(\sigma, r, \mathbf{u}, \theta, t), & V_0(\sigma) &= \max_{r \in [c_r, 1]} H_0(\sigma, r). \end{aligned}$$

Thus, in the notation of (98),

$$\text{val}(F; \Omega^*) = \min_{\sigma \in [0, C_\sigma]} V(\sigma), \quad \text{val}(F; \Omega) = \min_{\sigma \in [0, C_\sigma]} V_0(\sigma).$$

Proposition B.7 (Well-posedness and equivalent characterisations of the limiting AO). *The compactification constants may be chosen consistently with Propositions B.3 and B.5 so that the following statements hold.*

- (i) *The compact problems on Ω^* and Ω , and the natural-domain problem in (7), are all attained at the same unique saddle chain*

$$(\sigma^*, r^*, \mathbf{u}^*, \theta^*, t^*).$$

Their common value is $\bar{\phi}$. In particular,

$$\text{val}(F; \Omega) = \text{val}(F; \Omega^*) = \bar{\phi}.$$

The common saddle satisfies

$$0 < \sigma^* < C_\sigma, \quad c_r < r^* < 1, \quad 0 < t^* < C_t, \quad \|\mathbf{u}^*\|_2 < C_\beta, \quad |\theta^*| < C_\theta.$$

(ii) Set

$$\lambda^* = \frac{t^*}{r^*}.$$

Then $(\sigma^*, \lambda^*, \mathbf{u}^*, \theta^*)$ is the unique solution of [8] in

$$(0, \infty)^2 \times \text{range}(\mathbf{\Gamma}) \times \mathbb{R}.$$

The remaining saddle coordinates are recovered as

$$r^* = \frac{\sigma^* \sqrt{\kappa}}{\lambda^*}, \quad t^* = \sigma^* \sqrt{\kappa}.$$

Proof. The compatibility of the compactification constants is the choice made in the proof of Proposition [B.5] together with the preceding deterministic enlargements of C_σ , C_β , and C_θ .

Lemma [26] gives attainment, uniqueness, interiority, and equality of the compact problem on $\mathbf{\Omega}^*$ and the natural-domain problem. Lemma [27] shows that enlarging the compact signal domain to obtain $\mathbf{\Omega}$ leaves the value and the unique saddle chain unchanged. This proves part (i). Part (ii) follows from Lemma [28] $\square$

B.9 Restricted value comparison and PO localisation

We now use the saddle and ambient-domain properties of Proposition [B.7] to localise the PO through value comparisons similar to those of Thrampoulidis *et al.* (2018) by forcing either the radial variable or the signal and intercept variables away from the limiting saddle. Deterministic value gaps for the corresponding limiting AOs are transferred first to the finite AOs and then, through the conditional CGMT, to the restricted POs. The resulting separation rules out every PO optimiser outside the prescribed neighbourhood. Throughout this subsection, a minimisation over an empty set has value $+\infty$.

For $\epsilon > 0$, define the radial localising set

$$\mathcal{S}_w^\epsilon = \left\{ \mathbf{w} \in W_p : \left| \frac{\|\mathbf{w}\|_2}{\sqrt{p}} - \sigma^* \right| < \epsilon \right\},$$

and let $\mathcal{S}_w^{\epsilon,c}$ be its complement in W_p . Further, define the signal and intercept localising set

$$\mathcal{S}_{\mathbf{v},\theta}^\epsilon = \left\{ (\mathbf{v}, \theta) \in V_p \times [-C_\theta, C_\theta] : \left\| \mathbf{\Gamma}_p^{1/2} \mathbf{v} - \mathbf{u}^* \right\|_2 < \epsilon, |\theta - \theta^*| < \epsilon \right\},$$

and let $\mathcal{S}_{\mathbf{v},\theta}^{\epsilon,c}$ be its complement in $V_p \times [-C_\theta, C_\theta]$. Thus the target localising set for the PO variables is

$$\mathcal{S}^\epsilon = \mathcal{S}_w^\epsilon \times \mathcal{S}_{\mathbf{v},\theta}^\epsilon.$$

To prove localisation, we show that the PO value is strictly larger than the unrestricted value Ψ under the radial bad-set restriction and under every signal and intercept restriction used to cover $\mathcal{S}_{\mathbf{v},\theta}^{\epsilon,c}$. Both types of restriction can only increase the PO value and thus the purpose of the comparison argument is to show that this increase is bounded away from zero. The radial variable $\mathbf{w}$ is the outer minimisation variable in the PO of (63) and can therefore be restricted directly. By contrast, $(\mathbf{v}, \theta)$ are minimised inside the profile ψ , so restricting them changes the profile and produces a separate PO–AO pair rather than merely restricting the outer minimisation set as in Thrampoulidis *et al.* (2018, Theorem 3(ii)). Restricting the radial variable gives the value

$$\Psi_w^{\epsilon,c} = \min_{\mathbf{w} \in \mathcal{S}_w^{\epsilon,c}} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \left\{ -\frac{1}{n\sqrt{p}} \boldsymbol{\lambda}^\top \mathbf{H}_2 \mathbf{w} + \psi(\boldsymbol{\lambda}) \right\}. \quad (111)$$

To formulate the corresponding restrictions of the signal and intercept variables, condition on $\mathcal{C}_n$ and let $D \subseteq V_p \times [-C_\theta, C_\theta]$ be a nonempty compact convex set. Recall $L(\boldsymbol{\zeta}; \boldsymbol{\lambda})$ from Proposition [B.4], and define the D -restricted profile and PO value by

$$\Psi_D = \min_{\mathbf{w} \in W_p} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \left\{ -\frac{1}{n\sqrt{p}} \boldsymbol{\lambda}^\top \mathbf{H}_2 \mathbf{w} + \psi_D(\boldsymbol{\lambda}) \right\}, \quad \psi_D(\boldsymbol{\lambda}) = \min_{\boldsymbol{\zeta} \in Z_p(D)} L(\boldsymbol{\zeta}; \boldsymbol{\lambda}), \quad Z_p(D) = D \times \mathcal{B}_{C_\eta}^n. \quad (112)$$

Thus Ψ_D is the value of the same conditional PO with only $(\mathbf{v}, \theta)$ restricted to D . Moreover, since $Z_p(D)$ is contained in the unrestricted domain,

$$\psi_D(\boldsymbol{\lambda}) \geq \psi(\boldsymbol{\lambda}) \quad \text{for every } \boldsymbol{\lambda} \in \mathcal{B}_1^n, \quad \Psi_D \geq \Psi.$$

Conditional on $\mathcal{C}_n$, compactness of $Z_p(D)$ and joint continuity of L give attainment and continuity of ψ_D by Berge's maximum theorem (see, for example, [Aliprantis and Border 2006](#) Theorem 17.31.1). Since $L(\boldsymbol{\zeta}; \cdot)$ is affine for every fixed $\boldsymbol{\zeta}$, its pointwise infimum ψ_D is concave in $\boldsymbol{\lambda}$. Since $Z_p(D)$ is compact and convex, Lemma [10](#) applies with $Z = Z_p(D)$, and [\(112\)](#) is an admissible conditional CGMT instance of Proposition [B.4](#) with $D_0 = D$.

Since $\mathcal{S}_{\mathbf{v}, \theta}^{\epsilon, c}$ need not be convex, it cannot be used as a single restriction D while retaining this convex PO–AO structure. We therefore cover it by finitely many compact convex pieces, apply [\(112\)](#) to each piece, and combine the resulting comparisons over the fixed finite index set.

Fix a $1/2$ -net $\mathcal{A} \subset \mathbb{S}^1$ and enumerate it as

$$\mathcal{A} = \{\mathbf{a}_i \in \mathbb{S}^1 : i \in \mathcal{I}_{\mathcal{A}}\}, \quad \mathcal{I}_{\mathcal{A}} = \{1, \dots, |\mathcal{A}|\},$$

such that

$$\forall \mathbf{x} \in \mathbb{S}^1 \quad \exists i \in \mathcal{I}_{\mathcal{A}} : \|\mathbf{x} - \mathbf{a}_i\|_2 \leq \frac{1}{2}.$$

For instance, one may take $|\mathcal{A}| \leq 5^2$ ([Vershynin 2018](#), Corollary 4.2.13). For such a cover, define the index set $\mathcal{J} = \mathcal{I}_{\mathcal{A}} \cup \{+, -\}$, and define the intercept intervals

$$\Theta_+^\epsilon = [-C_\theta, C_\theta] \cap [\theta^* + \epsilon, \infty), \quad \Theta_-^\epsilon = [-C_\theta, C_\theta] \cap (-\infty, \theta^* - \epsilon].$$

For $j \in \mathcal{J}$, define

$$C_{p,j}^\epsilon = \begin{cases} \left\{ (\mathbf{v}, \theta) \in V_p \times [-C_\theta, C_\theta] : \mathbf{a}_j^\top (\mathbf{\Gamma}_p^{1/2} \mathbf{v} - \mathbf{u}^*) \geq \frac{\epsilon}{2} \right\}, & j \in \mathcal{I}_{\mathcal{A}}, \\ V_p \times \Theta_j^\epsilon, & j \in \{+, -\} \end{cases}. \quad (113)$$

Lemma [29](#) shows that

$$\mathcal{S}_{\mathbf{v}, \theta}^{\epsilon, c} \subseteq \bigcup_{j \in \mathcal{J}} C_{p,j}^\epsilon,$$

and that every nonempty piece is compact and convex. Set

$$Z_{p,j}^\epsilon = Z_p(C_{p,j}^\epsilon) = C_{p,j}^\epsilon \times \mathcal{B}_{C_\eta}^n, \quad \mathcal{J}_n^\epsilon = \{j \in \mathcal{J} : C_{p,j}^\epsilon \neq \emptyset\}.$$

For $j \in \mathcal{J}_n^\epsilon$, apply [\(112\)](#) with $D = C_{p,j}^\epsilon$ and write

$$\psi_j^\epsilon(\boldsymbol{\lambda}) = \psi_{C_{p,j}^\epsilon}(\boldsymbol{\lambda}) = \min_{\boldsymbol{\zeta} \in Z_{p,j}^\epsilon} L(\boldsymbol{\zeta}; \boldsymbol{\lambda}), \quad \Psi_j^\epsilon = \Psi_{C_{p,j}^\epsilon},$$

For $j \notin \mathcal{J}_n^\epsilon$, set $\Psi_j^\epsilon = +\infty$. For every $j \in \mathcal{J}_n^\epsilon$, the argument following [\(112\)](#) gives a valid conditional PO–AO pair. Consequently, it is enough to establish the required value separation for the finitely many values $\{\Psi_j^\epsilon : j \in \mathcal{J}_n^\epsilon\}$.

The scalarisation of Section [B.6](#) carries the restrictions above to the scalar variables $\boldsymbol{\omega} = (\sigma, r, \mathbf{u}, \theta, t)$ through

$$\frac{\|\mathbf{w}\|_2}{\sqrt{p}} \mapsto \sigma, \quad \mathbf{v} \mapsto \mathbf{u} = \mathbf{\Gamma}_p^{1/2} \mathbf{v}, \quad \theta \mapsto \theta.$$

Since $\mathbf{w} \mapsto \|\mathbf{w}\|_2/\sqrt{p}$ maps W_p onto $[0, C_\sigma]$, the radial restriction becomes

$$\mathcal{S}_\sigma^{\epsilon, c} = \{\sigma \in [0, C_\sigma] : |\sigma - \sigma^*| \geq \epsilon\},$$

which is nonempty whenever $\mathcal{S}_w^{\epsilon,c}$ is. For $U_0 \subseteq U$ and $j \in \mathcal{J}$, define, in correspondence with (113),

$$\mathbf{K}_j^\epsilon(U_0) = \begin{cases} \left\{ (\mathbf{u}, \theta, t) \in U_0 \times [-C_\theta, C_\theta] \times [0, C_t] : \mathbf{a}_j^\top (\mathbf{u} - \mathbf{u}^*) \geq \frac{\epsilon}{2} \right\}, & j \in \mathcal{I}_A, \\ U_0 \times \Theta_j^\epsilon \times [0, C_t], & j \in \{+, -\} \end{cases}.$$

By (91), the change of variables

$$\mathbf{u} = \mathbf{\Gamma}_p^{1/2} \mathbf{v},$$

maps V_p onto $U_n = U \cap \text{range}(\mathbf{\Gamma}_p)$. For $j \in \mathcal{I}_A$, it transforms the signal half-space in (113) into

$$\mathbf{a}_j^\top (\mathbf{u} - \mathbf{u}^*) \geq \frac{\epsilon}{2},$$

while for $j \in \{+, -\}$ it leaves the restriction $\theta \in \Theta_j^\epsilon$ unchanged. Hence the scalar image of $C_{p,j}^\epsilon \times [0, C_t]$ is exactly $\mathbf{K}_j^\epsilon(U_n)$ for every $j \in \mathcal{J}$. Recall $\mathbf{K}$ from (110) and set

$$\mathbf{K}_n = U_n \times [-C_\theta, C_\theta] \times [0, C_t].$$

For $j \in \mathcal{J}$, the finite and ambient restricted scalar domains are

$$\mathbf{\Omega}_{n,\sigma}^{\epsilon,c} = \mathcal{S}_\sigma^{\epsilon,c} \times [c_r, 1] \times \mathbf{K}_n, \quad \mathbf{\Omega}_\sigma^{\epsilon,c} = \mathcal{S}_\sigma^{\epsilon,c} \times [c_r, 1] \times \mathbf{K}, \quad \mathbf{\Omega}_{n,j}^\epsilon = [0, C_\sigma] \times [c_r, 1] \times \mathbf{K}_j^\epsilon(U_n), \quad \mathbf{\Omega}_j^\epsilon = [0, C_\sigma] \times [c_r, 1] \times \mathbf{K}_j^\epsilon(U). \quad (114)$$

With that, finally, define the restricted limiting AO values

$$\bar{\phi}_\sigma^{\epsilon,c} = \text{val}(F; \mathbf{\Omega}_\sigma^{\epsilon,c}), \quad \bar{\phi}_j^\epsilon = \text{val}(F; \mathbf{\Omega}_j^\epsilon), \quad j \in \mathcal{J}. \quad (115)$$

Lemma 30 proves strict gaps for all restricted limiting AOs. For the radial restriction, the claim is immediate when $\mathcal{S}_\sigma^{\epsilon,c}$ is empty. Otherwise, continuity of V_0 and uniqueness of its minimiser σ^* give

$$\bar{\phi}_\sigma^{\epsilon,c} = \min_{\sigma \in \mathcal{S}_\sigma^{\epsilon,c}} V_0(\sigma) > V_0(\sigma^*) = \bar{\phi},$$

since $\sigma^* \notin \mathcal{S}_\sigma^{\epsilon,c}$. For a signal or intercept piece, define the corresponding restricted inner and outer profiles by

$$H_j^\epsilon(\sigma, r) = \min_{(\mathbf{u}, \theta, t) \in \mathbf{K}_j^\epsilon(U)} F(\sigma, r, \mathbf{u}, \theta, t), \quad V_j^\epsilon(\sigma) = \max_{r \in [c_r, 1]} H_j^\epsilon(\sigma, r).$$

If $\bar{\phi}_j^\epsilon = \bar{\phi}$, then $V_j^\epsilon \geq V_0$ and uniqueness of σ^* first give $V_j^\epsilon(\sigma^*) = \bar{\phi}$. Consequently,

$$\bar{\phi} = H_0(\sigma^*, r^*) \leq H_j^\epsilon(\sigma^*, r^*) \leq V_j^\epsilon(\sigma^*) = \bar{\phi}.$$

Thus the restricted inner minimum at (σ^*, r^*) equals $\bar{\phi}$. Attainment and uniqueness of the unrestricted inner minimiser would require $(\mathbf{u}^*, \theta^*, t^*)$ to lie in a bad piece, which is impossible by construction. Lemma 30 thus shows that there is a deterministic $\eta_\epsilon > 0$, independent of n , such that

$$\bar{\phi}_\sigma^{\epsilon,c} \geq \bar{\phi} + 3\eta_\epsilon, \quad \bar{\phi}_j^\epsilon \geq \bar{\phi} + 3\eta_\epsilon, \quad j \in \mathcal{J}. \quad (116)$$

All that remains is to propagate the limiting AO value gaps through the CGMT pipeline back to the PO. The AO counterparts of (111) and Ψ_j^ϵ are

$$\phi_w^{\epsilon,c}(\mathbf{g}, \mathbf{h}) = \min_{\mathbf{w} \in \mathcal{S}_w^{\epsilon,c}} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \left\{ -\frac{1}{n\sqrt{p}} (\|\mathbf{w}\|_2 \mathbf{g}^\top \boldsymbol{\lambda} + \|\boldsymbol{\lambda}\|_2 \mathbf{w}^\top \mathbf{h}) + \psi(\boldsymbol{\lambda}) \right\},$$

and, for $j \in \mathcal{J}_n^\epsilon$,

$$\phi_j^\epsilon(\mathbf{g}, \mathbf{h}) = \min_{\mathbf{w} \in W_p} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \left\{ -\frac{1}{n\sqrt{p}} (\|\mathbf{w}\|_2 \mathbf{g}^\top \boldsymbol{\lambda} + \|\boldsymbol{\lambda}\|_2 \mathbf{w}^\top \mathbf{h}) + \psi_j^\epsilon(\boldsymbol{\lambda}) \right\}.$$

If $\mathcal{S}_w^{\epsilon,c} = \emptyset$, set $\phi_w^{\epsilon,c} = +\infty$, and set $\phi_j^\epsilon = +\infty$ for $j \notin \mathcal{J}_n^\epsilon$. The parametrisations below are understood on the events that the corresponding restricted domains are nonempty. In the notation of Proposition [B.4](#) the three PO–AO pairs are parameterised by

$$\begin{aligned} (\Psi, \phi) : \quad & W_0 = W_p, \quad D_0 = V_p \times [-C_\theta, C_\theta], \\ (\Psi_w^{\epsilon,c}, \phi_w^{\epsilon,c}) : \quad & W_0 = \mathcal{S}_w^{\epsilon,c}, \quad D_0 = V_p \times [-C_\theta, C_\theta], \\ (\Psi_j^\epsilon, \phi_j^\epsilon) : \quad & W_0 = W_p, \quad D_0 = C_{p,j}^\epsilon, \quad j \in \mathcal{J}_n^\epsilon. \end{aligned}$$

Lemma [32](#) shows that, on one event of conditional exponentially high probability, the unrestricted and restricted AOs are represented by the same objective F_n , with only the feasible domain changed:

$$\phi(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \mathbf{\Omega}_n), \quad \phi_w^{\epsilon,c}(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \mathbf{\Omega}_{n,\sigma}^{\epsilon,c}), \quad \phi_j^\epsilon(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \mathbf{\Omega}_{n,j}^\epsilon), \quad j \in \mathcal{J}_n^\epsilon.$$

Lemma [33](#) combines this simultaneous scalarisation with Proposition [B.6](#). In each restricted comparison, enlarging U_n to U enlarges only the innermost minimisation domain and can therefore only decrease the min–max–min value. Uniform control of $F_n - F$ on $\mathbf{\Omega}$ then gives the required lower bounds relative to [\(115\)](#), simultaneously over $j \in \mathcal{J}_n^\epsilon$.

Lemma [34](#) applies Proposition [B.4](#) in the required tail directions. The upper-tail comparison controls the unrestricted PO. The lower-tail comparison controls the radial restriction and every active signal or intercept piece. Since $\mathcal{J}_n^\epsilon \subseteq \mathcal{J}$ and $\mathcal{J}$ is deterministic and finite, a single uniform union bound covers all active pieces. Define

$$\mathcal{P}_n^\epsilon = \{\Psi \leq \bar{\phi} + \eta_\epsilon\} \cap \{\Psi_w^{\epsilon,c} \geq \bar{\phi}_\sigma^{\epsilon,c} - \eta_\epsilon\} \cap \bigcap_{j \in \mathcal{J}_n^\epsilon} \{\Psi_j^\epsilon \geq \bar{\phi}_j^\epsilon - \eta_\epsilon\}. \quad (117)$$

Lemma [34](#) gives $t_0 = t_0(\epsilon) > 0$, constants $C_\epsilon, c_\epsilon > 0$, and $N_\epsilon \in \mathbb{N}$ such that, for all $n > N_\epsilon$,

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(-\mathcal{P}_n^\epsilon \mid \mathcal{G}_n) \leq C_\epsilon \exp\{-c_\epsilon n\}. \quad (118)$$

Together with [\(116\)](#), this yields the local separation

$$\Psi \leq \bar{\phi} + \eta_\epsilon < \bar{\phi} + 2\eta_\epsilon \leq \min \left\{ \Psi_w^{\epsilon,c}, \min_{j \in \mathcal{J}_n^\epsilon} \Psi_j^\epsilon \right\} \quad \text{on } \mathcal{P}_n^\epsilon. \quad (119)$$

For all sufficiently large n , let $(\hat{\mathbf{v}}, \hat{\mathbf{w}}, \hat{\theta}, \hat{\boldsymbol{\eta}})$ be the minimising variables fixed in Proposition [B.3](#) and set

$$\hat{\boldsymbol{\zeta}} = (\hat{\mathbf{v}}, \hat{\theta}, \hat{\boldsymbol{\eta}}).$$

On $\mathcal{E}_n$, Lemma [9](#) shows that these minimising variables are unique and that $\hat{\mathbf{w}}$ is the unique minimiser of the profiled PO in [\(64\)](#).

Proposition B.8 (PO localisation). *For every $\epsilon > 0$, there are $t_0 = t_0(\epsilon) > 0$, constants $C_\epsilon, c_\epsilon > 0$, and $N_\epsilon \in \mathbb{N}$ such that, for all $n > N_\epsilon$,*

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr\left((\hat{\mathbf{w}}, \hat{\mathbf{v}}, \hat{\theta}) \notin \mathcal{S}^\epsilon \mid \mathcal{G}_n\right) \leq C_\epsilon \exp\{-c_\epsilon n\}.$$

Consequently, after adjusting the constants,

$$\Pr\left((\hat{\mathbf{w}}, \hat{\mathbf{v}}, \hat{\theta}) \notin \mathcal{S}^\epsilon\right) \leq C_\epsilon \exp\{-c_\epsilon n\}.$$

Proof. Let $\mathcal{P}_n^\epsilon$ be as in [\(117\)](#). By [\(118\)](#), and since $\Pr(-\mathcal{E}_n) = 0$ for $n \geq p + 1$, it is enough to prove

$$\mathcal{P}_n^\epsilon \cap \mathcal{E}_n \subseteq \left\{(\hat{\mathbf{w}}, \hat{\mathbf{v}}, \hat{\theta}) \in \mathcal{S}^\epsilon\right\}.$$

Work on $\mathcal{P}_n^\epsilon \cap \mathcal{E}_n$, where the compact PO has a unique primal minimiser by Lemma [9](#). If $\hat{\mathbf{w}} \in \mathcal{S}_w^{\epsilon,c}$, then, using $\hat{\mathbf{w}}$ as a feasible point in [\(111\)](#) and its optimality in the unrestricted profiled PO,

$$\Psi_w^{\epsilon,c} \leq \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^p} \left\{ -\frac{1}{n\sqrt{p}} \boldsymbol{\lambda}^\top \mathbf{H}_2 \hat{\mathbf{w}} + \psi(\boldsymbol{\lambda}) \right\} = \Psi.$$

This contradicts (119). Hence $\hat{\mathbf{w}} \in \mathcal{S}_{\mathbf{w}}^\epsilon$. If $(\hat{\mathbf{v}}, \hat{\theta}) \in \mathcal{S}_{\mathbf{v}, \theta}^{\epsilon, c}$, Lemma 29 gives $j \in \mathcal{J}$ such that $(\hat{\mathbf{v}}, \hat{\theta}) \in C_{p, j}^\epsilon$. Then $j \in \mathcal{J}_n^\epsilon$ and $\hat{\zeta} \in Z_{p, j}^\epsilon$. By (112),

$$\psi_j^\epsilon(\boldsymbol{\lambda}) \leq L(\hat{\zeta}; \boldsymbol{\lambda}) \quad \text{for every } \boldsymbol{\lambda} \in \mathcal{B}_1^n.$$

Using $\hat{\mathbf{w}}$ as a feasible point in the restricted PO therefore gives

$$\Psi_j^\epsilon \leq \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \left\{ -\frac{1}{n\sqrt{p}} \boldsymbol{\lambda}^\top \mathbf{H}_2 \hat{\mathbf{w}} + \psi_j^\epsilon(\boldsymbol{\lambda}) \right\} \leq \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \left\{ -\frac{1}{n\sqrt{p}} \boldsymbol{\lambda}^\top \mathbf{H}_2 \hat{\mathbf{w}} + L(\hat{\zeta}; \boldsymbol{\lambda}) \right\} = \Psi.$$

The final equality follows from the primal optimality of $(\hat{\mathbf{v}}, \hat{\mathbf{w}}, \hat{\theta}, \hat{\boldsymbol{\eta}})$ in (62) and the value identity in (63). This again contradicts (119). Thus $(\hat{\mathbf{v}}, \hat{\theta}) \in \mathcal{S}_{\mathbf{v}, \theta}^\epsilon$, and hence $(\hat{\mathbf{w}}, \hat{\mathbf{v}}, \hat{\theta}) \in \mathcal{S}^\epsilon$ on $\mathcal{P}_n^\epsilon \cap \mathcal{E}_n$. Consequently, using $\Pr(-\mathcal{E}_n) = 0$,

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr\left((\hat{\mathbf{w}}, \hat{\mathbf{v}}, \hat{\theta}) \notin \mathcal{S}^\epsilon \mid \mathcal{G}_n\right) \leq \mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\neg \mathcal{P}_n^\epsilon \mid \mathcal{G}_n) \leq C_\epsilon \exp\{-c_\epsilon n\}.$$

This proves the conditional bound. Taking expectations and splitting over $\mathcal{S}_n(t_0)$ gives

$$\Pr\left((\hat{\mathbf{w}}, \hat{\mathbf{v}}, \hat{\theta}) \notin \mathcal{S}^\epsilon\right) \leq \mathbb{E}\left[\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr\left((\hat{\mathbf{w}}, \hat{\mathbf{v}}, \hat{\theta}) \notin \mathcal{S}^\epsilon \mid \mathcal{G}_n\right)\right] + \Pr(\neg \mathcal{S}_n(t_0)).$$

The structural concentration bound for $\mathcal{S}_n(t_0)$ gives the unconditional estimate after adjusting constants. $\square$

B.10 Convergence on test functions

At last, we translate the localisation of the PO in (63) into the convergence on test function result of Theorem 1. The main argument relies on the conditional spherical representation of the signal-orthogonal component of the PO in (63), which extends the distributional equivalence of Zhao *et al.* (2022) Proposition 2.1 and which we subsequently replace by a Gaussian surrogate $\mathbf{z}_p$ and propagate through the pseudo-Lipschitz test functions.

For all sufficiently large n , let $(\hat{\mathbf{v}}, \hat{\mathbf{w}}, \hat{\theta}, \hat{\boldsymbol{\eta}})$ be the minimising variables fixed in Proposition B.3. By Lemma 9 they are unique on $\mathcal{E}_n$. Set

$$\hat{\boldsymbol{\beta}} = \mathbf{B}\hat{\mathbf{v}} + \mathbf{E}\hat{\mathbf{w}}. \quad (120)$$

Further, let $\hat{\sigma} = \|\hat{\mathbf{w}}\|_2/\sqrt{p}$ and introduce the notation $\mathbf{3} = (\hat{\mathbf{v}}, \hat{\theta}, \hat{\boldsymbol{\eta}}, \hat{\sigma})$. Since $s = \text{rank}(\mathbf{B}) \leq 2$ and $p > 2$, the space $\text{range}(\mathbf{B})^\perp$ is nontrivial. Let $\boldsymbol{\xi}_n \sim \mathcal{N}(\mathbf{0}_p, \mathbf{I}_p)$ be independent of the original variables, and define

$$\mathbf{d} = \begin{cases} \frac{\mathbf{E}\hat{\mathbf{w}}}{\|\hat{\mathbf{w}}\|_2}, & \hat{\sigma} > 0, \\ \frac{\mathbf{P}^\perp \boldsymbol{\xi}_n}{\|\mathbf{P}^\perp \boldsymbol{\xi}_n\|_2}, & \hat{\sigma} = 0, \quad \|\mathbf{P}^\perp \boldsymbol{\xi}_n\|_2 > 0, \\ \mathbf{E}\mathbf{e}_1, & \hat{\sigma} = 0, \quad \|\mathbf{P}^\perp \boldsymbol{\xi}_n\|_2 = 0, \end{cases} \quad (121)$$

where $\mathbf{e}_1$ is the first standard basis vector of $\mathbb{R}^{p-s}$. Then $\mathbf{d} \in \text{range}(\mathbf{B})^\perp$, $\|\mathbf{d}\|_2 = 1$, and

$$\hat{\boldsymbol{\beta}} = \mathbf{B}\hat{\mathbf{v}} + \hat{\sigma}\sqrt{p}\mathbf{d}. \quad (122)$$

Writing $\mathbb{S}_{\mathbf{B}}^\perp = \{\mathbf{x} \in \text{range}(\mathbf{B})^\perp : \|\mathbf{x}\|_2 = 1\}$, Lemma 35 shows that, conditionally on $\mathcal{C}_n$, the direction $\mathbf{d}$ is uniform on $\mathbb{S}_{\mathbf{B}}^\perp$ and independent of $\mathbf{3}$. Now, define the Gaussian surrogate $\mathbf{z}_p$ for the noise $\sqrt{p}\mathbf{d}$, namely

$$\mathbf{z}_p = \|\mathbf{P}^\perp \boldsymbol{\xi}_n\|_2 \mathbf{d} + \mathbf{P}\boldsymbol{\xi}_n. \quad (123)$$

On $\mathcal{E}_\sigma = \{\hat{\sigma} > 0\}$, Lemma 35 and the independence of $\boldsymbol{\xi}_n$ imply that, conditionally on $\sigma(\mathcal{C}_n, \mathbf{3})$, the vector $\mathbf{d}$ is uniform on the unit sphere of $\text{range}(\mathbf{B})^\perp$ and independent of $\boldsymbol{\xi}_n$. The Gaussian polar decomposition (Vershynin 2018 Exercise 3.3.7) gives $\|\mathbf{P}^\perp \boldsymbol{\xi}_n\|_2 \mathbf{d} \sim \mathcal{N}(\mathbf{0}_p, \mathbf{P}^\perp)$ independently of $\mathbf{P}\boldsymbol{\xi}_n \sim \mathcal{N}(\mathbf{0}_p, \mathbf{P})$ on $\mathcal{E}_\sigma$, while (121) gives $\mathbf{z}_p = \boldsymbol{\xi}_n$ on $\neg \mathcal{E}_\sigma$. Since $\mathcal{E}_\sigma \in \sigma(\mathbf{3})$, both branches have the same conditional law, so

$$\mathbf{z}_p \mid \mathcal{C}_n, \mathbf{3} \sim \mathcal{N}(\mathbf{0}_p, \mathbf{I}_p), \quad (124)$$

and $\mathbf{z}_p$ is independent of $\sigma(\mathcal{C}_n, \mathbf{3})$. In particular, it is standard Gaussian conditionally on $\mathcal{G}_n$.

Moreover, orthogonality in (123) gives

$$\frac{1}{p} \|\sqrt{p}\mathbf{d} - \mathbf{z}_p\|_2^2 = (1 - \|\mathbf{P}^\perp \boldsymbol{\xi}_n\|_2 / \sqrt{p})^2 + \|\mathbf{P} \boldsymbol{\xi}_n\|_2^2 / p.$$

Conditionally on $\mathcal{G}_n$, the two norms are Gaussian norms in dimensions $p - s$ and s , respectively, with the second equal to zero when $s = 0$. Since $s \leq 2$ and $|1 - \sqrt{(p-s)/p}| \leq 2/p$, Gaussian norm concentration (Vershynin 2018 Theorem 3.1.1) and a union bound give a universal constant $c > 0$ such that, for every $\delta \in (0, 1]$ and all sufficiently large n ,

$$\Pr(\|\sqrt{p}\mathbf{d} - \mathbf{z}_p\|_2 / \sqrt{p} > \delta \mid \mathcal{G}_n) \leq 4 \exp\{-c\delta^2 p\} \quad \text{almost surely.} \quad (125)$$

Now, fix a pseudo-Lipschitz function $\psi : \mathbb{R}^3 \rightarrow \mathbb{R}$ of order two with Lipschitz-constant L_ψ . With $\boldsymbol{\beta}^* = \mathbf{B}\mathbf{v}^*$ from (9), write $\mathbf{r}_p = \hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*$ and define the empirical averages

$$\begin{aligned} T_p^{\text{DY}} &= \frac{1}{p} \sum_{j=1}^p \psi(\hat{\boldsymbol{\beta}}_j^{\text{DY}} - \boldsymbol{\beta}_j^*, \boldsymbol{\beta}_{0,j}, \boldsymbol{\beta}_{P,j}), & T_p &= \frac{1}{p} \sum_{j=1}^p \psi([\mathbf{r}_p]_j, \boldsymbol{\beta}_{0,j}, \boldsymbol{\beta}_{P,j}), \\ T_p^z &= \frac{1}{p} \sum_{j=1}^p \psi(\sigma^*[\mathbf{z}_p]_j, \boldsymbol{\beta}_{0,j}, \boldsymbol{\beta}_{P,j}), & \bar{T}_p &= \frac{1}{p} \sum_{j=1}^p \mathbb{E}_G[\psi(\sigma^*G, \boldsymbol{\beta}_{0,j}, \boldsymbol{\beta}_{P,j})], \end{aligned}$$

together with the limit $T = \mathbb{E}[\psi(\sigma^*G, \bar{\boldsymbol{\beta}}_0, \bar{\boldsymbol{\beta}}_P)]$, where $(\bar{\boldsymbol{\beta}}_0, \bar{\boldsymbol{\beta}}_P) \sim \pi_{\{\boldsymbol{\beta}_0, \boldsymbol{\beta}_P\}}$ and $G \sim \mathcal{N}(0, 1)$ is independent of $(\bar{\boldsymbol{\beta}}_0, \bar{\boldsymbol{\beta}}_P)$. T_p^{DY} is the average from (51) that we wish to link to T . We shall do so by arguing that (i) $T_p^{\text{DY}} = T_p$ on the MDYPL containment event, (ii) $\mathbf{r}_p$ is close to $\sigma^*\mathbf{z}_p$ in normalised Euclidean norm, by PO localisation and the surrogate bound, (iii) this transfers to $|T_p - T_p^z|$ by pseudo-Lipschitz stability, while (124) identifies $\bar{T}_p$ as the $\mathcal{G}_n$ -conditional expectation of T_p^z , around which it concentrates, and (iv) $\bar{T}_p$ converges to T by the W_2 convergence of the signal array assumed in Section 2

Thus, for $\epsilon > 0$, define

$$\begin{aligned} \mathcal{E}_n^{\text{vec}}(\epsilon) &= \{\|\mathbf{r}_p - \sigma^*\mathbf{z}_p\|_2 / \sqrt{p} \leq \epsilon\}, \\ \mathcal{E}_n^{\text{PL}}(\epsilon) &= \{|T_p - T_p^z| \leq \epsilon\}, \\ \mathcal{E}_n^G(\epsilon) &= \{|\bar{T}_p - T_p^z| \leq \epsilon\}. \end{aligned}$$

We bound the four events in turn, conditionally on $\mathcal{G}_n$ where a conditional bound is available, and on the structural event $\mathcal{S}_n(t)$ of (83), on which all conditional constants are deterministic. Since $t_* \leq 1$, on $\mathcal{S}_n(t)$ with $t \in (0, t_*]$, we record that

$$\frac{1}{p} \sum_{j=1}^p (\boldsymbol{\beta}_{0,j}^2 + \boldsymbol{\beta}_{P,j}^2) = \text{trace}(\boldsymbol{\Gamma}_p) \leq \text{trace}(\boldsymbol{\Gamma}) + 2\|\boldsymbol{\Gamma}_p - \boldsymbol{\Gamma}\|_2 \leq \text{trace}(\boldsymbol{\Gamma}) + 2 = M_B, \quad (126)$$

which we will use repeatedly in our error expansions. By Proposition B.3 on $\mathcal{M}_n \cap \mathcal{E}_n$, $\hat{\theta} = \hat{\theta}^{\text{DY}}$, $\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^{\text{DY}}$, $\hat{\boldsymbol{\eta}} = \hat{\boldsymbol{\eta}}^{\text{DY}}$, so that $T_p^{\text{DY}} = T_p$ with $\Pr(\neg(\mathcal{M}_n \cap \mathcal{E}_n)) \leq C \exp\{-cn\}$.

Next, we work with the PO minimiser of (62) in the form of (122), where we replace the noise $\mathbf{d}$ by the Gaussian surrogate $\mathbf{z}_p$ from (123) to get

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^* - \sigma^*\mathbf{z}_p = \mathbf{B}(\hat{\mathbf{v}} - \mathbf{v}^*) + \sqrt{p}(\hat{\sigma} - \sigma^*)\mathbf{d} + \sigma^*(\sqrt{p}\mathbf{d} - \mathbf{z}_p).$$

Now, using that $\mathbf{B}^\top \mathbf{B} = p\boldsymbol{\Gamma}_p$, $\|\mathbf{d}\|_2 = 1$, and the operator Hölder continuity of the positive-semidefinite square root (Bhatia 1997 Theorem X.1.1), we obtain the upper bound

$$\frac{1}{\sqrt{p}} \|\mathbf{r}_p - \sigma^*\mathbf{z}_p\|_2 \leq \left\| \boldsymbol{\Gamma}_p^{1/2} \hat{\mathbf{v}} - \mathbf{u}^* \right\|_2 + \|\boldsymbol{\Gamma}_p - \boldsymbol{\Gamma}\|_2^{1/2} \|\mathbf{v}^*\|_2 + |\hat{\sigma} - \sigma^*| + \frac{\sigma^*}{\sqrt{p}} \|\sqrt{p}\mathbf{d} - \mathbf{z}_p\|_2. \quad (127)$$

For $\epsilon \in (0, 1]$, set

$$\epsilon' = \frac{\epsilon}{3 + \sigma^*}, \quad t_\epsilon = \min \left\{ t_0(\epsilon'), \frac{(\epsilon')^2}{(1 + \|\mathbf{v}^*\|_2)^2}, t_* \right\}, \quad (128)$$

with $t_0(\epsilon')$ as in Proposition B.8. On $\mathcal{S}_n(t_\epsilon)$, we have $\|\Gamma_p - \Gamma\|_2^{1/2} \|\mathbf{v}^*\|_2 \leq \epsilon'$ and on the localisation event $\{(\hat{\mathbf{w}}, \hat{\mathbf{v}}, \hat{\theta}) \in \mathcal{S}^{\epsilon'}\}$ of Proposition B.8, the first and third terms of (127) are each less than ϵ' . Finally, on $\{\|\sqrt{p}\mathbf{d} - \mathbf{z}_p\|_2/\sqrt{p} \leq \epsilon'\}$, the last term is at most $\sigma^* \epsilon'$. Hence

$$\mathcal{S}_n(t_\epsilon) \cap \{(\hat{\mathbf{w}}, \hat{\mathbf{v}}, \hat{\theta}) \in \mathcal{S}^{\epsilon'}\} \cap \{\|\sqrt{p}\mathbf{d} - \mathbf{z}_p\|_2/\sqrt{p} \leq \epsilon'\} \subseteq \mathcal{E}_n^{\text{vec}}(\epsilon).$$

Since $t_\epsilon \leq t_0(\epsilon')$, Proposition B.8 (125) with $\delta = \epsilon'$, and a union bound give for all n large enough,

$$\mathbb{1}_{\mathcal{S}_n(t_\epsilon)} \Pr(\neg \mathcal{E}_n^{\text{vec}}(\epsilon) \mid \mathcal{G}_n) \leq C_\epsilon \exp\{-c_\epsilon n\} + 4 \exp\{-c(\epsilon')^2 p\}. \quad (129)$$

Now let us translate this localisation and noise replacement into similarity of the averages T_p and $T_p^{\mathbf{z}}$. Fix a precision level $u > 0$ and with M_B from (126), set

$$M_z = 2(\sigma^*)^2 + M_B, \quad M_x = 2 + 4(\sigma^*)^2 + M_B, \quad L_0 = \{3(1 + M_x + M_z)\}^{1/2}, \quad \delta_u = \min \left\{ 1, \frac{u}{2(1 + L_\psi)L_0} \right\},$$

and define t_u as the t from (128) at the tolerance level $\epsilon = \delta_u$. By (124) and Gaussian norm concentration (Vershynin 2018 Theorem 3.1.1),

$$\Pr \left(\frac{1}{p} \|\mathbf{z}_p\|_2^2 > 2 \mid \mathcal{G}_n \right) \leq 2 \exp\{-cp\}.$$

Work on $\mathcal{S}_n(t_u) \cap \mathcal{E}_n^{\text{vec}}(\delta_u) \cap \{p^{-1} \|\mathbf{z}_p\|_2^2 \leq 2\}$, and collect the arguments of the two averages into

$$\mathbf{a}_j = ([\mathbf{r}_p]_j, \beta_{0,j}, \beta_{P,j}), \quad \mathbf{b}_j = (\sigma^*[\mathbf{z}_p]_j, \beta_{0,j}, \beta_{P,j}), \quad j = 1, \dots, p,$$

so that $T_p = p^{-1} \sum_{j=1}^p \psi(\mathbf{a}_j)$, $T_p^{\mathbf{z}} = p^{-1} \sum_{j=1}^p \psi(\mathbf{b}_j)$ and

$$\frac{1}{p} \sum_{j=1}^p \|\mathbf{a}_j - \mathbf{b}_j\|_2^2 = \frac{1}{p} \|\mathbf{r}_p - \sigma^* \mathbf{z}_p\|_2^2,$$

which is controlled by $\mathcal{E}_n^{\text{vec}}(\delta_u)$. It remains to bound the two empirical second moments. Splitting the squared norms into their surrogate and signal parts and using (126) together with $p^{-1} \|\mathbf{z}_p\|_2^2 \leq 2$,

$$\frac{1}{p} \sum_{j=1}^p \|\mathbf{b}_j\|_2^2 = (\sigma^*)^2 \frac{\|\mathbf{z}_p\|_2^2}{p} + \frac{1}{p} \sum_{j=1}^p (\beta_{0,j}^2 + \beta_{P,j}^2) \leq 2(\sigma^*)^2 + M_B = M_z.$$

The same split gives that, upon $\mathcal{E}_n^{\text{vec}}(\delta_u) \cap \{p^{-1} \|\mathbf{z}_p\|_2^2 \leq 2\}$,

$$\begin{aligned} \frac{1}{p} \sum_{j=1}^p \|\mathbf{a}_j\|_2^2 &= \frac{1}{p} \|\mathbf{r}_p\|_2^2 + \frac{1}{p} \sum_{j=1}^p (\beta_{0,j}^2 + \beta_{P,j}^2) \\ &\leq \frac{2}{p} \|\mathbf{r}_p - \sigma^* \mathbf{z}_p\|_2^2 + \frac{2(\sigma^*)^2}{p} \|\mathbf{z}_p\|_2^2 + \frac{1}{p} \sum_{j=1}^p (\beta_{0,j}^2 + \beta_{P,j}^2) \\ &\leq 2\delta_u^2 + 4(\sigma^*)^2 + M_B \\ &\leq 2 + 4(\sigma^*)^2 + M_B \\ &= M_x, \end{aligned} \quad (130)$$

where we used that $\mathbf{r}_p = (\mathbf{r}_p - \sigma^* \mathbf{z}_p) + \sigma^* \mathbf{z}_p$ and $(a + b)^2 \leq 2a^2 + 2b^2$. Consequently,

$$\begin{aligned}
|T_p - T_p^{\mathbf{z}}| &= \left| \frac{1}{p} \sum_{j=1}^p \psi(\mathbf{a}_j) - \psi(\mathbf{b}_j) \right| \\
&\leq \frac{1}{p} \sum_{j=1}^p |\psi(\mathbf{a}_j) - \psi(\mathbf{b}_j)| \\
&\leq L_\psi \frac{1}{p} \sum_{j=1}^p (1 + \|\mathbf{a}_j\|_2 + \|\mathbf{b}_j\|_2) \|\mathbf{a}_j - \mathbf{b}_j\|_2 \\
&\leq L_\psi \left[\frac{1}{p} \sum_{j=1}^p (1 + \|\mathbf{a}_j\|_2 + \|\mathbf{b}_j\|_2)^2 \right]^{1/2} \left[\frac{1}{p} \sum_{j=1}^p \|\mathbf{a}_j - \mathbf{b}_j\|_2^2 \right]^{1/2} \\
&\leq L_\psi \{3(1 + M_x + M_z)\}^{1/2} \frac{1}{\sqrt{p}} \|\mathbf{r}_p - \sigma^* \mathbf{z}_p\|_2 \\
&= L_\psi L_0 \frac{1}{\sqrt{p}} \|\mathbf{r}_p - \sigma^* \mathbf{z}_p\|_2 \leq L_\psi L_0 \delta_u \\
&\leq L_\psi L_0 \frac{u}{2(1 + L_\psi) L_0} \\
&= \frac{L_\psi}{1 + L_\psi} \frac{u}{2} \\
&\leq \frac{u}{2},
\end{aligned} \tag{131}$$

where we used triangle inequality, the pseudo-Lipschitz property of ψ termwise, followed by Cauchy-Schwarz (Rudin, 1976, Theorem 1.37(d)), then $(1 + a + b)^2 \leq 3(1 + a^2 + b^2)$ with the two preceding moment bounds, then the definition of L_0 and $\mathcal{E}_n^{\text{vec}}(\delta_u)$, and finally the definition of δ_u . Therefore,

$$\mathcal{S}_n(t_u) \cap \mathcal{E}_n^{\text{vec}}(\delta_u) \cap \left\{ \|\mathbf{z}_p\|_2^2 / p \leq 2 \right\} \subseteq \mathcal{E}_n^{\text{PL}}(u/2).$$

A union bound with (129) and $p/n \rightarrow \kappa \in (0, 1)$ gives, after adjusting constants, for all n large enough,

$$\mathbb{1}_{\mathcal{S}_n(t_u)} \Pr(-\mathcal{E}_n^{\text{PL}}(u/2) \mid \mathcal{G}_n) \leq C_u \exp\{-c_u n\}. \tag{132}$$

We next prove the concentration of $T_p^{\mathbf{z}}$ around $\bar{T}_p$. By (124), conditionally on $\mathcal{G}_n$ the coordinates of $\mathbf{z}_p$ are independent standard Gaussian variables, so the summands

$$Z_{p,j} = \psi(\sigma^*[\mathbf{z}_p]_j, \beta_{0,j}, \beta_{P,j}) - \mathbb{E}_G[\psi(\sigma^* G, \beta_{0,j}, \beta_{P,j})], \quad j = 1, \dots, p,$$

are independent with mean zero. The part of ψ that does not depend on the surrogate cancels under centring, i.e. for

$$D_{p,j}(z) = \psi(\sigma^* z, \beta_{0,j}, \beta_{P,j}) - \psi(0, \beta_{0,j}, \beta_{P,j}),$$

we have $Z_{p,j} = D_{p,j}([\mathbf{z}_p]_j) - \mathbb{E}_G[D_{p,j}(G)]$, and pseudo-Lipschitzness applied to $(\sigma^* z, \beta_{0,j}, \beta_{P,j})$ and $(0, \beta_{0,j}, \beta_{P,j})$ gives

$$|D_{p,j}(z)| \leq L_\psi \sigma^* \max\{\sigma^*, 2\} (|z| + z^2 + \|(\beta_{0,j}, \beta_{P,j})\|_2 |z|).$$

Sub-exponential centring (Vershynin, 2018, Exercise 2.7.10) and the subexponential bounds $\|G\|_{\psi_1} < \infty$ and $\|G^2\|_{\psi_1} < \infty$ (cf. Vershynin, 2018, Lemmas 2.7.6&2.7.7), give, conditional on $\mathcal{G}_n$,

$$\begin{aligned}
\|Z_{p,j}\|_{\psi_1} &= \|D_{p,j}([\mathbf{z}_p]_j) - \mathbb{E}_G[D_{p,j}(G)]\|_{\psi_1} \\
&\leq C \|D_{p,j}([\mathbf{z}_p]_j)\|_{\psi_1} \\
&\leq C L_\psi \sigma^* \max\{2, \sigma^*\} \left(\|[\mathbf{z}_p]_j\|_{\psi_1} + \|[\mathbf{z}_p]_j^2\|_{\psi_1} + \|(\beta_{0,j}, \beta_{P,j})\|_2 \|[\mathbf{z}_p]_j\|_{\psi_1} \right) \\
&\leq C_\psi (1 + \|(\beta_{0,j}, \beta_{P,j})\|_2) \\
&= K_{p,j},
\end{aligned}$$

where C_ψ depends only on L_ψ , σ^* and the subexponential norms of $|G|, G^2$.

Now, on $\mathcal{S}_n(t_u)$, the structural moment bound in (126) gives $\sum_{j=1}^p K_{p,j}^2 \leq C_{\psi, M_B} p$ and, since $\max_j \|(\beta_{0,j}, \beta_{P,j})\|_2^2 \leq M_B p$, also $\max_j K_{p,j} \leq C_{\psi, M_B} \sqrt{p}$. Thus, Bernstein's inequality for independent centred sub-exponential variables (Vershynin, 2018, Theorem 2.8.1), applied conditionally on $\mathcal{G}_n$, gives

$$\begin{aligned} \mathbb{1}_{\mathcal{S}_n(t_u)} \Pr(-\mathcal{E}_n^G(u/2) \mid \mathcal{G}_n) &= \mathbb{1}_{\mathcal{S}_n(t_u)} \Pr\left(|T_p^z - \bar{T}_p| > \frac{u}{2} \mid \mathcal{G}_n\right) \\ &= \mathbb{1}_{\mathcal{S}_n(t_u)} \Pr\left(\left|\sum_{j=1}^p Z_{p,j}\right| > \frac{pu}{2} \mid \mathcal{G}_n\right) \\ &\leq 2 \exp\left\{-c \min\left(\frac{(pu/2)^2}{\sum_{j=1}^p K_{p,j}^2}, \frac{pu/2}{\max_{1 \leq j \leq p} K_{p,j}}\right)\right\} \\ &\leq 2 \exp\left\{-c \min\left(C_{\psi, M_B}^{-1} pu^2, C_{\psi, M_B}^{-1} \sqrt{p} u\right)\right\} \\ &\leq C_u \exp\{-c_u \sqrt{n}\}. \end{aligned} \tag{133}$$

The remaining step involves no event: both $\bar{T}_p$ and T are $\mathcal{G}_n$ -measurable, so conditioning on $\mathcal{G}_n$ is vacuous for it. By Assumption (ii) the empirical measures $\pi_p = p^{-1} \sum_{j=1}^p \delta_{(\beta_{0,j}, \beta_{P,j})}$ converge almost surely in W_2 to $\pi_{\{\beta_0, \beta_P\}}$. Writing $\varphi(\mathbf{b}) = \mathbb{E}_G[\psi(\sigma^* G, \mathbf{b})]$ and $A_\psi = |\psi(\mathbf{0})| + 2L_\psi$, pseudo-Lipschitzness at $(0, \mathbf{0})$ with $t + t^2 \leq 1 + 2t^2$ gives $|\psi(z, \mathbf{b})| \leq A_\psi(1 + z^2 + \|\mathbf{b}\|_2^2)$ and hence, since $\mathbb{E}_G[(\sigma^* G)^2] = (\sigma^*)^2$, $|\varphi(\mathbf{b})| \leq A_\psi(1 + (\sigma^*)^2)(1 + \|\mathbf{b}\|_2^2)$, while dominated convergence gives continuity of φ . Continuous functions of at most quadratic growth are admissible test functions for W_2 convergence (Villani, 2009, Definition 6.8(iv) & Theorem 6.9), so that

$$\bar{T}_p = \int \varphi d\pi_p \xrightarrow{\text{a.s.}} \int \varphi d\pi_{\{\beta_0, \beta_P\}} = T. \tag{134}$$

The four bounds and (134) combine as follows.

Proposition B.9 (Convergence on test functions). *Assume the conditions of Section 2 and fix a pseudo-Lipschitz function $\psi : \mathbb{R}^3 \rightarrow \mathbb{R}$ of order two with constant L_ψ .*

(i) *For every $u > 0$ there exist $N \in \mathbb{N}$ and constants $C_u, c_u > 0$ such that, for $n > N$,*

$$\Pr(|T_p - \bar{T}_p| > u) \leq C_u \exp\{-c_u \sqrt{n}\},$$

where the constants c_u, C_u and N depend only on $u, L_\psi, \kappa, \alpha, \Gamma, \theta_0^, \theta_P^*$ and on the constants in Section 2*

(ii)

$$T_p^{DY} \xrightarrow{\text{a.s.}} T.$$

Proof. We prove each claim in turn.

(i) For part (i), on $\mathcal{E}_n^{\text{PL}}(u/2) \cap \mathcal{E}_n^G(u/2)$,

$$|T_p - \bar{T}_p| \leq |T_p - T_p^z| + |T_p^z - \bar{T}_p| \leq u,$$

the two summands being bounded by $u/2$ on $\mathcal{E}_n^{\text{PL}}(u/2)$ and $\mathcal{E}_n^G(u/2)$ respectively. Since $\mathcal{S}_n(t_u)$ is $\mathcal{G}_n$ -measurable, taking expectations in (132) and (133) and splitting over $\mathcal{S}_n(t_u)$ gives

$$\Pr(|T_p - \bar{T}_p| > u) \leq C_u \exp\{-c_u n\} + C'_u \exp\{-c'_u \sqrt{n}\} + \Pr(\neg \mathcal{S}_n(t_u)),$$

where the first term is the pseudo-Lipschitz bound (132) and the second the Gaussian bound (133). The structural concentration bound following (83) gives $\Pr(\neg \mathcal{S}_n(t_u)) \leq C \exp\{-cn\}$. Since $\exp\{-cn\} \leq \exp\{-c\sqrt{n}\}$, this proves part (i) after adjusting constants.

(ii) Part (ii) follows from two applications of the first Borel–Cantelli lemma (cf. [Shiryaev 2016](#), p. 308). First, the bound in part (i) is summable, so applying the lemma along $u = 1/m$ and intersecting over $m \in \mathbb{N}$ gives $T_p - \bar{T}_p \rightarrow 0$ almost surely. Second, $\Pr(\neg(\mathcal{M}_n \cap \mathcal{E}_n)) \leq C \exp\{-cn\}$ is summable, so $\mathcal{M}_n \cap \mathcal{E}_n$ holds eventually almost surely and hence $T_p^{\text{DY}} = T_p$ eventually almost surely. Combining the two with [\(134\)](#) proves part (ii). $\square$

We can finally complete the proof of Theorem [1](#)

Proof of Theorem [1](#). For each $m \in \mathbb{N}$, the unconditional bound in Proposition [B.8](#) applied with $\epsilon = 1/m$, is summable. The first Borel–Cantelli lemma (cf. [Shiryaev 2016](#), p. 308), followed by intersection over $m \in \mathbb{N}$, therefore gives

$$\hat{\sigma} \xrightarrow{\text{a.s.}} \sigma^*, \quad \hat{\mathbf{u}} = \mathbf{\Gamma}_p^{1/2} \hat{\mathbf{v}} \xrightarrow{\text{a.s.}} \mathbf{u}^*, \quad \hat{\theta} \xrightarrow{\text{a.s.}} \theta^*. \quad (135)$$

Assumption [\(i\)](#) and the first Borel–Cantelli lemma give $\mathbf{\Gamma}_p \rightarrow \mathbf{\Gamma}$ almost surely. Since $\mathbf{\Gamma}^{1/2} \mathbf{v}^* = \mathbf{u}^*$, the operator Hölder continuity of the positive-semidefinite square root ([Bhatia, 1997](#), Theorem X.1.1) gives

$$\mathbf{\Gamma}_p^{1/2} \mathbf{v}^* \xrightarrow{\text{a.s.}} \mathbf{u}^*. \quad (136)$$

By Proposition [B.3](#), $\Pr(\neg(\mathcal{M}_n \cap \mathcal{E}_n)) \leq C \exp\{-cn\}$ is summable, so the first Borel–Cantelli lemma gives that $\mathcal{M}_n \cap \mathcal{E}_n$ holds eventually almost surely, and hence

$$\hat{\theta} = \hat{\theta}^{\text{DY}}, \quad \hat{\beta} = \hat{\beta}^{\text{DY}} \quad \text{eventually almost surely.} \quad (137)$$

Part (i) is the intercept component of [\(135\)](#) combined with [\(137\)](#). Part (ii) is Proposition [B.9\(ii\)](#), because T_p^{DY} is the empirical average in [\(51\)](#). $\square$

The same argument records the following vector-level consequence.

Lemma 1 (Gaussian ℓ_2 -coupling for the MDYPL slope). *On the product extension of Section [B.10](#), the vector $\mathbf{z}_p$ in [\(123\)](#) satisfies $\mathbf{z}_p \sim \mathcal{N}(\mathbf{0}_p, \mathbf{I}_p)$, is independent of $\sigma(\mathcal{C}_n, \mathbf{3})$, and*

$$\frac{1}{\sqrt{p}} \left\| \hat{\beta}^{\text{DY}} - \mathbf{B} \mathbf{v}^* - \sigma^* \mathbf{z}_p \right\|_2 \xrightarrow{\text{a.s.}} 0. \quad (138)$$

Proof. The law and independence of $\mathbf{z}_p$ are [\(124\)](#). For every fixed $\epsilon \in (0, 1]$, with ϵ' as in [\(128\)](#), taking expectations in [\(129\)](#), splitting over $\mathcal{S}_n(t_\epsilon)$, and using the structural concentration bound following [\(83\)](#) gives

$$\Pr(\neg \mathcal{E}_n^{\text{vec}}(\epsilon)) \leq C_\epsilon \exp\{-c_\epsilon n\} + 4 \exp\{-c(\epsilon')^2 p\} + \Pr(\neg \mathcal{S}_n(t_\epsilon)).$$

Since $p/n \rightarrow \kappa \in (0, 1)$, the right-hand side is summable. The first Borel–Cantelli lemma, applied along $\epsilon = 1/m$ and followed by intersection over $m \in \mathbb{N}$, therefore gives

$$\frac{1}{\sqrt{p}} \left\| \hat{\beta} - \mathbf{B} \mathbf{v}^* - \sigma^* \mathbf{z}_p \right\|_2 \xrightarrow{\text{a.s.}} 0.$$

Finally, [\(137\)](#) gives $\hat{\beta} = \hat{\beta}^{\text{DY}}$ eventually almost surely, proving [\(138\)](#). $\square$

C Proofs for Section [4](#)

C.1 Proof of Theorem [2](#)

Theorem 2. *Assume the conditions of Section [2](#) and fix $\alpha \in (0, 1)$. The statements below hold for $j = \text{DY}$ and, when $v_{1,\alpha}^* \neq 0$, also for $j = \text{adj}$.*

(i) Conditionally on $\mathcal{F}_n$, almost surely,

$$(\eta_0(\mathbf{x}_{\text{new}}), \hat{\eta}_\alpha^j(\mathbf{x}_{\text{new}}), Y_{\text{new}}) \xrightarrow{d} (S_0, S_\alpha^j, Y). \quad (139)$$

The convergence also holds unconditionally.

(ii) Suppose that $\text{var}(S_\alpha^j) > 0$. Then, for every fixed $h \in \mathfrak{R}$ and $a \in \{-1, 1\}$,

$$\mathcal{E}_{n,\alpha}^{j,a}(h) \xrightarrow{\text{a.s.}} \mathcal{E}_\alpha^{j,a}(h), \quad (140)$$

where

$$\mathcal{E}_\alpha^{j,a}(h) = \mathbb{E} \left[\rho'(S_0) \Phi_N \left\{ \frac{a\{h - \bar{S}_\alpha^j(\mathbf{Q})\}}{\tau_\alpha^j} \right\} + \{1 - \rho'(S_0)\} \Phi_N \left\{ \frac{a\{\bar{S}_\alpha^j(\mathbf{Q}) - h\}}{\tau_\alpha^j} \right\} \right], \quad (141)$$

and which is understood by continuity if $\tau_\alpha = 0$. Moreover,

$$\Pr(Y_{\text{new}} \neq \hat{Y}_\alpha^{j,a}(\mathbf{x}_{\text{new}}; h)) = \mathbb{E} \{ \mathcal{E}_{n,\alpha}^{j,a}(h) \} \rightarrow \mathcal{E}_\alpha^{j,a}(h). \quad (142)$$

Proof. Fix $\alpha \in (0, 1)$ and assume throughout that $v_{1,\alpha}^* \neq 0$, so that $\check{\beta}_\alpha$ and $\hat{\eta}_\alpha^{\text{adj}}$ in Section 4.1 are defined. If $v_{1,\alpha}^* = 0$, the statements for $j = \text{adj}$ are void and the argument below applies after deleting every adj coordinate. Write

$$\beta_\alpha^* = v_{1,\alpha}^* \beta_0 + v_{2,\alpha}^* \beta_P = \mathbf{B} \mathbf{v}_\alpha^*, \quad \Delta_\alpha = \hat{\beta}_\alpha^{\text{DY}} - \beta_\alpha^*, \quad \mathbf{M}_\alpha = [\beta_0 \quad \hat{\beta}_\alpha^{\text{DY}} \quad \check{\beta}_\alpha] \in \mathbb{R}^{p \times 3}, \quad (143)$$

so that, by definition of the limiting logits in Section 4.1, $\hat{\beta}_\alpha^{\text{DY}} = \mathbf{B} \mathbf{v}_\alpha^* + \Delta_\alpha$ and $\check{\beta}_\alpha = \beta_0 + \Delta_\alpha / v_{1,\alpha}^*$.

We shall prove each claim of the Theorem in turn.

(i) Conditional on $\mathcal{F}_n$, the scores $\mathbf{M}_\alpha^\top \mathbf{x}_{\text{new}}$ are multivariate Gaussian with variance-covariance matrix $\mathbf{M}_\alpha^\top \mathbf{M}_\alpha / p$, whose limit Σ_α we determine below. Theorem 1(ii), applied to the test functions $\psi(d, b_0, b_P) \in \{db_0, db_P, d^2\}$, gives

$$\frac{\beta_0^\top \Delta_\alpha}{p} \xrightarrow{\text{a.s.}} 0, \quad \frac{\beta_P^\top \Delta_\alpha}{p} \xrightarrow{\text{a.s.}} 0, \quad \frac{\|\Delta_\alpha\|_2^2}{p} \xrightarrow{\text{a.s.}} (\sigma_\alpha^*)^2,$$

while Assumption (i) and the first Borel–Cantelli lemma (cf. Shiryaev [2016], p. 308) give $\Gamma_p \xrightarrow{\text{a.s.}} \Gamma$. Since $\mathbf{v}_\alpha^{*\top} \Gamma \mathbf{v}_\alpha^* = \|\mathbf{u}_\alpha^*\|_2^2$, expanding $\mathbf{M}_\alpha^\top \mathbf{M}_\alpha$ through (143) yields

$$\frac{\mathbf{M}_\alpha^\top \mathbf{M}_\alpha}{p} \xrightarrow{\text{a.s.}} \Sigma_\alpha = \begin{bmatrix} \gamma^2 & c_\alpha & \gamma^2 \\ c_\alpha & \|\mathbf{u}_\alpha^*\|_2^2 + (\sigma_\alpha^*)^2 & c_\alpha + \frac{(\sigma_\alpha^*)^2}{v_{1,\alpha}^*} \\ \gamma^2 & c_\alpha + \frac{(\sigma_\alpha^*)^2}{v_{1,\alpha}^*} & \gamma^2 + \frac{(\sigma_\alpha^*)^2}{(v_{1,\alpha}^*)^2} \end{bmatrix}, \quad c_\alpha = \gamma^2 v_{1,\alpha}^* + \varphi v_{2,\alpha}^*. \quad (144)$$

By (16), $\Sigma_\alpha = \text{cov}(S_0, S_\alpha^{\text{DY}}, S_\alpha^{\text{adj}})$. Conditionally on $\mathcal{F}_n$, by definition of the limiting logits in Section 4.1 and $\mathbf{x}_{\text{new}} \sim \mathcal{N}(\mathbf{0}_p, p^{-1} \mathbf{I}_p)$ independent of $\mathcal{F}_n$, the score triple

$$\boldsymbol{\eta}_{n,\alpha} = (\eta_0(\mathbf{x}_{\text{new}}), \hat{\eta}_\alpha^{\text{DY}}(\mathbf{x}_{\text{new}}), \hat{\eta}_\alpha^{\text{adj}}(\mathbf{x}_{\text{new}}))^\top,$$

is Gaussian with mean $(\theta_0, \hat{\theta}_\alpha^{\text{DY}}, \theta_0)^\top$ and covariance matrix $\mathbf{M}_\alpha^\top \mathbf{M}_\alpha / p$. Assumption (iii) with the first Borel–Cantelli lemma and Theorem 1(i) give $(\theta_0, \hat{\theta}_\alpha^{\text{DY}}, \theta_0)^\top \xrightarrow{\text{a.s.}} (\theta_0^*, \theta_\alpha^*, \theta_0^*)^\top$. Gaussian characteristic functions are continuous in the mean vector and the covariance matrix, so (144) and thus Shiryaev [2016] Chapter 3.3, Theorem 1), conditionally on $\mathcal{F}_n$, almost surely,

$$\boldsymbol{\eta}_{n,\alpha} \xrightarrow{d} (S_0, S_\alpha^{\text{DY}}, S_\alpha^{\text{adj}})^\top. \quad (145)$$

Moreover, $\varepsilon_{\text{new}} \sim \text{Unif}([0, 1])$ is independent of $(\mathcal{F}_n, \mathbf{x}_{\text{new}})$ and ε is independent of $(\mathbf{Q}, G)$, so (145) holds jointly with ε_{new} appended in the left-hand side and ε in the right-hand side. The map

$$(t, s_1, s_2, e) \mapsto (t, s_1, s_2, \mathbb{1}\{e < \rho'(t)\}) , \quad (146)$$

does not depend on n and is continuous except on $\{e = \rho'(t)\}$, which has probability zero under the limiting law because ε is uniform and independent of S_0 . The continuous mapping theorem for almost everywhere continuous functions (Shiryaev [2016], Chapter 3.8, Theorem 2), applied to (146), therefore gives, conditionally on $\mathcal{F}_n$,

$$(\eta_0(\mathbf{x}_{\text{new}}), \hat{\eta}_{\alpha}^{\text{DY}}(\mathbf{x}_{\text{new}}), \hat{\eta}_{\alpha}^{\text{adj}}(\mathbf{x}_{\text{new}}), Y_{\text{new}}) \xrightarrow{d} (S_0, S_{\alpha}^{\text{DY}}, S_{\alpha}^{\text{adj}}, Y) . \quad (147)$$

Retaining coordinates (1, 2, 4) and (1, 3, 4) in (147) proves (139) for $j = \text{DY}$ and $j = \text{adj}$, respectively. For the unconditional statement, let f be bounded and continuous on $\mathbb{R}^3$. Then, by the convergence in distribution (Shiryaev [2016], Chapter 2.10, Definition 4), almost surely,

$$\mathbb{E} [f(\eta_0(\mathbf{x}_{\text{new}}), \hat{\eta}_{\alpha}^j(\mathbf{x}_{\text{new}}), Y_{\text{new}}) \mid \mathcal{F}_n] \longrightarrow \mathbb{E} [f(S_0, S_{\alpha}^j, Y)] ,$$

and these conditional expectations are bounded by $\|f\|_{\infty}$, so dominated convergence (Shiryaev [2016], Chapter 2.6, Theorem 3) and the tower property (Shiryaev [2016], Chapter 3.8 Equation (4)) give (139) unconditionally.

(ii) Fix $h \in \mathbb{R}$, $a \in \{-1, 1\}$ and $j \in \{\text{DY}, \text{adj}\}$. By (14) and (15),

$$\begin{aligned} \mathcal{E}_{n,\alpha}^{j,a}(h) &= \Pr(\{a(\hat{\eta}_{\alpha}^j(\mathbf{x}_{\text{new}}) - h) \leq 0\} \cap \{Y_{\text{new}} = 1\} \mid \mathcal{F}_n) \\ &\quad + \Pr(\{a(\hat{\eta}_{\alpha}^j(\mathbf{x}_{\text{new}}) - h) > 0\} \cap \{Y_{\text{new}} = 0\} \mid \mathcal{F}_n) \\ &= \mathbb{E}[Y_{\text{new}} \mathbb{1}\{a(\hat{\eta}_{\alpha}^j(\mathbf{x}_{\text{new}}) - h) \leq 0\} \mid \mathcal{F}_n] \\ &\quad + \mathbb{E}[(1 - Y_{\text{new}}) \mathbb{1}\{a(\hat{\eta}_{\alpha}^j(\mathbf{x}_{\text{new}}) - h) > 0\} \mid \mathcal{F}_n] \\ &= \mathbb{E}[\rho'(\eta_0(\mathbf{x}_{\text{new}})) \mathbb{1}\{a(\hat{\eta}_{\alpha}^j(\mathbf{x}_{\text{new}}) - h) \leq 0\} \mid \mathcal{F}_n] \\ &\quad + \mathbb{E}[\rho'(-\eta_0(\mathbf{x}_{\text{new}})) \mathbb{1}\{a(\hat{\eta}_{\alpha}^j(\mathbf{x}_{\text{new}}) - h) > 0\} \mid \mathcal{F}_n] \\ &= \mathbb{E}[\rho'(\eta_0(\mathbf{x}_{\text{new}})) \Pr(a(\hat{\eta}_{\alpha}^j(\mathbf{x}_{\text{new}}) - h) \leq 0 \mid \eta_0(\mathbf{x}_{\text{new}}), \mathcal{F}_n) \mid \mathcal{F}_n] \\ &\quad + \mathbb{E}[\rho'(-\eta_0(\mathbf{x}_{\text{new}})) \Pr(a(\hat{\eta}_{\alpha}^j(\mathbf{x}_{\text{new}}) - h) > 0 \mid \eta_0(\mathbf{x}_{\text{new}}), \mathcal{F}_n) \mid \mathcal{F}_n] , \end{aligned} \quad (148)$$

where the third equality follows from the law of total expectation with respect to $\sigma(\mathcal{F}_n, \mathbf{x}_{\text{new}}) \supseteq \mathcal{F}_n$ and using that $\mathbb{E}[Y_{\text{new}} \mid \mathcal{F}_n, \mathbf{x}_{\text{new}}] = \rho'(\eta_0(\mathbf{x}_{\text{new}}))$ and the last equality follows from the law of total expectation with respect to $\mathcal{F}_n \subseteq \sigma(\mathcal{F}_n, \eta_0(\mathbf{x}_{\text{new}})) \subseteq \sigma(\mathcal{F}_n, \mathbf{x}_{\text{new}})$. As argued in (i), the pair $(\eta_0(\mathbf{x}_{\text{new}}), \hat{\eta}_{\alpha}^j(\mathbf{x}_{\text{new}}))$ is bivariate Gaussian conditionally on $\mathcal{F}_n$, so the inner conditional probabilities in (148) are normal distribution functions, and (141) is the limiting form of exactly this expression. Observe that the third equality in (148) is of the form

$$\mathcal{E}_{n,\alpha}^{j,a}(h) = \mathbb{E}[g(\eta_0(\mathbf{x}_{\text{new}}), \hat{\eta}_{\alpha}^j(\mathbf{x}_{\text{new}}); h, a) \mid \mathcal{F}_n] , \quad g(t, s; h, a) = \rho'(t) \mathbb{1}\{a(s - h) \leq 0\} + \rho'(-t) \mathbb{1}\{a(s - h) > 0\} ,$$

The function $g(\cdot, \cdot; h, a)$ is bounded by one and continuous except when $s = h$. Since $\text{var}(S_{\alpha}^j) > 0$, this is a set of measure zero for the law of (S_0, S_{α}^j) . The continuous mapping theorem (Shiryaev [2016], Chapter 3.8, Theorem 2), followed by dominated convergence (cf. (Shiryaev [2016], Chapter 2.6, Theorem 3)), therefore gives, almost surely,

$$\mathcal{E}_{n,\alpha}^{j,a}(h) \longrightarrow \mathbb{E}[g(S_0, S_{\alpha}^j; h, a)] . \quad (149)$$

Repeating the last equality of (148) in the limit, now conditioning on $\mathbf{Q}$ and using the conditional score moments of Section 4.1 gives, when $\tau_{\alpha}^j > 0$,

$$\Pr(a(S_{\alpha}^j - h) \leq 0 \mid \mathbf{Q}) = \Phi_N \left\{ \frac{a\{h - \bar{S}_{\alpha}^j(\mathbf{Q})\}}{\tau_{\alpha}^j} \right\} , \quad \Pr(a(S_{\alpha}^j - h) > 0 \mid \mathbf{Q}) = \Phi_N \left\{ \frac{a\{\bar{S}_{\alpha}^j(\mathbf{Q}) - h\}}{\tau_{\alpha}^j} \right\} ,$$

and substituting into (149) proves (140) with limit (141). If $\tau_\alpha^j = 0$, the same formula is understood as the corresponding pointwise limit of the normal distribution function. Finally, by (15) and the law of total expectation (Shiryaev 2016 Chapter 3.8 Equation (4)),

$$\Pr(Y_{\text{new}} \neq \hat{Y}_\alpha^{j,a}(\mathbf{x}_{\text{new}}; h)) = \mathbb{E}[\mathcal{E}_{n,\alpha}^{j,a}(h)] .$$

The integrand satisfies $0 \leq \mathcal{E}_{n,\alpha}^{j,a}(h) \leq 1$ and, by (140), converges almost surely to $\mathcal{E}_\alpha^{j,a}(h)$ and dominated convergence (Shiryaev 2016 Chapter 2.6, Theorem 3) gives (142).

This concludes the proof. $\square$

C.2 Proof of Proposition 1

Proposition 1. *Under the Gaussian-score setup in (21), suppose that $\chi \neq 0$. Then*

$$\mathbf{1}\{\chi(S - h_S) > 0\} , \quad (150)$$

is the unique minimiser, up to almost-sure equivalence, of $\Pr(Y \neq g(S))$ over all $\sigma(S)$ -measurable functions $g : \mathbb{R} \rightarrow \{0, 1\}$.

Proof. We recall that throughout $\chi \neq 0$. By (24), the prediction rule in (150) is

$$g^*(s) = \mathbf{1}\{\pi_S(s) > 1/2\} = \mathbf{1}\{\chi(s - h_S) > 0\} .$$

Let $g : \mathbb{R} \rightarrow \{0, 1\}$ be Borel measurable. Conditioning on S and using that $\pi_S(S) = \Pr(Y = 1 | S)$, we have that

$$\Pr(Y \neq g(S) | S) = \pi_S(S) + \{1 - 2\pi_S(S)\}g(S) .$$

Subtracting the same identity for g^* and taking expectations,

$$\begin{aligned} \Pr(Y \neq g(S)) - \Pr(Y \neq g^*(S)) &= \mathbb{E}[\{1 - 2\pi_S(S)\}\{g(S) - g^*(S)\}] \\ &= \mathbb{E}[|2\pi_S(S) - 1| \mathbf{1}\{g(S) \neq g^*(S)\}] \geq 0 . \end{aligned}$$

The second equality holds because the integrand is zero when $g(S) = g^*(S)$, and otherwise either $\pi_S(S) > 1/2$ which then requires $g(S) - g^*(S) = -1$, or $\pi_S(S) \leq 1/2$ and $g(S) - g^*(S) = 1$, so that in both cases the integrand is $|2\pi_S(S) - 1|$. This proves optimality of g^* .

To see uniqueness, note that the strict monotonicity used to obtain (24) also gives

$$\{\pi_S(S) = 1/2\} = \{S = h_S\} .$$

Since $\nu > 0$ in (21), S is a nondegenerate Gaussian random variable, and thus the event $\{S = h_S\}$ has probability zero and $|2\pi_S(S) - 1| > 0$ almost surely. Hence

$$\Pr(Y \neq g(S)) = \Pr(Y \neq g^*(S))$$

if and only if $g(S) = g^*(S)$ almost surely.

This concludes the proof. $\square$

C.3 Proof of Proposition 2

Proposition 2. *Assume the conditions of Theorem 2 with $\gamma^2 > 0$. For $j \in \{\text{DY}, \text{adj}\}$ and every $\alpha \in (0, 1)$ for which S_α^j is defined and informative, consider*

$$\min_{a \in \{-1, 1\}} \inf_{h \in \mathbb{R}} \mathcal{E}_\alpha^{j,a}(h) = \mathcal{E}^*((R_\alpha^j)^2), \quad \mathcal{E}^*(r) = \mathbb{E} \left[\rho'(S_0) \Phi_N \left(\frac{h_r - S_0}{\varsigma_r} \right) + \{1 - \rho'(S_0)\} \Phi_N \left(\frac{S_0 - h_r}{\varsigma_r} \right) \right] , \quad (151)$$

for $\varsigma_r^2 = \gamma^2(1-r)/r$ and $h_r = -\theta_0^*(1-r)/r$, $r \in (0, 1)$ and where for $r = 1$, we set $\mathcal{E}^*(1) = \mathbb{E}[\min\{\rho'(S_0), 1 - \rho'(S_0)\}]$.

Then $\mathcal{E}^$ is strictly decreasing on $(0, 1]$, and when the respective scores are defined and informative,*

(i) up to ties, and if attained,

$$\arg \min_{\alpha \in (0,1)} \mathcal{E}^*((R_\alpha^j)^2) = \arg \max_{\alpha \in (0,1)} (R_\alpha^j)^2,$$

which for the adjusted logit, is the α that minimises slope error, i.e.

$$\arg \min_{\alpha \in (0,1)} \mathcal{E}^*((R_\alpha^{\text{adj}})^2) = \arg \min_{\alpha \in (0,1)} \frac{(\sigma_\alpha^*)^2}{(v_{1,\alpha}^*)^2}.$$

(ii) oracle adjustment lowers the optimised error if and only if $(R_\alpha^{\text{adj}})^2 > (R_\alpha^{\text{DY}})^2$, which holds whenever $\varphi = 0$, $v_{1,\alpha}^* \neq 0$ and $\delta^2(v_{2,\alpha}^*)^2 > 0$.

Proof. Let S satisfy (21) with $\gamma^2 > 0$ and $\chi \neq 0$, write $r = R^2$, and set

$$W_r = \theta_0^* + \frac{\gamma^2}{\chi}(S - \mu).$$

Since $\text{cov}(S_0, W_r) = \gamma^2 = \text{var}(S_0)$ and $\text{var}(W_r) = \gamma^2/r$, joint Gaussianity gives

$$W_r = S_0 + \epsilon_r, \quad \epsilon_r \perp S_0, \quad \epsilon_r \sim \text{N}(0, \varsigma_r^2). \quad (152)$$

Moreover, the generic prediction threshold $h_S = \mu - (\nu^2/\chi)\theta_0^*$ gives

$$\theta_0^* + \frac{\gamma^2}{\chi}(h_S - \mu) = h_r, \quad \frac{\chi^2}{\gamma^2}(W_r - h_r) = \chi(S - h_S).$$

Thus the optimal rule in Proposition 1 is $\mathbb{1}(W_r > h_r)$, and

$$\min_{a \in \{-1,1\}} \inf_{h \in \mathbb{R}} \Pr[Y \neq \mathbb{1}\{a(S - h) > 0\}] = \Pr\{Y \neq \mathbb{1}(W_r > h_r)\}. \quad (153)$$

For $0 < r < 1$,

$$\Pr\{Y \neq \mathbb{1}(W_r > h_r)\} = \mathbb{E} \left[\rho'(S_0) \Phi_N \left(\frac{h_r - S_0}{\varsigma_r} \right) + \{1 - \rho'(S_0)\} \Phi_N \left(\frac{S_0 - h_r}{\varsigma_r} \right) \right] = \mathcal{E}^*(r).$$

At $r = 1$, $W_1 = S_0$ and $h_1 = 0$, so the left-hand side of (153) equals $\mathbb{E}[\min\{\rho'(S_0), 1 - \rho'(S_0)\}]$, because $\rho'(S_0) < 1/2$ if and only if $S_0 < 0$. Substituting the means, variances and covariances of S_α^{DY} and S_α^{adj} from (16) into $R^2 = \chi^2/(\gamma^2\nu^2)$ gives (27), and hence (151).

It remains to prove strict monotonicity. Fix $0 < r_1 < r_2 \leq 1$ and couple the corresponding channels as

$$W_2 = S_0 + \epsilon_2, \quad W_1 = W_2 + \xi,$$

where ϵ_2 and ξ are independent of S_0 and of each other, with

$$\text{var}(\epsilon_2) = \gamma^2 \left(\frac{1}{r_2} - 1 \right), \quad \text{var}(\xi) = \gamma^2 \left(\frac{1}{r_1} - \frac{1}{r_2} \right) > 0,$$

so that (S_0, W_k) has the law of (152) with $r = r_k$. For $\pi_k = \mathbb{E}[\rho'(S_0) \mid W_k]$, $k \in \{1, 2\}$, we have by construction $\pi_1 = \mathbb{E}[\pi_2 \mid W_1]$, and, since $\{W_k > h_{r_k}\} = \{\pi_k > 1/2\}$ by (24) applied to W_k , the error $\mathcal{E}^*(r_k)$ of $\mathbb{1}(W_k > h_{r_k})$ is $\mathbb{E}[\min\{\pi_k, 1 - \pi_k\}]$. By the strict monotonicity used to obtain (24), $\pi_2 - 1/2$ has the sign of $W_2 - h_{r_2}$, while

$$\text{var}(W_2 \mid W_1) = \frac{\gamma^2(r_2 - r_1)}{r_2^2} > 0,$$

so that, conditionally on W_1 , W_2 has full support and $\pi_2 - 1/2$ takes both signs with positive probability. The strict triangle inequality therefore gives, almost surely,

$$\begin{aligned} \min\{\pi_1, 1 - \pi_1\} &= \frac{1}{2} - |\mathbb{E}[\pi_2 - 1/2 \mid W_1]| \\ &> \frac{1}{2} - \mathbb{E}[|\pi_2 - 1/2| \mid W_1] = \mathbb{E}[\min\{\pi_2, 1 - \pi_2\} \mid W_1]. \end{aligned}$$

Taking expectations yields $\mathcal{E}^*(r_1) > \mathcal{E}^*(r_2)$, so $\mathcal{E}^*$ is strictly decreasing on $(0, 1]$.

Part (i) follows immediately. For the adjusted logit,

$$(R_\alpha^{\text{adj}})^2 = \frac{\gamma^2}{\gamma^2 + \tau_\alpha^2}, \quad \tau_\alpha^2 = \frac{(\sigma_\alpha^*)^2}{(v_{1,\alpha}^*)^2},$$

so maximising $(R_\alpha^{\text{adj}})^2$ is equivalent to minimising $(\sigma_\alpha^*)^2/(v_{1,\alpha}^*)^2$. For part (ii), strict monotonicity gives the stated comparison. If $\varphi = 0$, then $\|\mathbf{u}_\alpha^*\|_2^2 = \gamma^2(v_{1,\alpha}^*)^2 + \delta^2(v_{2,\alpha}^*)^2$, and therefore

$$(R_\alpha^{\text{DY}})^2 = \frac{\gamma^2(v_{1,\alpha}^*)^2}{\gamma^2(v_{1,\alpha}^*)^2 + \delta^2(v_{2,\alpha}^*)^2 + (\sigma_\alpha^*)^2}, \quad (R_\alpha^{\text{adj}})^2 = \frac{\gamma^2(v_{1,\alpha}^*)^2}{\gamma^2(v_{1,\alpha}^*)^2 + (\sigma_\alpha^*)^2},$$

so $(R_\alpha^{\text{adj}})^2 \geq (R_\alpha^{\text{DY}})^2$, strictly if and only if $\delta^2(v_{2,\alpha}^*)^2 > 0$. $\square$

D Proofs for Section 5

D.1 Finite-contrast Gaussian limit

Lemma 2 (Finite-contrast Gaussian limit). *Assume the conditions of Section 2. Let $\mathbf{D}_p \in \mathbb{R}^{k \times p}$ be $\mathcal{G}_n$ -measurable, with k fixed, and suppose that*

$$\mathbf{D}_p \mathbf{D}_p^\top = \mathbf{I}_k, \quad \left\| \left\| \mathbf{D}_p \mathbf{B} \Gamma_p^{+1/2} \right\| \right\|_2 = \mathcal{O}_p(1).$$

Let $(\hat{\mathbf{v}}, \hat{\mathbf{w}}, \hat{\theta}, \hat{\boldsymbol{\eta}})$ be the minimising variables fixed in Proposition B.3, and let $\mathbf{d}$ be as in (121). Then

$$\begin{aligned} \mathbf{D}_p \mathbf{B}(\hat{\mathbf{v}} - \mathbf{v}^*) &= o_p(1), \\ \left\| \left\| \mathbf{D}_p \mathbf{P} \mathbf{D}_p^\top \right\| \right\|_2 &\xrightarrow{p} 0. \end{aligned} \tag{154}$$

Moreover,

$$\sqrt{p} \mathbf{D}_p \mathbf{d} \xrightarrow{d} \mathcal{N}(\mathbf{0}_k, \mathbf{I}_k), \tag{155}$$

and

$$\frac{1}{\sigma^*} \mathbf{D}_p \left(\hat{\boldsymbol{\beta}}^{\text{DY}} - \mathbf{B} \mathbf{v}^* \right) \xrightarrow{d} \mathcal{N}(\mathbf{0}_k, \mathbf{I}_k). \tag{156}$$

Proof. By (135) and (136),

$$\left\| \left\| \Gamma_p^{1/2}(\hat{\mathbf{v}} - \mathbf{v}^*) \right\| \right\|_2 \xrightarrow{\text{a.s.}} 0.$$

Since $\text{null}(\Gamma_p) = \text{null}(\mathbf{B})$,

$$\mathbf{B} = \mathbf{B} \Gamma_p^{+1/2} \Gamma_p^{1/2}.$$

Consequently,

$$\|\mathbf{D}_p \mathbf{B}(\hat{\mathbf{v}} - \mathbf{v}^*)\|_2 \leq \left\| \left\| \mathbf{D}_p \mathbf{B} \Gamma_p^{+1/2} \right\| \right\|_2 \left\| \left\| \Gamma_p^{1/2}(\hat{\mathbf{v}} - \mathbf{v}^*) \right\| \right\|_2 = o_p(1).$$

Since $\mathbf{P} = \mathbf{B} \Gamma_p^+ \mathbf{B}^\top / p$,

$$\left\| \left\| \mathbf{D}_p \mathbf{P} \mathbf{D}_p^\top \right\| \right\|_2 \leq \frac{1}{p} \left\| \left\| \mathbf{D}_p \mathbf{B} \Gamma_p^{+1/2} \right\| \right\|_2^2 \xrightarrow{p} 0.$$

This proves (154).

Conditionally on $\mathcal{C}_n$, Lemma 35 gives

$$\sqrt{p} \mathbf{D}_p \mathbf{d} \stackrel{d}{=} \frac{\mathbf{D}_p \mathbf{P}^\perp \mathbf{g}_p}{\|\mathbf{P}^\perp \mathbf{g}_p\|_2 / \sqrt{p}}, \quad \mathbf{g}_p \sim \mathcal{N}(\mathbf{0}_p, \mathbf{I}_p),$$

where $\mathbf{g}_p$ may be taken independent of $\sigma(\mathcal{C}_n, \mathbf{3})$. Conditionally on $\mathcal{G}_n$,

$$\mathbf{D}_p \mathbf{P}^\perp \mathbf{g}_p \sim \mathcal{N}(\mathbf{0}_k, \mathbf{D}_p \mathbf{P}^\perp \mathbf{D}_p^\top),$$

and, by (154),

$$\mathbf{D}_p \mathbf{P}^\perp \mathbf{D}_p^\top = \mathbf{I}_k - \mathbf{D}_p \mathbf{P} \mathbf{D}_p^\top \xrightarrow{p} \mathbf{I}_k.$$

Moreover, with $s = \text{rank}(\mathbf{P}) \leq 2$,

$$\|\mathbf{P}^\perp \mathbf{g}_p\|_2^2 \mid \mathcal{G}_n \sim \chi_{p-s}^2, \quad \frac{\|\mathbf{P}^\perp \mathbf{g}_p\|_2}{\sqrt{p}} \xrightarrow{p} 1.$$

Cramér–Wold and Slutsky’s theorem (Serfling 1980 Sections 1.5.2 and 1.5.4) prove (155).

Set

$$\hat{\boldsymbol{\beta}} = \mathbf{B} \hat{\mathbf{v}} + \mathbf{E} \hat{\mathbf{w}}, \quad \hat{\sigma} = \frac{\|\hat{\mathbf{w}}\|_2}{\sqrt{p}}.$$

The exact PO decomposition is

$$\frac{1}{\sigma^*} \mathbf{D}_p (\hat{\boldsymbol{\beta}} - \mathbf{B} \mathbf{v}^*) = \frac{1}{\sigma^*} \mathbf{D}_p \mathbf{B} (\hat{\mathbf{v}} - \mathbf{v}^*) + \frac{\hat{\sigma}}{\sigma^*} \sqrt{p} \mathbf{D}_p \mathbf{d}. \quad (157)$$

Proposition B.7 gives $\sigma^* > 0$, and (135) gives $\hat{\sigma} \xrightarrow{p} \sigma^*$. Combining (154), (155), and (157) yields

$$\frac{1}{\sigma^*} \mathbf{D}_p (\hat{\boldsymbol{\beta}} - \mathbf{B} \mathbf{v}^*) \xrightarrow{d} \mathbf{N}(\mathbf{0}_k, \mathbf{I}_k).$$

Proposition B.3 gives $\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^{\text{DY}}$ with probability tending to one. This proves (156). $\square$

D.2 Proof of Theorem 3

Theorem 3. Assume the conditions of Section 2. Let $I \subseteq \{1, \dots, p\}$ be a deterministic sequence of index sets with $|I| = k$ fixed, write $\mathbf{B}_I = \mathbf{J}_I^\top \mathbf{B}$, assume the leverage condition

$$\|\mathbf{B}_I \boldsymbol{\Gamma}_p^{+1/2}\|_2 = \mathcal{O}_p(1), \quad (158)$$

and let $\mathbf{v}^* = [v_1^*, v_2^*]^\top$ be defined in (9). Then

$$\frac{1}{\sigma^*} (\hat{\boldsymbol{\beta}}_I^{\text{DY}} - \mathbf{B}_I \mathbf{v}^*) \xrightarrow{d} \mathbf{N}(\mathbf{0}_k, \mathbf{I}_k). \quad (159)$$

Consequently, for $v_1^* \neq 0$ and a deterministic $b_I^0 \in \mathfrak{R}^k$, under $H_0 : \beta_{0,I} = b_I^0$, we have

$$\mathbf{Z}_I^{\text{adj}}(b_I^0) = \frac{\check{\boldsymbol{\beta}}_I - b_I^0}{(\sigma^*/v_1^*)} \xrightarrow{d} \mathbf{N}(\mathbf{0}_k, \mathbf{I}_k). \quad (160)$$

Proof. Take $\mathbf{D}_p = \mathbf{J}_I^\top$. Then $\mathbf{D}_p \mathbf{D}_p^\top = \mathbf{I}_k$, and (158) is exactly the leverage assumption of Lemma 2. Hence (156) gives (159). If $v_1^* \neq 0$, then under H_0 ,

$$\begin{aligned} \hat{\boldsymbol{\beta}}_I^{\text{DY}} - \mathbf{B}_I \mathbf{v}^* &= \hat{\boldsymbol{\beta}}_I^{\text{DY}} - v_1^* \beta_{0,I} - v_2^* \beta_{P,I} \\ &= v_1^* (\check{\boldsymbol{\beta}}_I - b_I^0), \end{aligned}$$

which gives (160). $\square$

E Proofs for Section 6

E.1 Proof of Corollary 1

Corollary 1. Assume that the transformed parameters $\tilde{\theta}_0, \tilde{\theta}_P, \tilde{\beta}_0, \tilde{\beta}_P$ satisfy the conditions of Section 2, and let $\mathbf{v}^* = [v_1^*, v_2^*]^\top$, $\tilde{\theta}^*$ and σ^* denote the corresponding limiting state parameters of (8).

(i) Suppose that, almost surely,

$$\mathbf{m}_p \longrightarrow \mathbf{m}, \quad \limsup_{p \rightarrow \infty} \mathbf{m}_p^\top \tilde{\Gamma}_p^+ \mathbf{m}_p < \infty, \quad q_p \log p \longrightarrow 0, \quad (161)$$

for a deterministic $\mathbf{m} = (m_0, m_P)^\top \in \mathbb{R}^2$. Then

$$\hat{\theta}^{DY} \xrightarrow{\text{a.s.}} \tilde{\theta}^* - \mathbf{m}^\top \mathbf{v}^* = \tilde{\theta}^* - v_1^* m_0 - v_2^* m_P.$$

(ii) Suppose that $\sup_p \|\mathbf{R}_p\|_2 < \infty$, and, almost surely,

$$\frac{1}{p} \sum_{j=1}^p \delta_{(\sqrt{p}\tau_j \beta_{0,j}, \sqrt{p}\tau_j \beta_{P,j})} \xrightarrow{W_2} \pi_\Sigma, \quad (162)$$

for a deterministic probability law π_Σ . Then, for any pseudo-Lipschitz function ψ of order two,

$$\frac{1}{p} \sum_{j=1}^p \psi \left(\sqrt{p}\tau_j \{\hat{\beta}_j^{DY} - v_1^* \beta_{0,j} - v_2^* \beta_{P,j}\}, \sqrt{p}\tau_j \beta_{0,j}, \sqrt{p}\tau_j \beta_{P,j} \right) \xrightarrow{\text{a.s.}} \mathbb{E} [\psi(\sigma^* G, \bar{\beta}_{0,\Sigma}, \bar{\beta}_{P,\Sigma})],$$

where $(\bar{\beta}_{0,\Sigma}, \bar{\beta}_{P,\Sigma}) \sim \pi_\Sigma$, $G \sim N(0, 1)$, independent from each other.

Proof. Work throughout with the centred and whitened problem in (37). Let $(\hat{\beta}, \hat{\mathbf{v}}, \hat{\sigma}, \mathbf{d}, \mathbf{z}_p)$ denote its PO from (63) and Gaussian-surrogate quantities, and write $\tilde{\mathcal{C}}_n$ for the field $\mathcal{C}_n$ in (53) formed from the transformed problem.

We shall prove each claim in turn.

(i) By (38), write

$$\hat{\theta}^{DY} = \tilde{\theta}^{DY} - \left(p^{-1/2} \mathbf{L}_p^{-1} \boldsymbol{\mu}_p \right)^\top \tilde{\beta}^{DY},$$

where $\tilde{\theta}^{DY}, \tilde{\beta}^{DY}$ are the whitened, centred MDYPL estimates. By Theorem 1(i),

$$\tilde{\theta}^{DY} \xrightarrow{\text{a.s.}} \tilde{\theta}^*.$$

Now let $\hat{\beta}$ be the primal minimiser from the PO of (63) for the transformed parameters $\tilde{\mathbf{B}}_p, \tilde{\theta}_0, \tilde{\theta}_P$. Then on the containment event $\mathcal{M}_n \cap \mathcal{E}_n$ from (58) and (59), we have that

$$\tilde{\beta}^{DY} = \hat{\beta} = \tilde{\mathbf{B}}_p \hat{\mathbf{v}} + \hat{\sigma} \sqrt{p} \mathbf{d}, \quad (163)$$

where the second equality follows from (120) with $\mathbf{d} \in \text{range}(\tilde{\mathbf{B}}_p)^\perp$, $\|\mathbf{d}\|_2 = 1$. Define the whitened covariate mean

$$\mathbf{a}_p = p^{-1/2} \mathbf{L}_p^{-1} \boldsymbol{\mu}_p$$

and its component that is orthogonal to $\tilde{\mathbf{B}}_p$, i.e.

$$\mathbf{a}_p^\perp = \mathbf{P}_{\tilde{\mathbf{B}}_p}^\perp \mathbf{a}_p = p^{-1/2} \mathbf{L}_p^{-1} \boldsymbol{\mu}_p^\perp, \quad \boldsymbol{\mu}_p^\perp = \boldsymbol{\mu}_p - \Sigma_p \mathbf{B} \tilde{\Gamma}_p^+ \mathbf{m}_p.$$

Then, by (163) and since $\mathbf{d} \in \text{range}(\tilde{\mathbf{B}}_p)^\perp$,

$$\boldsymbol{\mu}_p^\top \hat{\beta}^{DY} = \mathbf{a}_p^\top \left(\tilde{\mathbf{B}}_p \hat{\mathbf{v}} + \hat{\sigma} \sqrt{p} \mathbf{d} \right) = \mathbf{m}_p^\top \hat{\mathbf{v}} + \hat{\sigma} \langle \mathbf{a}_p^\perp, \sqrt{p} \mathbf{d} \rangle.$$

To control the first term, note that

$$\begin{aligned} |\mathbf{m}_p^\top (\hat{\mathbf{v}} - \mathbf{v}^*)| &= |\mathbf{m}_p^\top \tilde{\Gamma}_p^{+1/2} \tilde{\Gamma}_p^{1/2} (\hat{\mathbf{v}} - \mathbf{v}^*)| \\ &\leq \left\| \mathbf{m}_p^\top \tilde{\Gamma}_p^{+1/2} \right\|_2 \left\{ \left\| \tilde{\Gamma}_p^{1/2} \hat{\mathbf{v}} - \mathbf{u}^* \right\|_2 + \left\| (\tilde{\Gamma}_p^{1/2} - \tilde{\Gamma}^{1/2}) \mathbf{v}^* \right\|_2 \right\} \\ &\leq \left\| \mathbf{m}_p^\top \tilde{\Gamma}_p^{+1/2} \right\|_2 \left\{ \left\| \tilde{\Gamma}_p^{1/2} \hat{\mathbf{v}} - \mathbf{u}^* \right\|_2 + \left\| \tilde{\Gamma}_p - \tilde{\Gamma} \right\|_2^{1/2} \left\| \mathbf{v}^* \right\|_2 \right\}, \end{aligned}$$

where we used the Cauchy–Schwarz inequality, the definition of $\mathbf{u}^*$ from Section B.7, the triangle inequality and operator Hölder continuity of the positive-semidefinite square root (Bhatia, 1997, Theorem X.1.1). Now, the term $\|\mathbf{m}_p^\top \tilde{\Gamma}_p^{+1/2}\|_2$ is $\mathcal{O}(1)$ almost surely by the assumption of (161). The second term $\|\tilde{\Gamma}_p^{1/2} \hat{\mathbf{v}} - \mathbf{u}^*\|_2 \rightarrow 0$ almost surely by the transformed versions of (135) and (136) with limiting signal matrix $\tilde{\Gamma}$. The transformed version of Assumption (i) and the first Borel–Cantelli lemma give $\tilde{\Gamma}_p \rightarrow \tilde{\Gamma}$ almost surely, and hence $\|\tilde{\Gamma}_p - \tilde{\Gamma}\|_2^{1/2} \rightarrow 0$. Since $\mathbf{v}^*$ is a fixed finite vector, this controls the remaining term.

It thus remains to control $\hat{\sigma}(\mathbf{a}_p^\perp, \sqrt{p}\mathbf{d})$. Since by (135), $\hat{\sigma} \rightarrow \sigma^*$ almost surely, it is sufficient to establish that $\langle \mathbf{a}_p^\perp, \sqrt{p}\mathbf{d} \rangle \rightarrow 0$ almost surely. Conditionally on $\tilde{\mathcal{C}}_n$, Lemma 35 gives that $\mathbf{d}$ is uniform on the unit sphere of $\text{range}(\tilde{\mathbf{B}}_p)^\perp$, whose dimension is $p-s$ for $s = \text{rank}(\tilde{\mathbf{B}}_p) \leq 2$. Since, conditionally on $\tilde{\mathcal{C}}_n$, $\sqrt{p}\mathbf{d}$ is subgaussian (Vershynin, 2018, Theorem 3.4.6) with constant subgaussian norm, and $\mathbf{a}_p^\perp \in \text{range}(\tilde{\mathbf{B}}_p)^\perp$ is $\tilde{\mathcal{C}}_n$ -measurable, with squared norm q_p , a Hoeffding bound (Vershynin, 2018, Theorem 2.6.3), therefore gives absolute constants $C, c > 0$ such that, for every $\epsilon > 0$ and all sufficiently large p ,

$$\mathbb{1}\{q_p > 0\} \Pr\left(|\langle \mathbf{a}_p^\perp, \sqrt{p}\mathbf{d} \rangle| > \epsilon \mid \tilde{\mathcal{C}}_n\right) \leq C \exp\left\{-\frac{c\epsilon^2}{q_p}\right\}. \quad (164)$$

Now, on the $\tilde{\mathcal{C}}_n$ -measurable event $\{q_p \log p \leq c\epsilon^2/3\}$, the right-hand side of (164) is at most Cp^{-3} . On $\{q_p = 0\}$, trivially, $\langle \mathbf{a}_p^\perp, \sqrt{p}\mathbf{d} \rangle = 0$. Taking expectations, applying the first Borel–Cantelli lemma (cf. Shiryaev, 2016, p. 308), and using $q_p \log p \rightarrow 0$ almost surely gives

$$\sqrt{p} |\langle \mathbf{a}_p^\perp, \mathbf{d} \rangle| \leq \epsilon$$

eventually almost surely. Intersecting over $\epsilon = 1/m$, $m \in \mathbb{N}$, proves $\langle \mathbf{a}_p^\perp, \sqrt{p}\mathbf{d} \rangle \rightarrow 0$ almost surely. Finally, the summability of the compact-containment failure probabilities in Lemma 3 and the first Borel–Cantelli lemma give

$$\mu_p^\top \hat{\beta}^{\text{DY}} \xrightarrow{\text{a.s.}} \mathbf{m}^\top \mathbf{v}^*,$$

and we conclude that

$$\lim_{n \rightarrow \infty} \hat{\theta}^{\text{DY}} = \tilde{\theta}^* - \mathbf{m}^\top \mathbf{v}^*,$$

almost surely as required.

- (ii) We start with the identity of (38). Let $\tilde{\mathbf{r}}_p = \tilde{\beta}^{\text{DY}} - \tilde{\mathbf{B}}_p \mathbf{v}^*$, $\mathbf{z}_p$ be the Gaussian surrogate of (123) from the transformed whitened problem, and for $\mathbf{A}_p = \Delta_p^{-1/2} \mathbf{L}_p^{-\top}$, write

$$\sqrt{p} \Delta_p^{-1/2} (\hat{\beta}^{\text{DY}} - \mathbf{B} \mathbf{v}^*) = \mathbf{A}_p \tilde{\mathbf{r}}_p = \sigma^* \mathbf{A}_p \mathbf{z}_p + \mathbf{A}_p (\tilde{\mathbf{r}}_p - \sigma^* \mathbf{z}_p). \quad (165)$$

Now, recall that $[\Delta_p^{-1/2}]_{jj} = \tau_j$, and set $\mathbf{b}_p = \sqrt{p} \Delta_p^{-1/2} \mathbf{B}$, $\mathbf{r}_p = \sqrt{p} \Delta_p^{-1/2} (\hat{\beta}^{\text{DY}} - \mathbf{B} \mathbf{v}^*)$, $\mathbf{x}_p = [\mathbf{r}_p, \mathbf{b}_p]$, and $\mathbf{y}_p = [\sigma^* \mathbf{A}_p \mathbf{z}_p, \mathbf{b}_p]$. Then (165), together with manipulations analogous to (130) and (131), yields

$$\begin{aligned} \left| \frac{1}{p} \sum_{j=1}^p \psi(\mathbf{x}_{p,j}) - \psi(\mathbf{y}_{p,j}) \right| &\leq L_\psi \frac{1}{p} \sum_{j=1}^p (1 + \|\mathbf{x}_{p,j}\|_2 + \|\mathbf{y}_{p,j}\|_2) |[\mathbf{A}_p (\tilde{\mathbf{r}}_p - \sigma^* \mathbf{z}_p)]_j| \\ &\leq C \left(1 + \|\Delta_p^{-1/2} \hat{\beta}^{\text{DY}}\|_2 + \|\Delta_p^{-1/2} \beta_0\|_2 + \|\Delta_p^{-1/2} \beta_P\|_2 + \frac{\|\mathbf{A}_p \mathbf{z}_p\|_2}{\sqrt{p}} \right) \frac{\|\mathbf{A}_p (\tilde{\mathbf{r}}_p - \sigma^* \mathbf{z}_p)\|_2}{\sqrt{p}}, \end{aligned} \quad (166)$$

for some constant C that depends on $L_\psi, \mathbf{v}^*, \sigma^*$. Now note that by (38),

$$\frac{\|\sqrt{p} \Delta_p^{-1/2} \hat{\beta}^{\text{DY}}\|_2^2}{p} = \frac{\|\Delta_p^{-1/2} \mathbf{L}_p^{-\top} \tilde{\beta}^{\text{DY}}\|_2^2}{p} \leq \|\mathbf{R}_p\| \frac{\|\tilde{\beta}^{\text{DY}}\|_2^2}{p} = \mathcal{O}(1), \quad (167)$$

almost surely by the assumption on $\mathbf{R}_p$, the containment event of Lemma 3 applied to $\tilde{\beta}^{\text{dy}}$ and the first Borel–Cantelli lemma. Similarly,

$$\left\| \Delta_p^{-1/2} \beta_0 \right\|_2 = \mathcal{O}(1), \quad \left\| \Delta_p^{-1/2} \beta_P \right\|_2 = \mathcal{O}(1),$$

almost surely by the W_2 -convergence assumption of their joint empirical distribution function (Villani 2009 Definition 6.8(iv) & Theorem 6.9). Finally, by standard Gaussian concentration (Vershynin 2018 Theorem 3.1.1) and the assumption in Corollary 1(ii),

$$\frac{\|\mathbf{A}_p \mathbf{z}_p\|_2}{\sqrt{p}} \leq \|\mathbf{R}_p\|_2 \frac{\|\mathbf{z}_p\|_2}{\sqrt{p}} = \mathcal{O}(1),$$

almost surely. Hence, by Lemma 1 applied to the whitened problem,

$$\frac{\|\mathbf{A}_p (\tilde{\mathbf{r}}_p - \sigma^* \mathbf{z}_p)\|_2}{\sqrt{p}} \leq \|\mathbf{R}_p\|_2^{1/2} \frac{\|\tilde{\mathbf{r}}_p - \sigma^* \mathbf{z}_p\|_2}{\sqrt{p}} \xrightarrow{\text{a.s.}} 0.$$

Consequently, since an almost surely bounded sequence times an almost surely null sequence is almost surely null, the last line in (166) goes to zero almost surely. It remains to control

$$T_p^{\mathbf{z}} = \frac{1}{p} \sum_{j=1}^p \psi(\mathbf{y}_{p,j}).$$

For that, condition on $\mathcal{G}_n = \sigma(\mathbf{B}, \theta_0, \theta_P)$ from (55). The matrix $\mathbf{A}_p$ is deterministic, the $\mathbf{b}_{p,j}$ are $\mathcal{G}_n$ -measurable, and $\mathcal{G}_n \subseteq \tilde{\mathcal{C}}_n$ (since $\boldsymbol{\mu}_p$ and $\mathbf{L}_p$ are deterministic and $\mathbf{L}_p$ is invertible, $(\mathbf{B}, \theta_0, \theta_P)$ are deterministic functions of the transformed signal and intercepts), so $\mathbf{z}_p$ is independent of $\mathcal{G}_n$ by Lemma 1. By assumption (ii) of Corollary 1, $\|\mathbf{R}_p\|_2 \leq K$ for a deterministic $K < \infty$. To get Lipschitz-control over the function ψ , for $M > 0$, introduce the clamp $c_M(x) = \text{sign}(x) \min\{|x|, M\}$ and define the clamped average

$$T_{p,M} = T_{p,M}(\mathbf{z}_p), \quad T_{p,M}(\mathbf{x}) = \frac{1}{p} \sum_{j=1}^p \psi(\sigma^* c_M\{[\mathbf{A}_p \mathbf{x}]_j\}, \mathbf{b}_{p,j}).$$

Pseudo-Lipschitzness, Cauchy–Schwarz and $\|\mathbf{A}_p\|_2^2 = \|\mathbf{R}_p\|_2 \leq K$ give

$$\begin{aligned} |T_{p,M}(\mathbf{x}) - T_{p,M}(\mathbf{y})| &\leq \frac{C}{p} \sum_{j=1}^p (1 + M + \|\mathbf{b}_{p,j}\|_2) |[\mathbf{A}_p(\mathbf{x} - \mathbf{y})]_j| \\ &\leq C \left(\frac{1 + M^2 + B_p}{p} \right)^{1/2} \|\mathbf{x} - \mathbf{y}\|_2, \end{aligned}$$

where C depends only on K , σ^* and the pseudo-Lipschitz constant of ψ and

$$B_p = \frac{1}{p} \sum_{j=1}^p \|\mathbf{b}_{p,j}\|_2^2.$$

Thus, on $\{B_p \leq B\}$, the clamped average $T_{p,M}$ is Lipschitz with Lipschitz constant upper bounded by $\sqrt{C_B(1 + M^2)/p}$ for some constant C_B large enough. Hence, conditionally on $\mathcal{G}_n$, Gaussian concentration (Vershynin 2018 Theorem 5.2.2) yields for every $\epsilon > 0$,

$$\mathbf{1}\{B_p \leq B\} \Pr(|T_{p,M} - \mathbb{E}[T_{p,M} | \mathcal{G}_n]| > \epsilon | \mathcal{G}_n) \leq 2 \exp \left\{ -\frac{c_B p \epsilon^2}{1 + M^2} \right\}.$$

For $p \geq 2$, set $M = 2\sqrt{\log p}$, so that

$$\mathbf{1}\{B_p \leq B\} \Pr(|T_{p,M} - \mathbb{E}[T_{p,M} | \mathcal{G}_n]| > \epsilon | \mathcal{G}_n) \leq 2 \exp \left\{ -\frac{c_B p \epsilon^2}{1 + 4 \log p} \right\}.$$

Thus, taking expectations, one gets

$$\Pr(|T_{p,M} - \mathbb{E}[T_{p,M} | \mathcal{G}_n]| > \epsilon \cap B_p \leq B) \leq 2 \exp \left\{ -\frac{c_B p \epsilon^2}{1 + 4 \log p} \right\},$$

which is summable in n and thus by the first Borel–Cantelli lemma, $\{|T_{p,M} - \mathbb{E}[T_{p,M} | \mathcal{G}_n]| > \epsilon\} \cap \{B_p \leq B\}$ occurs only finitely often almost surely. Separately, for B sufficiently large, by (167), $\{B_p > B\}$ finitely often almost surely. Hence, since

$$\{|T_{p,M} - \mathbb{E}[T_{p,M} | \mathcal{G}_n]| > \epsilon\} \subseteq \{|T_{p,M} - \mathbb{E}[T_{p,M} | \mathcal{G}_n]| > \epsilon\} \cap \{B_p \leq B\} \cup \{B_p > B\},$$

the first Borel–Cantelli lemma, applied along $\epsilon = 1/m$, $m \in \mathbb{N}$, gives

$$T_{p,M} - \mathbb{E}[T_{p,M} | \mathcal{G}_n] \xrightarrow{\text{a.s.}} 0.$$

Thus, finally, we need to control the truncation bias $T_p^z - T_{p,M}$. Pseudo-Lipschitzness gives

$$|T_p^z - T_{p,M}| \leq \sigma^* L_\psi \frac{1}{p} \sum_{j=1}^p (1 + 2\sigma^* |[\mathbf{A}_p \mathbf{z}_p]_j| + 2 \|\mathbf{b}_{p,j}\|_2) (|[\mathbf{A}_p \mathbf{z}_p]_j| - M)_+.$$

Now, consider the expectations of the j th summand

$$\begin{aligned} \mathbb{E}[(1 + 2\sigma^* |[\mathbf{A}_p \mathbf{z}_p]_j| + 2 \|\mathbf{b}_{p,j}\|_2) (|[\mathbf{A}_p \mathbf{z}_p]_j| - M)_+ | \mathcal{G}_n] &= (1 + 2 \|\mathbf{b}_{p,j}\|_2) \mathbb{E}[(|[\mathbf{A}_p \mathbf{z}_p]_j| - M)_+ | \mathcal{G}_n] \\ &\quad + 2\sigma^* \mathbb{E}[|[\mathbf{A}_p \mathbf{z}_p]_j| (|[\mathbf{A}_p \mathbf{z}_p]_j| - M)_+ | \mathcal{G}_n]. \end{aligned}$$

Now, since $\text{diag}(\mathbf{R}_p) = \mathbf{1}$, conditionally on $\mathcal{G}_n$ each $[\mathbf{A}_p \mathbf{z}_p]_j$ is standard normal. Thus,

$$\begin{aligned} \mathbb{E}[(|[\mathbf{A}_p \mathbf{z}_p]_j| - M)_+ | \mathcal{G}_n] &= 2 \int_M^\infty (x - M) \phi_N(x) dx \\ &= 2 \left(\int_M^\infty x \phi_N(x) dx - M \Phi_N(-M) \right) \\ &= 2(\phi_N(M) - M \Phi_N(-M)) \\ &\leq 2\phi_N(M), \end{aligned} \tag{168}$$

where ϕ_N, Φ_N denote the standard normal pdf and cdf function, respectively, and where the last equality follows from integration by parts. Similarly,

$$\begin{aligned} \mathbb{E}[|[\mathbf{A}_p \mathbf{z}_p]_j| (|[\mathbf{A}_p \mathbf{z}_p]_j| - M)_+ | \mathcal{G}_n] &= 2 \int_M^\infty x(x - M) \phi_N(x) dx \\ &= 2 \left(\int_M^\infty x^2 \phi_N(x) dx - M \phi_N(M) \right) \\ &= 2 \left([-x \phi_N(x)]_M^\infty + \int_M^\infty \phi_N(x) dx - M \phi_N(M) \right) \\ &= 2 \int_M^\infty \phi_N(x) dx \\ &\leq 2 \int_M^\infty \frac{x}{M} \phi_N(x) dx \\ &= 2 \frac{\phi_N(M)}{M} \\ &\leq 2\phi_N(M), \end{aligned}$$

where we used (168) and integration parts again and the last line holds for $M \geq 1$. Thus,

$$\begin{aligned} \mathbb{E} [|T_p^z - T_{p,M}| \mid \mathcal{G}_n] &\leq \sigma^* L_\psi \frac{1}{p} \sum_{j=1}^p (1 + 2 \|\mathbf{b}_{p,j}\|_2 + 2\sigma^*) 2\phi_N(M) \\ &= \exp\{-M^2/2\} \sqrt{\frac{2}{\pi}} \sigma^* L_\psi \frac{1}{p} \sum_{j=1}^p (1 + 2 \|\mathbf{b}_{p,j}\|_2 + 2\sigma^*) \\ &\leq \exp\{-M^2/2\} \sqrt{\frac{2}{\pi}} \sigma^* L_\psi (1 + 2\sigma^* + 2\sqrt{B_p}). \end{aligned} \quad (169)$$

Therefore, by Markov's inequality (Vershynin 2018 Proposition 1.2.4), for any $\epsilon > 0$,

$$\Pr (|T_p^z - T_{p,M}| > \epsilon \mid \mathcal{G}_n) \leq \frac{1}{\epsilon} \exp\{-M^2/2\} \sqrt{\frac{2}{\pi}} \sigma^* L_\psi (1 + 2\sigma^* + 2\sqrt{B_p}).$$

Hence, on $\{B_p \leq B\}$, for $M = 2\sqrt{\log p}$, this conditional expectation bound gives

$$\Pr (|T_p^z - T_{p,M}| > \epsilon \cap \{B_p \leq B\}) \leq C_B \epsilon^{-1} p^{-2},$$

for some large enough constant C_B depending on σ^*, L_ψ . This upper bound is summable, and thus by the first Borel–Cantelli lemma, $|T_p^z - T_{p,M}| > \epsilon \cap \{B_p \leq B\}$ finitely often almost surely. Now since also $\{B_p \leq B\}$ eventually almost surely for B large enough, we conclude that

$$|T_p^z - T_{p,M}| \xrightarrow{\text{a.s.}} 0.$$

Again, since $B_p \leq B$ eventually almost surely, it holds by (169),

$$|\mathbb{E}[T_p^z \mid \mathcal{G}_n] - \mathbb{E}[T_{p,M} \mid \mathcal{G}_n]| \leq \mathbb{E} [|T_p^z - T_{p,M}| \mid \mathcal{G}_n] \xrightarrow{\text{a.s.}} 0,$$

eventually almost surely. Moreover, $\text{diag}(\mathbf{R}_p) = \mathbf{1}$ implies

$$\bar{T}_p := \mathbb{E}[T_p^z \mid \mathcal{G}_n] = \frac{1}{p} \sum_{j=1}^p \mathbb{E}_G [\psi(\sigma^* G, \mathbf{b}_{p,j})].$$

Consequently,

$$T_p^z - \bar{T}_p \xrightarrow{\text{a.s.}} 0.$$

Finally, $\mathbb{E}_G[\psi(\sigma^* G, \mathbf{b})]$ is continuous and has at most quadratic growth. The W_2 convergence (162) and Villani (2009, Theorem 6.9), therefore gives

$$\bar{T}_p \xrightarrow{\text{a.s.}} \int \mathbb{E}_G[\psi(\sigma^* G, \mathbf{b})] \pi_\Sigma(d\mathbf{b}) = \mathbb{E} [\psi(\sigma^* G, \bar{\beta}_{0,\Sigma}, \bar{\beta}_{P,\Sigma})].$$

Therefore

$$T_p^z \xrightarrow{\text{a.s.}} \mathbb{E} [\psi(\sigma^* G, \bar{\beta}_{0,\Sigma}, \bar{\beta}_{P,\Sigma})].$$

Combining the last displays with (166) proves the claim. □

E.2 Proof of Corollary 2

Corollary 2. Assume that the centred and whitened triangular array satisfies the conditions of Section 2. Let $I \subseteq \{1, \dots, p\}$ be a deterministic sequence of index sets with $|I| = k$ fixed, and suppose that

$$\|\sqrt{p} \mathbf{C}_{I,p}^{1/2} \mathbf{J}_I^\top \mathbf{B} \tilde{\Gamma}_p^{+1/2}\|_2 = \mathcal{O}_p(1). \quad (170)$$

Let $\mathbf{v}^* = [v_1^*, v_2^*]^\top$ be the canonical alignment vector of the centred and whitened problem, and suppose that $v_1^* \neq 0$. For deterministic $b_I^0 \in \mathbb{R}^k$, under $H_0 : \beta_{0,I} = b_I^0$,

$$\mathbf{Z}_I^{\text{adj}}(b_I^0) = \sqrt{p} \mathbf{C}_{I,p}^{1/2} \frac{\check{\beta}_I - b_I^0}{(\sigma^*/v_1^*)} \xrightarrow{d} \mathbf{N}(\mathbf{0}_k, \mathbf{I}_k). \quad (171)$$

Proof. Let

$$\mathbf{D}_{I,p} = \mathbf{C}_{I,p}^{1/2} \mathbf{J}_I^\top \mathbf{L}_p^{-\top}.$$

Since $\Theta_p = \mathbf{L}_p^{-\top} \mathbf{L}_p^{-1}$,

$$\mathbf{D}_{I,p} \mathbf{D}_{I,p}^\top = \mathbf{C}_{I,p}^{1/2} \mathbf{J}_I^\top \Theta_p \mathbf{J}_I \mathbf{C}_{I,p}^{1/2} = \mathbf{I}_k.$$

Moreover,

$$\mathbf{D}_{I,p} \tilde{\mathbf{B}}_p \tilde{\mathbf{\Gamma}}_p^{+1/2} = \sqrt{p} \mathbf{C}_{I,p}^{1/2} \mathbf{J}_I^\top \mathbf{B} \tilde{\mathbf{\Gamma}}_p^{+1/2},$$

so (170) is exactly the leverage assumption of Lemma 2 in the centred and whitened coordinates. That lemma gives

$$\frac{1}{\sigma^*} \mathbf{D}_{I,p} \left(\tilde{\beta}^{\text{DY}} - \tilde{\mathbf{B}}_p \mathbf{v}^* \right) \xrightarrow{d} \mathbf{N}(\mathbf{0}_k, \mathbf{I}_k).$$

Using

$$\tilde{\beta}^{\text{DY}} = \sqrt{p} \mathbf{L}_p^\top \hat{\beta}^{\text{DY}}, \quad \tilde{\mathbf{B}}_p = \sqrt{p} \mathbf{L}_p^\top \mathbf{B},$$

the left-hand side equals

$$\frac{\sqrt{p}}{\sigma^*} \mathbf{C}_{I,p}^{1/2} \left(\hat{\beta}_I^{\text{DY}} - \mathbf{J}_I^\top \mathbf{B} \mathbf{v}^* \right).$$

Under H_0 ,

$$\begin{aligned} \mathbf{Z}_I^{\text{adj}}(b_I^0) &= \frac{\sqrt{p} v_1^*}{\sigma^*} \mathbf{C}_{I,p}^{1/2} (\check{\beta}_I - b_I^0) \\ &= \frac{\sqrt{p}}{\sigma^*} \mathbf{C}_{I,p}^{1/2} \left(\hat{\beta}_I^{\text{DY}} - \mathbf{J}_I^\top \mathbf{B} \mathbf{v}^* \right). \end{aligned}$$

This proves (171). $\square$

E.3 Proof of Proposition 3

Proposition 3. Assume the conditions of Section 2. Then

$$(\hat{\theta}_0, \hat{\gamma}^2, \hat{\varphi}) \xrightarrow{p} (\theta_0^*, \gamma^2, \varphi).$$

Proof. Let $\pi_p = \pi(\theta_0, \gamma_p)$, $\zeta_p = \zeta(\theta_0, \gamma_p)$, $c_p = c(\theta_0, \gamma_p)$, $b_p = \varphi_p c_p$. By (46) and (49), the conditional population targets of $\hat{\pi}$, $\hat{\zeta}^2$, and $\hat{b}$ are π_p , ζ_p^2 , and b_p , respectively. By the assumptions of Section 2 $\mathbb{E}[(\hat{\pi} - \pi_p)^2 | \mathcal{G}_n] = \pi_p(1 - \pi_p)/n \leq 1/(4n)$. The untruncated statistic $\hat{\zeta}^2$ is the order-two U -statistic with kernel $p Y_i Y_j \mathbf{x}_i^\top \mathbf{x}_j$ (Serfling 1980, Section 5.1), whose conditional expectation is $p \mathbb{E}[\mathbf{x}_i Y_i | \mathcal{G}_n]^\top \mathbb{E}[\mathbf{x}_i Y_i | \mathcal{G}_n] = \zeta_p^2$ by (46). The order-two Hoeffding variance decomposition (Serfling 1980, Section 5.2.1, Lemma A) gives

$$\text{var}(\hat{\zeta}^2 | \mathcal{G}_n) = \frac{4p^2}{n} \mathbb{E}[\mathbf{x}_i Y_i | \mathcal{G}_n]^\top \text{var}(\mathbf{x}_i Y_i | \mathcal{G}_n) \mathbb{E}[\mathbf{x}_i Y_i | \mathcal{G}_n] + \frac{2p^2}{n(n-1)} \text{trace} \{ \text{var}(\mathbf{x}_i Y_i | \mathcal{G}_n)^2 \}.$$

Since $Y_i \in \{0, 1\}$,

$$\mathbf{0} \preceq \text{var}(\mathbf{x}_i Y_i | \mathcal{G}_n) \preceq \mathbb{E}(\mathbf{x}_i \mathbf{x}_i^\top | \mathcal{G}_n) = \frac{1}{p} \mathbf{I}_p,$$

and hence

$$\text{var}(\hat{\zeta}^2 | \mathcal{G}_n) \leq \frac{4\zeta_p^2}{n} + \frac{2p}{n(n-1)}. \quad (172)$$

For the aligned moment, the sample covariance $\widehat{b}$ is the order-two U -statistic with kernel $(\mathbf{x}_i - \mathbf{x}_j)^\top \beta_P (Y_i - Y_j)/2$. Its conditional expectation is b_p , while, since $Y_i \in \{0, 1\}$ and $(\mathbf{x}_1 - \mathbf{x}_2)^\top \beta_P \mid \mathcal{G}_n \sim N(0, 2\delta_p^2)$, the conditional second moment of the kernel is at most

$$\frac{1}{4} \mathbb{E} \left[\{(\mathbf{x}_1 - \mathbf{x}_2)^\top \beta_P\}^2 \mid \mathcal{G}_n \right] = \frac{\delta_p^2}{2}.$$

The same variance decomposition therefore gives

$$\mathbb{E} \left[(\widehat{b} - b_p)^2 \mid \mathcal{G}_n \right] \leq \frac{3\delta_p^2}{n}. \quad (173)$$

By the assumptions of Section 2, γ_p^2 and δ_p^2 are bounded in probability and $p/n \rightarrow \kappa$. Since $\zeta_p = \gamma_p c_p$ and $0 < c_p \leq 1/4$, the right-hand sides of (172) and (173) converge to zero in probability. Chebyshev's inequality (Vershynin 2018 Corollary 1.2.5) therefore gives

$$(\widehat{\pi} - \pi_p, \widehat{\zeta}^2 - \zeta_p^2, \widehat{b} - b_p) \xrightarrow{P} \mathbf{0}_3.$$

Since $\zeta_p \geq 0$ and $\widehat{\zeta} = \{\max(\widehat{\zeta}^2, 0)\}^{1/2}$, we have $|\widehat{\zeta} - \zeta_p|^2 \leq |\widehat{\zeta}^2 - \zeta_p^2|$, and hence

$$(\widehat{\pi} - \pi_p, \widehat{\zeta} - \zeta_p, \widehat{b} - b_p) \xrightarrow{P} \mathbf{0}_3. \quad (174)$$

Now, to invert the relation of $\widehat{\pi}, \widehat{\zeta}$ to $\widehat{\gamma}, \widehat{\theta}_0$, note that $(\theta, g) \mapsto \{\pi(\theta, g), \zeta(\theta, g)\}$ is continuously differentiable on $\mathbb{R}^2$, with Jacobian determinant

$$\mathbb{E}\{\rho''(\theta + gZ)\} \mathbb{E}\{Z^2 \rho''(\theta + gZ)\} - \mathbb{E}\{Z \rho''(\theta + gZ)\}^2 > 0,$$

where the inequality is the weighted Cauchy-Schwarz inequality, and is strict because $\rho'' > 0$ and Z is nondegenerate. The inverse function theorem (Rudin 1976 Theorem 9.24) therefore applies at (θ_0^*, γ) and the assumptions of Section 2, continuity of π and ζ , and (174) give

$$(\widehat{\pi}, \widehat{\zeta}) \xrightarrow{P} \{\pi(\theta_0^*, \gamma), \zeta(\theta_0^*, \gamma)\}.$$

Hence, with probability tending to one, (48) has a local solution converging to (θ_0^*, γ) . Since $\zeta(\theta, g) = gc(\theta, g)$ has the same sign as g and $\widehat{\zeta} \geq 0$, this solution satisfies $g \geq 0$, and the identification argument in the main text makes it the unique solution in $\mathbb{R} \times [0, \infty)$. Thus

$$(\widehat{\theta}_0, \widehat{\gamma}) \xrightarrow{P} (\theta_0^*, \gamma).$$

By continuity and positivity of c ,

$$c(\widehat{\theta}_0, \widehat{\gamma}) \xrightarrow{P} c(\theta_0^*, \gamma) > 0.$$

Moreover,

$$b_p = \varphi_p c(\theta_0, \gamma_p) \xrightarrow{P} \varphi c(\theta_0^*, \gamma).$$

It follows from (174) that

$$\widetilde{\varphi} = \frac{\widehat{b}}{c(\widehat{\theta}_0, \widehat{\gamma})} \xrightarrow{P} \varphi.$$

Finally, $|\widehat{\varphi} - \widetilde{\varphi}| = (|\widetilde{\varphi}| - \widehat{\gamma}\delta_p)_+ \xrightarrow{P} (|\varphi| - \gamma\delta)_+ = 0$, because $\widehat{\gamma}\delta_p \xrightarrow{P} \gamma\delta$ and $|\varphi| \leq \gamma\delta$ by positive semidefiniteness of $\mathbf{\Gamma}$. Hence also $\widehat{\varphi} \xrightarrow{P} \varphi$, which concludes the proof. $\square$

F Auxiliary Lemmas

F.1 Lemmas for Section B.4

F.1.1 MDYPL containment

Lemma 3 (Joint MDYPL containment). *Let $\hat{\theta}^{DY}, \hat{\beta}^{DY}$ be the MDYPL estimates of (6), $\hat{\eta}^{DY} = \hat{\theta}^{DY}\mathbf{1} + \mathbf{X}\hat{\beta}^{DY}$, and let $\hat{\lambda}^{DY} = \tilde{\mathbf{y}} - \rho'(\hat{\eta}^{DY})$. Then there exist constants $C_{\beta,0}, C_{\theta,0}, C_{\eta,0}, C, c > 0$ and $N \in \mathbb{N}$ such that the following holds. For any constants $C_{\beta} \geq C_{\beta,0}$, $C_{\theta} \geq C_{\theta,0}$ and $C_{\eta} \geq C_{\eta,0}$, define*

$$\mathcal{M}_n = \left\{ \frac{\|\hat{\beta}^{DY}\|_2}{\sqrt{p}} \leq C_{\beta}, \left| \hat{\theta}^{DY} \right| \leq C_{\theta}, \frac{\|\hat{\eta}^{DY}\|_2}{\sqrt{n}} \leq C_{\eta}, \frac{\|\hat{\lambda}^{DY}\|_2}{\sqrt{n}} \leq 1 \right\}.$$

Then, for all $n > N$,

$$\Pr(\neg \mathcal{M}_n) \leq C \exp\{-cn\}.$$

The constants $C_{\beta,0}, C_{\theta,0}, C_{\eta,0}, C, c$ depend only on the fixed quantities $\kappa, \alpha, \mathbf{\Gamma}, \theta_0^*, \theta_P^*$ and the constants defined in Section 2.

Proof. Let K be the constant of Lemma 4. Take $C_{\beta,0} = K$, $C_{\theta,0} = K$, and $C_{\eta,0} = 3K$. Define

$$\widetilde{\mathbf{H}} = [\mathbf{1}, \mathbf{H}], \quad \mathcal{E}_{\beta,\theta} = \left\{ \frac{1}{\sqrt{p}} \left\| \begin{bmatrix} \sqrt{p} \hat{\theta}^{DY} \\ \hat{\beta}^{DY} \end{bmatrix} \right\|_2 \leq K \right\}, \quad \mathcal{E}_{\widetilde{\mathbf{H}}} = \left\{ \frac{\|\widetilde{\mathbf{H}}\|_2}{\sqrt{n}} \leq 3 \right\}.$$

We have

$$\hat{\eta}^{DY} = \frac{1}{\sqrt{p}} \widetilde{\mathbf{H}} \begin{bmatrix} \sqrt{p} \hat{\theta}^{DY} \\ \hat{\beta}^{DY} \end{bmatrix}.$$

On $\mathcal{E}_{\beta,\theta} \cap \mathcal{E}_{\widetilde{\mathbf{H}}}$,

$$\frac{\|\hat{\eta}^{DY}\|_2}{\sqrt{n}} \leq \frac{\|\widetilde{\mathbf{H}}\|_2}{\sqrt{n}} \frac{1}{\sqrt{p}} \left\| \begin{bmatrix} \sqrt{p} \hat{\theta}^{DY} \\ \hat{\beta}^{DY} \end{bmatrix} \right\|_2 \leq 3K.$$

For these baseline constants,

$$\neg \mathcal{M}_n \subseteq \neg \mathcal{E}_{\beta,\theta} \cup \neg \mathcal{E}_{\widetilde{\mathbf{H}}} \cup \left\{ \left\| \hat{\lambda}^{DY} \right\|_2 > \sqrt{n} \right\}.$$

Lemma 4 controls the first event. Since $p/n \rightarrow \kappa \in (0, 1)$, Lemma 36 applied for all sufficiently large n with $t = \sqrt{n} - \sqrt{p}$, controls the second by $C \exp\{-cn\}$. Since $\tilde{Y}_i \in (0, 1)$ and $\rho'(\hat{\eta}_i^{DY}) \in (0, 1)$, one has $|\hat{\lambda}_i^{DY}| < 1$ for every i , so the third event is empty. The claim follows by a union bound after adjusting N, C, c . Enlarging $C_{\beta}, C_{\theta}, C_{\eta}$ enlarges $\mathcal{M}_n$, so the same bound remains valid. $\square$

Lemma 4 (MDYPL boundedness). *Let $\hat{\theta}^{DY}, \hat{\beta}^{DY}$ be the MDYPL estimates of (6). Then, in the setup of Section 2, there exists an $N \in \mathbb{N}$ and constants K, C, c such that for all $n > N$,*

$$\Pr \left(\frac{1}{\sqrt{p}} \left\| \begin{bmatrix} \sqrt{p} \hat{\theta}^{DY} \\ \hat{\beta}^{DY} \end{bmatrix} \right\|_2 > K \right) \leq C \exp\{-cn\},$$

where K, C, c depend only on the constants $\kappa, \alpha, \mathbf{\Gamma}, \theta_0^*, \theta_P^*$ and on the constants in Assumptions (i) and (iii).

Proof. For $c = \alpha y + (1 - \alpha)\rho'(z)$, if $x \geq 0$, then $c \leq 1 - (1 - \alpha)\rho'(-z)$ and $\rho(x) > x$, so

$$\rho(x) - cx > (1 - \alpha)\rho'(-z)x.$$

If $x \leq 0$, then $c \geq (1 - \alpha)\rho'(z)$ and $\rho(x) > 0$, so

$$\rho(x) - cx > (1 - \alpha)\rho'(z)|x|.$$

Hence, in both cases,

$$\rho(x) - \{\alpha y + (1 - \alpha)\rho'(z)\}x > (1 - \alpha) \min\{\rho'(z), \rho'(-z)\} |x|.$$

Applying this bound coordinatewise with $x = \hat{\theta}^{\text{DY}} + \mathbf{x}_i^\top \hat{\beta}^{\text{DY}}$, $z = \theta_P + \mathbf{x}_i^\top \beta_P$, and $y = Y_i$, and using optimality of $\hat{\theta}^{\text{DY}}, \hat{\beta}^{\text{DY}}$, it holds that

$$\begin{aligned} n \log(2) &= -\ell(0, \mathbf{0}_p; \tilde{\mathbf{y}}, \mathbf{X}) \\ &\geq -\ell(\hat{\theta}^{\text{DY}}, \hat{\beta}^{\text{DY}}; \tilde{\mathbf{y}}, \mathbf{X}) \\ &\geq (1 - \alpha) \sum_{i=1}^n \omega(\theta_P + \mathbf{x}_i^\top \beta_P) \left| \hat{\theta}^{\text{DY}} + \mathbf{x}_i^\top \hat{\beta}^{\text{DY}} \right|, \end{aligned}$$

for $\omega(x) = \min\{\rho'(x), \rho'(-x)\} = \omega(|x|)$. Define

$$\mathcal{I} = \left\{ i \in [n] : |\theta_P + \mathbf{x}_i^\top \beta_P| \leq \frac{3}{\sqrt{\epsilon p}} \left\| \begin{bmatrix} \sqrt{p} \theta_P \\ \beta_P \end{bmatrix} \right\|_2 \right\}.$$

Let

$$\omega_\epsilon = \omega \left(\frac{3}{\sqrt{\epsilon p}} \left\| \begin{bmatrix} \sqrt{p} \theta_P \\ \beta_P \end{bmatrix} \right\|_2 \right).$$

Recall that $\mathbf{h}_i^\top$ is the i th row of $\mathbf{H} = \sqrt{p} \mathbf{X}$, so that $\mathbf{x}_i = \mathbf{h}_i / \sqrt{p}$ with $\mathbf{h}_i \sim \mathcal{N}(\mathbf{0}_p, \mathbf{I}_p)$, and that $\widetilde{\mathbf{H}} = [\mathbf{1}, \mathbf{H}]$ has rows $\tilde{\mathbf{h}}_i^\top = (1, \mathbf{h}_i^\top)$. Then, we get

$$\begin{aligned} \frac{\log(2)}{1 - \alpha} &\geq \frac{1}{n} \sum_{i=1}^n \omega(\theta_P + \mathbf{x}_i^\top \beta_P) \left| \hat{\theta}^{\text{DY}} + \mathbf{x}_i^\top \hat{\beta}^{\text{DY}} \right| \\ &\geq \frac{1}{n} \sum_{i \in \mathcal{I}} \omega(\theta_P + \mathbf{x}_i^\top \beta_P) \left| \hat{\theta}^{\text{DY}} + \mathbf{x}_i^\top \hat{\beta}^{\text{DY}} \right| \\ &\geq \frac{1}{n} \sum_{i \in \mathcal{I}} \omega_\epsilon \left| \hat{\theta}^{\text{DY}} + \mathbf{x}_i^\top \hat{\beta}^{\text{DY}} \right| \\ &= \omega_\epsilon \frac{1}{n} \sum_{i \in \mathcal{I}} \frac{1}{\sqrt{p}} \left| [1, \mathbf{h}_i^\top] \begin{bmatrix} \sqrt{p} \hat{\theta}^{\text{DY}} \\ \hat{\beta}^{\text{DY}} \end{bmatrix} \right| \\ &\geq \frac{\omega_\epsilon}{\sqrt{p}} \left\| \begin{bmatrix} \sqrt{p} \hat{\theta}^{\text{DY}} \\ \hat{\beta}^{\text{DY}} \end{bmatrix} \right\|_2 \inf_{\substack{\mathbf{u} \in \mathbb{R}^{p+1} \\ \|\mathbf{u}\|_2=1}} \frac{1}{n} \sum_{i \in \mathcal{I}} |\tilde{\mathbf{h}}_i^\top \mathbf{u}| \\ &= \frac{\omega_\epsilon}{\sqrt{p}} \left\| \begin{bmatrix} \sqrt{p} \hat{\theta}^{\text{DY}} \\ \hat{\beta}^{\text{DY}} \end{bmatrix} \right\|_2 \inf_{\substack{\mathbf{u} \in \mathbb{R}^{p+1} \\ \|\mathbf{u}\|_2=1}} \left(\frac{1}{n} \sum_{i=1}^n |\tilde{\mathbf{h}}_i^\top \mathbf{u}| - \frac{1}{n} \sum_{i \notin \mathcal{I}} |\tilde{\mathbf{h}}_i^\top \mathbf{u}| \right) \\ &\geq \frac{\omega_\epsilon}{\sqrt{p}} \left\| \begin{bmatrix} \sqrt{p} \hat{\theta}^{\text{DY}} \\ \hat{\beta}^{\text{DY}} \end{bmatrix} \right\|_2 \inf_{\substack{\mathbf{u} \in \mathbb{R}^{p+1} \\ \|\mathbf{u}\|_2=1}} \left(\frac{1}{n} \sum_{i=1}^n |\tilde{\mathbf{h}}_i^\top \mathbf{u}| - \frac{\sqrt{n - |\mathcal{I}|}}{n} \sqrt{\sum_{i \notin \mathcal{I}} (\tilde{\mathbf{h}}_i^\top \mathbf{u})^2} \right) \\ &\geq \frac{\omega_\epsilon}{\sqrt{p}} \left\| \begin{bmatrix} \sqrt{p} \hat{\theta}^{\text{DY}} \\ \hat{\beta}^{\text{DY}} \end{bmatrix} \right\|_2 \inf_{\substack{\mathbf{u} \in \mathbb{R}^{p+1} \\ \|\mathbf{u}\|_2=1}} \left(\frac{1}{n} \sum_{i=1}^n |\tilde{\mathbf{h}}_i^\top \mathbf{u}| - \frac{\sqrt{n - |\mathcal{I}|}}{n} \sqrt{\sum_{i=1}^n (\tilde{\mathbf{h}}_i^\top \mathbf{u})^2} \right) \\ &\geq \frac{\omega_\epsilon}{\sqrt{p}} \left\| \begin{bmatrix} \sqrt{p} \hat{\theta}^{\text{DY}} \\ \hat{\beta}^{\text{DY}} \end{bmatrix} \right\|_2 \left\{ \inf_{\substack{\mathbf{u} \in \mathbb{R}^{p+1} \\ \|\mathbf{u}\|_2=1}} \left(\frac{1}{n} \sum_{i=1}^n |\tilde{\mathbf{h}}_i^\top \mathbf{u}| \right) - \sqrt{\frac{n - |\mathcal{I}|}{n}} \sup_{\substack{\mathbf{u} \in \mathbb{R}^{p+1} \\ \|\mathbf{u}\|_2=1}} \left(\frac{1}{\sqrt{n}} \sqrt{\sum_{i=1}^n (\tilde{\mathbf{h}}_i^\top \mathbf{u})^2} \right) \right\} \\ &= \frac{\omega_\epsilon}{\sqrt{p}} \left\| \begin{bmatrix} \sqrt{p} \hat{\theta}^{\text{DY}} \\ \hat{\beta}^{\text{DY}} \end{bmatrix} \right\|_2 \left\{ \inf_{\substack{\mathbf{u} \in \mathbb{R}^{p+1} \\ \|\mathbf{u}\|_2=1}} \frac{1}{n} \|\widetilde{\mathbf{H}} \mathbf{u}\|_1 - \sqrt{\frac{n - |\mathcal{I}|}{n}} \frac{1}{\sqrt{n}} \|\widetilde{\mathbf{H}}\|_2 \right\}, \end{aligned} \tag{175}$$

The third line follows by the definition of $\mathcal{I}$ and since $\omega(x)$ is decreasing in $|x|$ and the seventh line follows by the Cauchy–Schwarz inequality (Rudin [1976], Theorem 1.37(d)).

Now let $c_{\ell_1}(\kappa)$ be the lower bound of Lemma [5], $\epsilon = (c_{\ell_1}(\kappa)/12)^2$ and $C_{\theta_P} = \max\{1, |\theta_P^*|\}$, $C_{\beta_P} = \max\{1, \delta\}$, and define the events

$$\begin{aligned}\mathcal{E}_1 &= \left\{ \inf_{\substack{\mathbf{u} \in \mathbb{R}^{p+1}; \\ \|\mathbf{u}\|_2=1}} \frac{1}{n} \|\widetilde{\mathbf{H}}\mathbf{u}\|_1 \geq \frac{c_{\ell_1}(\kappa)}{2} \right\} \\ \mathcal{E}_2 &= \left\{ \frac{\|\widetilde{\mathbf{H}}\|_2}{\sqrt{n}} \leq 3 \right\} \\ \mathcal{E}_3 &= \left\{ \frac{1}{\sqrt{p}} \left\| \begin{bmatrix} \sqrt{p}\theta_P \\ \beta_P \end{bmatrix} \right\|_2 \leq 2\sqrt{C_{\theta_P}^2 + C_{\beta_P}^2} \right\}.\end{aligned}$$

By Lemma [5] applied with the fixed gap $c_{\ell_1}(\kappa)/2$, there exist $n_1 \in \mathbb{N}$ and $c_1 = c_1(\kappa) > 0$ such that, for all $n > n_1$,

$$\Pr(\mathcal{E}_1) \geq 1 - 4\exp\{-c_1 n\}.$$

Since $1 - \sqrt{p/n} \rightarrow 1 - \sqrt{\kappa} > 0$, there exists $n_2 \in \mathbb{N}$ such that, for all $n > n_2$, $1 - \sqrt{p/n} \geq (1 - \sqrt{\kappa})/2$. Taking $t = \sqrt{n} - \sqrt{p}$ in Lemma [36] therefore gives

$$\Pr(\mathcal{E}_2) \geq 1 - 2\exp\left\{-\frac{c_{\text{sv}}}{4}(1 - \sqrt{\kappa})^2 n\right\}.$$

Finally, by the modelling assumptions (i) and (iii), there exists an $n_3 \in \mathbb{N}$ such that for all $n > n_3$,

$$\Pr(\mathcal{E}_3) \geq 1 - C_{\Theta} \exp\{-c_{\Theta} n\} - C_{\Gamma} \exp\{-c_{\Gamma} n\}.$$

Lemma [37] implies that, conditional on $\mathcal{E}_2$,

$$\sqrt{\frac{n - |\mathcal{I}|}{n}} \frac{1}{\sqrt{n}} \|\widetilde{\mathbf{H}}\|_2 \leq 3\sqrt{\epsilon} \leq c_{\ell_1}(\kappa)/4.$$

Conditionally on the event $\mathcal{E}_1 \cap \mathcal{E}_2 \cap \mathcal{E}_3$, ([175]) yields that

$$\frac{\log(2)}{1 - \alpha} \geq \omega\left(\frac{72\sqrt{C_{\theta_P}^2 + C_{\beta_P}^2}}{c_{\ell_1}(\kappa)}\right) \left\{ \frac{c_{\ell_1}(\kappa)}{2} - \frac{c_{\ell_1}(\kappa)}{4} \right\} \frac{1}{\sqrt{p}} \left\| \begin{bmatrix} \sqrt{p}\hat{\theta}^{\text{DY}} \\ \hat{\beta}^{\text{DY}} \end{bmatrix} \right\|_2.$$

From this we conclude that

$$\frac{1}{\sqrt{p}} \left\| \begin{bmatrix} \sqrt{p}\hat{\theta}^{\text{DY}} \\ \hat{\beta}^{\text{DY}} \end{bmatrix} \right\|_2 \leq \frac{4\log(2)}{1 - \alpha} \left\{ \omega\left(\frac{72\sqrt{C_{\theta_P}^2 + C_{\beta_P}^2}}{c_{\ell_1}(\kappa)}\right) c_{\ell_1}(\kappa) \right\}^{-1}.$$

Thus, for $n' = \max\{n_1, n_2, n_3\}$, it follows that for all $n > n'$,

$$\begin{aligned}\Pr(\mathcal{E}_1 \cap \mathcal{E}_2 \cap \mathcal{E}_3) &\geq 1 - \Pr(\neg\mathcal{E}_1) - \Pr(\neg\mathcal{E}_2) - \Pr(\neg\mathcal{E}_3) \\ &\geq 1 - 4\exp\{-c_1 n\} - 2\exp\{-c_2 n\} - 2C_3 \exp\{-c_3 n\},\end{aligned}$$

for $c_2 = c_{\text{sv}}(1 - \sqrt{\kappa})^2/4$, $C_3 = \max\{C_{\Theta}, C_{\Gamma}\}$, $c_3 = \min\{c_{\Theta}, c_{\Gamma}\}$. The claim follows by taking $N = \max\{n_1, n_2, n_3\}$ and $c = \min\{c_1, c_2, c_3\}$, $C = 6 + 2C_3$. $\square$

Lemma 5. *Let*

$$c_{\ell_1}(\kappa) = \inf_{r \in [0,1]} \mathbb{E}[(W(r) - \tau(r))_+], \quad (176)$$

with $(x)_+ = \max\{0, x\}$, $W(r) = |rZ - \sqrt{1 - \tau^2}|$, $Z \sim N(0, 1)$ and where $\tau(r)$ is the unique nonnegative root of

$$\mathbb{E}[\min\{W(r)^2, \tau(r)^2\}] = r^2 \kappa.$$

Then $c_{\ell_1}(\kappa) > 0$ for any $\kappa \in (0, 1)$.

Furthermore, let $\widetilde{\mathbf{H}} = [\mathbf{1} \quad \mathbf{G}]$, where $\mathbf{G}$ is an $n \times p$ matrix with i.i.d. $N(0, 1)$ entries and $p/n \rightarrow \kappa \in (0, 1)$ as $n \rightarrow \infty$. Then, for every fixed $\epsilon > 0$, there exist $N \in \mathbb{N}$ and $c = c(\kappa) > 0$ such that, for all $n > N$,

$$\Pr \left(\inf_{\substack{\mathbf{u} \in \mathbb{R}^{p+1}: \\ \|\mathbf{u}\|_2 = 1}} \frac{1}{n} \|\widetilde{\mathbf{H}}\mathbf{u}\|_1 < c_{\ell_1}(\kappa) - \epsilon \right) \leq 4 \exp\{-c n \epsilon^2 / 4\},$$

where $c_{\ell_1}(\kappa)$ is as defined in (176).

Proof. Consider the minimisation problem:

$$\inf_{\substack{\mathbf{u} \in \mathbb{R}^{p+1}: \\ \|\mathbf{u}\|_2 = 1}} \frac{1}{n} \|\widetilde{\mathbf{H}}\mathbf{u}\|_1.$$

We shall translate it into a min-max problem and employ the lower-tail comparison of Theorem B.2(i) to analyse its behaviour.

Constraint set For $\mathbf{u} \in \mathbb{R}^{p+1}$ with $\|\mathbf{u}\|_2 = 1$, write $\mathbf{u} = [\theta, \mathbf{w}^\top]^\top$, where $\|\mathbf{w}\|_2 \leq 1$ and $|\theta| = \sqrt{1 - \|\mathbf{w}\|_2^2}$. The minimisation may be restricted to vectors with nonpositive first coordinate. Indeed, define

$$\tilde{\mathbf{u}} = \begin{cases} \mathbf{u}, & \theta \leq 0, \\ -\mathbf{u}, & \theta > 0. \end{cases}$$

Then $\|\tilde{\mathbf{u}}\|_2 = 1$, the first coordinate of $\tilde{\mathbf{u}}$ is $-|\theta|$, and

$$\|\widetilde{\mathbf{H}}\tilde{\mathbf{u}}\|_1 = \|\widetilde{\mathbf{H}}\mathbf{u}\|_1.$$

Conversely, any such vector lies in the original constraint set. So it is equivalent to minimise over all unit vectors in $\mathbb{R}^{p+1}$ or over all vectors

$$\tilde{\mathbf{u}} = \begin{bmatrix} -\sqrt{1 - \|\mathbf{w}\|_2^2} \\ \mathbf{w} \end{bmatrix},$$

with $\mathbf{w} \in \mathbb{R}^p : \|\mathbf{w}\|_2 \leq 1$. Hence, rewrite the minimisation as

$$\frac{1}{n} \inf_{\substack{\mathbf{u} \in \mathbb{R}^{p+1}: \\ \|\mathbf{u}\|_2 = 1}} \|\widetilde{\mathbf{H}}\mathbf{u}\|_1 = \frac{1}{n} \inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2 \leq 1}} \left\| \mathbf{G}\mathbf{u} - \sqrt{1 - \|\mathbf{u}\|_2^2} \mathbf{1} \right\|_1.$$

Dual norm Now note that

$$\|\mathbf{a}\|_1 = \sup_{\|\mathbf{v}\|_\infty \leq 1} \mathbf{a}^\top \mathbf{v}.$$

Therefore,

$$\begin{aligned} \frac{1}{n} \inf_{\substack{\mathbf{u} \in \mathbb{R}^{p+1}: \\ \|\mathbf{u}\|_2 = 1}} \|\widetilde{\mathbf{H}}\mathbf{u}\|_1 &= \frac{1}{n} \inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2 \leq 1}} \left\| \mathbf{G}\mathbf{u} - \sqrt{1 - \|\mathbf{u}\|_2^2} \mathbf{1} \right\|_1 \\ &= \frac{1}{n} \inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2 \leq 1}} \sup_{\substack{\mathbf{v} \in \mathbb{R}^p: \\ \|\mathbf{v}\|_\infty \leq 1}} \mathbf{v}^\top \left(\mathbf{G}\mathbf{u} - \sqrt{1 - \|\mathbf{u}\|_2^2} \mathbf{1} \right) \\ &= \frac{1}{n} \inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2 \leq 1}} \sup_{\substack{\mathbf{v} \in \mathbb{R}^p: \\ \|\mathbf{v}\|_\infty \leq 1}} \mathbf{v}^\top \mathbf{G}\mathbf{u} - \sqrt{1 - \|\mathbf{u}\|_2^2} \mathbf{v}^\top \mathbf{1}. \end{aligned} \tag{177}$$

The constraint sets of (177) are compact, and the term

$$\psi(\mathbf{u}, \mathbf{v}) = -\sqrt{1 - \|\mathbf{u}\|_2^2} \mathbf{v}^\top \mathbf{1},$$

is continuous on the product of these constraint sets and thus the lower-tail part of Theorem B.2(i) applies. The outer factor n^{-1} is handled by applying the comparison to the unnormalised PO/AO values and rescaling the threshold, as recorded below.

PO & AO Define the PO

$$\Phi(\widetilde{\mathbf{H}}) = \frac{1}{n} \inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2 \leq 1}} \sup_{\substack{\mathbf{v} \in \mathbb{R}^n: \\ \|\mathbf{v}\|_\infty \leq 1}} \mathbf{v}^\top \mathbf{G} \mathbf{u} - \sqrt{1 - \|\mathbf{u}\|_2^2} \mathbf{v}^\top \mathbf{1},$$

and for two independent vectors $\mathbf{g} \in \mathbb{R}^n$, $\mathbf{h} \in \mathbb{R}^p$ with i.i.d. $N(0, 1)$, define the corresponding AO as

$$\phi(\mathbf{g}, \mathbf{h}) = \frac{1}{n} \inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2 \leq 1}} \sup_{\substack{\mathbf{v} \in \mathbb{R}^n: \\ \|\mathbf{v}\|_\infty \leq 1}} \|\mathbf{u}\|_2 \mathbf{g}^\top \mathbf{v} + \|\mathbf{v}\|_2 \mathbf{h}^\top \mathbf{u} - \sqrt{1 - \|\mathbf{u}\|_2^2} \mathbf{v}^\top \mathbf{1}.$$

Let $\bar{\Phi}(\widetilde{\mathbf{H}}) = n\Phi(\widetilde{\mathbf{H}})$ and $\bar{\phi}(\mathbf{g}, \mathbf{h}) = n\phi(\mathbf{g}, \mathbf{h})$ denote the corresponding unnormalised values. By the lower-tail part of Theorem B.2(i), for every $t \in \mathbb{R}$,

$$\Pr(\Phi(\widetilde{\mathbf{H}}) < t) = \Pr(\bar{\Phi}(\widetilde{\mathbf{H}}) < nt) \leq 2 \Pr(\bar{\phi}(\mathbf{g}, \mathbf{h}) \leq nt) = 2 \Pr(\phi(\mathbf{g}, \mathbf{h}) \leq t).$$

Radial decomposition Any $\mathbf{u} \in \mathbb{R}^p$ can be decomposed into a length r and a unit vector along its direction. Hence the AO is equivalent to

$$\phi(\mathbf{g}, \mathbf{h}) = \frac{1}{n} \inf_{\substack{r \in [0, 1] \\ \mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2 = 1}} \sup_{\substack{\mathbf{v} \in \mathbb{R}^n: \\ \|\mathbf{v}\|_\infty \leq 1}} r \mathbf{g}^\top \mathbf{v} + r \|\mathbf{v}\|_2 \mathbf{h}^\top \mathbf{u} - \sqrt{1 - r^2} \mathbf{v}^\top \mathbf{1}. \quad (178)$$

For every $\mathbf{u} \in \mathbb{R}^p$ with $\|\mathbf{u}\|_2 = 1$, the Cauchy–Schwarz inequality gives

$$\|\mathbf{v}\|_2 \mathbf{h}^\top \mathbf{u} \geq -\|\mathbf{v}\|_2 \|\mathbf{h}\|_2.$$

Equality is attained at $\mathbf{u} = -\mathbf{h}/\|\mathbf{h}\|_2$ when $\mathbf{h} \neq \mathbf{0}_p$, and by every feasible $\mathbf{u}$ when $\mathbf{h} = \mathbf{0}_p$. Hence, (178) becomes

$$\phi(\mathbf{g}, \mathbf{h}) = \frac{1}{n} \inf_{r \in [0, 1]} \sup_{\substack{\mathbf{v} \in \mathbb{R}^n: \\ \|\mathbf{v}\|_\infty \leq 1}} \mathbf{v}^\top \left(r \mathbf{g} - \sqrt{1 - r^2} \mathbf{1} \right) - r \|\mathbf{v}\|_2 \|\mathbf{h}\|_2. \quad (179)$$

Reparameterisation Define

$$\mathbf{b}(r) = r \mathbf{g} - \sqrt{1 - r^2} \mathbf{1}, \quad w_i(r) = |b_i(r)|, \quad \lambda(r) = r \frac{\|\mathbf{h}\|_2}{\sqrt{n}}, \quad s(\mathbf{a}) = \left(\frac{1}{n} \sum_{i=1}^n a_i^2 \right)^{1/2}.$$

For every $\mathbf{v}$ with $\|\mathbf{v}\|_\infty \leq 1$, setting $a_i = |v_i|$ gives

$$\mathbf{v}^\top \mathbf{b}(r) \leq \sum_{i=1}^n a_i w_i(r), \quad \|\mathbf{v}\|_2 = \|\mathbf{a}\|_2.$$

Conversely, for every $\mathbf{a} \in [0, 1]^n$, equality is attained by choosing $v_i = a_i \text{sign}\{b_i(r)\}$ when $b_i(r) \neq 0$ and $v_i = a_i$ when $b_i(r) = 0$. Therefore (179) is equivalent to

$$\phi(\mathbf{g}, \mathbf{h}) = \inf_{r \in [0, 1]} \sup_{\mathbf{a} \in [0, 1]^n} \frac{1}{n} \sum_{i=1}^n a_i w_i(r) - \lambda(r) s(\mathbf{a}).$$

For any fixed $r \in [0, 1]$ consider the inner maximisation, which is given by

$$\sup_{\mathbf{a} \in \mathbb{R}^n} \frac{1}{n} \sum_{i=1}^n a_i w_i(r) - \lambda(r) s(\mathbf{a}), \quad \text{such that } a_i \in [0, 1], \text{ for } i = 1, \dots, n.$$

Equivalently, define the box-constrained minimisation

$$m(r) = - \inf_{\mathbf{a} \in [0, 1]^n} - \frac{1}{n} \sum_{i=1}^n a_i w_i(r) + \lambda(r) s(\mathbf{a}). \quad (180)$$

The Lagrangian associated with (180) is given by

$$\mathcal{L}(\mathbf{a}, \boldsymbol{\mu}, \boldsymbol{\nu}) = -\frac{1}{n} \sum_{i=1}^n a_i w_i(r) + \lambda(r) s(\mathbf{a}) + \sum_{i=1}^n \mu_i (a_i - 1) - \sum_{i=1}^n \nu_i a_i.$$

For every fixed $r \in [0, 1]$, the objective in (180) is continuous and convex, and the feasible box $[0, 1]^n$ is compact. Hence an optimal solution exists. Slater's condition holds, for example, at every $\mathbf{a} \in (0, 1)^n$, so the KKT conditions are necessary and sufficient for optimality (see, for example, Beck [2017] Theorem 3.78).

KKT conditions Thus, consider the KKT conditions.

1. **Stationarity:** For $\mathbf{a} \neq \mathbf{0}_n$, a_i must satisfy:

$$\frac{\partial \mathcal{L}(\mathbf{a}, \boldsymbol{\mu}, \boldsymbol{\nu})}{\partial a_i} = -\frac{w_i(r)}{n} + \lambda(r) \frac{a_i}{n \cdot s(\mathbf{a})} + \mu_i - \nu_i = 0.$$

If $\mathbf{a} = \mathbf{0}_n$, then $\mathbf{0}_n$ must be in the subdifferential of $\mathcal{L}(\mathbf{0}_n, \boldsymbol{\mu}, \boldsymbol{\nu})$, denote it by $\partial \mathcal{L}(\mathbf{0}_n, \boldsymbol{\mu}, \boldsymbol{\nu})$. Now

$$\begin{aligned} \partial \mathcal{L}(\mathbf{0}_n, \boldsymbol{\mu}, \boldsymbol{\nu}) &= \left\{ \mathbf{s} \in \mathbb{R}^n : \mathcal{L}(\mathbf{a}, \boldsymbol{\mu}, \boldsymbol{\nu}) \geq \mathcal{L}(\mathbf{0}_n, \boldsymbol{\mu}, \boldsymbol{\nu}) + \mathbf{s}^\top \mathbf{a}, \forall \mathbf{a} \in \mathbb{R}^n \right\} \\ &= \left\{ \mathbf{s} \in \mathbb{R}^n : \frac{\lambda(r)}{\sqrt{n}} \|\mathbf{a}\|_2 \geq \mathbf{s}^\top \left(\mathbf{s} + \frac{1}{n} \mathbf{w}(r) + \boldsymbol{\nu} - \boldsymbol{\mu} \right), \forall \mathbf{a} \in \mathbb{R}^n \right\} \\ &= \left\{ \mathbf{s} \in \mathbb{R}^n : \frac{\lambda(r)}{\sqrt{n}} \geq \left\| \mathbf{s} + \frac{1}{n} \mathbf{w}(r) + \boldsymbol{\nu} - \boldsymbol{\mu} \right\|_2 \right\} \\ &= \left\{ \mathbf{s} \in \mathbb{R}^n : \mathbf{s} = \lambda(r) \mathbf{u} - \frac{1}{n} \mathbf{w}(r) - \boldsymbol{\nu} + \boldsymbol{\mu}, \mathbf{u} \in \mathbb{R}^n : \|\mathbf{u}\|_2 \leq \frac{1}{\sqrt{n}} \right\}. \end{aligned}$$

2. **Primal feasibility:** $a_i \in [0, 1]$ ($i = 1, \dots, n$)

3. **Dual Feasibility:** $\mu_i, \nu_i \geq 0$ ($i = 1, \dots, n$)

4. **Complementary slackness:** $\mu_i (a_i - 1) = 0, \nu_i a_i = 0$

Consider the following cases for candidate solutions:

(i) $a_i \in (0, 1)$: By complementary slackness, $\mu_i = \nu_i = 0$. Define $\tau_n(r) = \lambda(r)/s(\mathbf{a})$. Stationarity gives

$$\tau_n(r) a_i = w_i(r).$$

If $\lambda(r) = 0$, stationarity therefore requires $w_i(r) = 0$. In particular, when $r = 0$, $w_i(0) = 1$, so no interior coordinate can occur. If $\lambda(r) > 0$, then $\tau_n(r) > 0$ and

$$a_i = \frac{w_i(r)}{\tau_n(r)} < 1 \iff w_i(r) < \tau_n(r).$$

(ii) $a_i = 1$: By complementary slackness, $\nu_i = 0$. By stationarity: $\tau_n(r) + n\mu_i = w_i(r)$. If $r = 0$, $\tau_n(r) = 0$, $w_i(r) = 1$ so that $\mu_i = 1/n$ is feasible. If $r \neq 0$, then stationarity requires that $\mu_i = (w_i(r) - \tau_n(r))/n$. This is only dual feasible if $w_i(r) \geq \tau_n(r)$.

- (iii) $a_i = 0, \mathbf{a} \neq \mathbf{0}_n$: By complementary slackness, $\mu_i = 0$. By stationarity, $-n\nu_i = w_i(r)$. If $r = 0$, $w_i(r) = 1$ which requires that $\nu_i = -1/n$, which violates dual feasibility. If $r \neq 0$, then stationarity and dual feasibility can only hold simultaneously if $w_i(r) = 0$, which requires that $rg_i = \sqrt{1-r^2}$. By dual feasibility, this requires that $\nu_i = 0$.
- (iv) $\mathbf{a} = \mathbf{0}_n$: For every $\mathbf{a} \in [0, 1]^n$, the Cauchy-Schwarz inequality gives

$$\frac{1}{n} \sum_{i=1}^n a_i w_i(r) - \lambda(r) s(\mathbf{a}) \leq s(\mathbf{a}) \left\{ \frac{\|\mathbf{w}(r)\|_2}{\sqrt{n}} - r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} \right\}.$$

Hence $\mathbf{a} = \mathbf{0}_n$ is optimal if $\|\mathbf{w}(r)\|_2 \leq r\|\mathbf{h}\|_2$. Conversely, if $\|\mathbf{w}(r)\|_2 > r\|\mathbf{h}\|_2$, choose $t > 0$ sufficiently small that $t\mathbf{w}(r) \in [0, 1]^n$. At $\mathbf{a} = t\mathbf{w}(r)$, the fixed-radius objective equals

$$\frac{t}{n} \|\mathbf{w}(r)\|_2 \{ \|\mathbf{w}(r)\|_2 - r \|\mathbf{h}\|_2 \} > 0.$$

Therefore $\mathbf{a} = \mathbf{0}_n$ is optimal if and only if

$$\frac{\|\mathbf{w}(r)\|_2}{\sqrt{n}} - r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} \leq 0.$$

We now replace this possibility by a deterministic high-probability exclusion event. Fix $\kappa_+ \in (\kappa, 1)$. Choose $\eta_\kappa \in (0, 1 - \kappa_+)$, and set $\Delta_\kappa = \sqrt{1 - \eta_\kappa} - \sqrt{\kappa_+} > 0$. Define

$$\mathcal{E}_n^{\text{nz}} = \left\{ \sup_{r \in [0, 1]} \left| \frac{1}{n} \|\mathbf{w}(r)\|_2^2 - 1 \right| \leq \eta_\kappa \right\} \cap \left\{ \frac{1}{n} \|\mathbf{h}\|_2^2 \leq \kappa_+ \right\}.$$

For all sufficiently large n , there exist constants $C, c > 0$, depending only on κ , such that

$$\Pr(\neg \mathcal{E}_n^{\text{nz}}) \leq C \exp\{-cn\}.$$

Indeed,

$$\sup_{r \in [0, 1]} \left| \frac{1}{n} \|\mathbf{w}(r)\|_2^2 - 1 \right| = \sup_{r \in [0, 1]} \left| r^2 \left(\frac{1}{n} \sum_{i=1}^n g_i^2 - 1 \right) - 2r\sqrt{1-r^2} \frac{1}{n} \sum_{i=1}^n g_i \right| \leq \left| \frac{1}{n} \sum_{i=1}^n g_i^2 - 1 \right| + \left| \frac{1}{n} \sum_{i=1}^n g_i \right|,$$

The scalar Gaussian average is controlled by the standard one-dimensional Gaussian tail bound, while the two quadratic terms are controlled by chi-square concentration. (see, for example, [Vershynin 2018](#) Exercise 3.3.7 and Theorem 3.1.1). Hence the right-hand side has an exponentially small upper tail. The second event also has an exponentially small complement by the same chi-square concentration, since $p/n \rightarrow \kappa < \kappa_+$ and $\|\mathbf{h}\|_2^2 \sim \chi_p^2$.

On $\mathcal{E}_n^{\text{nz}}$,

$$\inf_{r \in [0, 1]} \left\{ \frac{1}{\sqrt{n}} \|\mathbf{w}(r)\|_2 - r \frac{1}{\sqrt{n}} \|\mathbf{h}\|_2 \right\} \geq \sqrt{1 - \eta_\kappa} - \sqrt{\kappa_+} = \Delta_\kappa > 0.$$

Consequently, on $\mathcal{E}_n^{\text{nz}}$, the case $\mathbf{a} = \mathbf{0}_n$ is not optimal for any $r \in [0, 1]$.

Set

$$A_{\ell_1, n}(r, \tau) = \frac{1}{n} \sum_{i=1}^n \min\{\tau^2, w_i(r)^2\}, \quad \lambda_n^2(r) = r^2 \frac{\|\mathbf{h}\|_2^2}{n},$$

and

$$A_{\ell_1, n}^{\max}(r) = \frac{1}{n} \sum_{i=1}^n w_i(r)^2.$$

For each $r \in [0, 1]$, define $\tau_n(r)$ as follows. If $\lambda_n^2(r) < A_{\ell_1, n}^{\max}(r)$, let $\tau_n(r)$ be the unique nonnegative solution of

$$A_{\ell_1, n}(r, \tau_n(r)) = \lambda_n^2(r).$$

If $\lambda_n^2(r) \geq A_{\ell_1, n}^{\max}(r)$, set $\tau_n(r) = 0$ as a harmless convention. Now define the extended fixed-radius value

$$\hat{\phi}_n(r; \mathbf{g}, \mathbf{h}) = \begin{cases} \frac{1}{n} \sum_{i=1}^n (w_i(r) - \tau_n(r))_+, & \lambda_n^2(r) < A_{\ell_1, n}^{\max}(r), \\ 0, & \lambda_n^2(r) \geq A_{\ell_1, n}^{\max}(r). \end{cases}$$

We now verify the fixed-radius value. If $\lambda_n^2(r) \geq A_{\ell_1, n}^{\max}(r)$, the characterisation above shows that $\mathbf{a} = \mathbf{0}_n$ is optimal and the value is zero. Suppose that

$$\lambda_n^2(r) < A_{\ell_1, n}^{\max}(r).$$

The map $\tau \mapsto A_{\ell_1, n}(r, \tau)$ is continuous and strictly increasing on $[0, \max_{1 \leq i \leq n} w_i(r)]$, with endpoint values zero and $A_{\ell_1, n}^{\max}(r)$. Hence the defining equation for $\tau_n(r)$ has a unique solution on this interval. If $\lambda(r) = 0$, then $\tau_n(r) = 0$ and the fixed-radius objective is maximised at $\mathbf{a}^* = \mathbf{1}_n$, with value

$$\frac{1}{n} \sum_{i=1}^n w_i(r) = \frac{1}{n} \sum_{i=1}^n \{w_i(r) - \tau_n(r)\}_+.$$

Suppose finally that $\lambda(r) > 0$. Then $\tau_n(r) > 0$. Define

$$a_i^* = \min \left\{ \frac{w_i(r)}{\tau_n(r)}, 1 \right\}, \quad \mu_i^* = \frac{\{w_i(r) - \tau_n(r)\}_+}{n}, \quad \nu_i^* = 0.$$

The threshold equation yields

$$s(\mathbf{a}^*)^2 = \frac{1}{n} \sum_{i=1}^n \min \left\{ \frac{w_i(r)^2}{\tau_n(r)^2}, 1 \right\} = \frac{A_{\ell_1, n}(r, \tau_n(r))}{\tau_n(r)^2} = \frac{\lambda(r)^2}{\tau_n(r)^2}.$$

Thus $\lambda(r)/s(\mathbf{a}^*) = \tau_n(r)$. The displayed $\mathbf{a}^*, \mu^*, \nu^*$ satisfy primal feasibility, dual feasibility, complementary slackness, and, for every i ,

$$-\frac{w_i(r)}{n} + \lambda(r) \frac{a_i^*}{ns(\mathbf{a}^*)} + \mu_i^* - \nu_i^* = 0.$$

They therefore satisfy the KKT conditions, so $\mathbf{a}^*$ is optimal. Moreover,

$$\frac{1}{n} \sum_{i=1}^n a_i^* w_i(r) - \lambda(r) s(\mathbf{a}^*) = \frac{1}{n} \sum_{i=1}^n \{w_i(r) - \tau_n(r)\}_+.$$

Thus $\hat{\phi}_n(r; \mathbf{g}, \mathbf{h})$ is exactly the value of the fixed- r inner maximisation for every $r \in [0, 1]$. Therefore

$$\phi(\mathbf{g}, \mathbf{h}) = \inf_{r \in [0, 1]} \hat{\phi}_n(r; \mathbf{g}, \mathbf{h}).$$

On $\mathcal{E}_n^{\text{nz}}$, the second branch in the definition of $\hat{\phi}_n$ never occurs. Hence, on $\mathcal{E}_n^{\text{nz}}$,

$$\hat{\phi}_n(r; \mathbf{g}, \mathbf{h}) = \frac{1}{n} \sum_{i=1}^n (w_i(r) - \tau_n(r))_+, \quad r \in [0, 1].$$

For later use, write

$$\phi(r, \tau; \mathbf{g}, \mathbf{h}) = \frac{1}{n} \sum_{i=1}^n (w_i(r) - \tau)_+.$$

Limiting AO We now concern ourselves with the limiting value of the AO. For this, first define the limiting counterparts to the AO quantities.

Let $Z \sim \mathcal{N}(0, 1)$ and define

$$W(r) = \left| rZ - \sqrt{1 - r^2} \right|.$$

For each $r \in [0, 1]$, define

$$A_{\ell_1}(r, \tau) = \mathbb{E} [\min\{\tau^2, W(r)^2\}],$$

and let $\tau(r) \geq 0$ be the unique solution of

$$A_{\ell_1}(r, \tau(r)) = r^2 \kappa. \quad (181)$$

In particular, $\tau(0) = 0$, whereas $\tau(r) > 0$ for $r > 0$. Existence, uniqueness, continuity, and uniform boundedness are established in Lemma 6(iv).

Now let

$$\varphi(r, \tau) = \mathbb{E} [(W(r) - \tau)_+].$$

Then define the limiting AO integrand at radius r as

$$\varphi(r) = \varphi(r, \tau(r)) = \mathbb{E} [(W(r) - \tau(r))_+].$$

Then, the limiting AO value is

$$\phi^* = \inf_{r \in [0, 1]} \varphi(r).$$

Convergence of the AO to the limiting AO We will show that

$$\phi(\mathbf{g}, \mathbf{h}) \xrightarrow{\mathbb{P}} \phi^*. \quad (182)$$

The threshold representation is used only on the event $\mathcal{E}_n^{\text{nz}}$, whose complement is exponentially small. Thus it is enough to prove uniform convergence of the extended fixed-radius values on this event. More precisely, for every fixed $\delta > 0$,

$$\Pr(|\phi(\mathbf{g}, \mathbf{h}) - \phi^*| > \delta) \leq \Pr(\neg \mathcal{E}_n^{\text{nz}}) + \Pr\left(\sup_{r \in [0, 1]} |\widehat{\phi}_n(r; \mathbf{g}, \mathbf{h}) - \varphi(r)| > \delta, \mathcal{E}_n^{\text{nz}}\right).$$

The first term is exponentially small, and the second term tends to zero by the uniform threshold argument below.

To establish (182), we consider the following steps:

1. For $T > 0$,

$$\begin{aligned} \sup_{\substack{r \in [0, 1] \\ \tau \in [0, T]}} |\phi(r, \tau; \mathbf{g}, \mathbf{h}) - \varphi(r, \tau)| &\xrightarrow{\mathbb{P}} 0 \\ \sup_{\substack{r \in [0, 1] \\ \tau \in [0, T]}} |A_{\ell_1, n}(r, \tau) - A_{\ell_1}(r, \tau)| &\xrightarrow{\mathbb{P}} 0. \end{aligned} \quad (183)$$

2. For any $r \in [0, 1]$, $\sup_{\tau \in [0, 1]} |\tau_n(r) - \tau(r)| \xrightarrow{\mathbb{P}} 0$

3. $\sup_{r \in [0, 1]} |\widehat{\phi}_n(r; \mathbf{g}, \mathbf{h}) - \varphi(r)| \xrightarrow{\mathbb{P}} 0$

4. $|\phi(\mathbf{g}, \mathbf{h}) - \phi^*| \xrightarrow{\mathbb{P}} 0$

5. $\mathbb{E} [\phi(\mathbf{g}, \mathbf{h})] \rightarrow \phi^*$

6. Tail bound based on Lipschitz Gaussian concentration

As a first step, we show that $\phi(r, \tau; \mathbf{g}, \mathbf{h})$ and $A_{\ell_1, n}(r, \tau)$ converge uniformly in probability to $\varphi(r, \tau)$ and $A_{\ell_1}(r, \tau)$, respectively, on $[0, 1] \times [0, T]$. For every fixed $z \in \mathfrak{R}$, the maps

$$(r, \tau) \mapsto \left(\left| rz - \sqrt{1 - r^2} \right| - \tau \right)_+,$$

$$(r, \tau) \mapsto \min \left\{ \tau^2, (rz - \sqrt{1 - r^2})^2 \right\}$$

are continuous on $[0, 1] \times [0, T]$. Uniformly over this parameter set,

$$\left(\left| rz - \sqrt{1 - r^2} \right| - \tau \right)_+ \leq |z| + 1,$$

$$\min \left\{ \tau^2, (rz - \sqrt{1 - r^2})^2 \right\} \leq 2z^2 + 2.$$

Both envelopes are integrable under the standard normal law. Dominated convergence gives continuity of φ and A_{ℓ_1} on the compact parameter set, and the uniform law of large numbers (see, for example, [Newey and McFadden 1994](#) Lemma 2.4) gives

$$\sup_{\substack{r \in [0, 1] \\ \tau \in [0, T]}} |\phi(r, \tau; \mathbf{g}, \mathbf{h}) - \varphi(r, \tau)| \xrightarrow{\mathbb{P}} 0,$$

$$\sup_{\substack{r \in [0, 1] \\ \tau \in [0, T]}} |A_{\ell_1, n}(r, \tau) - A_{\ell_1}(r, \tau)| \xrightarrow{\mathbb{P}} 0.$$

On $\mathcal{E}_n^{\text{nz}}$, for each $r \in [0, 1]$, the second branch in the definition of $\hat{\phi}_n$ never occurs, and $\tau_n(r)$ is the unique solution of

$$A_{\ell_1, n}(r, \tau_n(r)) = \lambda_n^2(r).$$

Similarly, $\tau(r)$ is the unique solution of [\(181\)](#) (see Lemma [6](#) (iv)). We now show that $\tau_n(r)$ converges uniformly over r to $\tau(r)$. Fix $\epsilon > 0$. Define

$$m_\epsilon^+ = \min_{r \in [0, 1]} \{A_{\ell_1}(r, \tau(r) + \epsilon) - r^2 \kappa\} > 0.$$

The minimum exists by continuity and compactness, and is strictly positive by strict monotonicity where applicable and by the extreme value theorem ([Rudin 1976](#) Theorem 4.16). Let $R_\epsilon = \{r \in [0, 1] : \tau(r) \geq \epsilon\}$. If $R_\epsilon \neq \emptyset$, define

$$m_\epsilon^- = \min_{r \in R_\epsilon} \{r^2 \kappa - A_{\ell_1}(r, \tau(r) - \epsilon)\} > 0.$$

This minimum exists because R_ϵ is compact, and it is strictly positive by strict monotonicity for $r > 0$ and the extreme value theorem ([Rudin 1976](#) Theorem 4.16). Set

$$m_\epsilon = \begin{cases} m_\epsilon^+, & R_\epsilon = \emptyset, \\ \min\{m_\epsilon^+, m_\epsilon^-\}, & R_\epsilon \neq \emptyset. \end{cases}$$

Since $\|\mathbf{h}\|_2^2 \sim \chi_p^2$ and $p/n \rightarrow \kappa$, chi-square concentration gives ([Vershynin 2018](#) Equation (3.1))

$$\frac{\|\mathbf{h}\|_2^2}{n} = \frac{p}{n} \frac{\|\mathbf{h}\|_2^2}{p} \xrightarrow{\mathbb{P}} \kappa.$$

Consequently,

$$\sup_{r \in [0, 1]} |\lambda_n^2(r) - r^2 \kappa| = \left| \frac{\|\mathbf{h}\|_2^2}{n} - \kappa \right| \xrightarrow{\mathbb{P}} 0.$$

Choose $T > \sup_{r \in [0, 1]} \tau(r) + \epsilon$, which is possible by Lemma [6](#) (iv). Combining the preceding convergence with [\(183\)](#), the intersection of the events

$$\sup_{r \in [0, 1]} |\lambda_n^2(r) - r^2 \kappa| \leq \frac{m_\epsilon}{4}$$

and

$$\sup_{\substack{r \in [0,1] \\ \tau \in [0,T]}} |A_{\ell_1,n}(r, \tau) - A_{\ell_1}(r, \tau)| \leq \frac{m_\epsilon}{4}$$

has probability tending to one. On this intersection, for every $r \in [0, 1]$,

$$A_{\ell_1,n}(r, \tau(r) + \epsilon) \geq A_{\ell_1}(r, \tau(r) + \epsilon) - \frac{m_\epsilon}{4} \geq r^2 \kappa + m_\epsilon^+ - \frac{m_\epsilon}{4} \geq r^2 \kappa + \frac{3m_\epsilon}{4} \geq \lambda_n^2(r) + \frac{m_\epsilon}{2}.$$

Since $\tau \mapsto A_{\ell_1,n}(r, \tau)$ is nondecreasing and $\tau_n(r)$ solves $A_{\ell_1,n}(r, \tau_n(r)) = \lambda_n^2(r)$, this gives $\tau_n(r) \leq \tau(r) + \epsilon$. If $r \in R_\epsilon$, then

$$A_{\ell_1,n}(r, \tau(r) - \epsilon) \leq A_{\ell_1}(r, \tau(r) - \epsilon) + \frac{m_\epsilon}{4} \leq r^2 \kappa - m_\epsilon^- + \frac{m_\epsilon}{4} \leq r^2 \kappa - \frac{3m_\epsilon}{4} \leq \lambda_n^2(r) - \frac{m_\epsilon}{2}.$$

and hence $\tau_n(r) \geq \tau(r) - \epsilon$. For $r \notin R_\epsilon$, the lower bound is automatic because $\tau_n(r) \geq 0 > \tau(r) - \epsilon$. Thus, on $\mathcal{E}_n^{\text{nz}}$, simultaneously for all $r \in [0, 1]$, $\tau_n(r) \in [\tau(r) - \epsilon, \tau(r) + \epsilon]$ on an event whose probability tends to one. Since $\Pr(\neg \mathcal{E}_n^{\text{nz}}) \rightarrow 0$, we obtain

$$\sup_{r \in [0,1]} |\tau_n(r) - \tau(r)| \xrightarrow{\text{p}} 0. \quad (184)$$

On $\mathcal{E}_n^{\text{nz}}$, recall that

$$\widehat{\phi}_n(r; \mathbf{g}, \mathbf{h}) = \phi(r, \tau_n(r); \mathbf{g}, \mathbf{h}), \quad \varphi(r) = \phi(r, \tau(r)).$$

Hence, decompose

$$\begin{aligned} |\widehat{\phi}_n(r; \mathbf{g}, \mathbf{h}) - \varphi(r)| &= |\phi(r, \tau_n(r); \mathbf{g}, \mathbf{h}) - \phi(r, \tau(r); \mathbf{g}, \mathbf{h}) + \phi(r, \tau(r); \mathbf{g}, \mathbf{h}) - \varphi(r, \tau(r))| \\ &\leq |\phi(r, \tau_n(r); \mathbf{g}, \mathbf{h}) - \phi(r, \tau(r); \mathbf{g}, \mathbf{h})| + |\phi(r, \tau(r); \mathbf{g}, \mathbf{h}) - \varphi(r, \tau(r))|. \end{aligned} \quad (185)$$

Now, note that

$$\phi(r, \tau_n(r); \mathbf{g}, \mathbf{h}) - \phi(r, \tau(r); \mathbf{g}, \mathbf{h}) = \frac{1}{n} \sum_{i=1}^n \{(w_i(r) - \tau_n(r))_+ - (w_i(r) - \tau(r))_+\}.$$

For fixed w , the map $\tau \mapsto (w - \tau)_+$ is 1-Lipschitz. Indeed, without loss of generality, let $\tau > \tau'$ and consider

1. $w > \tau > \tau'$: $|(w - \tau)_+ - (w - \tau')_+| = |\tau - \tau'|$.
2. $\tau \geq w > \tau'$: $|(w - \tau)_+ - (w - \tau')_+| = |w - \tau'| = w - \tau' \leq \tau - \tau' = |\tau - \tau'|$.
3. $\tau > \tau' \geq w$: $|(w - \tau)_+ - (w - \tau')_+| = 0 \leq |\tau - \tau'|$.

Thus

$$\begin{aligned} |\phi(r, \tau_n(r); \mathbf{g}, \mathbf{h}) - \phi(r, \tau(r); \mathbf{g}, \mathbf{h})| &\leq \frac{1}{n} \sum_{i=1}^n |(w_i(r) - \tau_n(r))_+ - (w_i(r) - \tau(r))_+| \\ &\leq \frac{1}{n} \sum_{i=1}^n |\tau_n(r) - \tau(r)| \\ &= |\tau_n(r) - \tau(r)|. \end{aligned}$$

Hence, by (183), (184), equation (185) becomes

$$\begin{aligned} \sup_{r \in [0,1]} |\widehat{\phi}_n(r; \mathbf{g}, \mathbf{h}) - \varphi(r)| &\leq \sup_{r \in [0,1]} |\phi(r, \tau_n(r); \mathbf{g}, \mathbf{h}) - \phi(r, \tau(r); \mathbf{g}, \mathbf{h})| + \sup_{r \in [0,1]} |\phi(r, \tau(r); \mathbf{g}, \mathbf{h}) - \varphi(r, \tau(r))| \\ &\leq \sup_{r \in [0,1]} |\tau_n(r) - \tau(r)| + \sup_{\substack{r \in [0,1] \\ \tau \in [0,T]}} |\phi(r, \tau; \mathbf{g}, \mathbf{h}) - \varphi(r, \tau)| \\ &\xrightarrow{\text{p}} 0. \end{aligned}$$

The convergence above holds unconditionally because $\Pr(\neg \mathcal{E}_n^{\text{pz}}) \rightarrow 0$. Finally, recall that the AO value is

$$\phi(\mathbf{g}, \mathbf{h}) = \inf_{r \in [0,1]} \widehat{\phi}_n(r; \mathbf{g}, \mathbf{h}),$$

and that the limiting AO value is

$$\phi^* = \inf_{r \in [0,1]} \varphi(r).$$

Let

$$\delta_n = \sup_{r \in [0,1]} \left| \widehat{\phi}_n(r; \mathbf{g}, \mathbf{h}) - \varphi(r) \right|.$$

Then

$$\varphi(r) = \widehat{\phi}_n(r; \mathbf{g}, \mathbf{h}) + (\varphi(r) - \widehat{\phi}_n(r; \mathbf{g}, \mathbf{h})) \leq \widehat{\phi}_n(r; \mathbf{g}, \mathbf{h}) + \delta_n,$$

and consequently $\phi^* \leq \phi(\mathbf{g}, \mathbf{h}) + \delta_n$. Similarly, $\phi^* \geq \phi(\mathbf{g}, \mathbf{h}) - \delta_n$, and in conclusion

$$|\phi(\mathbf{g}, \mathbf{h}) - \phi^*| \leq \delta_n \xrightarrow{p} 0. \quad (186)$$

We now show that $\mathbb{E}[\phi(\mathbf{g}, \mathbf{h})] \rightarrow \phi^*$. The raw AO values are uniformly integrable. Indeed, for every feasible radius, the inner supremum is nonnegative because $\mathbf{v} = \mathbf{0}_n$ is feasible. Hence $\phi(\mathbf{g}, \mathbf{h}) \geq 0$. On the other hand, taking $r = 0$ in (179) gives the value one, and therefore $0 \leq \phi(\mathbf{g}, \mathbf{h}) \leq 1$. Thus $\{\phi(\mathbf{g}, \mathbf{h})\}_{n \geq 1}$ is uniformly integrable. Together with (186), this implies $\mathbb{E}[\phi(\mathbf{g}, \mathbf{h})] \rightarrow \phi^*$. Finally, we establish a concentration inequality for $\phi(\mathbf{g}, \mathbf{h})$ around its mean. This step uses the original AO value, before the KKT threshold representation. Thus no exceptional event is needed here. Namely,

$$\phi(\mathbf{g}, \mathbf{h}) = \frac{1}{n} \inf_{\substack{\mathbf{u} \in \mathbb{R}^p: \\ \|\mathbf{u}\|_2 \leq 1}} \sup_{\substack{\mathbf{v} \in \mathbb{R}^n: \\ \|\mathbf{v}\|_\infty \leq 1}} \left\{ \|\mathbf{u}\|_2 \mathbf{g}^\top \mathbf{v} + \|\mathbf{v}\|_2 \mathbf{h}^\top \mathbf{u} - \sqrt{1 - \|\mathbf{u}\|_2^2} \mathbf{v}^\top \mathbf{1} \right\}.$$

Only the first two terms depend on $(\mathbf{g}, \mathbf{h})$. For fixed $(\mathbf{u}, \mathbf{v})$, define

$$F_n(\mathbf{g}, \mathbf{h}; \mathbf{u}, \mathbf{v}) = \frac{1}{n} \left\{ \|\mathbf{u}\|_2 \mathbf{g}^\top \mathbf{v} + \|\mathbf{v}\|_2 \mathbf{h}^\top \mathbf{u} - \sqrt{1 - \|\mathbf{u}\|_2^2} \mathbf{v}^\top \mathbf{1} \right\}.$$

For fixed $(\mathbf{u}, \mathbf{v})$, the gradients with respect to $\mathbf{g}$ and $\mathbf{h}$ are $\nabla_{\mathbf{g}} F_n = \|\mathbf{u}\|_2 \mathbf{v} / n$, $\nabla_{\mathbf{h}} F_n = \|\mathbf{v}\|_2 \mathbf{u} / n$. Hence $\|\nabla_{\mathbf{g}} F_n\|_2 \leq 1/\sqrt{n}$, $\|\nabla_{\mathbf{h}} F_n\|_2 \leq 1/\sqrt{n}$. Thus the joint gradient with respect to $(\mathbf{g}, \mathbf{h})$ is bounded by $\sqrt{2/n}$. Hence, for each fixed $(\mathbf{u}, \mathbf{v})$, the map $(\mathbf{g}, \mathbf{h}) \mapsto F_n(\mathbf{g}, \mathbf{h}; \mathbf{u}, \mathbf{v})$ is $\sqrt{2/n}$ -Lipschitz for the Euclidean norm on $\mathbb{R}^{n+p}$. Taking a supremum over $\mathbf{v}$ and an infimum over $\mathbf{u}$ cannot increase the Lipschitz constant. Indeed, if every f_α is L -Lipschitz, then

$$\left| \sup_{\alpha} f_{\alpha}(x) - \sup_{\alpha} f_{\alpha}(y) \right| \leq \sup_{\alpha} |f_{\alpha}(x) - f_{\alpha}(y)| \leq L \|x - y\|_2,$$

and the same argument applies to infima. Therefore the AO value map $(\mathbf{g}, \mathbf{h}) \mapsto \phi(\mathbf{g}, \mathbf{h})$ is also $\sqrt{2/n}$ -Lipschitz.

By Gaussian concentration for Lipschitz functions (see, e.g., [Vershynin 2018](#) Theorem 5.2.2), there exists a constant $c > 0$ such that, for all $t > 0$,

$$\Pr(|\phi(\mathbf{g}, \mathbf{h}) - \mathbb{E}[\phi(\mathbf{g}, \mathbf{h})]| > t) \leq 2 \exp\{-cnt^2\}.$$

Since $\mathbb{E}[\phi(\mathbf{g}, \mathbf{h})] \rightarrow \phi^*$, for every fixed $\epsilon > 0$ and all sufficiently large n , $|\mathbb{E}[\phi(\mathbf{g}, \mathbf{h})] - \phi^*| \leq \epsilon/2$. Hence

$$\Pr(\phi(\mathbf{g}, \mathbf{h}) \leq \phi^* - \epsilon) \leq \Pr(\phi(\mathbf{g}, \mathbf{h}) - \mathbb{E}[\phi(\mathbf{g}, \mathbf{h})] \leq -\epsilon/2) \leq 2 \exp\{-cne^2/4\}.$$

Tail bound for the PO Combining the lower-tail comparison from Theorem B.2 with the AO concentration bound gives, for every fixed $\epsilon > 0$ and all sufficiently large n ,

$$\Pr\left(\Phi(\widetilde{\mathbf{H}}) < \phi^* - \epsilon\right) \leq 2\Pr(\phi(\mathbf{g}, \mathbf{h}) \leq \phi^* - \epsilon) \leq 4\exp\{-cne^2/4\}.$$

Now let $c_{\ell_1}(\kappa) = \phi^*$. Strict positivity is shown in Lemma 6 (v). This concludes the proof. $\square$

Lemma 6. For $r \in [0, 1]$, $\tau \in \mathfrak{R}_{\geq 0}$, let

$$A_{\ell_1}(r, \tau) = \mathbb{E}[\min\{\tau^2, W(r)^2\}],$$

where

$$W(r) = \left| rZ - \sqrt{1 - r^2} \right|,$$

and $Z \sim \mathcal{N}(0, 1)$. Then

- (i) For any $r \in (0, 1]$, the map $\tau \mapsto A_{\ell_1}(r, \tau)$ is strictly increasing on $\mathfrak{R}_{\geq 0}$. For $r = 0$, $A_{\ell_1}(0, \tau) = \min\{\tau^2, 1\}$, so it is strictly increasing on $[0, 1]$ and constant on $[1, \infty)$.
- (ii) $\lim_{\tau \rightarrow 0} A_{\ell_1}(r, \tau) = 0$, $\lim_{\tau \rightarrow \infty} A_{\ell_1}(r, \tau) = 1$
- (iii) For any $\tau > 0$, $r \mapsto A_{\ell_1}(r, \tau)$ is strictly decreasing in r on $[0, 1]$
- (iv) For $\kappa \in (0, 1)$, $\tau(r)$, defined as the solution to

$$A_{\ell_1}(r, \tau) = r^2\kappa, \tag{187}$$

exists, is unique, is continuous in r , and bounded uniformly over $r \in [0, 1]$.

(v) Let

$$c_{\ell_1}(\kappa) = \inf_{r \in [0, 1]} \mathbb{E}[(W(r) - \tau(r))_+],$$

with $(x)_+ = \max\{0, x\}$ and $\tau(r)$ defined by (187). Then for any $\kappa \in (0, 1)$,

$$c_{\ell_1}(\kappa) > 0.$$

Proof. We shall prove each claim in turn.

- (i) For $r = 0$, $W(0) = 1$, so $A_{\ell_1}(0, \tau) = \min\{\tau^2, 1\}$, which is strictly increasing on $[0, 1]$ and constant on $[1, \infty)$. For $r > 0$, fix $0 \leq \tau' < \tau$. The difference $\min\{W(r)^2, \tau^2\} - \min\{W(r)^2, (\tau')^2\}$ is nonnegative and is strictly positive on $\{W(r) > \tau'\}$. Since $W(r)$ is the absolute value of a nondegenerate normal random variable, $\Pr(W(r) > \tau') > 0$. Hence $A_{\ell_1}(r, \tau) > A_{\ell_1}(r, \tau')$, as required.
- (ii) By dominated convergence,

$$\begin{aligned} \lim_{\tau \rightarrow \infty} \mathbb{E}[\min\{W(r)^2, \tau^2\}] &= \mathbb{E}[\lim_{\tau \rightarrow \infty} \min\{W(r)^2, \tau^2\}] = \mathbb{E}[W(r)^2] = 1 \\ \lim_{\tau \rightarrow 0} \mathbb{E}[\min\{W(r)^2, \tau^2\}] &= \mathbb{E}[\lim_{\tau \rightarrow 0} \min\{W(r)^2, \tau^2\}] = 0. \end{aligned}$$

- (iii) Fix $\tau > 0$. For $r \in (0, 1)$, set $s = \sqrt{1 - r^2}$, $Y = rZ - s$. Fix $r_1 \in (0, 1)$ and a compact interval $I \subset (0, 1)$ containing r_1 . For every fixed Z , the map

$$u \mapsto \min\left\{\tau^2, \left(uZ - \sqrt{1 - u^2}\right)^2\right\}$$

is absolutely continuous on I , and, wherever its derivative exists,

$$\left| \frac{\partial}{\partial u} \min\left\{\tau^2, \left(uZ - \sqrt{1 - u^2}\right)^2\right\} \right| \leq C_I(1 + Z^2)$$

for a finite deterministic constant C_I . Since $\Pr(|Y| = \tau) = 0$, differentiation under the expectation gives

$$\frac{\partial}{\partial r} A_{\ell_1}(r, \tau) = 2 \mathbb{E} \left[\mathbb{1}\{|Y| < \tau\} Y \left(Z + \frac{r}{s} \right) \right]. \quad (188)$$

Let

$$f_r(y) = \frac{1}{r} \phi \left(\frac{y+s}{r} \right)$$

be the density of Y . Since $f'_r(y) = -\frac{y+s}{r^2} f_r(y)$, integration by parts in (188) gives

$$\frac{\partial}{\partial r} A_{\ell_1}(r, \tau) = \frac{2}{r} \int_{-\tau}^{\tau} y(y+s) f_r(y) dy + \frac{2r}{s} \int_{-\tau}^{\tau} y f_r(y) dy = -2r \left[\tau \{f_r(\tau) + f_r(-\tau)\} + \frac{r^2}{s} \{f_r(\tau) - f_r(-\tau)\} \right].$$

Writing $k = \frac{\tau s}{r^2} > 0$ we have

$$\frac{f_r(-\tau) - f_r(\tau)}{f_r(-\tau) + f_r(\tau)} = \tanh(k) < k = \frac{\tau s}{r^2},$$

where $\tanh(k) < k$ follows because $k - \tanh(k)$ vanishes at zero and has derivative $\tanh^2(k) > 0$ for $k > 0$. Hence

$$\tau \{f_r(\tau) + f_r(-\tau)\} - \frac{r^2}{s} \{f_r(-\tau) - f_r(\tau)\} > 0,$$

and therefore

$$\frac{\partial}{\partial r} A_{\ell_1}(r, \tau) < 0, \quad r \in (0, 1).$$

Finally, $(r, \tau) \mapsto A_{\ell_1}(r, \tau)$ is jointly continuous on $[0, 1] \times \mathbb{R}_{\geq 0}$ by dominated convergence, since

$$\min\{\tau^2, W(r)^2\} \leq W(r)^2 \leq Z^2 + 1.$$

The strict decrease on $(0, 1)$ therefore extends to $[0, 1]$ by continuity at the endpoints.

- (iv) For $r = 0$, the equation is $A_{\ell_1}(0, \tau) = 0$, whose unique solution is $\tau(0) = 0$. For $r > 0$, continuity of $\tau \mapsto A_{\ell_1}(r, \tau)$ follows from the joint continuity established in part (iii). Hence the map $\tau \mapsto A_{\ell_1}(r, \tau) - r^2 \kappa$ is continuous. It is negative at $\tau = 0$ and has limit $1 - r^2 \kappa \geq 1 - \kappa > 0$ as $\tau \rightarrow \infty$. By the intermediate value theorem (Rudin, 1976 Theorem 4.23), there must be a value $\tau \geq 0$ such that $A_{\ell_1}(r, \tau) - r^2 \kappa = 0$. Uniqueness follows by strict monotonicity of A_{ℓ_1} in τ for $r > 0$. To establish uniform boundedness of $\tau(r)$, for any $r \in [0, 1]$, let $\tau(r)$ be the solution to $A_{\ell_1}(r, \tau) - r^2 \kappa = 0$. Then, since $r \mapsto A_{\ell_1}(r, \tau)$ is decreasing in r ,

$$A_{\ell_1}(1, \tau(r)) - \kappa \leq A_{\ell_1}(1, \tau(r)) - r^2 \kappa \leq A_{\ell_1}(r, \tau(r)) - r^2 \kappa = 0.$$

Now since $\tau \mapsto A_{\ell_1}(1, \tau)$ is strictly increasing in τ , it follows that $\tau(1) \geq \tau(r)$ for any $r \in [0, 1]$.

Finally to show that $\tau(r)$ is continuous, consider any convergent sequence $\{r_n\}_{n \in \mathbb{N}}$, $r_n \in [0, 1]$ with limit $r^* \in [0, 1]$. Concurrently, let τ_n be the unique solution to $A_{\ell_1}(r_n, \tau) - r_n^2 \kappa = 0$. Since $\tau_n \in [0, \tau(1)]$, by Bolzano–Weierstrass (Rudin, 1976 Theorem 3.6(b)), there must exist a convergent subsequence, call it τ_{n_k} with limit say $\bar{\tau}$. Then by joint continuity of $A_{\ell_1}(r, \tau)$, it follows that

$$0 = \lim_{k \rightarrow \infty} A_{\ell_1}(r_{n_k}, \tau_{n_k}) - r_{n_k}^2 \kappa = A_{\ell_1}(r^*, \bar{\tau}) - (r^*)^2 \kappa.$$

By uniqueness of $\tau(r)$, $\bar{\tau} = \tau(r^*)$ and thus every convergent subsequence must converge to $\tau(r^*)$. But then also $\tau_n \rightarrow \tau(r^*)$. If this were not the case, then τ_n would have to have a non-convergent subsequence for which $|\tau_{n_k} - \bar{\tau}| > \epsilon$. For τ_{n_k} , which is still bounded, another convergent subsequence must exist and thus converge to $\bar{\tau}$, which contradicts $|\tau_{n_k} - \bar{\tau}| > \epsilon$.

- (v) For existence, note that the map $(w, \tau) \mapsto (w - \tau)_+$ is continuous and that $(W(r) - \tau(r))_+ \leq W(r) \leq |Z| + 1$ is integrable. Hence, by dominated convergence $(w, \tau) \mapsto \mathbb{E}[(w - \tau)_+]$ is continuous. Further $r \mapsto \tau(r)$ is continuous. Thus, $r \mapsto \mathbb{E}[(W(r) - \tau(r))_+]$ is continuous and by the extreme value theorem, there exists a $r^* \in [0, 1]$ such that

$$\mathbb{E}[(W(r^*) - \tau(r^*))_+] = \inf_{r \in [0, 1]} \mathbb{E}[(W(r) - \tau(r))_+].$$

Now assume that for this r^* , $c_{\ell_1}(\kappa) = 0$. Then it must hold that $\tau(r^*) \geq W(r^*)$ Z -almost everywhere. In that case $A_{\ell_1}(r^*, \tau(r^*)) = \mathbb{E}[W(r^*)^2] = 1$. But then $A_{\ell_1}(r^*, \tau(r^*)) - (r^*)^2 \kappa > 0$, so that $\tau(r^*)$ is no solution to (187), a contradiction. Thus, $c_{\ell_1}(\kappa) > 0$ for all $\kappa \in (0, 1)$.

This concludes the proof. $\square$

F.1.2 Product-domain constraints

Lemma 7 (Orthogonal coordinate decomposition). *Let $\mathbf{B} \in \mathbb{R}^{p \times 2}$, $\text{rank}(\mathbf{B}) = s$ and*

$$\mathbf{P} = \mathbf{B}(\mathbf{B}^\top \mathbf{B})^+ \mathbf{B}^\top,$$

and $\mathbf{E} \in \mathbb{R}^{p \times (p-s)}$ has orthonormal columns and satisfies

$$\mathbf{E}\mathbf{E}^\top = \mathbf{P}^\perp = \mathbf{I}_p - \mathbf{P}.$$

Then, for every $\boldsymbol{\beta} \in \mathbb{R}^p$ there exist unique $\mathbf{v} \in \text{range}(\mathbf{B}^\top)$, $\mathbf{w} \in \mathbb{R}^{p-s} : \mathbf{E}\mathbf{w} \in \ker(\mathbf{B}^\top)$ such that

$$\boldsymbol{\beta} = \mathbf{B}\mathbf{v} + \mathbf{E}\mathbf{w}.$$

Proof. Since $\mathbf{P}$ is the orthogonal projector onto $\text{range}(\mathbf{B})$ and $\mathbf{E}\mathbf{E}^\top = \mathbf{P}^\perp$ with orthonormal columns, $\text{range}(\mathbf{E}) = \ker(\mathbf{B}^\top)$ and hence $\mathbf{B}^\top \mathbf{E} = \mathbf{0}$. For any $\boldsymbol{\beta} \in \mathbb{R}^p$, take $\mathbf{v} = \mathbf{B}^+ \boldsymbol{\beta} \in \text{range}(\mathbf{B}^\top)$ and $\mathbf{w} = \mathbf{E}^\top \boldsymbol{\beta}$. Then $\mathbf{B}\mathbf{v} = \mathbf{P}\boldsymbol{\beta}$ and $\mathbf{E}\mathbf{w} = \mathbf{P}^\perp \boldsymbol{\beta}$, so $\boldsymbol{\beta} = \mathbf{B}\mathbf{v} + \mathbf{E}\mathbf{w}$. If two such representations exist, orthogonality gives $\mathbf{E}(\mathbf{w}_1 - \mathbf{w}_2) = \mathbf{0}$ and $\mathbf{B}(\mathbf{v}_1 - \mathbf{v}_2) = \mathbf{0}$. Since $\mathbf{E}$ has full column rank and $\text{range}(\mathbf{B}^\top) \cap \ker(\mathbf{B}) = \{\mathbf{0}\}$, both differences vanish. $\square$

Lemma 8 (Product-domain relaxation preserves the compact MDYPL saddle). *Let $\mathbf{B} \in \mathbb{R}^{p \times 2}$, set $s = \text{rank}(\mathbf{B})$, and let $\mathbf{E} \in \mathbb{R}^{p \times (p-s)}$ satisfy*

$$\mathbf{E}^\top \mathbf{E} = \mathbf{I}_{p-s}, \quad \text{range}(\mathbf{E}) = \text{range}(\mathbf{B})^\perp.$$

Let $\mathbf{H}_1 = \mathbf{H}\mathbf{B}$ and $\mathbf{H}_2 = \mathbf{H}\mathbf{E}$. For constants $C_\sigma \geq C_\beta > 0$, consider the sets

$$V_p = \{\mathbf{v} \in \text{range}(\mathbf{B}^\top) : \|\mathbf{B}\mathbf{v}\|_2 \leq C_\beta \sqrt{p}\}, \quad W_p = \{\mathbf{w} \in \mathbb{R}^{p-s} : \|\mathbf{w}\|_2 \leq C_\sigma \sqrt{p}\},$$

and

$$\mathcal{K}_{C_\beta}^p = \{(\mathbf{v}, \mathbf{w}) : \mathbf{v} \in \text{range}(\mathbf{B}^\top), \mathbf{w} \in \mathbb{R}^{p-s}, \|\mathbf{B}\mathbf{v} + \mathbf{E}\mathbf{w}\|_2 \leq C_\beta \sqrt{p}\}.$$

Let

$$d(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}) = \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \frac{1}{\sqrt{p}} \mathbf{H}_2 \mathbf{w}, \quad L(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}, \boldsymbol{\lambda}) = \frac{1}{n} \{\mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} + \boldsymbol{\lambda}^\top d(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta})\},$$

and define the coupled-domain and product-domain saddle problems, respectively, as

$$\Psi_{\mathcal{K}} = \min_{(\mathbf{v}, \mathbf{w}) \in \mathcal{K}_{C_\beta}^p} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} L(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}, \boldsymbol{\lambda}), \quad \Psi_{V \times W} = \min_{\substack{\mathbf{v} \in V_p, \mathbf{w} \in W_p \\ |\theta| \leq C_\theta \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n}} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} L(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}, \boldsymbol{\lambda}).$$

On the event $\mathcal{E}_n \cap \mathcal{M}_n$ from (58) and (59) respectively, the two values are equal, i.e. $\Psi_{\mathcal{K}} = \Psi_{V \times W}$. Moreover, every optimiser of the product-domain problem satisfies $d(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}) = \mathbf{0}$, and consequently, every product-domain optimiser has the same $(\theta, \boldsymbol{\beta}, \boldsymbol{\eta})$ coordinates as the original MDYPL optimiser of (6), where $\boldsymbol{\beta} = \mathbf{B}\mathbf{v} + \mathbf{E}\mathbf{w}$.

Proof. Work on $\mathcal{M}_n \cap \mathcal{E}_n$. On $\mathcal{E}_n$, the intercept augmented design matrix $\widetilde{\mathbf{H}} = [\mathbf{1}, \mathbf{H}]$ has full column rank and the minimiser of (6) exists and is unique (Rigon and Aliverti 2023 Theorem 1). Let $\hat{\boldsymbol{\eta}}^{\text{DY}} = \hat{\theta}^{\text{DY}} \mathbf{1} + \mathbf{H} \hat{\boldsymbol{\beta}}^{\text{DY}} / \sqrt{p}$ and write the orthogonal coordinate representation of the original MDYPL solution as $\hat{\boldsymbol{\beta}}^{\text{DY}} = \mathbf{B} \hat{\mathbf{v}} + \mathbf{E} \hat{\mathbf{w}}$. On $\mathcal{M}_n \cap \mathcal{E}_n$, $\|\hat{\boldsymbol{\beta}}^{\text{DY}}\|_2 \leq C_\beta \sqrt{p}$, $|\hat{\theta}^{\text{DY}}| \leq C_\theta$, $\hat{\boldsymbol{\eta}}^{\text{DY}} \in \mathcal{B}_{C_\eta}^n$. Hence $(\hat{\mathbf{v}}, \hat{\mathbf{w}}) \in \mathcal{K}_{C_\beta}^p$. Since $\mathbf{B}^\top \mathbf{E} = \mathbf{0}$, $\mathbf{E}^\top \mathbf{E} = \mathbf{I}_{p-s}$, and $C_\sigma \geq C_\beta$, this also gives $(\hat{\mathbf{v}}, \hat{\mathbf{w}}) \in V_p \times W_p$. Thus the original MDYPL solution is feasible for both the coupled-domain and product-domain saddle problems.

Let $\hat{\boldsymbol{\lambda}}^{\text{DY}} = \tilde{\mathbf{y}} - \rho'(\hat{\boldsymbol{\eta}}^{\text{DY}})$. Since $\tilde{Y}_i \in (0, 1)$ and $\rho'((\hat{\boldsymbol{\eta}}^{\text{DY}})_i) \in (0, 1)$ for every i , $|\hat{\lambda}_i^{\text{DY}}| < 1$, $i = 1, \dots, n$. Therefore $\|\hat{\boldsymbol{\lambda}}^{\text{DY}}\|_2 < \sqrt{n}$, so $\hat{\boldsymbol{\lambda}}^{\text{DY}}$ lies in the relative interior of $\mathcal{B}_1^n$.

Recall the empirical loss G_n from (67), so that the Lagrangian of the statement is

$$L(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}, \boldsymbol{\lambda}) = G_n(\boldsymbol{\eta}) + \frac{1}{n} \boldsymbol{\lambda}^\top d(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}).$$

Because $\hat{\boldsymbol{\eta}}^{\text{DY}}$ is the fitted predictor of the original MDYPL solution, $d(\hat{\mathbf{v}}, \hat{\mathbf{w}}, \hat{\theta}^{\text{DY}}, \hat{\boldsymbol{\eta}}^{\text{DY}}) = \mathbf{0}$. On $\mathcal{E}_n$, $(\hat{\theta}^{\text{DY}}, \hat{\boldsymbol{\beta}}^{\text{DY}})$ is the genuine MDYPL optimiser. The KKT equations for (56) identify $\hat{\boldsymbol{\lambda}}^{\text{DY}} = \tilde{\mathbf{y}} - \rho'(\hat{\boldsymbol{\eta}}^{\text{DY}})$ as its equality-constraint multiplier and give

$$\mathbf{1}^\top \hat{\boldsymbol{\lambda}}^{\text{DY}} = 0, \quad \mathbf{H}^\top \hat{\boldsymbol{\lambda}}^{\text{DY}} = \mathbf{0}_p.$$

Since $\mathbf{H}_1 = \mathbf{H}\mathbf{B}$ and $\mathbf{H}_2 = \mathbf{H}\mathbf{E}$, it follows that $\mathbf{H}_1^\top \hat{\boldsymbol{\lambda}}^{\text{DY}} = \mathbf{0}$, $\mathbf{H}_2^\top \hat{\boldsymbol{\lambda}}^{\text{DY}} = \mathbf{0}$. Consequently, for every $(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta})$,

$$L(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}, \hat{\boldsymbol{\lambda}}^{\text{DY}}) = \frac{1}{n} \left\{ \mathbf{1}^\top \rho(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} + (\hat{\boldsymbol{\lambda}}^{\text{DY}})^\top \boldsymbol{\eta} \right\}.$$

Using $\hat{\boldsymbol{\lambda}}^{\text{DY}} = \tilde{\mathbf{y}} - \rho'(\hat{\boldsymbol{\eta}}^{\text{DY}})$, this becomes

$$L(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}, \hat{\boldsymbol{\lambda}}^{\text{DY}}) = \frac{1}{n} \sum_{i=1}^n \{ \rho(\boldsymbol{\eta}_i) - \rho'((\hat{\boldsymbol{\eta}}^{\text{DY}})_i) \boldsymbol{\eta}_i \}.$$

The right-hand side is convex in $\boldsymbol{\eta}$ and is minimised at $\boldsymbol{\eta} = \hat{\boldsymbol{\eta}}^{\text{DY}}$. Hence

$$L(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}, \hat{\boldsymbol{\lambda}}^{\text{DY}}) \geq L(\hat{\mathbf{v}}, \hat{\mathbf{w}}, \hat{\theta}^{\text{DY}}, \hat{\boldsymbol{\eta}}^{\text{DY}}, \hat{\boldsymbol{\lambda}}^{\text{DY}}) = G_n(\hat{\boldsymbol{\eta}}^{\text{DY}}),$$

for every $(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta})$.

Now fix any point in the product domain. Since $\mathcal{B}_1^n = \{\boldsymbol{\lambda} \in \mathbb{R}^n : \|\boldsymbol{\lambda}\|_2 \leq \sqrt{n}\}$, we have

$$\max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} L(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}, \boldsymbol{\lambda}) = G_n(\boldsymbol{\eta}) + \frac{1}{n} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} \boldsymbol{\lambda}^\top d(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}) = G_n(\boldsymbol{\eta}) + \frac{1}{\sqrt{n}} \|d(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta})\|_2.$$

In particular,

$$\max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} L(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}, \boldsymbol{\lambda}) \geq L(\mathbf{v}, \mathbf{w}, \theta, \boldsymbol{\eta}, \hat{\boldsymbol{\lambda}}^{\text{DY}}) \geq G_n(\hat{\boldsymbol{\eta}}^{\text{DY}}).$$

At the original MDYPL solution, because the equality constraint is satisfied,

$$\max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} L(\hat{\mathbf{v}}, \hat{\mathbf{w}}, \hat{\theta}^{\text{DY}}, \hat{\boldsymbol{\eta}}^{\text{DY}}, \boldsymbol{\lambda}) = G_n(\hat{\boldsymbol{\eta}}^{\text{DY}}).$$

Therefore the product-domain value is $\Psi_{V \times W} = G_n(\hat{\boldsymbol{\eta}}^{\text{DY}})$. The same argument applies to the coupled-domain problem, since the original MDYPL solution belongs to $\mathcal{K}_{C_\beta}^p$, and hence $\Psi_{\mathcal{K}} = G_n(\hat{\boldsymbol{\eta}}^{\text{DY}}) = \Psi_{V \times W}$.

It remains to show that the product-domain problem introduces no infeasible optimiser. Let $(\tilde{\mathbf{v}}, \tilde{\mathbf{w}}, \tilde{\theta}, \tilde{\boldsymbol{\eta}})$ be a product-domain optimiser. Then

$$\max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} L(\tilde{\mathbf{v}}, \tilde{\mathbf{w}}, \tilde{\theta}, \tilde{\boldsymbol{\eta}}, \boldsymbol{\lambda}) = G_n(\hat{\boldsymbol{\eta}}^{\text{DY}}).$$

Towards a contradiction, assume that $d(\tilde{\mathbf{v}}, \tilde{\mathbf{w}}, \tilde{\theta}, \tilde{\boldsymbol{\eta}}) \neq \mathbf{0}$. Then

$$\begin{aligned} \max_{\boldsymbol{\lambda} \in \mathcal{B}_1^n} L(\tilde{\mathbf{v}}, \tilde{\mathbf{w}}, \tilde{\theta}, \tilde{\boldsymbol{\eta}}, \boldsymbol{\lambda}) - L(\tilde{\mathbf{v}}, \tilde{\mathbf{w}}, \tilde{\theta}, \tilde{\boldsymbol{\eta}}, \hat{\boldsymbol{\lambda}}^{\text{DY}}) &= \frac{1}{\sqrt{n}} \|d(\tilde{\mathbf{v}}, \tilde{\mathbf{w}}, \tilde{\theta}, \tilde{\boldsymbol{\eta}})\|_2 - \frac{1}{n} (\hat{\boldsymbol{\lambda}}^{\text{DY}})^\top d(\tilde{\mathbf{v}}, \tilde{\mathbf{w}}, \tilde{\theta}, \tilde{\boldsymbol{\eta}}) \\ &\geq \left(\frac{1}{\sqrt{n}} - \frac{\|\hat{\boldsymbol{\lambda}}^{\text{DY}}\|_2}{n} \right) \|d(\tilde{\mathbf{v}}, \tilde{\mathbf{w}}, \tilde{\theta}, \tilde{\boldsymbol{\eta}})\|_2 > 0. \end{aligned}$$

The final inequality uses $\|\hat{\lambda}^{\text{DY}}\|_2 < \sqrt{n}$. Since $L(\tilde{\mathbf{v}}, \tilde{\mathbf{w}}, \tilde{\theta}, \tilde{\eta}, \hat{\lambda}^{\text{DY}}) \geq G_n(\hat{\eta}^{\text{DY}})$, we obtain

$$\max_{\lambda \in \mathcal{B}_1^n} L(\tilde{\mathbf{v}}, \tilde{\mathbf{w}}, \tilde{\theta}, \tilde{\eta}, \lambda) > G_n(\hat{\eta}^{\text{DY}}),$$

contradicting optimality of $(\tilde{\mathbf{v}}, \tilde{\mathbf{w}}, \tilde{\theta}, \tilde{\eta})$. Therefore $d(\tilde{\mathbf{v}}, \tilde{\mathbf{w}}, \tilde{\theta}, \tilde{\eta}) = \mathbf{0}$. Thus every product-domain optimiser is feasible for the equality-constrained MDYPL problem and attains the original MDYPL objective value. Since, on $\mathcal{M}_n \cap \mathcal{E}_n$, the original MDYPL optimiser is unique, $\tilde{\theta} = \hat{\theta}^{\text{DY}}$, $\tilde{\eta} = \hat{\eta}^{\text{DY}}$, $\mathbf{B}\tilde{\mathbf{v}} + \mathbf{E}\tilde{\mathbf{w}} = \hat{\beta}^{\text{DY}}$. This proves the claim. $\square$

Lemma 9 (Uniqueness of the compact PO). *On $\mathcal{E}_n$, the compact problem in (62) has a saddle point, and all its saddle points have the same minimising variables, denoted by $(\hat{\mathbf{v}}, \hat{\mathbf{w}}, \hat{\theta}, \hat{\eta})$. Consequently, $\hat{\mathbf{w}}$ is the unique minimiser of the profiled PO in the final line of (63), and the inner minimising variables at every saddle point of the corresponding fixed- $\hat{\mathbf{w}}$ problem are $(\hat{\mathbf{v}}, \hat{\theta}, \hat{\eta})$.*

Proof. Maximising over $\lambda \in \mathcal{B}_1^n$ reduces (62) to minimising, over its compact convex primal domain, the continuous convex function

$$\frac{1}{n} \{ \mathbf{1}^\top \rho(\eta) - \tilde{\mathbf{y}}^\top \eta \} + \frac{1}{\sqrt{n}} \left\| \eta - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \frac{1}{\sqrt{p}} \mathbf{H}_2 \mathbf{w} \right\|_2.$$

Hence a minimiser exists. Let $(\mathbf{v}, \mathbf{w}, \theta, \eta)$ and $(\mathbf{v}', \mathbf{w}', \theta', \eta')$ be minimisers, with residuals $\mathbf{r}$ and $\mathbf{r}'$, respectively. Their midpoint is feasible and, by convexity, is also a minimiser. Strict convexity of $\mathbf{1}^\top \rho(\eta) - \tilde{\mathbf{y}}^\top \eta$ in η then forces $\eta = \eta'$. The endpoint objective values give $\|\mathbf{r}\|_2 = \|\mathbf{r}'\|_2$, while midpoint optimality gives

$$\left\| \frac{\mathbf{r} + \mathbf{r}'}{2} \right\|_2 = \frac{\|\mathbf{r}\|_2 + \|\mathbf{r}'\|_2}{2}.$$

Equality in the Euclidean triangle inequality, together with equality of the norms, gives $\mathbf{r} = \mathbf{r}'$. Hence

$$\theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \frac{1}{\sqrt{p}} \mathbf{H}_2 \mathbf{w} = \theta' \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v}' + \frac{1}{\sqrt{p}} \mathbf{H}_2 \mathbf{w}'.$$

On $\mathcal{E}_n$, injectivity of

$$(\theta, \beta) \mapsto \theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H} \beta$$

followed by Lemma 7 gives

$$(\mathbf{v}, \mathbf{w}, \theta, \eta) = (\mathbf{v}', \mathbf{w}', \theta', \eta').$$

The Lagrangian in (62) is continuous, convex in its minimising variables and affine in λ on compact convex domains. Sion's theorem and attainment therefore give a saddle point. The minimising variables of every saddle point minimise the primal objective above and hence equal $(\hat{\mathbf{v}}, \hat{\mathbf{w}}, \hat{\theta}, \hat{\eta})$.

For each fixed $\mathbf{w}$, the Sion identity in (63) identifies the profiled value with the attained minimum of the same primal objective over $(\mathbf{v}, \theta, \eta)$. Thus the profiled minimisers are exactly the $\mathbf{w}$ -coordinates of minimisers of the primal objective, so $\hat{\mathbf{w}}$ is unique. At $\hat{\mathbf{w}}$, the inner minimising variables at every saddle point attain this fixed- $\hat{\mathbf{w}}$ primal minimum and therefore equal $(\hat{\mathbf{v}}, \hat{\theta}, \hat{\eta})$. $\square$

F.2 Lemmas for Section B.5

Lemma 10 (Concave ψ and ball-to-radius minimax reduction). *Let*

$$Z \subseteq \mathbb{R}^d \times \mathbb{R} \times \mathbb{R}^n,$$

be a nonempty compact convex set, with coordinates $\zeta = (\mathbf{v}, \theta, \eta)$. Let $S_\lambda \subseteq \mathbb{R}^n$ be nonempty, compact, and convex, and let $S_{\mathbf{w}}$ be a nonempty compact subset of a finite-dimensional Euclidean space. Define

$$\psi(\lambda) = \min_{\zeta = (\mathbf{v}, \theta, \eta) \in Z} \left\{ \frac{1}{n} \{ \mathbf{1}^\top \rho(\eta) - \tilde{\mathbf{y}}^\top \eta \} + \frac{1}{n} \lambda^\top \eta - \frac{1}{n} \theta \mathbf{1}^\top \lambda - \frac{1}{n\sqrt{p}} \lambda^\top \mathbf{H}_1 \mathbf{v} \right\}.$$

Then the following hold.

- (i) The function ψ is concave in $\boldsymbol{\lambda}$ and continuous on $S_{\boldsymbol{\lambda}}$. Moreover, the minimisation defining ψ is over a compact convex set, its objective is convex in $\boldsymbol{\zeta}$ for fixed $\boldsymbol{\lambda}$, affine in $\boldsymbol{\lambda}$ for fixed $\boldsymbol{\zeta}$, and continuous in all variables. Consequently, the map

$$(\boldsymbol{w}, \boldsymbol{\lambda}) \mapsto -\frac{1}{n\sqrt{p}} \boldsymbol{\lambda}^\top \boldsymbol{H}_2 \boldsymbol{w} + \psi(\boldsymbol{\lambda}), \quad (189)$$

is continuous on $S_{\boldsymbol{w}} \times S_{\boldsymbol{\lambda}}$, affine in $\boldsymbol{w}$, and concave in $\boldsymbol{\lambda}$. In particular, it is convex-concave whenever $S_{\boldsymbol{w}}$ is convex.

- (ii) For $\boldsymbol{\zeta} = (\boldsymbol{v}, \theta, \boldsymbol{\eta}) \in Z$ and $\boldsymbol{b} \in \mathbb{R}^n$, recall $G_n(\boldsymbol{\eta})$ from (67) and define

$$\boldsymbol{a}_b(\boldsymbol{\zeta}) = \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \boldsymbol{H}_1 \boldsymbol{v} - \boldsymbol{b}.$$

Then, for every $R > 0$, every $c \geq 0$, and every $\boldsymbol{b} \in \mathbb{R}^n$,

$$\max_{\boldsymbol{\lambda} \in \mathbb{R}^n: \|\boldsymbol{\lambda}\|_2 \leq R} \min_{\boldsymbol{\zeta} \in Z} \left\{ G_n(\boldsymbol{\eta}) + \frac{1}{n} \boldsymbol{\lambda}^\top \boldsymbol{a}_b(\boldsymbol{\zeta}) - \frac{c}{n} \|\boldsymbol{\lambda}\|_2 \right\} = \max_{\tilde{r} \in [0, R]} \min_{\boldsymbol{\zeta} \in Z} \left\{ G_n(\boldsymbol{\eta}) + \frac{\tilde{r}}{n} \|\boldsymbol{a}_b(\boldsymbol{\zeta})\|_2 - \frac{c\tilde{r}}{n} \right\}. \quad (190)$$

Proof. We prove each part in turn.

- (i) Define

$$L(\boldsymbol{\zeta}; \boldsymbol{\lambda}) = \frac{1}{n} \{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\boldsymbol{y}}^\top \boldsymbol{\eta} \} + \frac{1}{n} \boldsymbol{\lambda}^\top \boldsymbol{\eta} - \frac{1}{n} \theta \mathbf{1}^\top \boldsymbol{\lambda} - \frac{1}{n\sqrt{p}} \boldsymbol{\lambda}^\top \boldsymbol{H}_1 \boldsymbol{v}.$$

For fixed $\boldsymbol{\lambda}$, the map $\boldsymbol{\zeta} \mapsto L(\boldsymbol{\zeta}; \boldsymbol{\lambda})$ is convex and continuous on Z , because $\mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta})$ is convex in $\boldsymbol{\eta}$ and all remaining dependence on $\boldsymbol{\zeta}$ is affine. For fixed $\boldsymbol{\zeta}$, the map $\boldsymbol{\lambda} \mapsto L(\boldsymbol{\zeta}; \boldsymbol{\lambda})$ is affine. Hence ψ is the pointwise infimum of affine functions of $\boldsymbol{\lambda}$, and is therefore concave.

Since Z is compact and L is continuous, the minimum defining ψ is attained. By Berge's maximum theorem (see, for example, Aliprantis and Border [2006] Theorem 17.31.1), applied to the constant compact correspondence $\boldsymbol{\lambda} \mapsto Z$, the function ψ is continuous on $S_{\boldsymbol{\lambda}}$.

The map in (189) is affine in $\boldsymbol{w}$ for fixed $\boldsymbol{\lambda}$. For fixed $\boldsymbol{w}$, it is the sum of an affine function of $\boldsymbol{\lambda}$ and the concave function $\psi(\boldsymbol{\lambda})$, and hence is concave in $\boldsymbol{\lambda}$. Continuity follows from continuity of the bilinear term and continuity of ψ . If $S_{\boldsymbol{w}}$ is convex, its affine dependence on $\boldsymbol{w}$ gives the stated convex-concave property.

- (ii) For $\boldsymbol{\zeta} \in Z$ and $\boldsymbol{\lambda} \in \mathbb{R}^n$, define

$$K(\boldsymbol{\zeta}, \boldsymbol{\lambda}) = G_n(\boldsymbol{\eta}) + \frac{1}{n} \boldsymbol{\lambda}^\top \boldsymbol{a}_b(\boldsymbol{\zeta}) - \frac{c}{n} \|\boldsymbol{\lambda}\|_2.$$

For fixed $\boldsymbol{\lambda}$, the map $\boldsymbol{\zeta} \mapsto K(\boldsymbol{\zeta}, \boldsymbol{\lambda})$ is convex and continuous on Z . For fixed $\boldsymbol{\zeta}$, the map $\boldsymbol{\lambda} \mapsto K(\boldsymbol{\zeta}, \boldsymbol{\lambda})$ is concave and continuous on the ball $\{\boldsymbol{\lambda} \in \mathbb{R}^n : \|\boldsymbol{\lambda}\|_2 \leq R\}$, because it is the sum of an affine function of $\boldsymbol{\lambda}$ and $-c\|\boldsymbol{\lambda}\|_2/n$, with $c \geq 0$. Since both constraint sets are compact and convex, Sion's min-max theorem (see, for example, Simons [1995] Theorem 3) gives

$$\max_{\boldsymbol{\lambda} \in \mathbb{R}^n: \|\boldsymbol{\lambda}\|_2 \leq R} \min_{\boldsymbol{\zeta} \in Z} K(\boldsymbol{\zeta}, \boldsymbol{\lambda}) = \min_{\boldsymbol{\zeta} \in Z} \max_{\boldsymbol{\lambda} \in \mathbb{R}^n: \|\boldsymbol{\lambda}\|_2 \leq R} K(\boldsymbol{\zeta}, \boldsymbol{\lambda}). \quad (191)$$

For fixed $\boldsymbol{\zeta} \in Z$, write $\boldsymbol{\lambda} = \tilde{r} \boldsymbol{u}$, where $\tilde{r} \in [0, R]$ and $\boldsymbol{u} \in \mathbb{S}^{n-1}$. Then

$$\max_{\boldsymbol{\lambda} \in \mathbb{R}^n: \|\boldsymbol{\lambda}\|_2 \leq R} \left\{ \frac{1}{n} \boldsymbol{\lambda}^\top \boldsymbol{a}_b(\boldsymbol{\zeta}) - \frac{c}{n} \|\boldsymbol{\lambda}\|_2 \right\} = \max_{\tilde{r} \in [0, R]} \left\{ \frac{\tilde{r}}{n} \|\boldsymbol{a}_b(\boldsymbol{\zeta})\|_2 - \frac{c\tilde{r}}{n} \right\},$$

where we used

$$\max_{\boldsymbol{u} \in \mathbb{S}^{n-1}} \boldsymbol{u}^\top \boldsymbol{a}_b(\boldsymbol{\zeta}) = \|\boldsymbol{a}_b(\boldsymbol{\zeta})\|_2.$$

Thus

$$\max_{\boldsymbol{\lambda} \in \mathbb{R}^n: \|\boldsymbol{\lambda}\|_2 \leq R} \min_{\boldsymbol{\zeta} \in Z} K(\boldsymbol{\zeta}, \boldsymbol{\lambda}) = \min_{\boldsymbol{\zeta} \in Z} \max_{\tilde{r} \in [0, R]} \left\{ G_n(\boldsymbol{\eta}) + \frac{\tilde{r}}{n} \|\mathbf{a}_b(\boldsymbol{\zeta})\|_2 - \frac{c\tilde{r}}{n} \right\}.$$

For fixed $\tilde{r} \in [0, R]$, the map

$$\boldsymbol{\zeta} \mapsto G_n(\boldsymbol{\eta}) + \frac{\tilde{r}}{n} \|\mathbf{a}_b(\boldsymbol{\zeta})\|_2 - \frac{c\tilde{r}}{n},$$

is convex and continuous on Z , because $\tilde{r} \geq 0$ and $\mathbf{a}_b$ is affine. For fixed $\boldsymbol{\zeta}$, the same expression is affine, hence concave and continuous, in $\tilde{r}$. Applying the same theorem on $Z \times [0, R]$ gives

$$\min_{\boldsymbol{\zeta} \in Z} \max_{\tilde{r} \in [0, R]} \left\{ G_n(\boldsymbol{\eta}) + \frac{\tilde{r}}{n} \|\mathbf{a}_b(\boldsymbol{\zeta})\|_2 - \frac{c\tilde{r}}{n} \right\} = \max_{\tilde{r} \in [0, R]} \min_{\boldsymbol{\zeta} \in Z} \left\{ G_n(\boldsymbol{\eta}) + \frac{\tilde{r}}{n} \|\mathbf{a}_b(\boldsymbol{\zeta})\|_2 - \frac{c\tilde{r}}{n} \right\}. \quad (192)$$

Combining (191)–(192) proves (190). □

F.3 Lemmas for Section B.6

F.3.1 Small- r environment and exclusion

Lemma 11 (Conditional small- r environment). *Fix $t_0 > 0$. There exist deterministic constants*

$$d_r > 0, \quad a \in (0, 1/2), \quad K, C_L \in (0, \infty),$$

and constants $C, c > 0$ and $N \in \mathbb{N}$ such that, for all $n > N$,

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\neg \mathcal{E}_n^{\text{sr}}(d_r, a, K, C_L) \mid \mathcal{G}_n) \leq C \exp\{-cn\}. \quad (193)$$

The constants depend only on

$$\kappa, \alpha, \boldsymbol{\Gamma}, \theta_0^*, \theta_P^*, t_0, C_\beta, C_\theta, C_\sigma,$$

and on the constants in the probability bounds of Section 2. In particular, they do not depend on C_η . Moreover, deterministically, for every $\sigma \in [0, C_\sigma]$ and every $\mathbf{b} \in \sigma \mathbf{g} + \mathcal{R}_0$,

$$|b_i - \eta_{\tilde{Y}, i}| \leq L_i, \quad i = 1, \dots, n. \quad (194)$$

Proof. Let $s = \text{rank}(\mathbf{B}) \in \{0, 1, 2\}$ and choose, by a fixed $\mathcal{G}_n$ -measurable orthonormalisation rule, an orthonormal basis $\{\mathbf{b}_1, \dots, \mathbf{b}_s\}$ of $\text{col}(\mathbf{B})$. With $\mathbf{h}_i^\top$ denoting the i th row of $\mathbf{H}$, set

$$\mathbf{r}_i = (1, \mathbf{h}_i^\top \mathbf{b}_1, \dots, \mathbf{h}_i^\top \mathbf{b}_s)^\top \in \mathbb{R}^{s+1},$$

with $\mathbf{r}_i = 1$ when $s = 0$. Conditional on $\mathcal{G}_n$, the last s coordinates of $\mathbf{r}_i$ are independent standard Gaussian variables, so

$$\mathbb{E}[\mathbf{r}_i \mathbf{r}_i^\top \mid \mathcal{G}_n] = \mathbf{I}_{s+1}.$$

Define

$$\boldsymbol{\Sigma}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{r}_i \mathbf{r}_i^\top, \quad \boldsymbol{\mu}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{r}_i \eta_{\tilde{Y}, i}, \quad s_n = \frac{1}{n} \sum_{i=1}^n \eta_{\tilde{Y}, i}^2,$$

and

$$\bar{\boldsymbol{\mu}}_n = \mathbb{E}[\mathbf{r}_i \eta_{\tilde{Y}, i} \mid \mathcal{G}_n], \quad \bar{s}_n = \mathbb{E}[\eta_{\tilde{Y}, i}^2 \mid \mathcal{G}_n].$$

On $\mathcal{S}_n(t_0)$, both $|\theta_P|$ and $\|\boldsymbol{\beta}_P\|_2/\sqrt{p}$ are bounded by deterministic constants. Thus

$$\eta_{P, i} = \theta_P + \frac{1}{\sqrt{p}} \mathbf{h}_i^\top \boldsymbol{\beta}_P$$

is conditionally subgaussian with a deterministic norm bound. If $q = \rho'(x)$, the two possible values $(1 - \alpha)q$ and $\alpha + (1 - \alpha)q$ of the pseudo-response satisfy

$$|\text{logit}((1 - \alpha)q)| \vee |\text{logit}(\{\alpha + (1 - \alpha)q\})| \leq |x| + C_\alpha$$

for a finite constant C_α depending only on α . Consequently, conditional on $\mathcal{G}_n$ and on $\mathcal{S}_n(t_0)$,

$$\|\eta_{\tilde{Y},i}\|_{\psi_2} \leq K_\eta \quad (195)$$

for a deterministic $K_\eta < \infty$, uniformly in i and n .

By [Vershynin \[2018\]](#) Lemmas 2.7.6 and 2.7.7, squares and products of subgaussian variables are sub-exponential. Since $s + 1 \leq 3$, there is a deterministic constant $K_0 < \infty$ such that, conditionally on $\mathcal{G}_n$ and on $\mathcal{S}_n(t_0)$,

$$\|\eta_{\tilde{Y},i}^2 - \bar{s}_n\|_{\psi_1} \leq K_0, \quad \|\mathbf{r}_i \eta_{\tilde{Y},i} - \bar{\boldsymbol{\mu}}_n\|_{\psi_1} \leq K_0$$

coordinatewise, and

$$\|\mathbf{r}_i \mathbf{r}_i^\top - \mathbf{I}_{s+1}\|_{\psi_1} \leq K_0$$

entrywise. Bernstein's inequality for sums of independent sub-exponential variables ([Vershynin \[2018\]](#), Theorem 2.8.1), applied conditionally on $\mathcal{G}_n$, and a union bound over the fixed number of coordinates give, for every fixed $u > 0$,

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr \left(\max \left\{ |s_n - \bar{s}_n|, \|\boldsymbol{\mu}_n - \bar{\boldsymbol{\mu}}_n\|_2, \|\boldsymbol{\Sigma}_n - \mathbf{I}_{s+1}\|_2 \right\} > u \mid \mathcal{G}_n \right) \leq C \exp\{-cn \min(u^2, u)\}. \quad (196)$$

We next obtain a population residual that is bounded away from zero. Let

$$q_{0,i} = \rho' \left(\theta_0 + \frac{1}{\sqrt{p}} \mathbf{h}_i^\top \boldsymbol{\beta}_0 \right), \quad q_{P,i} = \rho' \left(\theta_P + \frac{1}{\sqrt{p}} \mathbf{h}_i^\top \boldsymbol{\beta}_P \right).$$

Because $\boldsymbol{\beta}_0, \boldsymbol{\beta}_P \in \text{col}(\mathbf{B})$, both $q_{0,i}$ and $q_{P,i}$ are measurable with respect to $\sigma(\mathbf{r}_i, \mathcal{G}_n)$. Conditional on this field,

$$B_i = \mathbb{1} \{ \varepsilon_i < q_{0,i} \} \sim \text{Bernoulli}(q_{0,i}), \quad \tilde{Y}_i = (1 - \alpha)q_{P,i} + \alpha B_i.$$

The two corresponding values of $\eta_{\tilde{Y},i}$ differ by

$$\int_{(1-\alpha)q_{P,i}}^{(1-\alpha)q_{P,i}+\alpha} \frac{du}{u(1-u)} \geq 4\alpha.$$

It follows that

$$\text{var}(\eta_{\tilde{Y},i} \mid \mathbf{r}_i, \mathcal{G}_n) \geq 16\alpha^2 q_{0,i}(1 - q_{0,i}).$$

On $\mathcal{S}_n(t_0)$, the pair

$$\left(\theta_0, \frac{\|\boldsymbol{\beta}_0\|_2^2}{p} \right)$$

ranges over a deterministic compact subset of $\mathbb{R} \times [0, \infty)$. The map

$$(\theta, v) \mapsto \mathbb{E} [\rho'(\theta + \sqrt{v}Z) \{1 - \rho'(\theta + \sqrt{v}Z)\}], \quad Z \sim \mathcal{N}(0, 1),$$

is continuous and strictly positive on this compact set. Hence there is a deterministic $\underline{\mu}_0 > 0$ such that, on $\mathcal{S}_n(t_0)$,

$$\mathbb{E}[q_{0,i}(1 - q_{0,i}) \mid \mathcal{G}_n] \geq \underline{\mu}_0.$$

With

$$\tau_0 = 4\alpha \sqrt{\underline{\mu}_0},$$

for every $\mathbf{a} \in \mathbb{R}^{s+1}$,

$$\begin{aligned} \mathbb{E} \left[(\eta_{\tilde{Y},i} - \mathbf{r}_i^\top \mathbf{a})^2 \mid \mathcal{G}_n \right] &\geq \mathbb{E} \left[\text{var}(\eta_{\tilde{Y},i} \mid \mathbf{r}_i, \mathcal{G}_n) \mid \mathcal{G}_n \right] \\ &\geq \tau_0^2. \end{aligned}$$

Since $\mathbb{E}[\mathbf{r}_i \mathbf{r}_i^\top \mid \mathcal{G}_n] = \mathbf{I}_{s+1}$, this is equivalent to

$$\bar{s}_n - \|\bar{\boldsymbol{\mu}}_n\|_2^2 \geq \tau_0^2. \quad (197)$$

Let $\mathbf{U} = [\mathbf{b}_1, \dots, \mathbf{b}_s]$. Since $\text{col}(\mathbf{U}) = \text{col}(\mathbf{B})$,

$$\text{col}(\mathbf{H}\mathbf{U}) = \text{col}(\mathbf{H}\mathbf{B}) = \text{col}(\mathbf{H}_1).$$

Thus the matrix with rows $\mathbf{r}_i^\top$ has column space $\text{span}\{\mathbf{1}, \text{col}(\mathbf{H}_1)\}$. Let $\boldsymbol{\Pi}_n^\perp$ denote the orthogonal projector onto the complement of this space. Then

$$\frac{1}{n} \|\boldsymbol{\Pi}_n^\perp \boldsymbol{\eta}_{\tilde{Y}}\|_2^2 = \inf_{\mathbf{a} \in \mathbb{R}^{s+1}} \frac{1}{n} \sum_{i=1}^n (\eta_{\tilde{Y},i} - \mathbf{r}_i^\top \mathbf{a})^2.$$

By (195), there is a deterministic $B < \infty$ such that $\bar{s}_n \leq B^2$ on $\mathcal{S}_n(t_0)$. Conditional Cauchy-Schwarz gives

$$\|\bar{\boldsymbol{\mu}}_n\|_2 \leq \mathbb{E}[\|\mathbf{r}_i\|_2^2 \mid \mathcal{G}_n]^{1/2} \bar{s}_n^{1/2} \leq \sqrt{3}B.$$

Fix $\delta_0 \in (0, 1/2]$ and suppose that the maximum in (196) is at most δ_0 . Then, for every $\mathbf{a} \in \mathbb{R}^{s+1}$,

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (\eta_{\tilde{Y},i} - \mathbf{r}_i^\top \mathbf{a})^2 &= s_n - 2\mathbf{a}^\top \boldsymbol{\mu}_n + \mathbf{a}^\top \boldsymbol{\Sigma}_n \mathbf{a} \\ &\geq s_n - 2\mathbf{a}^\top \boldsymbol{\mu}_n + (1 - \delta_0) \|\mathbf{a}\|_2^2 \\ &\geq s_n - \frac{\|\boldsymbol{\mu}_n\|_2^2}{1 - \delta_0}. \end{aligned}$$

Writing $m = \|\bar{\boldsymbol{\mu}}_n\|_2 \leq \sqrt{3}B$ and using

$$\frac{(m + \delta_0)^2 - (1 - \delta_0)m^2}{1 - \delta_0} \leq 2\delta_0(m + 1)^2,$$

we obtain from (197) that

$$\frac{1}{n} \|\boldsymbol{\Pi}_n^\perp \boldsymbol{\eta}_{\tilde{Y}}\|_2^2 \geq \tau_0^2 - \delta_0 \left\{ 1 + 2(\sqrt{3}B + 1)^2 \right\}.$$

Choose δ_0 so that the subtracted term is at most $\tau_0^2/2$, and set

$$a_0 = \frac{\tau_0}{2}.$$

By (196),

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr \left(\frac{\|\boldsymbol{\Pi}_n^\perp \boldsymbol{\eta}_{\tilde{Y}}\|_2}{\sqrt{n}} < a_0 \mid \mathcal{G}_n \right) \leq C \exp\{-cn\}. \quad (198)$$

Choose $\delta \in (0, 1/2)$ sufficiently small that

$$q_\delta = \frac{\sqrt{\kappa} + \delta}{1 - \delta} : \quad q_\delta^2 < 1 - \delta,$$

which is possible because $\kappa < 1$, and define

$$d_r = a_0 \sqrt{1 - \delta - q_\delta^2} > 0. \quad (199)$$

We now establish the envelope bounds. By (93), on the common completed AO probability space,

$$\frac{1}{\sqrt{p}} \mathbf{H}_1 = \mathbf{Z} \mathbf{\Gamma}_p^{1/2},$$

where the rows $\mathbf{z}_i^\top$ of $\mathbf{Z}$ are conditionally independent standard Gaussian vectors. If $\mathbf{v} \in V_p$ and $\mathbf{u} = \mathbf{\Gamma}_p^{1/2} \mathbf{v}$, then

$$\|\mathbf{u}\|_2^2 = \mathbf{v}^\top \mathbf{\Gamma}_p \mathbf{v} = \frac{1}{p} \|\mathbf{B}\mathbf{v}\|_2^2 \leq C_\beta^2.$$

Consequently,

$$\sup_{\mathbf{v} \in V_p} \left| \frac{1}{\sqrt{p}} (\mathbf{H}_1 \mathbf{v})_i \right| \leq C_\beta \|\mathbf{z}_i\|_2.$$

Together with (195) and the Gaussianity of g_i , this shows that L_i is conditionally subgaussian and L_i^2 is conditionally sub-exponential, with deterministic norm bounds on $\mathcal{S}_n(t_0)$. Choose C_L larger than twice a uniform upper bound for $\mathbb{E}[L_i^2 \mid \mathcal{G}_n]$. Bernstein's inequality (Vershynin, 2018, Theorem 2.8.1) then gives

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr \left(\frac{1}{n} \sum_{i=1}^n L_i^2 > C_L \mid \mathcal{G}_n \right) \leq C \exp\{-cn\}. \quad (200)$$

Let

$$Q_{P,i} = \frac{1}{\sqrt{p}} (\mathbf{H} \mathbf{\beta}_P)_i.$$

On $\mathcal{S}_n(t_0)$, $Q_{P,i}$ is conditionally Gaussian with uniformly bounded variance. Fix $M_0 < \infty$. On $\{|Q_{P,i}| \leq M_0\} \cap \mathcal{S}_n(t_0)$,

$$\tilde{Y}_i \geq (1 - \alpha) \rho'(\theta_P^* - t_0 - M_0),$$

and

$$1 - \tilde{Y}_i \geq (1 - \alpha) \{1 - \rho'(\theta_P^* + t_0 + M_0)\}.$$

Set

$$a = \min \left\{ (1 - \alpha) \rho'(\theta_P^* - t_0 - M_0), (1 - \alpha) \{1 - \rho'(\theta_P^* + t_0 + M_0)\}, \frac{1}{4} \right\}.$$

Then $a \in (0, 1/2)$ and

$$\{\tilde{Y}_i \notin [a, 1 - a]\} \subseteq \{|Q_{P,i}| > M_0\}. \quad (201)$$

The uniform conditional subgaussian bound for L_i gives a uniform fourth moment, and conditional Cauchy-Schwarz therefore yields

$$\begin{aligned} \mathbb{E} [L_i^2 \mathbb{1}\{|Q_{P,i}| > M_0\} \mid \mathcal{G}_n] &\leq \mathbb{E}[L_i^4 \mid \mathcal{G}_n]^{1/2} \\ &\quad \times \Pr(|Q_{P,i}| > M_0 \mid \mathcal{G}_n)^{1/2}. \end{aligned}$$

The right-hand side tends to zero uniformly on $\mathcal{S}_n(t_0)$ as $M_0 \rightarrow \infty$. The same subgaussian bound gives

$$\sup_{i,n} \mathbb{1}_{\mathcal{S}_n(t_0)} \mathbb{E} [L_i^2 \mathbb{1}\{L_i > K\} \mid \mathcal{G}_n] \rightarrow 0$$

as $K \rightarrow \infty$. Choose first M_0 and then K so that

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \mathbb{E} [L_i^2 \mathbb{1}\{|Q_{P,i}| > M_0 \text{ or } L_i > K\} \mid \mathcal{G}_n] \leq \frac{d_r^2}{128}.$$

The variables in this expectation are conditionally independent across i , and their centred versions have uniformly bounded conditional sub-exponential norms because they are dominated by L_i^2 . Another application of the Bernstein inequality (Vershynin, 2018, Theorem 2.8.1), together with (201), gives

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr \left(\frac{1}{n} \sum_{i \notin \mathcal{I}_n(a,K)} L_i^2 > \frac{d_r^2}{64} \mid \mathcal{G}_n \right) \leq C \exp\{-cn\}. \quad (202)$$

It remains to turn the projected residual into the uniform gap. Conditional on $\mathcal{C}_n$, the vector $\boldsymbol{\eta}_{\tilde{Y}}$ and the projector $\boldsymbol{\Pi}_n^\perp$ are fixed, whereas $\mathbf{g}$ and $\mathbf{h}$ are independent standard Gaussian vectors. Set

$$a_n = \frac{\|\boldsymbol{\Pi}_n^\perp \boldsymbol{\eta}_{\tilde{Y}}\|_2}{\sqrt{n}}, \quad b_n = \frac{\|\boldsymbol{\Pi}_n^\perp \mathbf{g}\|_2}{\sqrt{n}}, \quad c_n = \frac{\|\mathbf{h}\|_2}{\sqrt{n}},$$

and

$$d_n = \frac{1}{n} \langle \boldsymbol{\Pi}_n^\perp \boldsymbol{\eta}_{\tilde{Y}}, \boldsymbol{\Pi}_n^\perp \mathbf{g} \rangle.$$

The rank m of $\boldsymbol{\Pi}_n^\perp$ satisfies $n - 3 \leq m \leq n$. Since $\|\boldsymbol{\Pi}_n^\perp \mathbf{g}\|_2^2 \sim \chi_m^2$, Gaussian norm concentration (Vershynin 2018, Theorem 3.1.1) and

$$0 \leq 1 - \sqrt{m/n} \leq \frac{3}{n}$$

give, for all sufficiently large n ,

$$\Pr(|b_n - 1| > \delta \mid \mathcal{C}_n) \leq C \exp\{-cn\}. \quad (203)$$

Similarly, since $p/n \rightarrow \kappa$ and $\mathbf{h} \sim \mathcal{N}(\mathbf{0}_{p-s}, \mathbf{I}_{p-s})$ conditionally on $\mathcal{C}_n$,

$$\Pr(c_n > \sqrt{\kappa} + \delta \mid \mathcal{C}_n) \leq C \exp\{-cn\}.$$

Moreover,

$$d_n = \frac{1}{n} \langle \boldsymbol{\Pi}_n^\perp \boldsymbol{\eta}_{\tilde{Y}}, \mathbf{g} \rangle \sim \mathcal{N}\left(0, \frac{a_n^2}{n}\right)$$

conditionally on $\mathcal{C}_n$. Hence, on $\{a_n \geq a_0\}$,

$$\Pr(|d_n| > \delta(1 - \delta)a_n \mid \mathcal{C}_n) \leq 2 \exp\left\{-\frac{n\delta^2(1 - \delta)^2}{2}\right\}. \quad (204)$$

On $\{b_n \geq 1 - \delta\}$, the complementary event in (204) implies $|d_n| \leq \delta a_n b_n$.

For every $\sigma \geq 0$, projection onto $\text{span}\{\mathbf{1}, \text{col}(\mathbf{H}_1)\}^\perp$ gives

$$\begin{aligned} M_n(\sigma, \boldsymbol{\eta}_{\tilde{Y}}) - \sigma \frac{\|\mathbf{h}\|_2}{\sqrt{n}} &\geq \frac{1}{\sqrt{n}} \|\boldsymbol{\Pi}_n^\perp (\boldsymbol{\eta}_{\tilde{Y}} - \sigma \mathbf{g})\|_2 - \sigma c_n \\ &= \sqrt{a_n^2 + \sigma^2 b_n^2 - 2\sigma d_n} - \sigma c_n. \end{aligned}$$

Suppose that

$$a_n \geq a_0, \quad b_n \geq 1 - \delta, \quad c_n \leq \sqrt{\kappa} + \delta, \quad |d_n| \leq \delta a_n b_n.$$

With $t = \sigma b_n$ and $q_n = c_n/b_n \leq q_\delta$, one has

$$0 \leq q_n^2 \leq q_\delta^2 < 1 - \delta,$$

so the square root below is real. Moreover,

$$\sqrt{a_n^2 + \sigma^2 b_n^2 - 2\sigma d_n} - \sigma c_n \geq \sqrt{1 - \delta} \sqrt{a_n^2 + t^2} - q_n t.$$

By the Cauchy–Schwarz inequality,

$$q_n t + a_n \sqrt{1 - \delta - q_n^2} \leq \sqrt{1 - \delta} \sqrt{a_n^2 + t^2}.$$

Consequently, uniformly over $\sigma \geq 0$,

$$M_n(\sigma, \boldsymbol{\eta}_{\tilde{Y}}) - \sigma \frac{\|\mathbf{h}\|_2}{\sqrt{n}} \geq a_n \sqrt{1 - \delta - q_n^2} \geq d_r,$$

where the last step uses (199). Combining (198) and (203)–(204) by the tower property shows that the gap inequality fails with $\mathcal{G}_n$ -conditional probability at most $C \exp\{-cn\}$ on $\mathcal{S}_n(t_0)$.

Finally, if $\mathbf{b} \in \sigma\mathbf{g} + \mathcal{R}_0$, then

$$b_i = \sigma g_i + \theta + \frac{1}{\sqrt{p}}(\mathbf{H}_1 \mathbf{v})_i$$

for some $|\theta| \leq C_\theta$ and $\mathbf{v} \in V_p$, so (194) follows directly from the definition of L_i . The gap bound, (200), and (202), combined by a union bound, prove (193). $\square$

Proposition F.1 (Deterministic small- r exclusion). *Assume that $\mathcal{E}_n^{\text{sr}}(d_r, a, K, C_L)$ holds and that $C_\eta \geq \sqrt{C_L}$. Set*

$$c_{a,K} = \rho''(\text{logit}(1-a) + K), \quad r_0 = \min \left\{ \frac{1}{2}, \frac{c_{a,K} d_r}{16} \right\}.$$

Let $\mathcal{R} \subseteq \mathcal{R}_0$ be nonempty, compact, and convex. For $\sigma \in [0, C_\sigma]$, $r \geq 0$, and $\boldsymbol{\eta} \in \mathbb{R}^n$, define

$$M_{\mathcal{R}}(\sigma, \boldsymbol{\eta}) = \frac{1}{\sqrt{n}} \text{dist}_2(\boldsymbol{\eta} - \sigma\mathbf{g}, \mathcal{R}),$$

$$\tilde{\boldsymbol{\eta}}_{\mathcal{R}, \sigma, r} = \arg \min_{\boldsymbol{\eta} \in \mathbb{R}^n} \{G_n(\boldsymbol{\eta}) + r M_{\mathcal{R}}(\sigma, \boldsymbol{\eta})\},$$

and

$$A_{\mathcal{R}}(\sigma, r) = -\sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \min_{\boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n} \{G_n(\boldsymbol{\eta}) + r M_{\mathcal{R}}(\sigma, \boldsymbol{\eta})\}.$$

Then, uniformly over $\sigma \in [0, C_\sigma]$ and over all such $\mathcal{R}$, the following hold.

(i) For every $r \geq 0$, the unconstrained minimiser exists uniquely and

$$\frac{\|\tilde{\boldsymbol{\eta}}_{\mathcal{R}, \sigma, r}\|_2}{\sqrt{n}} \leq \sqrt{C_L} \leq C_\eta.$$

Hence it is also the unique minimiser over $\mathcal{B}_{C_\eta}^n$.

(ii) For every $r \geq 0$,

$$\frac{1}{\sqrt{n}} \|\tilde{\boldsymbol{\eta}}_{\mathcal{R}, \sigma, r} - \boldsymbol{\eta}_{\tilde{Y}}\|_2 \leq \frac{2r}{c_{a,K}} + \frac{d_r}{8}. \quad (205)$$

In particular, the left-hand side is at most $d_r/4$ whenever $r \in [0, r_0]$.

(iii) For $0 \leq r < r' \leq r_0$,

$$A_{\mathcal{R}}(\sigma, r') - A_{\mathcal{R}}(\sigma, r) \geq \frac{3d_r}{4}(r' - r). \quad (206)$$

Consequently, for every $c \in (0, r_0]$,

$$\max_{r \in [0, 1]} A_{\mathcal{R}}(\sigma, r) = \max_{r \in [c, 1]} A_{\mathcal{R}}(\sigma, r). \quad (207)$$

All conclusions remain valid after any further deterministic enlargement of C_η .

Proof. Fix σ and $\mathcal{R}$, and write

$$M(\boldsymbol{\eta}) = M_{\mathcal{R}}(\sigma, \boldsymbol{\eta}), \quad \tilde{\boldsymbol{\eta}}_r = \tilde{\boldsymbol{\eta}}_{\mathcal{R}, \sigma, r}.$$

For $i = 1, \dots, n$, set

$$f_i(x) = \rho(x) - \tilde{Y}_i x.$$

Since $\tilde{Y}_i \in (0, 1)$, f_i is strictly convex and coercive, with unique minimiser $\eta_{\tilde{Y}, i}$. Hence G_n is strictly convex and coercive. The function M is convex, continuous, nonnegative, and $1/\sqrt{n}$ -Lipschitz. Therefore $G_n + rM$ is strictly convex and coercive for every $r \geq 0$, so $\tilde{\boldsymbol{\eta}}_r$ exists uniquely.

Let

$$\mathbf{b} = \sigma \mathbf{g} + P_{\mathcal{R}}(\tilde{\boldsymbol{\eta}}_r - \sigma \mathbf{g}).$$

Then $\mathbf{b} \in \sigma \mathbf{g} + \mathcal{R} \subseteq \sigma \mathbf{g} + \mathcal{R}_0$ and

$$M(\tilde{\boldsymbol{\eta}}_r) = \frac{1}{\sqrt{n}} \|\tilde{\boldsymbol{\eta}}_r - \mathbf{b}\|_2.$$

For each i , let J_i be the closed interval with endpoints $\eta_{\tilde{Y},i}$ and b_i , and define

$$\eta'_i = P_{J_i}(\tilde{\eta}_{r,i}).$$

If $\tilde{\eta}_{r,i} \notin J_i$, then, for some $t_i \in [0, 1)$,

$$\eta'_i = (1 - t_i)\eta_{\tilde{Y},i} + t_i\tilde{\eta}_{r,i}.$$

Since $\tilde{\eta}_{r,i} \neq \eta_{\tilde{Y},i}$ and $\eta_{\tilde{Y},i}$ is the unique minimiser of f_i , convexity, with strict inequality when $t_i \in (0, 1)$, gives

$$\begin{aligned} f_i(\eta'_i) &\leq (1 - t_i)f_i(\eta_{\tilde{Y},i}) + t_i f_i(\tilde{\eta}_{r,i}) \\ &< f_i(\tilde{\eta}_{r,i}). \end{aligned}$$

Moreover, since $b_i \in J_i$ and projection onto an interval is nonexpansive,

$$|\eta'_i - b_i| \leq |\tilde{\eta}_{r,i} - b_i|.$$

Thus, if the vector $\boldsymbol{\eta}'$ differs from $\tilde{\boldsymbol{\eta}}_r$, then

$$G_n(\boldsymbol{\eta}') < G_n(\tilde{\boldsymbol{\eta}}_r)$$

and

$$M(\boldsymbol{\eta}') \leq \frac{1}{\sqrt{n}} \|\boldsymbol{\eta}' - \mathbf{b}\|_2 \leq \frac{1}{\sqrt{n}} \|\tilde{\boldsymbol{\eta}}_r - \mathbf{b}\|_2 = M(\tilde{\boldsymbol{\eta}}_r),$$

contradicting optimality. Hence, for every i , $\tilde{\eta}_{r,i}$ lies between $\eta_{\tilde{Y},i}$ and b_i . By (194),

$$|b_i - \eta_{\tilde{Y},i}| \leq L_i.$$

The definition of L_i also gives $|\eta_{\tilde{Y},i}| \leq L_i$ and $|b_i| \leq L_i$. Therefore

$$|\tilde{\eta}_{r,i} - \eta_{\tilde{Y},i}| \leq L_i, \quad |\tilde{\eta}_{r,i}| \leq L_i. \quad (208)$$

The environment event now gives

$$\frac{\|\tilde{\boldsymbol{\eta}}_r\|_2}{\sqrt{n}} \leq \left(\frac{1}{n} \sum_{i=1}^n L_i^2 \right)^{1/2} \leq \sqrt{C_L} \leq C_\eta.$$

Thus the unconstrained minimiser is feasible for $\mathcal{B}_{C_\eta}^n$ and is the unique constrained minimiser. This proves (i).

Set

$$\Delta_r = \frac{1}{\sqrt{n}} \|\tilde{\boldsymbol{\eta}}_r - \boldsymbol{\eta}_{\tilde{Y}}\|_2.$$

Optimality against $\boldsymbol{\eta}_{\tilde{Y}}$ and the Lipschitz property of M give

$$\begin{aligned} G_n(\tilde{\boldsymbol{\eta}}_r) - G_n(\boldsymbol{\eta}_{\tilde{Y}}) &\leq r \{M(\boldsymbol{\eta}_{\tilde{Y}}) - M(\tilde{\boldsymbol{\eta}}_r)\} \\ &\leq r \Delta_r. \end{aligned} \quad (209)$$

Since $f'_i(\eta_{\tilde{Y},i}) = 0$ and $f''_i = \rho''$,

$$G_n(\tilde{\boldsymbol{\eta}}_r) - G_n(\boldsymbol{\eta}_{\tilde{Y}}) = \frac{1}{n} \sum_{i=1}^n D_i,$$

where

$$D_i = \int_{\eta_{\tilde{Y},i}}^{\tilde{\eta}_{r,i}} (\tilde{\eta}_{r,i} - u) \rho''(u) du \geq 0.$$

If $i \in \mathcal{I}_n(a, K)$, then $|\eta_{\tilde{Y},i}| \leq \text{logit}(1-a)$ and (208) gives $|\tilde{\eta}_{r,i} - \eta_{\tilde{Y},i}| \leq K$. Every point between $\eta_{\tilde{Y},i}$ and $\tilde{\eta}_{r,i}$ therefore belongs to $[-\text{logit}(1-a) - K, \text{logit}(1-a) + K]$. Since ρ'' is even and nonincreasing on $[0, \infty)$,

$$D_i \geq \frac{c_{a,K}}{2} (\tilde{\eta}_{r,i} - \eta_{\tilde{Y},i})^2, \quad i \in \mathcal{I}_n(a, K).$$

Using this inequality on $\mathcal{I}_n(a, K)$, (208) on its complement, and (209), we obtain

$$\begin{aligned} \Delta_r^2 &\leq \frac{2}{c_{a,K}} \frac{1}{n} \sum_{i=1}^n D_i + \frac{1}{n} \sum_{i \notin \mathcal{I}_n(a, K)} L_i^2 \\ &\leq \frac{2r}{c_{a,K}} \Delta_r + \frac{d_r^2}{64}. \end{aligned}$$

If $x^2 \leq ux + v^2$ with $x, u, v \geq 0$, then $x \leq u + v$. Hence (205) follows. If $r \leq r_0$, then $2r/c_{a,K} \leq d_r/8$, and therefore $\Delta_r \leq d_r/4$. This proves (ii).

Let $0 \leq r < r' \leq r_0$ and write

$$p(r) = G_n(\tilde{\eta}_r) + rM(\tilde{\eta}_r).$$

Optimality of $\tilde{\eta}_r$, tested at $\tilde{\eta}_{r'}$, gives

$$p(r') - p(r) \geq (r' - r)M(\tilde{\eta}_{r'}).$$

By (i),

$$A_{\mathcal{R}}(\sigma, r) = -\sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + p(r).$$

Using the Lipschitz property of M , part (ii), the inclusion $\mathcal{R} \subseteq \mathcal{R}_0$, and the gap component of $\mathcal{E}_n^{\text{sr}}(d_r, a, K, C_L)$, we obtain

$$\begin{aligned} A_{\mathcal{R}}(\sigma, r') - A_{\mathcal{R}}(\sigma, r) &\geq (r' - r) \left\{ M(\tilde{\eta}_{r'}) - \sigma \frac{\|\mathbf{h}\|_2}{\sqrt{n}} \right\} \\ &\geq (r' - r) \left\{ M(\eta_{\tilde{Y}}) - \Delta_{r'} - \sigma \frac{\|\mathbf{h}\|_2}{\sqrt{n}} \right\} \\ &\geq (r' - r) \left\{ M_n(\sigma, \eta_{\tilde{Y}}) - \sigma \frac{\|\mathbf{h}\|_2}{\sqrt{n}} - \frac{d_r}{4} \right\} \\ &\geq \frac{3d_r}{4} (r' - r), \end{aligned}$$

which is (206).

For fixed σ and $\mathcal{R}$, continuity of M and compactness of $\mathcal{B}_{C_\eta}^n$ give

$$B_{\sigma, \mathcal{R}} = \sup_{\boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n} M_{\mathcal{R}}(\sigma, \boldsymbol{\eta}) < \infty.$$

Value comparison in the constrained minimum yields, for $r, r' \in [0, 1]$,

$$|A_{\mathcal{R}}(\sigma, r') - A_{\mathcal{R}}(\sigma, r)| \leq \left\{ \sigma \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + B_{\sigma, \mathcal{R}} \right\} |r' - r|.$$

Thus the maxima in (207) are attained. By (206), every $r < c$ satisfies $A_{\mathcal{R}}(\sigma, r) \leq A_{\mathcal{R}}(\sigma, c)$, which proves (207). The proof uses C_η only through $C_\eta \geq \sqrt{C_L}$, so every conclusion is preserved by enlargement of C_η . $\square$

F.3.2 Upper bounds and Moreau-envelope feasibility

Lemma 12 (*t*-upper bound). *Fix $c_r \in (0, 1]$. Define the event*

$$\mathcal{I}_n^t = \{ \|\mathbf{g}\|_2 \leq 2\sqrt{n} \}.$$

Set

$$K_t = \frac{1}{2} (C_\theta + 3C_\beta + 2C_\sigma)^2 + \frac{1}{2} + \log 2,$$

and choose

$$C_t > \frac{2(K_t + \log 2)}{c_r}.$$

Then, on $\mathcal{I}_n^t$, $\mathcal{E}_n^t(c_r, C_t)$ holds, where

$$\mathcal{E}_n^t(c_r, C_t) = \{ \phi^{r,t}(c_r, C_t; \mathbf{g}, \mathbf{h}) = \phi^{r,t}(c_r, \infty; \mathbf{g}, \mathbf{h}) = \phi^r(c_r; \mathbf{g}, \mathbf{h}) \}.$$

More precisely, on $\mathcal{I}_n^t$, for every $\sigma \in [0, C_\sigma]$ and every $r \in [c_r, 1]$, the inner infimum in (79), before the outer σ -minimum and r -maximum, is unchanged when $t \in (0, \infty)$ is replaced by $t \in (0, C_t]$. Moreover, there exist constants $C, c > 0$ and a $N \in \mathbb{N}$ such that, for all $n > N$,

$$\Pr(\neg \mathcal{I}_n^t \mid \mathcal{G}_n) \leq C \exp\{-cn\}.$$

Consequently,

$$\Pr(\neg \mathcal{E}_n^t(c_r, C_t) \mid \mathcal{G}_n) \leq C \exp\{-cn\}.$$

The constants C, c, N are universal (they come from the Gaussian norm concentration for $\mathbf{g}$ alone); K_t depends only on $C_\theta, C_\beta, C_\sigma$.

Proof. For fixed $\sigma \in [0, C_\sigma]$ and $r \in [c_r, 1]$, consider the inner optimisation after the norm identity (78):

$$\min_{\substack{\mathbf{v} \in V_p \\ |\theta| \leq C_\theta \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n}} \inf_{t > 0} \left[\frac{r}{2t} \frac{1}{n} \left\| \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} \right\|_2^2 + \frac{rt}{2} + \frac{1}{n} \{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} \} \right].$$

We first give a uniform upper bound on this value. Since $0 \in V_p$, $0 \in [-C_\theta, C_\theta]$, and $0 \in \mathcal{B}_{C_\eta}^n$, we may test the feasible point

$$\mathbf{v} = 0, \quad \theta = 0, \quad \boldsymbol{\eta} = 0, \quad t = 1.$$

At this point,

$$\frac{r}{2t} \frac{1}{n} \left\| \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} \right\|_2^2 + \frac{rt}{2} + \frac{1}{n} \{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} \} = \frac{r}{2} \sigma^2 \frac{\|\mathbf{g}\|_2^2}{n} + \frac{r}{2} + \log 2.$$

On $\mathcal{I}_n^t$, this is bounded above by

$$2C_\sigma^2 + \frac{1}{2} + \log 2 \leq K_t,$$

since $2C_\sigma^2 = \frac{1}{2}(2C_\sigma)^2 \leq \frac{1}{2}(C_\theta + 3C_\beta + 2C_\sigma)^2$. Thus the inner infimum is at most K_t , uniformly over $\sigma \in [0, C_\sigma]$ and $r \in [c_r, 1]$.

Next, for every $y \in (0, 1)$ and every $x \in \mathbb{R}$,

$$\rho(x) - yx \geq \inf_{z \in \mathbb{R}} \{ \rho(z) - yz \} = -y \log y - (1 - y) \log(1 - y) \geq 0 \geq -\log 2.$$

Therefore, for every $\boldsymbol{\eta} \in \mathbb{R}^n$,

$$\frac{1}{n} \{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} \} \geq -\log 2.$$

Since the square term is nonnegative and $r \geq c_r$, any feasible point with $t > C_t$ has objective value at least

$$\frac{c_r t}{2} - \log 2 > K_t,$$

by the choice of C_t . Such a point cannot attain the inner infimum, because the feasible point above has value at most K_t . Hence, for every $\sigma \in [0, C_\sigma]$ and $r \in [c_r, 1]$, the infimum over $t > 0$ is unchanged when restricted to $t \in (0, C_t]$.

It follows that

$$\phi^{r,t}(c_r, C_t; \mathbf{g}, \mathbf{h}) = \phi^{r,t}(c_r, \infty; \mathbf{g}, \mathbf{h}) = \phi^r(c_r; \mathbf{g}, \mathbf{h}),$$

on $\mathcal{I}_n^t$, which proves the deterministic implication.

It remains to bound the probability of $\mathcal{I}_n^t$. Conditional on $\mathcal{G}_n$, the vector $\mathbf{g}$ is independent of $\mathcal{G}_n$ and has i.i.d. standard normal entries. Hence $\|\mathbf{g}\|_2^2 \sim \chi_n^2$, and the standard chi-square concentration bound (see, for example, [Vershynin 2018](#) Theorem 3.1.1) gives

$$\Pr(\|\mathbf{g}\|_2 > 2\sqrt{n} \mid \mathcal{G}_n) \leq C \exp\{-cn\},$$

for all sufficiently large n . This proves the conditional bound. $\square$

Lemma 13 (Moreau feasibility). *Fix $c_r \in (0, 1]$ and $C_t \in (0, \infty)$, and set $L = C_t/c_r$. Let $C_M < \infty$ and define the centre-control event*

$$\mathcal{E}_n^{\text{ctr}}(C_M) = \sup_{\substack{|\theta| \leq C_\theta \\ \mathbf{v} \in V_p \\ \sigma \in [0, C_\sigma] \\ r \in [c_r, 1] \\ t \in (0, C_t]}} \frac{1}{\sqrt{n}} \left\| \theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \sigma \mathbf{g} + \frac{t}{r} \tilde{\mathbf{y}} \right\|_2 \leq C_M.$$

If $C_\eta \geq C_M + L$, then, on $\mathcal{E}_n^{\text{ctr}}(C_M)$, for every admissible $(\theta, \mathbf{v}, \sigma, r, t)$, set

$$\lambda = \frac{t}{r}, \quad \mathbf{b} = \theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \sigma \mathbf{g} + \lambda \tilde{\mathbf{y}}.$$

The unconstrained coordinatewise Moreau minimiser belongs to $\mathcal{B}_{C_\eta}^n$ and

$$M_\rho^{\text{const}}(\mathbf{b}, \lambda) = M_\rho(\mathbf{b}, \lambda).$$

Consequently,

$$\mathcal{E}_n^{\text{ctr}}(C_M) \implies \mathcal{E}_n^{\text{M}}(c_r, C_t, C_\eta).$$

Proof. Fix $|\theta| \leq C_\theta$, $\mathbf{v} \in V_p$, $\sigma \in [0, C_\sigma]$, $r \in [c_r, 1]$, and $t \in (0, C_t]$, and set

$$\lambda = \frac{t}{r}, \quad \mathbf{b} = \theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \sigma \mathbf{g} + \lambda \tilde{\mathbf{y}}.$$

Then $\lambda \in (0, L]$. For each coordinate, let

$$\eta_i^* = \arg \min_{\eta \in \mathbb{R}} \left\{ \rho(\eta) + \frac{1}{2\lambda} (\eta - b_i)^2 \right\}.$$

Since the objective is strictly convex and differentiable, η_i^* is unique and satisfies

$$\rho'(\eta_i^*) + \frac{1}{\lambda} (\eta_i^* - b_i) = 0.$$

Therefore,

$$\eta_i^* = b_i - \lambda \rho'(\eta_i^*).$$

Because $0 < \rho'(\eta_i^*) < 1$, we have

$$|\eta_i^*| \leq |b_i| + \lambda.$$

Thus, for $\boldsymbol{\eta}^* = (\eta_1^*, \dots, \eta_n^*)^\top$,

$$\frac{\|\boldsymbol{\eta}^*\|_2}{\sqrt{n}} \leq \frac{\|\mathbf{b}\|_2}{\sqrt{n}} + \lambda \leq C_M + L \leq C_\eta,$$

on $\mathcal{E}_n^{\text{ctr}}(C_M)$. Hence the unconstrained coordinatewise Moreau minimiser is feasible for the constraint $\boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n$. Therefore,

$$M_\rho^{\text{const}}(\mathbf{b}, \lambda) = M_\rho(\mathbf{b}, \lambda),$$

for every point in the scalar AO domain. Substituting this equality into the definitions of $\phi^{M, \text{const}}(c_r, C_t; \mathbf{g}, \mathbf{h})$ and $\phi^M(c_r, C_t; \mathbf{g}, \mathbf{h})$ gives

$$\phi^{M, \text{const}}(c_r, C_t; \mathbf{g}, \mathbf{h}) = \phi^M(c_r, C_t; \mathbf{g}, \mathbf{h}).$$

This is exactly $\mathcal{E}_n^M(c_r, C_t, C_\eta)$. $\square$

Lemma 14 (Probability of the Moreau centre-control event). *Fix $c_r \in (0, 1]$ and $C_t \in (0, \infty)$, and set*

$$L = \frac{C_t}{c_r}.$$

There exist constants $C_M < \infty$, $C, c > 0$ and $N \in \mathbb{N}$ such that, for all $n > N$,

$$\Pr(\neg \mathcal{E}_n^{\text{ctr}}(C_M) \mid \mathcal{G}_n) \leq C \exp\{-cn\}.$$

The constants depend only on $C_\beta, C_\theta, C_\sigma, c_r, C_t$ and universal Gaussian concentration constants. In particular, they do not depend on the realised values of $\mathcal{G}_n$.

Proof. For every admissible $(\theta, \mathbf{v}, \sigma, r, t)$, we have

$$\frac{1}{\sqrt{n}} \left\| \theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \sigma \mathbf{g} + \frac{t}{r} \tilde{\mathbf{y}} \right\|_2 \leq |\theta| + \frac{1}{\sqrt{n}} \left\| \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} \right\|_2 + \sigma \frac{\|\mathbf{g}\|_2}{\sqrt{n}} + \frac{t}{r} \frac{\|\tilde{\mathbf{y}}\|_2}{\sqrt{n}}.$$

Since $0 < \tilde{Y}_i < 1$, it holds that $\|\tilde{\mathbf{y}}\|_2/\sqrt{n} \leq 1$. Moreover, $|\theta| \leq C_\theta$, $\sigma \leq C_\sigma$, $t/r \leq L$. It remains to control the signal term and the fresh Gaussian term.

By (93), on the common extension, $\mathbf{H}_1/\sqrt{p} = \mathbf{Z}\boldsymbol{\Gamma}_p^{1/2}$ almost surely, where $\mathbf{Z} \in \mathbb{R}^{n \times 2}$ has i.i.d. $\mathcal{N}(0, 1)$ entries conditional on $\mathcal{G}_n$. For $\mathbf{v} \in V_p$, set $\mathbf{u} = \boldsymbol{\Gamma}_p^{1/2} \mathbf{v}$. Then $\|\mathbf{u}\|_2^2 = \mathbf{v}^\top \boldsymbol{\Gamma}_p \mathbf{v} = \|\mathbf{B}\mathbf{v}\|_2^2/p \leq C_\beta^2$. Therefore, on the common extension,

$$\sup_{\mathbf{v} \in V_p} \frac{1}{\sqrt{n}} \left\| \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} \right\|_2 = \sup_{\mathbf{u} \in U \cap \text{range}(\boldsymbol{\Gamma}_p)} \frac{1}{\sqrt{n}} \|\mathbf{Z}\mathbf{u}\|_2 \leq C_\beta \frac{\|\mathbf{Z}\|_2}{\sqrt{n}}.$$

By the standard Gaussian operator-norm bound for the $n \times 2$ matrix $\mathbf{Z}$ (see, for example, Rudelson and Vershynin [2010] Theorem 2.6), there exist constants $C, c > 0$ and $N_1 \in \mathbb{N}$ such that, for all $n > N_1$,

$$\Pr\left(\frac{\|\mathbf{Z}\|_2}{\sqrt{n}} > 3 \mid \mathcal{G}_n\right) \leq C \exp\{-cn\}.$$

Similarly, since $\mathbf{g} \sim \mathcal{N}(\mathbf{0}_n, \mathbf{I}_n)$ independently of $\mathcal{G}_n$, there exist constants $C, c > 0$ and $N_2 \in \mathbb{N}$ such that, for all $n > N_2$,

$$\Pr\left(\frac{\|\mathbf{g}\|_2}{\sqrt{n}} > 2 \mid \mathcal{G}_n\right) \leq C \exp\{-cn\}.$$

On the intersection of the two events

$$\left\{ \frac{\|\mathbf{Z}\|_2}{\sqrt{n}} \leq 3 \right\} \cap \left\{ \frac{\|\mathbf{g}\|_2}{\sqrt{n}} \leq 2 \right\},$$

we have

$$\sup_{\substack{|\theta| \leq C_\theta \\ \mathbf{v} \in V_p \\ \sigma \in [0, C_\sigma] \\ r \in [c_r, 1] \\ t \in (0, C_t]}} \frac{1}{\sqrt{n}} \left\| \theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} + \sigma \mathbf{g} + \frac{t}{r} \tilde{\mathbf{y}} \right\|_2 \leq C_\theta + 3C_\beta + 2C_\sigma + L.$$

Thus take $C_M = C_\theta + 3C_\beta + 2C_\sigma + L$. A union bound gives

$$\Pr(-\mathcal{E}_n^{\text{ctr}}(C_M) \mid \mathcal{G}_n) \leq C \exp\{-cn\},$$

after adjusting C, c, N . □

F.4 Lemmas for Section B.7

F.4.1 Nullspace and moving-domain control

For $y \in (0, 1)$ and $\lambda \geq 0$ write

$$e_\lambda(x, y) = m_\rho(x + \lambda y, \lambda) - yx - \frac{\lambda}{2} y^2, \quad x \in \mathfrak{R},$$

with $m_\rho(x, 0) = \rho(x)$; this is the function used again in Lemma 23 and for $\lambda = t/r$ the expectation term of $\bar{F}_n$ (conditionally on $\mathcal{G}_n$) and of F is $\mathbb{E}[e_\lambda(\theta + \mathbf{z}^\top \mathbf{u} + \sigma G, \tilde{Y}^{(n)}) \mid \mathcal{G}_n]$ and $\mathbb{E}[e_\lambda(\theta + \mathbf{z}^\top \mathbf{u} + \sigma G, \tilde{Y})]$, respectively. For each y , $x \mapsto e_\lambda(x, y)$ is strictly convex: for $\lambda = 0$, $e_0(x, y) = \rho(x) - yx$ and $\partial_x^2 e_0(x, y) = \rho''(x) > 0$; for $\lambda > 0$, the Moreau-envelope derivative formula of Salehi et al. (2019, Lemma 2, equation (29)) and the proximal-map derivative formula of Donoho and Montanari (2016, Proposition 6.3) give $\partial_x^2 e_\lambda(x, y) = \rho''(\eta) / \{1 + \lambda \rho''(\eta)\} > 0$ with $\eta = \text{prox}_{\lambda \rho}(x + \lambda y)$. In particular $e_\lambda(\cdot, y)$ is convex (see also Beck (2017, Chapter 6)).

Lemma 15 (Nullspace never helps for the conditional mean objective). *For every $\boldsymbol{\omega} = (\sigma, r, \mathbf{u}, \theta, t) \in \boldsymbol{\Omega}$, let $\boldsymbol{\omega}_p = (\sigma, r, \mathbf{P}_{\Gamma_p} \mathbf{u}, \theta, t)$. Then, conditional on $\mathcal{G}_n$, $\bar{F}_n(\boldsymbol{\omega}_p) \leq \bar{F}_n(\boldsymbol{\omega})$. Consequently, $\text{val}(\bar{F}_n; \boldsymbol{\Omega}_n) = \text{val}(\bar{F}_n; \boldsymbol{\Omega})$.*

Proof. Fix σ, r, θ, t and write $\lambda = t/r$. Let $\mathbf{u}_R = \mathbf{P}_{\Gamma_p} \mathbf{u}$, $\mathbf{u}_N = \mathbf{u} - \mathbf{u}_R$. Then $\mathbf{u}_R \in \text{range}(\Gamma_p)$ and $\mathbf{u}_N \in \ker(\Gamma_p)$. Since $\mathbf{P}_{\Gamma_p}$ is an orthogonal projector, $\|\mathbf{u}_R\|_2 \leq \|\mathbf{u}\|_2$, so if $\mathbf{u} \in U$ then $\mathbf{u}_R \in U \cap \text{range}(\Gamma_p)$.

Conditional on $\mathcal{G}_n$, let $\mathbf{z} \sim \mathcal{N}(\mathbf{0}_2, \mathbf{I}_2)$. Because $\text{range}(\Gamma_p) = \text{range}(\Gamma_p^{1/2})$, there exists $\mathbf{a} \in \mathfrak{R}^2$ such that $\mathbf{u}_R = \Gamma_p^{1/2} \mathbf{a}$. Hence $\mathbf{z}^\top \mathbf{u}_R = \mathbf{a}^\top \Gamma_p^{1/2} \mathbf{z}$ is measurable with respect to $\Gamma_p^{1/2} \mathbf{z}$. The pseudo-response variable $\tilde{Y}^{(n)}$ in $\bar{F}_n$ is a measurable function of $(\Gamma_p^{1/2} \mathbf{z}, \varepsilon)$ and of the fixed parameters (θ_0, θ_P) . Since $\Gamma_p^{1/2} \mathbf{u}_N = \mathbf{0}$, the scalar $\mathbf{z}^\top \mathbf{u}_N$ is independent of $(\Gamma_p^{1/2} \mathbf{z}, \tilde{Y}^{(n)}, G)$ and has mean zero conditional on $\mathcal{G}_n$.

Write $\xi_R = \theta + \mathbf{z}^\top \mathbf{u}_R + \sigma G$. Conditioning on $(\Gamma_p^{1/2} \mathbf{z}, \tilde{Y}^{(n)}, G)$, under which ξ_R and $\tilde{Y}^{(n)}$ are fixed, convexity of $e_\lambda(\cdot, \tilde{Y}^{(n)})$ and Jensen's inequality (Shiryaev, 2016, Chapter 2.7) give

$$\mathbb{E} \left[e_\lambda \left(\xi_R + \mathbf{z}^\top \mathbf{u}_N, \tilde{Y}^{(n)} \right) \mid \Gamma_p^{1/2} \mathbf{z}, \tilde{Y}^{(n)}, G, \mathcal{G}_n \right] \geq e_\lambda \left(\xi_R, \tilde{Y}^{(n)} \right).$$

Taking expectations gives $\bar{F}_n(\sigma, r, \mathbf{u}_R, \theta, t) \leq \bar{F}_n(\sigma, r, \mathbf{u}, \theta, t)$. The terms $-\sigma r \mu_{\mathbf{h}} / \sqrt{n} + rt/2$, with $\mu_{\mathbf{h}} = \mathbb{E}[\|\mathbf{h}\|_2 \mid \mathcal{G}_n]$, do not depend on $\mathbf{u}$, so they are unaffected.

It remains to pass from the pointwise inequality to the value. Since $\boldsymbol{\Omega}_n \subseteq \boldsymbol{\Omega}$, we always have $\text{val}(\bar{F}_n; \boldsymbol{\Omega}) \leq \text{val}(\bar{F}_n; \boldsymbol{\Omega}_n)$, because the innermost operation over $\mathbf{u}$ is a minimisation and $\boldsymbol{\Omega}$ has the larger $\mathbf{u}$ -domain. Conversely, the projection argument above shows that every $\mathbf{u} \in U$ can be replaced by $\mathbf{P}_{\Gamma_p} \mathbf{u} \in U \cap \text{range}(\Gamma_p)$ without increasing $\bar{F}_n$. Hence enlarging the $\mathbf{u}$ -domain from $\boldsymbol{\Omega}_n$ to $\boldsymbol{\Omega}$ cannot lower the value. Thus, $\text{val}(\bar{F}_n; \boldsymbol{\Omega}_n) = \text{val}(\bar{F}_n; \boldsymbol{\Omega})$. □

Lemma 16 (Strict nullspace penalty for the limiting objective). *Let $\mathbf{P}_\Gamma$ be the orthogonal projector onto $\text{range}(\Gamma)$. For $\mathbf{u} \in U$, write $\mathbf{u}_R = \mathbf{P}_\Gamma \mathbf{u}$, $\mathbf{u}_N = \mathbf{u} - \mathbf{u}_R$. Then, for every $(\sigma, r, \mathbf{u}, \theta, t) \in \boldsymbol{\Omega}$, we have $F(\sigma, r, \mathbf{u}_R, \theta, t) \leq F(\sigma, r, \mathbf{u}, \theta, t)$. Moreover, if $\mathbf{u}_N \neq \mathbf{0}$, then the inequality is strict: $F(\sigma, r, \mathbf{u}_R, \theta, t) < F(\sigma, r, \mathbf{u}, \theta, t)$. Consequently, $\text{val}(F; \boldsymbol{\Omega}) = \text{val}(F; \boldsymbol{\Omega}^*)$. In particular, for every fixed (σ, r) , every minimiser of $(\mathbf{u}, \theta, t) \mapsto F(\sigma, r, \mathbf{u}, \theta, t)$ over $U \times [-C_\theta, C_\theta] \times [0, C_t]$ has $\mathbf{u} \in \text{range}(\Gamma)$.*

Proof. Fix $(\sigma, r, \mathbf{u}, \theta, t) \in \Omega$ and set $\lambda = t/r$. Since $r \in [c_r, 1]$, we have $\lambda \geq 0$. Also, $\mathbf{u}_R \in \text{range}(\mathbf{\Gamma})$, $\mathbf{u}_N \in \ker(\mathbf{\Gamma})$, $\|\mathbf{u}_R\|_2 \leq \|\mathbf{u}\|_2$. Thus $\mathbf{u}_R \in U \cap \text{range}(\mathbf{\Gamma})$.

Let

$$\mathbf{Q} = \mathbf{\Gamma}^{1/2} \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}_2, \mathbf{I}_2).$$

The limiting pseudo-response $\tilde{Y}$ is a measurable function of $(\mathbf{Q}, \varepsilon)$ and of the fixed limiting parameters (θ_0^*, θ_P^*) . Since $\mathbf{u}_N \in \ker(\mathbf{\Gamma})$, $\mathbf{\Gamma}^{1/2} \mathbf{u}_N = \mathbf{0}_2$. Hence

$$\text{Cov}(\mathbf{z}^\top \mathbf{u}_N, \mathbf{Q}) = \mathbf{\Gamma}^{1/2} \mathbf{u}_N = \mathbf{0}_2.$$

Because all variables are jointly Gaussian before applying the response map, $\mathbf{z}^\top \mathbf{u}_N$ is independent of $\mathbf{Q}$. It is also independent of ε and G . Therefore

$$\mathbf{z}^\top \mathbf{u}_N \text{ is independent of } (\mathbf{Q}, \tilde{Y}, G),$$

and

$$\mathbb{E}[\mathbf{z}^\top \mathbf{u}_N] = 0.$$

If $\mathbf{u}_N \neq \mathbf{0}$, then

$$\mathbf{z}^\top \mathbf{u}_N \sim \mathcal{N}\left(0, \|\mathbf{u}_N\|_2^2\right),$$

is nondegenerate.

Recall that $x \mapsto e_\lambda(x, y)$ is strictly convex for every $y \in (0, 1)$ and $\lambda \geq 0$ (see the paragraph preceding Lemma 15).

Because $\mathbf{u}_R \in \text{range}(\mathbf{\Gamma}) = \text{range}(\mathbf{\Gamma}^{1/2})$, there exists $a \in \mathbb{R}^2$ such that $\mathbf{u}_R = \mathbf{\Gamma}^{1/2} a$. Hence $\mathbf{z}^\top \mathbf{u}_R = a^\top \mathbf{\Gamma}^{1/2} \mathbf{z} = a^\top \mathbf{Q}$, so $\mathbf{z}^\top \mathbf{u}_R$ is measurable with respect to $\mathbf{Q}$. Write $\xi_R = \theta + \mathbf{z}^\top \mathbf{u}_R + \sigma G$. Conditionally on $(\mathbf{Q}, \tilde{Y}, G)$, the variable ξ_R is fixed, while $\mathbf{z}^\top \mathbf{u}_N$ is independent of $(\mathbf{Q}, \tilde{Y}, G)$ and has mean zero. Jensen's inequality gives

$$\mathbb{E}\left[e_\lambda(\xi_R + \mathbf{z}^\top \mathbf{u}_N, \tilde{Y}) \mid \mathbf{Q}, \tilde{Y}, G\right] \geq e_\lambda(\xi_R, \tilde{Y}).$$

If $\mathbf{u}_N \neq \mathbf{0}$, then $\mathbf{z}^\top \mathbf{u}_N$ is nondegenerate and $e_\lambda(\cdot, \tilde{Y})$ is strictly convex, so the inequality is strict almost surely. Taking expectations gives

$$\mathbb{E}\left[e_\lambda\left(\theta + \mathbf{z}^\top \mathbf{u} + \sigma G, \tilde{Y}\right)\right] \geq \mathbb{E}\left[e_\lambda\left(\theta + \mathbf{z}^\top \mathbf{u}_R + \sigma G, \tilde{Y}\right)\right],$$

with strict inequality if $\mathbf{u}_N \neq \mathbf{0}$.

By the definition of e_λ , this is exactly the comparison of the expectation terms in F . The remaining terms $-\sigma r \sqrt{\kappa} + rt/2$ do not depend on $\mathbf{u}$. Therefore $F(\sigma, r, \mathbf{u}_R, \theta, t) \leq F(\sigma, r, \mathbf{u}, \theta, t)$, with strict inequality whenever $\mathbf{u}_N \neq \mathbf{0}$.

It remains to pass from the pointwise comparison to the value identity. For every fixed (σ, r) , the inner minimum over the full signal ball is no larger than the inner minimum over its intersection with $\text{range}(\mathbf{\Gamma})$. Conversely, projecting every feasible $\mathbf{u}$ onto $\text{range}(\mathbf{\Gamma})$ preserves feasibility and cannot increase F , so the two inner minima are equal. Taking the remaining operations in the profiled order gives $\text{val}(F; \Omega) = \text{val}(F; \Omega^*)$.

Finally, suppose that $(\tilde{\mathbf{u}}, \tilde{\theta}, \tilde{t})$ minimises $F(\sigma, r, \cdot)$ over the ambient inner domain and that $\tilde{\mathbf{u}} \notin \text{range}(\mathbf{\Gamma})$. Then $\tilde{\mathbf{u}}_N = \tilde{\mathbf{u}} - \mathbf{P}_\Gamma \tilde{\mathbf{u}} \neq \mathbf{0}$, so the strict part gives $F(\sigma, r, \mathbf{P}_\Gamma \tilde{\mathbf{u}}, \tilde{\theta}, \tilde{t}) < F(\sigma, r, \tilde{\mathbf{u}}, \tilde{\theta}, \tilde{t})$, contradicting minimality. Hence every ambient inner minimiser has $\tilde{\mathbf{u}} \in \text{range}(\mathbf{\Gamma})$. $\square$

These deterministic comparisons are combined with the uniform-convergence ingredients below in Proposition B.6

F.4.2 Uniform-convergence ingredients

Lemma 17 (Conditional pointwise concentration of F_n). *Fix $\omega = (\sigma, r, \mathbf{u}, \theta, t) \in \Omega$, and let $F_n(\omega)$ and $\bar{F}_n(\omega)$ be defined in (95) and (96). Then there exist constants $C, c > 0$ such that, for all $s > 0$ and all n sufficiently large,*

$$\Pr(|F_n(\omega) - \bar{F}_n(\omega)| > s \mid \mathcal{G}_n) \leq C \exp\{-cns^2\}.$$

The constants depend only on $C_\sigma, C_\beta, C_\theta, C_t, c_r$ and on universal Gaussian concentration constants. In particular, they do not depend on the realised value of $\mathcal{G}_n$.

Proof. Let $\lambda = t/r$. Since $\boldsymbol{\omega} \in \boldsymbol{\Omega}$, we have $0 \leq \sigma \leq C_\sigma$, $c_r \leq r \leq 1$, $\|\mathbf{u}\|_2 \leq C_\beta$, $|\theta| \leq C_\theta$, and $0 \leq t \leq C_t$. Thus, $0 \leq \lambda \leq L$, $L = C_t/c_r$.

Conditional on $\mathcal{G}_n$, the rows $(\mathbf{z}_i, g_i, \varepsilon_i)$ are independent and identically distributed, and are independent of $\mathbf{h}$. Define $\xi_i = \theta + \mathbf{z}_i^\top \mathbf{u} + \sigma g_i$ and $f_i(\boldsymbol{\omega}) = m_\rho(\xi_i + \lambda \tilde{Y}_i, \lambda) - \tilde{Y}_i \xi_i - (\lambda/2) \tilde{Y}_i^2$. Then

$$F_n(\boldsymbol{\omega}) = \frac{1}{n} \sum_{i=1}^n f_i(\boldsymbol{\omega}) - \sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \frac{rt}{2},$$

and $\bar{F}_n(\boldsymbol{\omega}) = \mathbb{E}[F_n(\boldsymbol{\omega}) \mid \mathcal{G}_n]$. Hence

$$F_n(\boldsymbol{\omega}) - \bar{F}_n(\boldsymbol{\omega}) = \frac{1}{n} \sum_{i=1}^n \{f_i(\boldsymbol{\omega}) - \mathbb{E}[f_i(\boldsymbol{\omega}) \mid \mathcal{G}_n]\} - \sigma r \frac{\|\mathbf{h}\|_2 - \mu_{\mathbf{h}}}{\sqrt{n}},$$

where $\mu_{\mathbf{h}} = \mathbb{E}[\|\mathbf{h}\|_2 \mid \mathcal{G}_n]$

We first bound the empirical average. Since $0 < \tilde{Y}_i < 1$ and $0 \leq \lambda \leq L$, and since

$$0 \leq m_\rho(b, \lambda) \leq \rho(b) \leq \log 2 + |b|,$$

for all $b \in \Re$ and $\lambda \geq 0$, we have

$$|f_i(\boldsymbol{\omega})| \leq \log 2 + |\xi_i + \lambda \tilde{Y}_i| + |\tilde{Y}_i \xi_i| + \frac{\lambda \tilde{Y}_i^2}{2} \leq \log 2 + 2|\xi_i| + \frac{3L}{2} \leq K_f (1 + \|\mathbf{z}_i\|_2 + |g_i|),$$

where $K_f < \infty$ depends only on $C_\sigma, C_\beta, C_\theta, C_t, c_r$. Thus, conditional on $\mathcal{G}_n$,

$$\|f_i(\boldsymbol{\omega})\|_{\psi_2} \leq K,$$

for a deterministic constant $K < \infty$ with the same parameter dependence. Centering changes the subgaussian norm by at most an absolute factor, so

$$\|f_i(\boldsymbol{\omega}) - \mathbb{E}[f_i(\boldsymbol{\omega}) \mid \mathcal{G}_n]\|_{\psi_2} \leq K,$$

after increasing K . Since the summands are independent conditional on $\mathcal{G}_n$, the general Hoeffding inequality (Vershynin [2018] Theorem 2.6.3) gives

$$\Pr \left(\left| \frac{1}{n} \sum_{i=1}^n \{f_i(\boldsymbol{\omega}) - \mathbb{E}[f_i(\boldsymbol{\omega}) \mid \mathcal{G}_n]\} \right| > \frac{s}{2} \mid \mathcal{G}_n \right) \leq 2 \exp\{-c_1 n s^2\},$$

for a constant $c_1 > 0$ depending only on K .

It remains to control the $\mathbf{h}$ term. Since $\sigma r \leq C_\sigma$, we have

$$\Pr \left(\sigma r \frac{\|\mathbf{h}\|_2 - \mu_{\mathbf{h}}}{\sqrt{n}} > \frac{s}{2} \mid \mathcal{G}_n \right) \leq \Pr \left(\|\mathbf{h}\|_2 - \mu_{\mathbf{h}} > \frac{s\sqrt{n}}{2C_\sigma} \mid \mathcal{G}_n \right).$$

Conditional on $\mathcal{G}_n$, Gaussian concentration for Lipschitz functions (Vershynin [2018] Theorem 5.2.2) gives

$$\Pr (\|\mathbf{h}\|_2 - \mu_{\mathbf{h}} > u \mid \mathcal{G}_n) \leq 2 \exp\{-cu^2\},$$

for a universal constant $c > 0$. Taking $u = s\sqrt{n}/(2C_\sigma)$ gives

$$\Pr \left(\sigma r \frac{\|\mathbf{h}\|_2 - \mu_{\mathbf{h}}}{\sqrt{n}} > \frac{s}{2} \mid \mathcal{G}_n \right) \leq 2 \exp\{-c_2 n s^2\}.$$

A union bound yields

$$\Pr (|F_n(\boldsymbol{\omega}) - \bar{F}_n(\boldsymbol{\omega})| > s \mid \mathcal{G}_n) \leq C \exp\{-c n s^2\},$$

after adjusting $C, c > 0$. □

Lemma 18 (Structural bias of the conditional AO mean). *Let $\bar{F}_n(\boldsymbol{\omega})$ and $F(\boldsymbol{\omega})$ be defined in (96) and (97). There exists a constant $K < \infty$, depending only on $\kappa, \alpha, \boldsymbol{\Gamma}, \theta_0^*, \theta_P^*, C_\sigma, C_\beta, C_\theta, C_t, c_r$ such that, for all $t_0 \in (0, 1]$ and all sufficiently large n ,*

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \sup_{\boldsymbol{\omega} \in \boldsymbol{\Omega}} |\bar{F}_n(\boldsymbol{\omega}) - F(\boldsymbol{\omega})| \leq K \left(t_0 + \sqrt{t_0} + \left| \frac{\mu_{\mathbf{h}}}{\sqrt{n}} - \sqrt{\kappa} \right| \right), \quad \mu_{\mathbf{h}} = \mathbb{E}[\|\mathbf{h}\|_2 \mid \mathcal{G}_n].$$

Consequently, for every $s > 0$, there exist $t_0 = t_0(s) > 0$ and $N_s \in \mathbb{N}$ such that, for all $n > N_s$,

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \sup_{\boldsymbol{\omega} \in \boldsymbol{\Omega}} |\bar{F}_n(\boldsymbol{\omega}) - F(\boldsymbol{\omega})| \leq \frac{s}{3}.$$

Proof. Fix $\boldsymbol{\omega} = (\sigma, r, \mathbf{u}, \theta, t) \in \boldsymbol{\Omega}$ and set $\lambda = t/r$. Since $\boldsymbol{\omega} \in \boldsymbol{\Omega}$, we have $0 \leq \lambda \leq L$, $L = C_t/c_r$. Let $\mathbf{q}_n = \boldsymbol{\Gamma}_p^{1/2} \mathbf{z}$, $\mathbf{Q} = \boldsymbol{\Gamma}^{1/2} \mathbf{z}$, $\mathbf{z} \sim \mathcal{N}(\mathbf{0}_2, \mathbf{I}_2)$, and define

$$\tilde{Y}^{(n)} = \alpha \mathbb{1}\{\varepsilon < \rho'(\theta_0 + q_{n,1})\} + (1 - \alpha) \rho'(\theta_P + q_{n,2}),$$

and

$$\tilde{Y} = \alpha \mathbb{1}\{\varepsilon < \rho'(\theta_0^* + Q_1)\} + (1 - \alpha) \rho'(\theta_P^* + Q_2).$$

Also write $\xi = \theta + \mathbf{z}^\top \mathbf{u} + \sigma G$, $G \sim \mathcal{N}(0, 1)$, where $(\mathbf{z}, G, \varepsilon)$ are mutually independent. Then

$$\bar{F}_n(\boldsymbol{\omega}) - F(\boldsymbol{\omega}) = \mathbb{E} \left[m_\rho(\xi + \lambda \tilde{Y}^{(n)}, \lambda) - m_\rho(\xi + \lambda \tilde{Y}, \lambda) \right] - \mathbb{E} \left[(\tilde{Y}^{(n)} - \tilde{Y}) \xi \right] - \frac{\lambda}{2} \mathbb{E} \left[(\tilde{Y}^{(n)})^2 - \tilde{Y}^2 \right] - \sigma r \left(\frac{\mu_{\mathbf{h}}}{\sqrt{n}} - \sqrt{\kappa} \right).$$

We control the difference between $\tilde{Y}^{(n)}$ and $\tilde{Y}$. On $\mathcal{S}_n(t_0)$, $|\theta_0 - \theta_0^*| \leq t_0$, $|\theta_P - \theta_P^*| \leq t_0$, and $\|\boldsymbol{\Gamma}_p - \boldsymbol{\Gamma}\|_2 \leq t_0$. By the Hölder continuity of the positive semidefinite square-root map (Bhatia, 1997, Theorem X.1.1),

$$\left\| \boldsymbol{\Gamma}_p^{1/2} - \boldsymbol{\Gamma}^{1/2} \right\|_2 \leq \|\boldsymbol{\Gamma}_p - \boldsymbol{\Gamma}\|_2^{1/2} \leq \sqrt{t_0}.$$

Using $\sup_x \rho''(x) \leq 1/4$, we have

$$|\rho'(\theta_P + q_{n,2}) - \rho'(\theta_P^* + Q_2)| \leq \frac{1}{4} (t_0 + \sqrt{t_0} \|\mathbf{z}\|_2).$$

For the Bernoulli part, conditional on $\mathbf{z}$, the uniform variable ε gives

$$\mathbb{E}_\varepsilon [|\mathbb{1}\{\varepsilon < \rho'(\theta_0 + q_{n,1})\} - \mathbb{1}\{\varepsilon < \rho'(\theta_0^* + Q_1)\}| \mid \mathbf{z}] = |\rho'(\theta_0 + q_{n,1}) - \rho'(\theta_0^* + Q_1)| \leq \frac{1}{4} (t_0 + \sqrt{t_0} \|\mathbf{z}\|_2).$$

Combining the two bounds, there is a constant $K_1 < \infty$, depending only on α , such that

$$\mathbb{E}_\varepsilon \left[\left| \tilde{Y}^{(n)} - \tilde{Y} \right| \mid \mathbf{z} \right] \leq K_1 (t_0 + \sqrt{t_0} \|\mathbf{z}\|_2).$$

Therefore,

$$\mathbb{E} \left[\left| \tilde{Y}^{(n)} - \tilde{Y} \right| \right] \leq K_2 (t_0 + \sqrt{t_0}).$$

Moreover, since

$$|\xi| \leq C_\theta + C_\beta \|\mathbf{z}\|_2 + C_\sigma |G|,$$

we also get

$$\mathbb{E} \left[(1 + |\xi|) \left| \tilde{Y}^{(n)} - \tilde{Y} \right| \right] \leq K_3 (t_0 + \sqrt{t_0}).$$

Next, by Lemma 38, the map $b \mapsto m_\rho(b, \lambda)$ is 1-Lipschitz for every $\lambda \geq 0$. Hence

$$\left| m_\rho(\xi + \lambda \tilde{Y}^{(n)}, \lambda) - m_\rho(\xi + \lambda \tilde{Y}, \lambda) \right| \leq \lambda \left| \tilde{Y}^{(n)} - \tilde{Y} \right| \leq L \left| \tilde{Y}^{(n)} - \tilde{Y} \right|.$$

Also, since $0 < \tilde{Y}^{(n)}, \tilde{Y} < 1$, $|(\tilde{Y}^{(n)})^2 - \tilde{Y}^2| \leq 2|\tilde{Y}^{(n)} - \tilde{Y}|$. Thus,

$$\begin{aligned} |\bar{F}_n(\boldsymbol{\omega}) - F(\boldsymbol{\omega})| &\leq L \mathbb{E} \left[\left| \tilde{Y}^{(n)} - \tilde{Y} \right| \right] + \mathbb{E} \left[|\xi| \left| \tilde{Y}^{(n)} - \tilde{Y} \right| \right] + L \mathbb{E} \left[\left| \tilde{Y}^{(n)} - \tilde{Y} \right| \right] + C_\sigma \left| \frac{\mu_{\mathbf{h}}}{\sqrt{n}} - \sqrt{\kappa} \right| \\ &\leq K \left(t_0 + \sqrt{t_0} + \left| \frac{\mu_{\mathbf{h}}}{\sqrt{n}} - \sqrt{\kappa} \right| \right). \end{aligned}$$

The constant K is uniform over $\boldsymbol{\omega} \in \boldsymbol{\Omega}$, because all coordinates of $\boldsymbol{\omega}$ are restricted to compact sets with deterministic radii. Taking the supremum over $\boldsymbol{\Omega}$ proves the first claim.

Finally, for each $d \geq 1$, let $\tilde{\mathbf{h}}_d \sim \mathcal{N}(\mathbf{0}_d, \mathbf{I}_d)$. Since $\text{rank}(\mathbf{B}) \in \{0, 1, 2\}$ and $p/n \rightarrow \kappa$,

$$\sup_{j \in \{0, 1, 2\}} \left| \frac{\mathbb{E} \left[\left\| \tilde{\mathbf{h}}_{p-j} \right\|_2 \right]}{\sqrt{n}} - \sqrt{\kappa} \right| \rightarrow 0.$$

Consequently,

$$\frac{\mu_{\mathbf{h}}}{\sqrt{n}} \rightarrow \sqrt{\kappa},$$

uniformly over the possible values of $\text{rank}(\mathbf{B})$. Given $s > 0$, choose $t_0 = t_0(s) > 0$ small enough so that

$$K(t_0 + \sqrt{t_0}) \leq \frac{s}{6},$$

and then choose N_s large enough so that, for all $n > N_s$,

$$K \left| \frac{\mu_{\mathbf{h}}}{\sqrt{n}} - \sqrt{\kappa} \right| \leq \frac{s}{6}.$$

This gives

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \sup_{\boldsymbol{\omega} \in \boldsymbol{\Omega}} |\bar{F}_n(\boldsymbol{\omega}) - F(\boldsymbol{\omega})| \leq \frac{s}{3},$$

for all $n > N_s$. □

Lemma 19 (Lipschitz bound for the finite AO objective). *Let $\boldsymbol{\omega} = (\sigma, r, \mathbf{u}, \theta, t)$, $\boldsymbol{\omega}' = (\sigma', r', \mathbf{u}', \theta', t')$ be two points in $\boldsymbol{\Omega}$. Let $\lambda = t/r$, $\lambda' = t'/r'$. Then*

$$|F_n(\boldsymbol{\omega}) - F_n(\boldsymbol{\omega}')| \leq L_{\sigma,n} |\sigma - \sigma'| + L_{r,n} |r - r'| + L_{u,n} \|\mathbf{u} - \mathbf{u}'\|_2 + L_{\theta,n} |\theta - \theta'| + L_{t,n} |t - t'|,$$

where

$$L_{\theta,n} = 1 + \frac{\|\tilde{\mathbf{y}}^\top \mathbf{1}\|}{n}, \quad L_{u,n} = \frac{\|\mathbf{Z}\|_2}{\sqrt{n}} + \frac{\|\mathbf{Z}^\top \tilde{\mathbf{y}}\|_2}{n}, \quad L_{\sigma,n} = \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \frac{\|\mathbf{g}\|_2}{\sqrt{n}} + \frac{|\tilde{\mathbf{y}}^\top \mathbf{g}|}{n}, \quad L_{t,n} = \frac{1}{2} + \frac{1}{c_r} \left(\frac{\|\tilde{\mathbf{y}}\|_2}{\sqrt{n}} + \frac{1}{2} \right) + \frac{1}{2c_r} \frac{\|\tilde{\mathbf{y}}\|_2^2}{n},$$

and

$$L_{r,n} = C_\sigma \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \frac{C_t}{2} + \frac{C_t}{c_r^2} \left(\frac{\|\tilde{\mathbf{y}}\|_2}{\sqrt{n}} + \frac{1}{2} \right) + \frac{C_t}{2c_r^2} \frac{\|\tilde{\mathbf{y}}\|_2^2}{n}.$$

Proof. Write

$$\Delta_\sigma = \sigma - \sigma', \quad \Delta_r = r - r', \quad \Delta_{\mathbf{u}} = \mathbf{u} - \mathbf{u}', \quad \Delta_\theta = \theta - \theta', \quad \Delta_t = t - t'.$$

Since $r, r' \in [c_r, 1]$ and $t, t' \in [0, C_t]$,

$$\left| \frac{t}{r} - \frac{t'}{r'} \right| = \left| \frac{(t - t')r' + t'(r' - r)}{rr'} \right| \leq \frac{|\Delta_t|}{c_r} + \frac{C_t}{c_r^2} |\Delta_r|.$$

Similarly,

$$\left| \frac{t}{2r} - \frac{t'}{2r'} \right| \leq \frac{|\Delta_t|}{2c_r} + \frac{C_t}{2c_r^2} |\Delta_r|.$$

Define

$$\mathbf{x} = \theta \mathbf{1} + \mathbf{Z}\mathbf{u} + \sigma \mathbf{g} + \lambda \tilde{\mathbf{y}}, \quad \mathbf{x}' = \theta' \mathbf{1} + \mathbf{Z}\mathbf{u}' + \sigma' \mathbf{g} + \lambda' \tilde{\mathbf{y}}.$$

By Lemma 38

$$\left| \frac{1}{n} M_\rho(\mathbf{x}, \lambda) - \frac{1}{n} M_\rho(\mathbf{x}', \lambda') \right| \leq \frac{1}{\sqrt{n}} \|\mathbf{x} - \mathbf{x}'\|_2 + \frac{1}{2} |\lambda - \lambda'|.$$

Furthermore,

$$\mathbf{x} - \mathbf{x}' = \Delta_\theta \mathbf{1} + \mathbf{Z}\Delta_{\mathbf{u}} + \Delta_\sigma \mathbf{g} + (\lambda - \lambda') \tilde{\mathbf{y}},$$

and hence

$$\frac{1}{\sqrt{n}} \|\mathbf{x} - \mathbf{x}'\|_2 \leq |\Delta_\theta| + \frac{\|\mathbf{Z}\|_2}{\sqrt{n}} \|\Delta_{\mathbf{u}}\|_2 + \frac{\|\mathbf{g}\|_2}{\sqrt{n}} |\Delta_\sigma| + \frac{\|\tilde{\mathbf{y}}\|_2}{\sqrt{n}} |\lambda - \lambda'|.$$

Combining the last three displays gives the Moreau part of the Lipschitz bound.

Next,

$$\left| -\sigma r \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \sigma' r' \frac{\|\mathbf{h}\|_2}{\sqrt{n}} \right| \leq \frac{\|\mathbf{h}\|_2}{\sqrt{n}} (|\Delta_\sigma| + C_\sigma |\Delta_r|).$$

Also,

$$\left| \frac{rt}{2} - \frac{r't'}{2} \right| \leq \frac{C_t}{2} |\Delta_r| + \frac{1}{2} |\Delta_t|.$$

For the linear term,

$$\left| \frac{1}{n} \tilde{\mathbf{y}}^\top [(\theta - \theta') \mathbf{1} + \mathbf{Z}(\mathbf{u} - \mathbf{u}') + (\sigma - \sigma') \mathbf{g}] \right| \leq \frac{\|\tilde{\mathbf{y}}^\top \mathbf{1}\|}{n} |\Delta_\theta| + \frac{\|\mathbf{Z}^\top \tilde{\mathbf{y}}\|_2}{n} \|\Delta_{\mathbf{u}}\|_2 + \frac{\|\tilde{\mathbf{y}}^\top \mathbf{g}\|}{n} |\Delta_\sigma|.$$

Finally,

$$\left| \frac{t}{2rn} \|\tilde{\mathbf{y}}\|_2^2 - \frac{t'}{2r'n} \|\tilde{\mathbf{y}}\|_2^2 \right| \leq \frac{\|\tilde{\mathbf{y}}\|_2^2}{n} \left(\frac{|\Delta_t|}{2c_r} + \frac{C_t}{2c_r^2} |\Delta_r| \right).$$

Collecting the coefficients of $|\Delta_\sigma|$, $|\Delta_r|$, $\|\Delta_{\mathbf{u}}\|_2$, $|\Delta_\theta|$, and $|\Delta_t|$ gives the stated bound. $\square$

Lemma 20 (Bounded Lipschitz constants). *Let*

$$L_n = \max \{L_{\sigma,n}, L_{r,n}, L_{u,n}, L_{\theta,n}, L_{t,n}\}.$$

There exist constants $\bar{C} < \infty$, $C, c > 0$ and $N \in \mathbb{N}$ such that, for all $n > N$,

$$\Pr(L_n > \bar{C} \mid \mathcal{G}_n) \leq C \exp\{-cn\}.$$

In particular,

$$\mathbf{1}_{S_n(t_0)} \Pr(L_n > \bar{C} \mid \mathcal{G}_n) \leq C \exp\{-cn\}.$$

The constants depend only on $\kappa, C_\sigma, C_\beta, C_\theta, C_t, c_r$ and on universal Gaussian concentration constants.

Proof. Since $0 < \tilde{Y}_i < 1$, we have $\|\tilde{\mathbf{y}}^\top \mathbf{1}\|/n \leq 1$, $\|\tilde{\mathbf{y}}\|_2/\sqrt{n} \leq 1$, and $\|\tilde{\mathbf{y}}\|_2^2/n \leq 1$. Therefore, $L_{\theta,n} \leq 2$ and

$$L_{t,n} \leq \frac{1}{2} + \frac{1}{c_r} \left(1 + \frac{1}{2} \right) + \frac{1}{2c_r}.$$

Also,

$$L_{r,n} \leq C_\sigma \frac{\|\mathbf{h}\|_2}{\sqrt{n}} + \frac{C_t}{2} + \frac{3C_t}{2c_r^2} + \frac{C_t}{2c_r^2}.$$

We next control the Gaussian terms. Conditional on $\mathcal{G}_n$, the vector $\mathbf{g}$ and the matrix $\mathbf{Z}$ have standard Gaussian entries, while

$$\mathbf{h} \sim \mathcal{N}(\mathbf{0}_{p-\text{rank}(\mathbf{B})}, \mathbf{I}_{p-\text{rank}(\mathbf{B})}).$$

These Gaussian objects are conditionally independent. Standard chi-square concentration (see, for example, [Vershynin 2018](#) Theorem 3.1.1) gives constants $C, c > 0$ such that, for all sufficiently large n ,

$$\Pr\left(\frac{\|\mathbf{g}\|_2}{\sqrt{n}} > 2 \mid \mathcal{G}_n\right) \leq Ce^{-cn},$$

and

$$\Pr\left(\frac{\|\mathbf{h}\|_2}{\sqrt{n}} > 2\sqrt{\kappa} + 1 \mid \mathcal{G}_n\right) \leq Ce^{-cn}.$$

Moreover, since $\mathbf{Z}$ is $n \times 2$ with i.i.d. standard Gaussian entries, the Gaussian singular-value bound gives

$$\Pr\left(\frac{\|\mathbf{Z}\|_2}{\sqrt{n}} > 2 \mid \mathcal{G}_n\right) \leq Ce^{-cn},$$

for all sufficiently large n . Finally,

$$\frac{\|\mathbf{Z}^\top \tilde{\mathbf{y}}\|_2}{n} \leq \frac{\|\mathbf{Z}\|_2 \|\tilde{\mathbf{y}}\|_2}{n} \leq \frac{\|\mathbf{Z}\|_2}{\sqrt{n}},$$

and

$$\frac{|\tilde{\mathbf{y}}^\top \mathbf{g}|}{n} \leq \frac{\|\tilde{\mathbf{y}}\|_2 \|\mathbf{g}\|_2}{n} \leq \frac{\|\mathbf{g}\|_2}{\sqrt{n}}.$$

On the intersection of the above Gaussian events, all quantities $L_{\sigma,n}$, $L_{r,n}$, $L_{u,n}$, $L_{\theta,n}$, and $L_{t,n}$ are bounded by a deterministic constant $\bar{C} < \infty$ depending only on $\kappa, C_\sigma, C_t, c_r$. A union bound gives

$$\Pr(L_n > \bar{C} \mid \mathcal{G}_n) \leq C \exp\{-cn\}.$$

This proves the claim. $\square$

Lemma 21 (Lipschitz bound for the limiting AO objective). *There exists a constant $\bar{C}_F < \infty$, depending only on*

$$\kappa, \alpha, \mathbf{\Gamma}, \theta_0^*, \theta_P^*, C_\sigma, C_\beta, C_\theta, C_t, c_r,$$

such that, for all $\boldsymbol{\omega}, \boldsymbol{\omega}' \in \boldsymbol{\Omega}$,

$$|F(\boldsymbol{\omega}) - F(\boldsymbol{\omega}')| \leq \bar{C}_F \|\boldsymbol{\omega} - \boldsymbol{\omega}'\|_2.$$

Proof. Let $\boldsymbol{\omega} = (\sigma, r, \mathbf{u}, \theta, t)$, $\boldsymbol{\omega}' = (\sigma', r', \mathbf{u}', \theta', t')$ be points in $\boldsymbol{\Omega}$, and write $\lambda = t/r$, $\lambda' = t'/r'$. Since $r, r' \in [c_r, 1]$ and $t, t' \in [0, C_t]$,

$$|\lambda - \lambda'| \leq \frac{|t - t'|}{c_r} + \frac{C_t}{c_r^2} |r - r'|.$$

Also, if

$$\xi = \theta + \mathbf{z}^\top \mathbf{u} + \sigma G, \quad \xi' = \theta' + \mathbf{z}^\top \mathbf{u}' + \sigma' G,$$

then

$$|\xi - \xi'| \leq |\theta - \theta'| + \|\mathbf{z}\|_2 \|\mathbf{u} - \mathbf{u}'\|_2 + |G| |\sigma - \sigma'|.$$

Because $0 < \tilde{Y} < 1$, Lemma [38](#) gives

$$\left| m_\rho(\xi + \lambda \tilde{Y}, \lambda) - \tilde{Y} \xi - \frac{\lambda}{2} \tilde{Y}^2 - m_\rho(\xi' + \lambda' \tilde{Y}, \lambda') + \tilde{Y} \xi' + \frac{\lambda'}{2} \tilde{Y}^2 \right| \leq 2 |\xi - \xi'| + 2 |\lambda - \lambda'|.$$

Furthermore,

$$|\sigma r - \sigma' r'| \leq |\sigma - \sigma'| + C_\sigma |r - r'|,$$

and

$$|rt - r't'| \leq |t - t'| + C_t |r - r'|.$$

Since

$$\mathbb{E}[\|\mathbf{z}\|_2] + \mathbb{E}[|G|] < \infty,$$

taking expectations and collecting the coordinatewise coefficients gives a deterministic constant $\bar{C}_F < \infty$ with the stated dependence such that

$$|F(\boldsymbol{\omega}) - F(\boldsymbol{\omega}')| \leq \bar{C}_F \|\boldsymbol{\omega} - \boldsymbol{\omega}'\|_2.$$

□

F.5 Lemmas for Section B.8

F.5.1 Well-posedness and natural-domain presentation

Lemma 22 (Population residual gap for the limiting pseudo-response). *Let $\mathbf{Q} = [Q_1, Q_2]^\top$ as in Section 3.1 and set*

$$q_0 = \rho'(\theta_0^* + Q_1), \quad q_P = \rho'(\theta_P^* + Q_2), \quad y = \alpha \mathbf{1}\{\varepsilon < q_0\} + (1 - \alpha)q_P = \tilde{Y},$$

so that y is the limiting pseudo-response of Section 4.1. Let

$$y_0 = (1 - \alpha)q_P, \quad y_1 = \alpha + (1 - \alpha)q_P,$$

and define $\eta_y = \text{logit}(y)$. Then $0 < y < 1$ almost surely, $\eta_y \in L^2$, and, with

$$d_0^2 = \inf_{\substack{\theta \in \mathfrak{R} \\ \mathbf{u} \in \text{range}(\mathbf{\Gamma})}} \mathbb{E} \left[(\eta_y - \theta - \mathbf{z}^\top \mathbf{u})^2 \right],$$

one has

$$d_0^2 \geq 16\alpha^2 \mathbb{E}[q_0(1 - q_0)] > 0. \quad (210)$$

Moreover, for every $\sigma \geq 0$,

$$\inf_{\substack{\theta \in \mathfrak{R} \\ \mathbf{u} \in \text{range}(\mathbf{\Gamma})}} \|\eta_y - \theta - \mathbf{z}^\top \mathbf{u} - \sigma G\|_{L^2} = \sqrt{d_0^2 + \sigma^2}, \quad (211)$$

and consequently

$$\inf_{\sigma \geq 0} \left\{ \sqrt{d_0^2 + \sigma^2} - \sigma \sqrt{\kappa} \right\} = d_0 \sqrt{1 - \kappa} > 0. \quad (212)$$

Proof. Conditional on $\mathbf{z}$, the pseudo-response y equals y_1 with probability q_0 and y_0 with probability $1 - q_0$. Since $0 < q_0, q_P < 1$ almost surely and $\alpha \in (0, 1)$, both conditional atoms have positive probability and belong to $(0, 1)$.

Put $A = \theta_P^* + Q_2$. Direct calculation gives $\text{logit}(y_0) = A + \log(1 - \alpha) - \log(1 + \alpha e^A)$ and $\text{logit}(y_1) = \log(\alpha + e^A) - \log(1 - \alpha)$. Hence there is a constant $C_\alpha < \infty$ such that

$$\max \{ |\text{logit}(y_0)|, |\text{logit}(y_1)| \} \leq |A| + C_\alpha.$$

Since A is Gaussian, $\eta_y \in L^2$.

Conditional on $\mathbf{z}$,

$$\text{var}(\eta_y \mid \mathbf{z}) = q_0(1 - q_0) (\text{logit}(y_1) - \text{logit}(y_0))^2.$$

Because $y_1 - y_0 = \alpha$ and $d\text{logit}(v)/dv = 1/\{v(1 - v)\} \geq 4$, $v \in (0, 1)$, one has $\text{logit}(y_1) - \text{logit}(y_0) \geq 4\alpha$. For every $\theta \in \mathfrak{R}$ and $\mathbf{u} \in \text{range}(\mathbf{\Gamma})$, the variable $\theta + \mathbf{z}^\top \mathbf{u}$ is $\sigma(\mathbf{z})$ -measurable. Therefore

$$\mathbb{E} \left[(\eta_y - \theta - \mathbf{z}^\top \mathbf{u})^2 \right] \geq \mathbb{E}[\text{var}(\eta_y \mid \mathbf{z})] \geq 16\alpha^2 \mathbb{E}[q_0(1 - q_0)].$$

The final expectation is strictly positive because $q_0(1 - q_0) > 0$ almost surely. Taking the infimum proves (210).

Finally, G is independent of $(\eta_y, \mathbf{z})$, has mean zero and variance one. Thus, for every admissible $(\theta, \mathbf{u})$,

$$\mathbb{E} \left[(\eta_y - \theta - \mathbf{z}^\top \mathbf{u} - \sigma G)^2 \right] = \mathbb{E} \left[(\eta_y - \theta - \mathbf{z}^\top \mathbf{u})^2 \right] + \sigma^2.$$

Taking the infimum gives (211). A direct one-dimensional minimisation gives

$$\inf_{\sigma \geq 0} \left\{ \sqrt{d_0^2 + \sigma^2} - \sigma \sqrt{\kappa} \right\} = d_0 \sqrt{1 - \kappa},$$

which proves (212). $\square$

Lemma 23 (Geometry and smoothness of the limiting AO). *For $y \in (0, 1)$, write*

$$\ell_y(\eta) = \rho(\eta) - y\eta$$

and, for $\lambda \geq 0$, recall from the paragraph preceding Lemma 15 that

$$e_\lambda(x, y) = m_\rho(x + \lambda y, \lambda) - yx - \frac{\lambda}{2}y^2,$$

with $m_\rho(x, 0) = \rho(x)$. For $r > 0$ and $t > 0$,

$$e_{t/r}(x, y) = \inf_{\eta \in \mathbb{R}} \left\{ \ell_y(\eta) + \frac{r}{2t}(\eta - x)^2 \right\}. \quad (213)$$

At $t = 0$, give the quadratic perspective its closed extension

$$\left[\frac{r}{2t}(\eta - x)^2 \right]_{t=0} = \begin{cases} 0, & \eta = x, \\ +\infty, & \eta \neq x. \end{cases}$$

Then the right-hand side of (213) equals $\ell_y(x)$ at $t = 0$, consistently with the continuous extension of the compact limiting objective. The resulting closed function F is jointly continuous on

$$[0, \infty) \times (0, \infty) \times \text{range}(\mathbf{\Gamma}) \times \mathbb{R} \times [0, \infty),$$

where the coordinates are ordered as $(\sigma, r, \mathbf{u}, \theta, t)$.

For every fixed $r > 0$, the closed map $(\sigma, \mathbf{u}, \theta, t) \mapsto F(\sigma, r, \mathbf{u}, \theta, t)$ is strictly convex on

$$[0, \infty) \times \text{range}(\mathbf{\Gamma}) \times \mathbb{R} \times [0, \infty).$$

For every fixed $(\sigma, \mathbf{u}, \theta, t)$ with $t > 0$, the map $r \mapsto F(\sigma, r, \mathbf{u}, \theta, t)$ is strictly concave on $(0, \infty)$.

On the relative positive domain $r, t > 0$, the function F is C^2 . Put

$$\lambda = \frac{t}{r}, \quad \xi = \theta + \mathbf{z}^\top \mathbf{u} + \sigma G,$$

$$\eta = \text{prox}_{\lambda\rho}(\xi + \lambda y), \quad R = \rho'(\eta) - y, \quad W = \frac{\rho''(\eta)}{1 + \lambda\rho''(\eta)}.$$

Then

$$\begin{aligned} \partial_\theta F &= \mathbb{E}[R], & \nabla_{\mathbf{u}} F &= \mathbb{E}[\mathbf{z}R], \\ \partial_\sigma F &= \mathbb{E}[GR] - r\sqrt{\kappa}, & \partial_t F &= -\frac{1}{2r} \mathbb{E}[R^2] + \frac{r}{2}, \\ \partial_r F &= \frac{t}{2r^2} \mathbb{E}[R^2] - \sigma\sqrt{\kappa} + \frac{t}{2}. \end{aligned} \quad (214)$$

For a direction $h = (h_\sigma, h_{\mathbf{u}}, h_\theta, h_t)$ in the minimising coordinates, set $A_h = h_\theta + \mathbf{z}^\top h_{\mathbf{u}} + Gh_\sigma$. Then

$$D_x^2 F(x, r)[h, h] = \mathbb{E} \left[W \left(A_h - \frac{R}{r} h_t \right)^2 \right], \quad (215)$$

and

$$\partial_{rr} F(x, r) = -\frac{t}{r^3} \mathbb{E} \left[\frac{R^2}{1 + (t/r)\rho''(\eta)} \right] < 0. \quad (216)$$

Proof. Expanding the square in the Moreau definition gives (213). Since $\ell'_y(\eta) = \rho'(\eta) - y$ has absolute value at most one, ℓ_y is one-Lipschitz. Testing $\eta = x$ gives $e_\lambda(x, y) \leq \ell_y(x)$, whereas $\ell_y(\eta) \geq \ell_y(x) - |\eta - x|$ gives

$$e_\lambda(x, y) \geq \ell_y(x) - \frac{\lambda}{2}.$$

Thus $e_\lambda(x, y) \rightarrow \ell_y(x)$ as $\lambda \rightarrow 0$, locally uniformly when r is bounded away from zero. Together with the one-Lipschitz property of ℓ_y , dominated convergence gives joint continuity of the closed expectation and hence of F .

For fixed $r > 0$, the map

$$(\eta, x, t) \mapsto \ell_y(\eta) + \frac{r}{2t}(\eta - x)^2,$$

with its closed perspective at $t = 0$ is jointly convex. Partial minimisation over η preserves convexity in (x, t) . Since $x = \theta + \mathbf{z}^\top \mathbf{u} + \sigma G$ is affine in $(\sigma, \mathbf{u}, \theta)$, expectation preserves convexity, and the remaining terms

$$-\sigma r \sqrt{\kappa} + \frac{rt}{2},$$

are affine in the minimising coordinates. For fixed minimising coordinates, the expression inside the infimum in (213) is affine in r . Its infimum is therefore concave in r , and the remaining r -dependence is affine.

Fix $\lambda > 0$, $x \in \mathfrak{R}$, and $y \in (0, 1)$, and set

$$\eta = \text{prox}_{\lambda\rho}(x + \lambda y), \quad R = \rho'(\eta) - y, \quad W = \frac{\rho''(\eta)}{1 + \lambda\rho''(\eta)}.$$

The Moreau-envelope derivatives of Salehi *et al.* (2019) Lemma 2, equation (29)) and the proximal-map derivatives of Donoho and Montanari (2016, Proposition 6.3), together with the proximal first-order condition and the chain rule, give

$$\begin{aligned} \eta &= x - \lambda R, & \partial_x e_\lambda(x, y) &= R, & \partial_\lambda e_\lambda(x, y) &= -\frac{1}{2}R^2, \\ \partial_x R &= W, & \partial_\lambda R &= -WR, & \partial_y R &= -\frac{1}{1 + \lambda\rho''(\eta)}. \end{aligned} \quad (217)$$

Here the λ -derivatives are taken with x and y fixed. Since $0 < \rho'(\eta) < 1$, $0 < y < 1$, and $0 < \rho''(\eta) \leq 1/4$,

$$|R| < 1, \quad 0 < W \leq \frac{1}{4}, \quad 1 - \lambda W = \frac{1}{1 + \lambda\rho''(\eta)} > 0. \quad (218)$$

Consequently,

$$D_{(x, \lambda)}^2 e_\lambda(x, y)[(a, b), (a, b)] = W(a - Rb)^2 \quad (219)$$

for every $(a, b) \in \mathfrak{R}^2$.

Applying (217) pointwise at $x = \xi$ and using the chain rule gives (214). On every compact subset of the relative domain $r, t > 0$, the first derivatives are dominated by a constant times

$$1 + \|\mathbf{z}\|_2 + |G|,$$

and the second derivatives by a constant times

$$(1 + \|\mathbf{z}\|_2 + |G|)^2.$$

The bounds in (218) make these envelopes integrable. Dominated convergence therefore justifies differentiation under the expectation and gives continuous second derivatives.

For fixed r , an increment h changes the two arguments (ξ, λ) by

$$d\xi = A_h, \quad d\lambda = \frac{h_t}{r}.$$

Equation (219) gives (215). Also,

$$\frac{\partial R}{\partial r} = \frac{t}{r^2} W R.$$

Differentiating the last line of (214) gives

$$\partial_{rr} F = -\frac{t}{r^3} \mathbb{E}[R^2] + \frac{t^2}{r^4} \mathbb{E}[W R^2] = -\frac{t}{r^3} \mathbb{E}[R^2(1 - \lambda W)],$$

which is (216) by (218).

It remains to prove strictness. Suppose first that $t > 0$ and that the quadratic form in (215) vanishes for a direction h . Since $W > 0$ almost surely,

$$A_h = \frac{R}{r} h_t \quad \text{almost surely.}$$

If $h_t = 0$, then the affine Gaussian variable A_h vanishes almost surely, which implies

$$h_\sigma = 0, \quad h_{\mathbf{u}} = \mathbf{0}, \quad h_\theta = 0.$$

If $h_t \neq 0$, boundedness of R implies that A_h is bounded almost surely. Hence again $h_\sigma = 0$ and $h_{\mathbf{u}} = \mathbf{0}$, so R would have to be almost surely constant. This is impossible: conditional on $(\mathbf{z}, G)$, the response takes the two distinct values y_0 and y_1 with positive probabilities, while (217) gives

$$\partial_y R = -\frac{1}{1 + \lambda \rho''(\eta)} < 0.$$

Thus the Hessian is positive definite in every nonzero minimising direction when $t > 0$.

To extend strict convexity to the closed domain, take two distinct points in the minimising domain and restrict $F(\cdot, r)$ to the line segment joining them. If at least one endpoint has positive t , every interior point of the segment has positive t , and the preceding positive second directional derivative gives strict convexity along the segment. If both endpoints have $t = 0$, the restriction is

$$\mathbb{E}[\ell_y(\theta + \mathbf{z}^\top \mathbf{u} + \sigma G)] - \sigma r \sqrt{\kappa},$$

and its second directional derivative is

$$\mathbb{E}[\rho''(\xi) A_h^2] > 0,$$

for every nonzero coefficient direction. This proves strict convexity on the closed minimising domain.

Finally, the expectation in (216) is strictly positive. Conditional on $(\mathbf{z}, G)$, the two possible residual values are distinct, so they cannot both be zero: both atoms have positive conditional probability. Hence (216) is strictly negative, proving strict concavity in r when $t > 0$. $\square$

Lemma 24 (Coercivity and existence of a closed limiting-AO saddle). *There exist deterministic constants $\bar{r} \in (0, 1)$, $a > 0$, $B_a < \infty$, and $B_t < \infty$ such that*

$$F(\sigma, \bar{r}, \mathbf{u}, \theta, t) \geq a \sqrt{\theta^2 + \|\mathbf{u}\|_2^2 + \sigma^2}, \quad (220)$$

for every $\sigma \geq 0$, $\mathbf{u} \in \text{range}(\mathbf{\Gamma})$, $\theta \in \mathfrak{R}$, and $t \geq 0$. For every $c \in (0, \bar{r}]$, define

$$\mathcal{X}_{\text{cl}} = [0, \infty) \times \text{range}(\mathbf{\Gamma}) \times \mathfrak{R} \times [0, \infty),$$

and

$$J_c(x) = \max_{r \in [c, 1]} F(x, r), \quad x = (\sigma, \mathbf{u}, \theta, t) \in \mathcal{X}_{\text{cl}}.$$

Then J_c is continuous and coercive, and

$$\min_{x \in \mathcal{X}_{\text{cl}}} \max_{r \in [c, 1]} F(x, r) = \max_{r \in [c, 1]} \inf_{x \in \mathcal{X}_{\text{cl}}} F(x, r). \quad (221)$$

The common value is attained by a saddle point (x_c, r_c) . Every minimiser $x = (\sigma, \mathbf{u}, \theta, t)$ of J_c satisfies

$$\sqrt{\theta^2 + \|\mathbf{u}\|_2^2 + \sigma^2} \leq B_a, \quad (222)$$

and

$$t \leq B_t. \quad (223)$$

The bounds are uniform over $c \in (0, \bar{r}]$.

Proof. Choose $\delta \in (0, 1 - \sqrt{\kappa})$. For a unit coefficient direction

$$d = (d_\theta, d_{\mathbf{u}}, d_\sigma), \quad d_\theta^2 + \|d_{\mathbf{u}}\|_2^2 + d_\sigma^2 = 1,$$

write $D_d = d_\theta + \mathbf{z}^\top d_{\mathbf{u}} + d_\sigma G$ and set $m(y) = \min(y, 1 - y) > 0$ almost surely. For every unit direction, $\mathbb{E}[D_d^2] = 1$, and the family $\{D_d : \|d\|_2 = 1\}$ has uniformly bounded fourth moments. For $a > 0$ and $r > 0$,

$$\mathbb{E}[D_d^2 \mathbf{1}\{r|D_d| > m(y)\}] \leq (\mathbb{E}[D_d^4])^{1/2} \Pr(m(y) < a)^{1/2} + \frac{r^2}{a^2} \mathbb{E}[D_d^4].$$

The bound is uniform in d . First choose a small and then choose $\bar{r} \in (0, 1)$ small enough that

$$\inf_{\|d\|_2=1} \mathbb{E}[D_d^2 \mathbf{1}\{\bar{r}|D_d| \leq m(y)\}] \geq 1 - \delta. \quad (224)$$

For $v \in [0, 1]$, let $\rho^*(v) = v \log v + (1 - v) \log(1 - v)$, with $0 \log 0 = 0$. Thus $\rho^*(v) \leq 0$. Fenchel's inequality (Beck [2017], Chapter 4) gives

$$\ell_y(\eta) \geq w\eta - \rho^*(y + w),$$

whenever $-y \leq w \leq 1 - y$. For $X \in L^2$, define

$$K_r(X) = \inf_{\eta \in L^2} \{\mathbb{E}[\ell_y(\eta)] + r \|\eta - X\|_{L^2}\}.$$

For $t > 0$, apply (213) at the pointwise proximal minimiser and use $a^2/(2t) + t/2 \geq a$ for $a \geq 0$ to obtain

$$F(\sigma, r, \mathbf{u}, \theta, t) \geq K_r(\theta + \mathbf{z}^\top \mathbf{u} + \sigma G) - r\sigma\sqrt{\kappa}. \quad (225)$$

At $t = 0$, the same bound follows by taking $\eta = \theta + \mathbf{z}^\top \mathbf{u} + \sigma G$ in the definition of K_r .

Put $s = \sqrt{\theta^2 + \|\mathbf{u}\|_2^2 + \sigma^2}$. If $s = 0$, the desired lower bound follows from $\ell_y \geq -\rho^*(y) \geq 0$. If $s > 0$, set

$$d = \frac{(\theta, \mathbf{u}, \sigma)}{s},$$

and define

$$w = \bar{r} D_d \mathbf{1}\{\bar{r}|D_d| \leq m(y)\}.$$

Then $-y \leq w \leq 1 - y$ almost surely and $\|w\|_{L^2} \leq \bar{r}$. For every $\eta \in L^2$,

$$\mathbb{E}[\ell_y(\eta)] + \bar{r} \|\eta - s D_d\|_{L^2} \geq \mathbb{E}[w\eta] - \mathbb{E}[\rho^*(y + w)] + \bar{r} \|\eta - s D_d\|_{L^2} \geq s \mathbb{E}[w D_d] - \mathbb{E}[\rho^*(y + w)] \geq s \bar{r} (1 - \delta),$$

where the second inequality uses Cauchy–Schwarz (Rudin [1976] Theorem 1.37(d)) and $\|w\|_{L^2} \leq \bar{r}$, and the final inequality uses (224) and $\rho^* \leq 0$. Taking the infimum over η and using $\sigma \leq s$ in (225) gives

$$F(\sigma, \bar{r}, \mathbf{u}, \theta, t) \geq \bar{r} (1 - \delta - \sqrt{\kappa}) s.$$

Thus (220) holds with

$$a = \bar{r} (1 - \delta - \sqrt{\kappa}) > 0.$$

Since $J_c \geq F(\cdot, \bar{r})$, the coefficient block is coercively controlled. If that block remains bounded while $t \rightarrow \infty$, then

$$e_{t/\bar{r}}(\xi, y) \geq \inf_{\eta \in \mathbb{R}} \ell_y(\eta) = -\rho^*(y) \geq 0,$$

so

$$F(\sigma, \bar{r}, \mathbf{u}, \theta, t) \geq -\bar{r}\sigma\sqrt{\kappa} + \frac{\bar{r}t}{2} \rightarrow +\infty.$$

Hence J_c is coercive. Continuity follows from Lemma 23. Therefore J_c attains its minimum.

The interval $[c, 1]$ is compact and convex, the closed natural minimising domain is convex, and Lemma 23 gives continuity, convexity in x , and concavity in r . Sion's min-max theorem (see, for example, Simons 1995 Theorem 3) gives (221). Define

$$h_c(r) = \inf_{x \in \mathcal{X}_{c1}} F(x, r).$$

As an infimum of continuous functions, h_c is upper semicontinuous and therefore attains its maximum on $[c, 1]$. Let x_c minimise J_c , let r_c maximise h_c , and let v_c denote the common minimax value. Then

$$v_c = h_c(r_c) \leq F(x_c, r_c) \leq J_c(x_c) = v_c.$$

Thus equality holds throughout and (x_c, r_c) is a saddle point.

Finally, the feasible point $(\sigma, \mathbf{u}, \theta, t) = (0, \mathbf{0}, 0, 0)$ has value $\log 2$ for every r , so every minimiser of J_c satisfies $J_c(x) \leq \log 2$. The coercive bound gives

$$\sqrt{\theta^2 + \|\mathbf{u}\|_2^2 + \sigma^2} \leq \frac{\log 2}{a} = B_a.$$

Also,

$$F(\sigma, \bar{r}, \mathbf{u}, \theta, t) \geq -\bar{r}\sigma\sqrt{\kappa} + \frac{\bar{r}t}{2}.$$

Since $F(\sigma, \bar{r}, \mathbf{u}, \theta, t) \leq J_c(x) \leq \log 2$ and $\sigma \leq B_a$,

$$t \leq \frac{2}{\bar{r}} \{\log 2 + \bar{r}B_a\sqrt{\kappa}\} = B_t.$$

This proves the uniform bounds. $\square$

Lemma 25 (Exclusion of the lower-radius boundary). *Let $\bar{r}$ be as in Lemma 24. There exists $c_0 \in (0, \bar{r})$ such that, for every $c \in (0, c_0]$, every saddle point (x_c, r_c) of (221) satisfies $r_c > c$.*

Proof. Suppose otherwise. Then there are $c_j \downarrow 0$ and saddle points $(\sigma_j, c_j, \mathbf{u}_j, \theta_j, t_j)$ of the closed games over $[c_j, 1]$. By Lemma 24, their minimising coordinates are uniformly bounded.

Set

$$\xi_j = \theta_j + \mathbf{z}^\top \mathbf{u}_j + \sigma_j G.$$

Let η_j be the minimiser in the perspective representation of Lemma 23 at $(\sigma_j, c_j, \mathbf{u}_j, \theta_j, t_j)$, and set

$$M_j = \|\eta_j - \xi_j\|_{L^2}.$$

Joint minimisation in $(\theta, \mathbf{u}, t)$ gives

$$t_j = M_j = \inf_{\substack{\theta \in \mathbb{R} \\ \mathbf{u} \in \text{range}(\Gamma)}} \|\eta_j - \theta - \mathbf{z}^\top \mathbf{u} - \sigma_j G\|_{L^2}.$$

By Lemma 22, strict convexity and compactness of the uniformly bounded predictor family give $\Delta_B > 0$ such that

$$\mathbb{E}[\ell_y(\xi_j)] \geq \mathbb{E}[\ell_y(\eta_y)] + \Delta_B.$$

Testing the joint minimisation with $\eta = \eta_y$, its L^2 projection, and $t = \sqrt{d_0^2 + \sigma_j^2}$, using (211), gives

$$\mathbb{E}[\ell_y(\eta_j)] + c_j M_j \leq \mathbb{E}[\ell_y(\eta_y)] + c_j \sqrt{d_0^2 + \sigma_j^2}.$$

Since ℓ_y is one-Lipschitz,

$$\mathbb{E}[\ell_y(\eta_j)] \geq \mathbb{E}[\ell_y(\eta_y)] + \Delta_B - M_j,$$

and therefore

$$(1 - c_j)M_j \geq \Delta_B - c_j \sqrt{d_0^2 + \sigma_j^2}.$$

Thus $M_j \geq \Delta_B/2 > 0$ for all sufficiently large j ; we restrict attention to such j , so that $t_j = M_j$ is interior and the first-order condition in η below applies. The first-order condition in η is

$$\rho'(\eta_j) - y + \frac{c_j}{M_j}(\eta_j - \xi_j) = 0 \quad \text{almost surely}.$$

Since ρ' is increasing and $\rho'(\eta_y) = y$,

$$\{\rho'(\eta_j) - y\}(\eta_j - \eta_y) \geq 0 \quad \text{almost surely}.$$

Substituting the first-order condition gives

$$(\eta_j - \xi_j)(\eta_j - \eta_y) \leq 0 \quad \text{almost surely},$$

so η_j lies between ξ_j and η_y . In either case,

$$|\eta_j - \eta_y| \vee |\eta_j - \xi_j| \leq |\xi_j - \eta_y| \quad \text{almost surely}.$$

Consequently, the same first-order condition gives

$$|\rho'(\eta_j) - y| \leq \frac{c_j}{M_j} |\xi_j - \eta_y|.$$

The coefficient bounds imply

$$|\xi_j - \eta_y| \leq B_a (1 + \|\mathbf{z}\|_2 + |G|) + |\eta_y| \in L^2.$$

Since $c_j/M_j \rightarrow 0$, it follows that $\eta_j \rightarrow \eta_y$ almost surely and, by dominated convergence, in L^2 . Now M_j and $\sqrt{d_0^2 + \sigma_j^2}$ are the distances of η_j and η_y , respectively, from the same closed affine set. Hence

$$\left| M_j - \sqrt{d_0^2 + \sigma_j^2} \right| \leq \|\eta_j - \eta_y\|_{L^2} = o(1).$$

By (212),

$$M_j - \sigma_j \sqrt{\kappa} \geq d_0 \sqrt{1 - \kappa} + o(1) > 0$$

for all sufficiently large j . The first-order condition and $t_j = M_j$ also give

$$\mathbb{E} \left[\{\rho'(\eta_j) - y\}^2 \right] = c_j^2.$$

Therefore, by (214),

$$\partial_r F(\sigma_j, r, \mathbf{u}_j, \theta_j, t_j)|_{r=c_j} = M_j - \sigma_j \sqrt{\kappa} > 0.$$

This contradicts maximality at the left endpoint $r = c_j$. □

Lemma 26 (Compact and natural limiting-AO saddle). *Let $\bar{r}, B_a, B_t$ be the deterministic constants from Lemma 24, and let c_0 be the constant from Lemma 25. Suppose that*

$$C_\sigma > B_a, \quad C_\beta > B_a, \quad C_\theta > B_a, \quad C_t > B_t, \quad 0 < c_r \leq \min\{\bar{r}, c_0\}.$$

Then the compact limiting AO has a unique optimiser chain

$$(\sigma^*, r^*, \mathbf{u}^*, \theta^*, t^*) \in \Omega^*,$$

in the sense that

$$\arg \min_{\sigma \in [0, C_\sigma]} V(\sigma) = \{\sigma^*\},$$

$$\arg \max_{r \in [c_r, 1]} H(\sigma^*, r) = \{r^*\},$$

and

$$\arg \min_{(\mathbf{u}, \theta, t) \in \mathbf{K}^*} F(\sigma^*, r^*, \mathbf{u}, \theta, t) = \{(\mathbf{u}^*, \theta^*, t^*)\}.$$

Moreover,

$$0 < \sigma^* < C_\sigma, \quad c_r < r^* < 1, \quad 0 < t^* < C_t, \quad \|\mathbf{u}^*\|_2 < C_\beta, \quad |\theta^*| < C_\theta.$$

In particular, $\mathbf{u}^*$ lies in the relative interior of U^* within $\text{range}(\mathbf{\Gamma})$. The same tuple is the unique saddle point of the natural-domain problem in (7), whose value is $\bar{\phi}$.

Proof. For $x = (\sigma, \mathbf{u}, \theta, t)$, write

$$F(x, r) = F(\sigma, r, \mathbf{u}, \theta, t).$$

Apply Lemma 24 with $c = c_r$. It gives a saddle (x^*, r^*) , where

$$x^* = (\sigma^*, \mathbf{u}^*, \theta^*, t^*),$$

such that

$$F(x^*, r) \leq F(x^*, r^*) \leq F(x, r^*) \quad (226)$$

for every $r \in [c_r, 1]$ and every $x \in \mathcal{X}_{\text{cl}}$. Moreover, Lemma 25 gives $r^* > c_r$. Suppose that $\sigma^* = 0$. If $t^* > 0$, let R^* denote the residual appearing in (214). Since R^* is independent of G when $\sigma^* = 0$,

$$\partial_\sigma^+ F(\sigma, r^*, \mathbf{u}^*, \theta^*, t^*)|_{\sigma=0} = -r^* \sqrt{\kappa} < 0.$$

If $t^* = 0$, direct differentiation of the closed extension gives the same right derivative. Both conclusions contradict the minimality in (226). Hence $\sigma^* > 0$ if $t^* = 0$, then

$$F(x^*, r) = \mathbb{E} [\ell_{\tilde{Y}} (\theta^* + \mathbf{z}^\top \mathbf{u}^* + \sigma^* G)] - \sigma^* r \sqrt{\kappa}$$

is strictly decreasing in r . Its maximiser on $[c_r, 1]$ would therefore be c_r , contradicting $r^* > c_r$. Thus $t^* > 0$. The t -coordinate is now interior in the closed minimising domain. Hence $\partial_t F(x^*, r^*) = 0$, and (214) gives

$$(r^*)^2 = \mathbb{E}[(R^*)^2].$$

Since $\tilde{Y} \in (0, 1)$ and $\rho'(\eta) \in (0, 1)$ almost surely, $|R^*| < 1$ almost surely. Therefore $r^* < 1$. The bounds (222)–(223) and the strict choices of the compactification constants give

$$\sigma^* < C_\sigma, \quad \|\mathbf{u}^*\|_2 < C_\beta, \quad |\theta^*| < C_\theta, \quad t^* < C_t.$$

Thus (x^*, r^*) belongs to the stated relative interior. Set

$$v^* = F(x^*, r^*).$$

The right inequality in (226) gives

$$H(\sigma, r^*) \geq v^*$$

for every $\sigma \in [0, C_\sigma]$, and hence $V(\sigma) \geq v^*$. Conversely, the left inequality gives

$$H(\sigma^*, r) \leq F(x^*, r) \leq v^*$$

for every $r \in [c_r, 1]$, while $H(\sigma^*, r^*) = v^*$. Therefore

$$V(\sigma^*) = v^* = \bar{\phi}.$$

Suppose that $V(\sigma) = v^*$. Then $H(\sigma, r^*) = v^*$. By compactness and continuity, the inner minimum is attained at some $k = (\mathbf{u}, \theta, t) \in \mathbf{K}^*$, so

$$F(\sigma, r^*, k) = v^*.$$

Lemma 23 makes the map $x \mapsto F(x, r^*)$ strictly convex, while (226) shows that x^* is its minimiser. Hence $(\sigma, k) = x^*$, and therefore $\sigma = \sigma^*$. This proves uniqueness of the outer minimiser. If $H(\sigma^*, r) = v^*$, then the preceding inequalities imply $F(x^*, r) = v^*$. Since $t^* > 0$, Lemma 23 makes $r \mapsto F(x^*, r)$ strictly concave, so $r = r^*$. Thus r^* is the unique maximiser of $H(\sigma^*, \cdot)$. Strict convexity of $x \mapsto F(x, r^*)$ also gives the stated uniqueness of $(\mathbf{u}^*, \theta^*, t^*)$.

It remains to pass to the natural domain. Since $r^* \in (c_r, 1)$,

$$\partial_r F(x^*, r^*) = 0.$$

Because $t^* > 0$, the map $r \mapsto F(x^*, r)$ is strictly concave on $(0, \infty)$. Hence

$$F(x^*, r) \leq F(x^*, r^*)$$

for every $r > 0$. The right inequality in (226) already holds for every $x \in \mathcal{X}_{c1}$. Consequently, (x^*, r^*) is a saddle of the natural-domain problem in (7), with value $\bar{\phi}$. Finally, let $(\tilde{x}, \tilde{r})$ be any other natural-domain saddle. The two saddle inequalities give

$$F(x^*, r^*) \leq F(\tilde{x}, r^*) \leq F(\tilde{x}, \tilde{r}) \leq F(x^*, \tilde{r}) \leq F(x^*, r^*).$$

Equality therefore holds throughout. Strict convexity in x gives $\tilde{x} = x^*$, and strict concavity in r gives $\tilde{r} = r^*$. This proves uniqueness on the natural domain. $\square$

F.5.2 Ambient-domain relaxation

Lemma 27 (Ambient-domain relaxation of the limiting AO). *Let $(\sigma^*, r^*, \mathbf{u}^*, \theta^*, t^*)$ be the unique compact saddle chain from Lemma 26. For every $(\sigma, r) \in [0, C_\sigma] \times [c_r, 1]$,*

$$H_0(\sigma, r) = H(\sigma, r).$$

For every $\sigma \in [0, C_\sigma]$,

$$V_0(\sigma) = V(\sigma).$$

Every minimiser defining $H_0(\sigma, r)$ belongs to $\text{range}(\mathbf{\Gamma})$ and is therefore a minimiser defining $H(\sigma, r)$. Consequently,

$$\text{val}(F; \mathbf{\Omega}) = \text{val}(F; \mathbf{\Omega}^*) = \bar{\phi}.$$

Moreover, the ambient profiles have the same unique saddle chain: V_0 has the unique minimiser σ^* , $H_0(\sigma^*, \cdot)$ has the unique maximiser r^* , and $(\mathbf{u}^*, \theta^*, t^*)$ is the unique minimiser of

$$(\mathbf{u}, \theta, t) \mapsto F(\sigma^*, r^*, \mathbf{u}, \theta, t),$$

over $\mathbf{K}$.

Proof. By Lemma 21 the function F is continuous on the ambient compact domain $\mathbf{\Omega}$. Hence all inner minima and outer maxima defining H , V , H_0 , and V_0 are attained.

Fix $(\sigma, r) \in [0, C_\sigma] \times [c_r, 1]$. Since $\mathbf{K}^* \subseteq \mathbf{K}$, the ambient inner minimisation is over a larger set, and therefore $H_0(\sigma, r) \leq H(\sigma, r)$.

For $(\mathbf{u}, \theta, t) \in \mathbf{K}$, write

$$\mathbf{u}_R = \mathbf{P}_\Gamma \mathbf{u}, \quad \mathbf{u}_N = \mathbf{u} - \mathbf{u}_R,$$

where $\mathbf{P}_\Gamma$ is the orthogonal projector onto $\text{range}(\mathbf{\Gamma})$. Since $\mathbf{P}_\Gamma$ is an orthogonal projector, $\|\mathbf{u}_R\|_2 \leq \|\mathbf{u}\|_2 \leq C_\beta$, so $(\mathbf{u}_R, \theta, t) \in \mathbf{K}^*$. By Lemma 16

$$F(\sigma, r, \mathbf{u}_R, \theta, t) \leq F(\sigma, r, \mathbf{u}, \theta, t).$$

Taking the infimum over $\mathbf{K}$ gives $H(\sigma, r) \leq H_0(\sigma, r)$, and hence $H_0(\sigma, r) = H(\sigma, r)$. Taking the maximum over $r \in [c_r, 1]$ gives $V_0(\sigma) = V(\sigma)$, and taking the minimum over $\sigma \in [0, C_\sigma]$ gives the displayed equality of values. Now let $(\tilde{\mathbf{u}}, \tilde{\theta}, \tilde{t})$ be any minimiser defining $H_0(\sigma, r)$ and write

$$\tilde{\mathbf{u}}_R = \mathbf{P}_\Gamma \tilde{\mathbf{u}}, \quad \tilde{\mathbf{u}}_N = \tilde{\mathbf{u}} - \tilde{\mathbf{u}}_R.$$

If $\tilde{\mathbf{u}}_N \neq \mathbf{0}$, the final assertion of Lemma 16 gives

$$F(\sigma, r, \tilde{\mathbf{u}}_R, \tilde{\theta}, \tilde{t}) < F(\sigma, r, \tilde{\mathbf{u}}, \tilde{\theta}, \tilde{t}),$$

contradicting ambient minimality. Thus $\tilde{\mathbf{u}} \in \text{range}(\mathbf{\Gamma})$, so the ambient minimiser belongs to $\mathbf{K}^*$ and, because $H_0(\sigma, r) = H(\sigma, r)$, also minimises the starred profile. Since $V_0 = V$ pointwise and V has the unique minimiser σ^* by Lemma 26, V_0 has the same unique minimiser. Likewise, $H_0(\sigma^*, \cdot) = H(\sigma^*, \cdot)$, so $H_0(\sigma^*, \cdot)$ has the unique maximiser r^* . Finally, every ambient inner minimiser at (σ^*, r^*) is a starred inner minimiser by the preceding argument, and the latter is uniquely $(\mathbf{u}^*, \theta^*, t^*)$ by Lemma 26. This proves the ambient uniqueness conclusions. $\square$

F.5.3 First-order equations

Lemma 28 (First-order equations for the limiting AO). *Let $(\sigma^*, r^*, \mathbf{u}^*, \theta^*, t^*)$ be the unique interior saddle given by Lemma 26, and set*

$$\lambda^* = \frac{t^*}{r^*}.$$

For $\lambda > 0$ and $(\sigma, \mathbf{u}, \theta) \in (0, \infty) \times \text{range}(\mathbf{\Gamma}) \times \mathfrak{R}$, define $\xi = \theta + \mathbf{z}^\top \mathbf{u} + \sigma G$. With q_0 and q_P as in Lemma 22 set

$$\tilde{Y}^{(0)} = (1 - \alpha)q_P, \quad \tilde{Y}^{(1)} = \alpha + (1 - \alpha)q_P.$$

For $b \in \{0, 1\}$, let $\eta_b = \text{prox}_{\lambda\rho}(\xi + \lambda\tilde{Y}^{(b)})$, and define $R_b = \rho'(\eta_b) - \tilde{Y}^{(b)}$ and

$$W_b = \frac{\rho''(\eta_b)}{1 + \lambda\rho''(\eta_b)}.$$

Define $\bar{R} = q_0 R_1 + (1 - q_0) R_0$, $\bar{R}_2 = q_0 R_1^2 + (1 - q_0) R_0^2$, and $\bar{W} = q_0 W_1 + (1 - q_0) W_0$. Then $(\sigma^, \lambda^*, \mathbf{u}^*, \theta^*)$ satisfies*

$$\begin{aligned} \mathbb{E}[\bar{R}] &= 0, \\ \mathbf{P}_\Gamma \mathbb{E}[\mathbf{z}\bar{R}] &= \mathbf{0}, \\ \lambda \mathbb{E}[\bar{W}] &= \kappa, \\ \frac{\sigma^2 \kappa}{\lambda^2} &= \mathbb{E}[\bar{R}_2], \end{aligned} \tag{227}$$

where all expectations in (227) are with respect to $(\mathbf{z}, G)$, and the equations are evaluated at $(\sigma, \lambda, \mathbf{u}, \theta) = (\sigma^, \lambda^*, \mathbf{u}^*, \theta^*)$. Moreover,*

$$r^* = \frac{\sigma^* \sqrt{\kappa}}{\lambda^*}, \quad t^* = r^* \lambda^* = \sigma^* \sqrt{\kappa}.$$

Conversely, let

$$(\sigma, \lambda, \mathbf{u}, \theta) \in (0, \infty)^2 \times \text{range}(\mathbf{\Gamma}) \times \mathfrak{R},$$

and suppose that it satisfies the four equations in (8), with the starred quantities replaced by $(\sigma, \lambda, \mathbf{u}, \theta)$. Then

$$0 < \frac{\sigma \sqrt{\kappa}}{\lambda} < 1.$$

Setting

$$r = \frac{\sigma \sqrt{\kappa}}{\lambda}, \quad t = r \lambda = \sigma \sqrt{\kappa},$$

recovers the unique natural-domain saddle, and hence

$$(\sigma, r, \mathbf{u}, \theta, t) = (\sigma^*, r^*, \mathbf{u}^*, \theta^*, t^*).$$

In particular, $(\sigma^, \lambda^*, \mathbf{u}^*, \theta^*)$ is the unique solution of the four equations in (8) in*

$$(0, \infty)^2 \times \text{range}(\mathbf{\Gamma}) \times \mathfrak{R}.$$

Proof. By Lemma 26 all first derivatives of F vanish at the limiting saddle, with the $\mathbf{u}$ -derivative understood relative to $\text{range}(\mathbf{\Gamma})$. By (97), after setting

$$\lambda = \frac{t}{r}, \quad t = r\lambda,$$

the limiting AO objective can be written in the variables

$$a = (\sigma, \mathbf{u}, \theta), \quad r, \quad \lambda,$$

as

$$\mathcal{F}(a, r, \lambda) = \mathbb{E}[e_\lambda(\xi, \tilde{Y})] - \sigma r \sqrt{\kappa} + \frac{r^2 \lambda}{2},$$

where $\xi = \theta + \mathbf{z}^\top \mathbf{u} + \sigma G$ and

$$e_\lambda(x, v) = m_\rho(x + \lambda v, \lambda) - vx - \frac{\lambda}{2} v^2.$$

Because $r^* > 0$, the change of variables $(r, t) \mapsto (r, \lambda = t/r)$ is smooth and invertible in a neighbourhood of the saddle. Hence,

$$\partial_\lambda \mathcal{F} = r \partial_t F, \quad \partial_r \mathcal{F} = \partial_r F + \lambda \partial_t F,$$

and the first-order conditions in (r, t) obtained above are equivalent to the first-order conditions in (r, λ) at the limiting saddle. The conditions in σ, θ , and the admissible $\mathbf{u}$ -directions are unchanged by this coordinate transformation. Let $\eta = \text{prox}_{\lambda\rho}(\xi + \lambda\tilde{Y})$ and set $R = \rho'(\eta) - \tilde{Y}$. By (217),

$$\eta = \xi - \lambda R, \quad \partial_x e_\lambda(\xi, \tilde{Y}) = R.$$

Therefore $\partial_\theta \mathcal{F} = \mathbb{E}[R]$. The first-order condition in θ gives

$$\mathbb{E}[R] = 0. \tag{228}$$

For a direction $\mathbf{d} \in \text{range}(\mathbf{\Gamma})$,

$$\left. \frac{d}{ds} \mathcal{F}(\sigma, r, \mathbf{u} + s\mathbf{d}, \theta, \lambda) \right|_{s=0} = \mathbb{E}[(\mathbf{z}^\top \mathbf{d}) R] = \mathbf{d}^\top \mathbb{E}[\mathbf{z} R].$$

Since this derivative vanishes for every $\mathbf{d} \in \text{range}(\mathbf{\Gamma})$,

$$\mathbf{P}_\Gamma \mathbb{E}[\mathbf{z} R] = \mathbf{0}. \tag{229}$$

The first-order condition in σ is

$$\partial_\sigma \mathcal{F} = \mathbb{E}[GR] - r\sqrt{\kappa} = 0. \tag{230}$$

Conditional on $(\mathbf{z}, \varepsilon)$, the random variable $\tilde{Y}$ is fixed and independent of G , and the dependence of R on G is only through $\xi = \theta + \mathbf{z}^\top \mathbf{u} + \sigma G$. By (217),

$$\frac{\partial R}{\partial \xi} = \frac{\rho''(\eta)}{1 + \lambda \rho''(\eta)}.$$

Hence

$$\frac{\partial R}{\partial G} = \sigma \frac{\rho''(\eta)}{1 + \lambda \rho''(\eta)}.$$

Since $0 \leq \tilde{Y} \leq 1$, $0 < \rho'(\eta) < 1$, and $0 \leq \rho''(\eta) \leq 1/4$, one has $|R| \leq 1$ and

$$0 \leq \frac{\partial R}{\partial \xi} = \frac{\rho''(\eta)}{1 + \lambda \rho''(\eta)} \leq \frac{1}{4}.$$

Thus, Stein's lemma (Stein [1981] Lemma 1) yields

$$\mathbb{E}[GR] = \sigma \mathbb{E} \left[\frac{\rho''(\eta)}{1 + \lambda \rho''(\eta)} \right].$$

Substituting into (230) gives

$$\sigma \mathbb{E} \left[\frac{\rho''(\eta)}{1 + \lambda \rho''(\eta)} \right] = r \sqrt{\kappa}. \quad (231)$$

Next, (217) gives

$$\partial_\lambda e_\lambda(\xi, \tilde{Y}) = -\frac{1}{2} R^2.$$

Thus

$$\partial_\lambda \mathcal{F} = -\frac{1}{2} \mathbb{E}[R^2] + \frac{r^2}{2}.$$

The first-order condition in λ gives

$$r^2 = \mathbb{E}[R^2]. \quad (232)$$

The first-order condition in r gives

$$\partial_r \mathcal{F} = -\sigma \sqrt{\kappa} + r \lambda = 0,$$

and therefore

$$r \lambda = \sigma \sqrt{\kappa}. \quad (233)$$

Since $\sigma > 0$ by Lemma 26 combining (231) and (233) yields

$$\lambda \mathbb{E} \left[\frac{\rho''(\eta)}{1 + \lambda \rho''(\eta)} \right] = \kappa. \quad (234)$$

Combining (232) and (233) gives

$$\frac{\sigma^2 \kappa}{\lambda^2} = \mathbb{E}[R^2]. \quad (235)$$

Also (233) gives

$$r = \frac{\sigma \sqrt{\kappa}}{\lambda},$$

and, since $t = r \lambda$, $t = \sigma \sqrt{\kappa}$. It remains to integrate out the Bernoulli randomness. Conditional on $(\mathbf{z}, G)$, the pseudo-response is $\tilde{Y}^{(1)}$ with probability q_0 and $\tilde{Y}^{(0)}$ with probability $1 - q_0$. Hence

$$\mathbb{E}[R \mid \mathbf{z}, G] = q_0 R_1 + (1 - q_0) R_0 = \bar{R},$$

$$\mathbb{E}[R^2 \mid \mathbf{z}, G] = q_0 R_1^2 + (1 - q_0) R_0^2 = \bar{R}_2,$$

and

$$\mathbb{E} \left[\frac{\rho''(\eta)}{1 + \lambda \rho''(\eta)} \mid \mathbf{z}, G \right] = q_0 W_1 + (1 - q_0) W_0 = \bar{W}.$$

Substituting these conditional expectations into (228), (229), (234), and (235) gives exactly (227). Evaluating at $(\sigma, \lambda, \mathbf{u}, \theta) = (\sigma^*, \lambda^*, \mathbf{u}^*, \theta^*)$ proves the necessity statements.

For the converse, let

$$(\sigma, \lambda, \mathbf{u}, \theta) \in (0, \infty)^2 \times \text{range}(\mathbf{\Gamma}) \times \mathfrak{R},$$

satisfy the four equations in [8], with the starred quantities replaced by $(\sigma, \lambda, \mathbf{u}, \theta)$. Let

$$\xi = \theta + \mathbf{z}^\top \mathbf{u} + \sigma G, \quad \eta = \text{prox}_{\lambda \rho} \left(\xi + \lambda \tilde{Y} \right),$$

and set

$$R = \rho'(\eta) - \tilde{Y}, \quad W = \frac{\rho''(\eta)}{1 + \lambda \rho''(\eta)}.$$

Since $\alpha \in (0, 1)$, the definition of the limiting pseudo-response in Section 3.1 gives $0 < \tilde{Y} < 1$ and since $0 < \rho'(\eta) < 1$, we have $|R| < 1$. The fourth equation in (8) therefore gives

$$0 < \frac{\sigma^2 \kappa}{\lambda^2} = \mathbb{E}[R^2] < 1.$$

Consequently, $r = \sigma\sqrt{\kappa}/\lambda \in (0, 1)$. Setting $t = r\lambda = \sigma\sqrt{\kappa}$ we see that $(\sigma, r, \mathbf{u}, \theta, t)$ belongs to the natural domain. The first two equations in (8) and (214) give

$$\partial_\theta F = 0, \quad \mathbf{P}_\Gamma \nabla_{\mathbf{u}} F = \mathbf{0}.$$

Stein's identity used above gives

$$\mathbb{E}[GR] = \sigma \mathbb{E}[W].$$

Using the third equation in (8) and the definition of r ,

$$\mathbb{E}[GR] - r\sqrt{\kappa} = \sigma \mathbb{E}[W] - \frac{\sigma\sqrt{\kappa}}{\lambda} \sqrt{\kappa} = \frac{\sigma\kappa}{\lambda} - \frac{\sigma\kappa}{\lambda} = 0.$$

Hence $\partial_\sigma F = 0$. The fourth equation in (8) gives

$$\mathbb{E}[R^2] = \frac{\sigma^2 \kappa}{\lambda^2} = r^2.$$

The formula for $\partial_t F$ in (214) therefore yields

$$\partial_t F = -\frac{1}{2r} \mathbb{E}[R^2] + \frac{r}{2} = 0.$$

Finally, (214) gives

$$\partial_r F = \frac{t}{2r^2} \mathbb{E}[R^2] - \sigma\sqrt{\kappa} + \frac{t}{2} = t - \sigma\sqrt{\kappa} = 0.$$

Thus all relative first-order conditions in the minimising coordinates and the first-order condition in the maximising coordinate are satisfied.

By Lemma 23, for this fixed r the objective is strictly convex in $(\sigma, \mathbf{u}, \theta, t)$, while for the fixed minimising coordinates it is strictly concave in r . Hence $(\sigma, r, \mathbf{u}, \theta, t)$ is the unique natural-domain saddle. Lemma 26 identifies that saddle with

$$(\sigma^*, r^*, \mathbf{u}^*, \theta^*, t^*).$$

It follows that

$$(\sigma, \lambda, \mathbf{u}, \theta) = (\sigma^*, \lambda^*, \mathbf{u}^*, \theta^*),$$

which proves the converse and uniqueness on the stated positive domain. $\square$

F.6 Lemmas for Section B.9

F.6.1 Finite convex cover and limiting gaps

Lemma 29 (Finite convex cover of the signal-intercept bad set). *For every $\epsilon > 0$,*

$$\mathcal{S}_{\mathbf{v}, \theta}^{\epsilon, c} \subseteq \bigcup_{j \in \mathcal{J}} C_{p, j}^\epsilon.$$

Moreover, each nonempty $C_{p, j}^\epsilon$ is compact and convex.

Proof. For $i \in \mathcal{I}_A$, the set $C_{p, i}^\epsilon$ is the intersection of $V_p \times [-C_\theta, C_\theta]$ with a closed affine half-space. For $j \in \{+, -\}$, $C_{p, j}^\epsilon = V_p \times \Theta_j^\epsilon$. The set V_p is compact and convex, and the intervals are closed, possibly empty. Hence every nonempty $C_{p, j}^\epsilon$ is compact and convex.

Let $(\mathbf{v}, \theta) \in \mathcal{S}_{\mathbf{v}, \theta}^{\epsilon, c}$. Then either

$$\left\| \mathbf{\Gamma}_p^{1/2} \mathbf{v} - \mathbf{u}^* \right\|_2 \geq \epsilon,$$

or $|\theta - \theta^*| \geq \epsilon$. If $\theta - \theta^* \geq \epsilon$, then $(\mathbf{v}, \theta) \in C_{p, +}^\epsilon$. If $\theta^* - \theta \geq \epsilon$, then $(\mathbf{v}, \theta) \in C_{p, -}^\epsilon$. Otherwise, set

$$\mathbf{a}_0 = \frac{\mathbf{\Gamma}_p^{1/2} \mathbf{v} - \mathbf{u}^*}{\left\| \mathbf{\Gamma}_p^{1/2} \mathbf{v} - \mathbf{u}^* \right\|_2} \in \mathbb{S}^1.$$

Choose $i \in \mathcal{I}_A$ such that $\|\mathbf{a}_i - \mathbf{a}_0\|_2 \leq 1/2$. Then

$$\mathbf{a}_i^\top (\mathbf{\Gamma}_p^{1/2} \mathbf{v} - \mathbf{u}^*) \geq \mathbf{a}_0^\top (\mathbf{\Gamma}_p^{1/2} \mathbf{v} - \mathbf{u}^*) - \|\mathbf{a}_i - \mathbf{a}_0\|_2 \left\| \mathbf{\Gamma}_p^{1/2} \mathbf{v} - \mathbf{u}^* \right\|_2 \geq \frac{\epsilon}{2}.$$

Thus $(\mathbf{v}, \theta) \in C_{p, i}^\epsilon$, proving the cover. $\square$

Lemma 30 (Finite-piece limiting AO value gaps). *Let $(\sigma^*, r^*, \mathbf{u}^*, \theta^*, t^*)$ be the unique compact saddle chain from Proposition B.7. Then, for every $\epsilon > 0$, there exists a deterministic $\eta_\epsilon > 0$, independent of n , such that*

$$\bar{\phi}_\sigma^{\epsilon, c} \geq \bar{\phi} + 3\eta_\epsilon, \quad \bar{\phi}_j^\epsilon \geq \bar{\phi} + 3\eta_\epsilon, \quad j \in \mathcal{J}.$$

Proof. We first prove a strict gap for every nonempty restricted domain. If $\mathbf{\Omega}_\sigma^{\epsilon, c}$ is empty, then $\bar{\phi}_\sigma^{\epsilon, c} = +\infty$. Otherwise, (115) gives

$$\bar{\phi}_\sigma^{\epsilon, c} = \min_{\sigma \in \mathcal{S}_\sigma^{\epsilon, c}} V_0(\sigma).$$

The minimum is attained because $V_0 = V$ by Lemma 27 and V is continuous by Lemma 26. Suppose that $\bar{\phi}_\sigma^{\epsilon, c} = \bar{\phi}$. Then there is $\tilde{\sigma} \in \mathcal{S}_\sigma^{\epsilon, c}$ such that $V_0(\tilde{\sigma}) = \bar{\phi}$. Since

$$\bar{\phi} = \min_{\sigma \in [0, C_\sigma]} V_0(\sigma),$$

and V_0 has the unique minimiser σ^* by Lemma 27 we obtain $\tilde{\sigma} = \sigma^*$, contradicting $\tilde{\sigma} \in \mathcal{S}_\sigma^{\epsilon, c}$. Hence $\bar{\phi}_\sigma^{\epsilon, c} > \bar{\phi}$. Now fix $j \in \mathcal{J}$. If $\mathbf{\Omega}_j^\epsilon$ is empty, then $\bar{\phi}_j^\epsilon = +\infty$. Otherwise, define

$$H_j^\epsilon(\sigma, r) = \min_{(\mathbf{u}, \theta, t) \in \mathbf{K}_j^\epsilon(U)} F(\sigma, r, \mathbf{u}, \theta, t), \quad V_j^\epsilon(\sigma) = \max_{r \in [c_r, 1]} H_j^\epsilon(\sigma, r).$$

The set $\mathbf{K}_j^\epsilon(U)$ is nonempty and compact. Since F is continuous, Berge's maximum theorem gives continuity and attainment of H_j^ϵ on $[0, C_\sigma] \times [c_r, 1]$. A second application gives continuity and attainment of V_j^ϵ on $[0, C_\sigma]$. Because $\mathbf{K}_j^\epsilon(U) \subseteq \mathbf{K}$,

$$H_j^\epsilon(\sigma, r) \geq H_0(\sigma, r), \quad V_j^\epsilon(\sigma) \geq V_0(\sigma).$$

Suppose that $\bar{\phi}_j^\epsilon = \bar{\phi}$. By attainment, there is $\tilde{\sigma} \in [0, C_\sigma]$ such that $V_j^\epsilon(\tilde{\sigma}) = \bar{\phi}$. Therefore

$$\bar{\phi} = V_j^\epsilon(\tilde{\sigma}) \geq V_0(\tilde{\sigma}) \geq \bar{\phi}.$$

Thus $V_0(\tilde{\sigma}) = \bar{\phi}$, and Lemma 27 gives $\tilde{\sigma} = \sigma^*$. Hence $V_j^\epsilon(\sigma^*) = \bar{\phi}$. At $r = r^*$,

$$H_j^\epsilon(\sigma^*, r^*) \geq H_0(\sigma^*, r^*) = \bar{\phi},$$

by Lemma 27, while

$$H_j^\epsilon(\sigma^*, r^*) \leq V_j^\epsilon(\sigma^*) = \bar{\phi}.$$

Consequently, $H_j^\epsilon(\sigma^*, r^*) = \bar{\phi}$. The inner minimum is attained, so there is $(\tilde{\mathbf{u}}, \tilde{\theta}, \tilde{t}) \in \mathbf{K}_j^\epsilon(U)$ with

$$F(\sigma^*, r^*, \tilde{\mathbf{u}}, \tilde{\theta}, \tilde{t}) = \bar{\phi}.$$

By Lemma 27 the unique minimiser of $F(\sigma^*, r^*, \cdot)$ over $\mathbf{K}$ is $(\mathbf{u}^*, \theta^*, t^*)$. Hence $(\tilde{\mathbf{u}}, \tilde{\theta}, \tilde{t}) = (\mathbf{u}^*, \theta^*, t^*)$. This point lies in none of the bad pieces: if $j = i \in \mathcal{I}_A$, then

$$\mathbf{a}_i^\top (\mathbf{u}^* - \mathbf{u}^*) = 0 < \frac{\epsilon}{2},$$

while $\theta^* \notin \Theta_+^\epsilon$ and $\theta^* \notin \Theta_-^\epsilon$. This is a contradiction. Therefore $\bar{\phi}_j^\epsilon > \bar{\phi}$ for every nonempty piece. Define

$$\mathfrak{D}_\epsilon = \{\bar{\phi}_\sigma^{\epsilon, c} - \bar{\phi} : \boldsymbol{\Omega}_\sigma^{\epsilon, c} \neq \emptyset\} \cup \{\bar{\phi}_j^\epsilon - \bar{\phi} : j \in \mathcal{J}, \boldsymbol{\Omega}_j^\epsilon \neq \emptyset\}.$$

If $\mathfrak{D}_\epsilon = \emptyset$, set $\eta_\epsilon = 1$. Otherwise set

$$\eta_\epsilon = \frac{1}{3} \min \mathfrak{D}_\epsilon.$$

Every element of $\mathfrak{D}_\epsilon$ is strictly positive, and $\mathcal{J}$ is finite. Hence $\eta_\epsilon > 0$, and the stated bounds follow. Empty restricted domains also satisfy them by convention. $\square$

F.6.2 Restricted scalarisation and AO barriers

Lemma 31 (Large- t exclusion on restricted convex pieces). *Fix $\epsilon > 0$, $c_r > 0$, and $C_t > 0$. For $\boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n$, $(\mathbf{v}, \theta) \in V_p \times [-C_\theta, C_\theta]$, $\sigma \in [0, C_\sigma]$, $r \in [c_r, 1]$, and $t > 0$, define*

$$\mathcal{T}_n(\sigma, r, \mathbf{v}, \theta, \boldsymbol{\eta}, t) = \frac{r}{2t} \frac{1}{n} \left\| \boldsymbol{\eta} - \theta \mathbf{1} - \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} - \sigma \mathbf{g} \right\|_2^2 + \frac{rt}{2} + \frac{1}{n} \{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} \}.$$

Let $\mathcal{E}_{n, \text{res}}^{t, \epsilon}(c_r, C_t)$ be the event that, simultaneously for every $j \in \mathcal{J}_n^\epsilon$, every $\sigma \in [0, C_\sigma]$, and every $r \in [c_r, 1]$,

$$\inf_{\substack{(\mathbf{v}, \theta) \in C_{p, j}^\epsilon \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n \\ t > 0}} \mathcal{T}_n(\sigma, r, \mathbf{v}, \theta, \boldsymbol{\eta}, t) = \inf_{\substack{(\mathbf{v}, \theta) \in C_{p, j}^\epsilon \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n \\ t \in (0, C_t]}} \mathcal{T}_n(\sigma, r, \mathbf{v}, \theta, \boldsymbol{\eta}, t).$$

Recall K_t from Lemma 12. Suppose that

$$C_t > \max \left\{ 1, \frac{2(K_t + \log 2)}{c_r} \right\}.$$

Then there exist constants $C, c > 0$ and $N \in \mathbb{N}$ such that, for all $n > N$,

$$\Pr(\neg \mathcal{E}_{n, \text{res}}^{t, \epsilon}(c_r, C_t) \mid \mathcal{G}_n) \leq C \exp\{-cn\}.$$

In particular, for every fixed $t_0 > 0$,

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\neg \mathcal{E}_{n, \text{res}}^{t, \epsilon}(c_r, C_t) \mid \mathcal{G}_n) \leq C \exp\{-cn\}.$$

The constants depend only on c_r , C_β , C_θ , C_σ and universal Gaussian concentration constants.

Proof. Define the event

$$\mathcal{I}_n^{t, \text{res}} = \left\{ \frac{\|\mathbf{g}\|_2}{\sqrt{n}} \leq 2 \right\} \cap \left\{ \sup_{\mathbf{v} \in V_p} \frac{1}{\sqrt{n}} \left\| \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} \right\|_2 \leq 3C_\beta \right\}.$$

We first show that

$$\Pr(\neg \mathcal{I}_n^{t, \text{res}} \mid \mathcal{G}_n) \leq C \exp\{-cn\}.$$

The first event follows from chi-square concentration (Vershynin 2018, Equation (3.1)) for $\mathbf{g} \sim N(0, I_n)$.

For the second event, use (93). On the common extension,

$$\frac{1}{\sqrt{p}} \mathbf{H}_1 = \mathbf{Z} \boldsymbol{\Gamma}_p^{1/2},$$

where $\mathbf{Z} \in \mathbb{R}^{n \times 2}$ has i.i.d. standard normal entries conditional on $\mathcal{G}_n$. For $\mathbf{v} \in V_p$, set $\mathbf{u} = \mathbf{\Gamma}_p^{1/2} \mathbf{v}$. By the definition of V_p ,

$$\|\mathbf{u}\|_2^2 = \mathbf{v}^\top \mathbf{\Gamma}_p \mathbf{v} = \frac{1}{p} \|\mathbf{B}\mathbf{v}\|_2^2 \leq C_\beta^2.$$

Therefore, on the common extension,

$$\sup_{\mathbf{v} \in V_p} \frac{1}{\sqrt{n}} \left\| \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} \right\|_2 \leq C_\beta \frac{\|\mathbf{Z}\|_2}{\sqrt{n}}.$$

The standard Gaussian operator-norm bound for an $n \times 2$ matrix gives

$$\Pr \left(\frac{\|\mathbf{Z}\|_2}{\sqrt{n}} > 3 \right) \leq C \exp\{-cn\},$$

for all large enough n . This proves the probability bound for $\mathcal{I}_n^{t,\text{res}}$. Now work on $\mathcal{I}_n^{t,\text{res}}$. Fix $j \in \mathcal{J}_n^\epsilon$, $\sigma \in [0, C_\sigma]$, and $r \in [c_r, 1]$. Since $j \in \mathcal{J}_n^\epsilon$, the set $C_{p,j}^\epsilon$ is nonempty. Choose any $(\mathbf{v}_j, \theta_j) \in C_{p,j}^\epsilon$. Then $|\theta_j| \leq C_\theta$, $\mathbf{v}_j \in V_p$. Hence, on $\mathcal{I}_n^{t,\text{res}}$,

$$\frac{1}{\sqrt{n}} \left\| \theta_j \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v}_j + \sigma \mathbf{g} \right\|_2 \leq C_\theta + 3C_\beta + 2C_\sigma.$$

Test the feasible point

$$\mathbf{v} = \mathbf{v}_j, \quad \theta = \theta_j, \quad \boldsymbol{\eta} = 0, \quad t = 1.$$

Since $C_t > 1$, this point is feasible also for the truncated problem. At this point,

$$\frac{1}{n} \{ \mathbf{1}^\top \boldsymbol{\rho}(0) - \tilde{\mathbf{y}}^\top 0 \} = \log 2.$$

Using $r \leq 1$, we obtain

$$\begin{aligned} \mathcal{T}_n(\sigma, r, \mathbf{v}_j, \theta_j, 0, 1) &= \frac{r}{2} \frac{1}{n} \left\| \theta_j \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v}_j + \sigma \mathbf{g} \right\|_2^2 + \frac{r}{2} + \log 2 \\ &\leq \frac{1}{2} (C_\theta + 3C_\beta + 2C_\sigma)^2 + \frac{1}{2} + \log 2 \\ &= K_t. \end{aligned}$$

Thus the truncated infimum is at most K_t . On the other hand, for every $y \in (0, 1)$ and every $x \in \mathbb{R}$,

$$\rho(x) - yx \geq \inf_{z \in \mathbb{R}} \{ \rho(z) - yz \} = -y \log y - (1-y) \log(1-y) \in [0, \log(2)].$$

Therefore, for every $\boldsymbol{\eta} \in \mathbb{R}^n$,

$$\frac{1}{n} \{ \mathbf{1}^\top \boldsymbol{\rho}(\boldsymbol{\eta}) - \tilde{\mathbf{y}}^\top \boldsymbol{\eta} \} \geq 0.$$

Since the square term in $\mathcal{T}_n$ is nonnegative and $r \geq c_r$, every feasible point with $t > C_t$ satisfies

$$\mathcal{T}_n(\sigma, r, \mathbf{v}, \theta, \boldsymbol{\eta}, t) \geq \frac{c_r t}{2} \geq \frac{c_r C_t}{2} > K_t.$$

Such a point cannot contribute to the infimum, because the feasible point $(\mathbf{v}_j, \theta_j, 0, 1)$ has value at most K_t . Consequently

$$\inf_{\substack{(\mathbf{v}, \theta) \in C_{p,j}^\epsilon \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n \\ t > 0}} \mathcal{T}_n(\sigma, r, \mathbf{v}, \theta, \boldsymbol{\eta}, t) = \inf_{\substack{(\mathbf{v}, \theta) \in C_{p,j}^\epsilon \\ \boldsymbol{\eta} \in \mathcal{B}_{C_\eta}^n \\ t \in (0, C_t]}} \mathcal{T}_n(\sigma, r, \mathbf{v}, \theta, \boldsymbol{\eta}, t).$$

The argument was uniform in $j \in \mathcal{J}_n^\epsilon$, $\sigma \in [0, C_\sigma]$, and $r \in [c_r, 1]$. Hence

$$\mathcal{I}_n^{t,\text{res}} \subseteq \mathcal{E}_{n,\text{res}}^{t,\epsilon}(c_r, C_t).$$

The claimed conditional probability bound follows from the probability bound for $\mathcal{I}_n^{t,\text{res}}$. $\square$

Lemma 32 (Simultaneous restricted scalarisation). *Fix $\epsilon > 0$ and $t_0 \in (0, t_\star]$. Let $d_r, a, K, C_L, c_r, C_t, C_M, C_\eta$ be the constants fixed from Proposition B.5 with structural tolerance $t_\star$, and write*

$$\mathcal{E}_n^{\text{sr}} = \mathcal{E}_n^{\text{sr}}(d_r, a, K, C_L).$$

For every $j \in \mathcal{J}_n^\epsilon$, define the restricted predictor set

$$R_j^\epsilon = \left\{ \theta \mathbf{1} + \frac{1}{\sqrt{p}} \mathbf{H}_1 \mathbf{v} : (\mathbf{v}, \theta) \in C_{p,j}^\epsilon \right\}.$$

Then there exists an event $\mathcal{R}_{n,\text{res}}^\epsilon$ and constants $C_\epsilon, c_\epsilon > 0$ and $N_\epsilon \in \mathbb{N}$ such that, for all $n > N_\epsilon$,

$$\mathbf{1}_{\mathcal{S}_n(t_0)} \Pr(-\mathcal{R}_{n,\text{res}}^\epsilon \mid \mathcal{G}_n) \leq C_\epsilon \exp\{-c_\epsilon n\}.$$

On $\mathcal{R}_{n,\text{res}}^\epsilon$,

$$\phi(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \boldsymbol{\Omega}_n),$$

$$\phi_{\mathbf{w}}^{\epsilon,c}(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \boldsymbol{\Omega}_{n,\sigma}^{\epsilon,c}),$$

with both sides interpreted as $+\infty$ when $\mathcal{S}_{\mathbf{w}}^{\epsilon,c}$ is empty, and, simultaneously for every $j \in \mathcal{J}_n^\epsilon$,

$$\phi_j^\epsilon(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \boldsymbol{\Omega}_{n,j}^\epsilon).$$

Proof. Let

$$\mathcal{R}_{n,\text{res}}^\epsilon = \mathcal{R}_n \cap \mathcal{E}_n^{\text{sr}} \cap \mathcal{I}_n^t \cap \mathcal{E}_{n,\text{res}}^{t,\epsilon}(c_r, C_t) \cap \mathcal{E}_n^{\text{ctr}}(C_M).$$

The bounds of Proposition B.5, obtained with structural tolerance $t_\star$, remain valid after multiplication by $\mathbf{1}_{\mathcal{S}_n(t_0)}$ because $\mathcal{S}_n(t_0) \subseteq \mathcal{S}_n(t_\star)$. The probability bounds for the remaining three events follow from Lemmas 12, 31 and 14. A union bound therefore gives the claimed conditional estimate.

The unrestricted identity follows from $\mathcal{R}_n$. For the $\mathbf{w}$ -restricted AO, the radial parametrisation $\mathbf{w} = \sqrt{p} \sigma \mathbf{q}$ changes only the outer σ -domain from $[0, C_\sigma]$ to $\mathcal{S}_\sigma^{\epsilon,c}$. On $\mathcal{E}_n^{\text{sr}}$, Proposition F.1 provides the pointwise small- r reduction for $\mathcal{R} = \mathcal{R}_0$; on $\mathcal{I}_n^t$, Lemma 12 provides the pointwise t -reduction uniformly in σ ; and $\mathcal{E}_n^{\text{ctr}}(C_M)$ gives pointwise Moreau feasibility. Hence the same scalarisation argument gives

$$\phi_{\mathbf{w}}^{\epsilon,c}(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \boldsymbol{\Omega}_{n,\sigma}^{\epsilon,c}).$$

Fix $j \in \mathcal{J}_n^\epsilon$. By Lemma 29 the set $Z_{p,j}^\epsilon = C_{p,j}^\epsilon \times \mathcal{B}_{C_\eta}^n$ is nonempty, compact, and convex. Since the restricted predictor set R_j^ϵ defined in the statement is the image of $C_{p,j}^\epsilon$ under an affine map, it is also nonempty, compact, and convex, and its definition gives $R_j^\epsilon \subseteq \mathcal{R}_0$. Lemma 10, applied on this restricted set, therefore gives the same ball-to-radius reduction as in the unrestricted scalarisation. The resulting distance profile is $M_{R_j^\epsilon}$.

On $\mathcal{E}_n^{\text{sr}}$, Proposition F.1 applies simultaneously to every R_j^ϵ . Since the final radius c_r fixed in Proposition B.5 satisfies $c_r \leq r_0$, it permits the restriction $r \in [c_r, 1]$ for every piece. The event $\mathcal{E}_{n,\text{res}}^{t,\epsilon}(c_r, C_t)$ then permits the restriction $t \in (0, C_t]$ simultaneously over all pieces.

Finally, on $\mathcal{E}_n^{\text{ctr}}(C_M)$, the condition

$$C_\eta \geq C_M + \frac{C_t}{c_r}$$

and Lemma 13 show that the unconstrained coordinatewise Moreau minimisers are feasible for $\mathcal{B}_{C_\eta}^n$. This pointwise feasibility is inherited by every restricted predictor set. Hence the completion-of-squares and Moreau-envelope calculation of Section B.6 applies without change.

Under the Gaussian representation in 93, the map $\mathbf{u} = \Gamma_p^{1/2} \mathbf{v}$ sends $C_{p,j}^\epsilon \times [0, C_t]$ exactly onto $\mathbf{K}_j^\epsilon(U_n)$. The resulting objective is F_n on $\boldsymbol{\Omega}_{n,j}^\epsilon$, and therefore

$$\phi_j^\epsilon(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \boldsymbol{\Omega}_{n,j}^\epsilon).$$

The construction is simultaneous over $j \in \mathcal{J}_n^\epsilon$. □

Lemma 33 (Restricted AO value bounds on finite convex pieces). *Recall the restricted scalar domains in (114), the restricted limiting values in (115), and the restricted AO values defined in Section B.9. For every $\epsilon > 0$ and $s > 0$, define*

$$\mathcal{A}_n^\epsilon(s) = \{|\phi(\mathbf{g}, \mathbf{h}) - \bar{\phi}| \leq s\} \cap \{\phi_{\mathbf{w}}^{\epsilon,c}(\mathbf{g}, \mathbf{h}) \geq \bar{\phi}_\sigma^{\epsilon,c} - s\} \cap \bigcap_{j \in \mathcal{J}_n^\epsilon} \{\phi_j^\epsilon(\mathbf{g}, \mathbf{h}) \geq \bar{\phi}_j^\epsilon - s\}.$$

Then there exist $t_0 = t_0(\epsilon, s) > 0$, constants $C_{\epsilon,s}, c_{\epsilon,s} > 0$, and $N_{\epsilon,s} \in \mathbb{N}$ such that, for all $n > N_{\epsilon,s}$,

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\neg \mathcal{A}_n^\epsilon(s) \mid \mathcal{G}_n) \leq C_{\epsilon,s} \exp\{-c_{\epsilon,s}n\}.$$

Proof. Set

$$\delta = \frac{s}{3}.$$

Let $\mathcal{R}_{n,\text{res}}^\epsilon$ be the simultaneous scalarisation event from Lemma 32. On this event,

$$\phi(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \mathbf{\Omega}_n),$$

$$\phi_{\mathbf{w}}^{\epsilon,c}(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \mathbf{\Omega}_{n,\sigma}^{\epsilon,c}),$$

whenever $\mathcal{S}_{\mathbf{w}}^{\epsilon,c}$ is nonempty, and

$$\phi_j^\epsilon(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \mathbf{\Omega}_{n,j}^\epsilon), \quad j \in \mathcal{J}_n^\epsilon.$$

By Lemma 32 after possibly decreasing t_0 and adjusting constants,

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\neg \mathcal{R}_{n,\text{res}}^\epsilon \mid \mathcal{G}_n) \leq C \exp\{-cn\}.$$

By Proposition B.6 after replacing t_0 by the minimum of its current value and the structural tolerance furnished there, and after adjusting constants,

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\{\neg \mathcal{U}_n(\delta)\} \cup \{\neg \mathcal{N}_n(\delta)\} \mid \mathcal{G}_n) \leq C_\delta \exp\{-c_\delta n\}.$$

Work on

$$\mathcal{R}_{n,\text{res}}^\epsilon \cap \mathcal{U}_n(\delta) \cap \mathcal{N}_n(\delta).$$

For the unrestricted AO value, the standard implication from (99), (101), and (102) gives $|\phi(\mathbf{g}, \mathbf{h}) - \bar{\phi}| \leq 3\delta = s$. In particular, $\phi(\mathbf{g}, \mathbf{h}) \leq \bar{\phi} + s$.

Next consider the $\mathbf{w}$ -restricted AO. If $\mathcal{S}_{\mathbf{w}}^{\epsilon,c}$ is empty, there is nothing to prove. Otherwise,

$$\phi_{\mathbf{w}}^{\epsilon,c}(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \mathbf{\Omega}_{n,\sigma}^{\epsilon,c}).$$

Since

$$\mathbf{\Omega}_{n,\sigma}^{\epsilon,c} \subseteq \mathbf{\Omega}_\sigma^{\epsilon,c},$$

only through the innermost minimisation domain in $\mathbf{u}$, enlarging the $\mathbf{u}$ -domain can only decrease the min-max-min value. Hence

$$\text{val}(F_n; \mathbf{\Omega}_{n,\sigma}^{\epsilon,c}) \geq \text{val}(F_n; \mathbf{\Omega}_\sigma^{\epsilon,c}).$$

On $\mathcal{U}_n(\delta)$, (101) gives

$$\text{val}(F_n; \mathbf{\Omega}_{n,\sigma}^{\epsilon,c}) \geq \text{val}(F; \mathbf{\Omega}_\sigma^{\epsilon,c}) - \delta = \bar{\phi}_\sigma^{\epsilon,c} - \delta.$$

Therefore

$$\phi_{\mathbf{w}}^{\epsilon,c}(\mathbf{g}, \mathbf{h}) \geq \bar{\phi}_\sigma^{\epsilon,c} - \delta \geq \bar{\phi}_\sigma^{\epsilon,c} - s.$$

Now fix $j \in \mathcal{J}_n^\epsilon$. On the same event,

$$\phi_j^\epsilon(\mathbf{g}, \mathbf{h}) = \text{val}(F_n; \mathbf{\Omega}_{n,j}^\epsilon).$$

Since $\Omega_{n,j}^\epsilon \subseteq \Omega_j^\epsilon$ only through the innermost minimisation domain in $(\mathbf{u}, \theta, t)$,

$$\text{val}(F_n; \Omega_{n,j}^\epsilon) \geq \text{val}(F_n; \Omega_j^\epsilon).$$

On $\mathcal{U}_n(\delta)$, (101) gives

$$\text{val}(F_n; \Omega_j^\epsilon) \geq \text{val}(F; \Omega_j^\epsilon) - \delta = \bar{\phi}_j^\epsilon - \delta.$$

Hence

$$\phi_j^\epsilon(\mathbf{g}, \mathbf{h}) \geq \bar{\phi}_j^\epsilon - s,$$

for every $j \in \mathcal{J}_n^\epsilon$.

By the definition of $\mathcal{A}_n^\epsilon(s)$ in the statement, we have shown that

$$\mathcal{R}_{n,\text{res}}^\epsilon \cap \mathcal{U}_n(\delta) \cap \mathcal{N}_n(\delta) \subseteq \mathcal{A}_n^\epsilon(s).$$

Taking complements and using the three conditional probability bounds above gives the claim, after adjusting constants. $\square$

F.6.3 Restricted PO barriers and value concentration

Lemma 34 (Restricted PO value barriers). *For every $\epsilon > 0$, with η_ϵ from Lemma 30, there are $t_0 = t_0(\epsilon) > 0$, constants $C_\epsilon, c_\epsilon > 0$, and $N_\epsilon \in \mathbb{N}$ such that, for all $n > N_\epsilon$,*

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\Psi > \bar{\phi} + \eta_\epsilon \mid \mathcal{G}_n) \leq C_\epsilon \exp\{-c_\epsilon n\},$$

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\Psi_{\mathbf{w}}^{\epsilon,c} < \bar{\phi}_\sigma^{\epsilon,c} - \eta_\epsilon \mid \mathcal{G}_n) \leq C_\epsilon \exp\{-c_\epsilon n\},$$

where the second display is trivial when $\mathcal{S}_{\mathbf{w}}^{\epsilon,c} = \emptyset$, and

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\exists j \in \mathcal{J}_n^\epsilon : \Psi_j^\epsilon < \bar{\phi}_j^\epsilon - \eta_\epsilon \mid \mathcal{G}_n) \leq C_\epsilon \exp\{-c_\epsilon n\}.$$

Consequently, the event $\mathcal{P}_n^\epsilon$ in (117) satisfies

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(-\mathcal{P}_n^\epsilon \mid \mathcal{G}_n) \leq C_\epsilon \exp\{-c_\epsilon n\}.$$

Proof. Set $s_\epsilon = \eta_\epsilon/2$. By Lemma 33 with $s = s_\epsilon$, after possibly decreasing t_0 and adjusting constants,

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(|\phi(\mathbf{g}, \mathbf{h}) - \bar{\phi}| > s_\epsilon \mid \mathcal{G}_n) \leq C_\epsilon \exp\{-c_\epsilon n\},$$

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\phi_{\mathbf{w}}^{\epsilon,c}(\mathbf{g}, \mathbf{h}) < \bar{\phi}_\sigma^{\epsilon,c} - s_\epsilon \mid \mathcal{G}_n) \leq C_\epsilon \exp\{-c_\epsilon n\},$$

and

$$\mathbb{1}_{\mathcal{S}_n(t_0)} \Pr(\exists j \in \mathcal{J}_n^\epsilon : \phi_j^\epsilon(\mathbf{g}, \mathbf{h}) < \bar{\phi}_j^\epsilon - s_\epsilon \mid \mathcal{G}_n) \leq C_\epsilon \exp\{-c_\epsilon n\}.$$

For the unrestricted value, apply Proposition B.4 with $W_0 = W_p$, $D_0 = V_p \times [-C_\theta, C_\theta]$. Since W_p is convex, the upper-tail comparison gives, almost surely,

$$\Pr(\Psi > \bar{\phi} + \eta_\epsilon \mid \mathcal{C}_n) \leq 2 \Pr(\phi(\mathbf{g}, \mathbf{h}) \geq \bar{\phi} + \eta_\epsilon \mid \mathcal{C}_n).$$

Because $s_\epsilon < \eta_\epsilon$,

$$\{\phi(\mathbf{g}, \mathbf{h}) \geq \bar{\phi} + \eta_\epsilon\} \subseteq \{|\phi(\mathbf{g}, \mathbf{h}) - \bar{\phi}| > s_\epsilon\}.$$

Taking conditional expectations with respect to $\mathcal{G}_n$ gives the first bound. For the radial comparison, define the $\mathcal{C}_n$ -measurable nonempty compact-valued fallback set

$$\widetilde{W}_\epsilon = \begin{cases} \mathcal{S}_{\mathbf{w}}^{\epsilon,c}, & \mathcal{S}_{\mathbf{w}}^{\epsilon,c} \neq \emptyset, \\ W_p, & \mathcal{S}_{\mathbf{w}}^{\epsilon,c} = \emptyset \end{cases}.$$

Apply Proposition B.4 with $W_0 = \widetilde{W}_\epsilon$ and $D_0 = V_p \times [-C_\theta, C_\theta]$. On the event $\{\mathcal{S}_{\mathbf{w}}^{\epsilon,c} \neq \emptyset\}$, the resulting PO and AO are exactly $\Psi_{\mathbf{w}}^{\epsilon,c}$ and $\phi_{\mathbf{w}}^{\epsilon,c}$. On the complementary event the desired lower-tail bound is trivial

because $\Psi_{\mathbf{w}}^{\epsilon,c} = +\infty$. Since $\mathcal{S}_{\mathbf{w}}^{\epsilon,c}$ is compact when nonempty and no convexity is required for the lower-tail comparison, almost surely,

$$\mathbb{1}_{\{\mathcal{S}_{\mathbf{w}}^{\epsilon,c} \neq \emptyset\}} \Pr(\Psi_{\mathbf{w}}^{\epsilon,c} < \bar{\phi}_{\sigma}^{\epsilon,c} - \eta_{\epsilon} \mid \mathcal{C}_n) \leq 2\mathbb{1}_{\{\mathcal{S}_{\mathbf{w}}^{\epsilon,c} \neq \emptyset\}} \Pr(\phi_{\mathbf{w}}^{\epsilon,c}(\mathbf{g}, \mathbf{h}) \leq \bar{\phi}_{\sigma}^{\epsilon,c} - \eta_{\epsilon} \mid \mathcal{C}_n).$$

The AO event on the right is contained in

$$\{\phi_{\mathbf{w}}^{\epsilon,c}(\mathbf{g}, \mathbf{h}) < \bar{\phi}_{\sigma}^{\epsilon,c} - s_{\epsilon}\}.$$

Taking conditional expectations with respect to $\mathcal{G}_n$ gives the radial bound.

For each fixed $j \in \mathcal{J}$, define the $\mathcal{C}_n$ -measurable nonempty compact convex fallback set

$$\tilde{C}_{p,j}^{\epsilon} = \begin{cases} C_{p,j}^{\epsilon}, & j \in \mathcal{J}_n^{\epsilon}, \\ V_p \times [-C_{\theta}, C_{\theta}], & j \notin \mathcal{J}_n^{\epsilon}. \end{cases}$$

Apply Proposition B.4 with $W_0 = W_p$ and $D_0 = \tilde{C}_{p,j}^{\epsilon}$. On the event $\{j \in \mathcal{J}_n^{\epsilon}\}$, the resulting PO and AO are exactly Ψ_j^{ϵ} and ϕ_j^{ϵ} . Hence, almost surely,

$$\mathbb{1}_{\{j \in \mathcal{J}_n^{\epsilon}\}} \Pr(\Psi_j^{\epsilon} < \bar{\phi}_j^{\epsilon} - \eta_{\epsilon} \mid \mathcal{C}_n) \leq 2\mathbb{1}_{\{j \in \mathcal{J}_n^{\epsilon}\}} \Pr(\phi_j^{\epsilon}(\mathbf{g}, \mathbf{h}) \leq \bar{\phi}_j^{\epsilon} - \eta_{\epsilon} \mid \mathcal{C}_n).$$

The AO event on the right is contained in

$$\{\phi_j^{\epsilon}(\mathbf{g}, \mathbf{h}) < \bar{\phi}_j^{\epsilon} - s_{\epsilon}\}.$$

Summing over the deterministic finite set $\mathcal{J}$, taking conditional expectations with respect to $\mathcal{G}_n$, and using $|\mathcal{J}_n^{\epsilon}| \leq |\mathcal{J}|$ gives

$$\Pr(\exists j \in \mathcal{J}_n^{\epsilon} : \Psi_j^{\epsilon} < \bar{\phi}_j^{\epsilon} - \eta_{\epsilon} \mid \mathcal{G}_n) \leq 2|\mathcal{J}| \Pr(\exists j \in \mathcal{J}_n^{\epsilon} : \phi_j^{\epsilon}(\mathbf{g}, \mathbf{h}) < \bar{\phi}_j^{\epsilon} - s_{\epsilon} \mid \mathcal{G}_n).$$

The simultaneous AO bound yields the piecewise PO estimate after absorbing $2|\mathcal{J}|$ into C_{ϵ} . A final union bound proves the estimate for $\neg \mathcal{P}_n^{\epsilon}$. $\square$

F.7 Lemmas for Section B.10

Lemma 35 (Spherical representation of the PO solution). *Let $\mathbf{d}$ be as in (121). Conditionally on $\mathcal{C}_n$, the direction $\mathbf{d}$ is uniform on $\mathbb{S}_{\mathbf{B}}^{\perp}$ and independent of $\mathbf{3}$.*

Proof. Condition on $\mathcal{C}_n$ and work on a fixed rank stratum. Write $s = \text{rank}(\mathbf{B})$, $k = p - s \geq 1$, $\mathcal{E}_{\sigma} = \{\hat{\sigma} > 0\}$, $\mathbf{d}' = \mathbf{E}^{\top} \mathbf{d}$. By Lemma 9 outside a conditional null event the compact problem in (62) has a unique primal minimiser. Let $\mathbf{R}$ be any $k \times k$ orthogonal matrix. Replacing $\mathbf{H}_2$ by $\mathbf{H}_2 \mathbf{R}^{\top}$ maps the primal minimiser to $(\hat{\mathbf{v}}, \mathbf{R} \hat{\mathbf{w}}, \hat{\theta}, \hat{\eta})$. Indeed, $\mathbf{H}_2 \mathbf{R}^{\top} \mathbf{R} \hat{\mathbf{w}} = \mathbf{H}_2 \hat{\mathbf{w}}$, $\|\mathbf{R} \hat{\mathbf{w}}\|_2 = \|\hat{\mathbf{w}}\|_2$, so both the objective and the feasible domain are preserved. Consequently, $\mathbf{3}$ and $\mathcal{E}_{\sigma}$ are unchanged, while on $\mathcal{E}_{\sigma}$, $\mathbf{d}' = \frac{\mathbf{w}}{\|\mathbf{w}\|_2}$ is mapped to $\mathbf{R} \mathbf{d}'$. Conditional on $\mathcal{C}_n$, the matrix $\mathbf{H}_2$ has independent standard Gaussian entries, and hence

$$\mathbf{H}_2 \mathbf{R}^{\top} \stackrel{\text{d}}{=} \mathbf{H}_2,$$

conditionally on $\mathcal{C}_n$. It follows that, for every bounded Borel function $f : \mathbb{S}^{k-1} \rightarrow \mathfrak{R}$ and every bounded Borel function g of $\mathbf{3}$,

$$\mathbb{E}[f(\mathbf{d}')g(\mathbf{3})\mathbb{1}_{\{\mathcal{E}_{\sigma}\}} \mid \mathcal{C}_n] = \mathbb{E}[f(\mathbf{R} \mathbf{d}')g(\mathbf{3})\mathbb{1}_{\{\mathcal{E}_{\sigma}\}} \mid \mathcal{C}_n]. \quad (236)$$

Let μ be the conditional law of $\mathbf{d}'$ given $\sigma(\mathcal{C}_n, \mathbf{3})$. Since $\mathcal{E}_{\sigma} \in \sigma(\mathbf{3})$, (236) and the almost-sure uniqueness of regular conditional distributions (Shiryaev, 2016, Chapter 2, Section 7) imply that, for each fixed $k \times k$ orthogonal matrix $\mathbf{R}$,

$$\mu(\mathbf{R}^{-1}A) = \mu(A), \quad (237)$$

for every Borel set $A \subseteq \mathbb{S}^{k-1}$, on $\mathcal{E}_{\sigma}$, outside a null set that may depend on $\mathbf{R}$.

The collection of $k \times k$ orthogonal matrices is a compact metric subset of $\mathbb{R}^{k \times k}$ and therefore has a countable dense subset D (Aliprantis and Border [2006] Lemma 3.26). Intersect the probability-one events in (237) over $\mathbf{R} \in D$. On the resulting probability-one event, μ is invariant under every $\mathbf{R} \in D$ simultaneously on $\mathcal{E}_\sigma$. Now let $\mathbf{R}$ be any $k \times k$ orthogonal matrix. Since D is dense, there exists a sequence $(\mathbf{R}_m)_{m \in \mathbb{N}} \subseteq D$ such that

$$\|\mathbf{R}_m - \mathbf{R}\|_2 \rightarrow 0.$$

Therefore, on $\mathcal{E}_\sigma$, and continuous f ,

$$\int_{\mathbb{S}^{k-1}} f(\mathbf{R}x) \mu(dx) = \lim_{m \rightarrow \infty} \int_{\mathbb{S}^{k-1}} f(\mathbf{R}_m x) \mu(dx) = \int_{\mathbb{S}^{k-1}} f(x) \mu(dx).$$

Since continuous functions determine Borel probability measures on the compact sphere (Dudley [2018] Lemma 9.3.2), μ is invariant under every $k \times k$ orthogonal matrix. The unique probability measure with this invariance is the uniform probability measure on $\mathbb{S}^{k-1}$ (see Eaton [1989] Theorem 4.4). Hence, on $\mathcal{E}_\sigma$, $\mathbf{d}'$ is conditionally uniform on $\mathbb{S}^{k-1}$. Since $\mathbf{E}^\top \mathbf{E} = \mathbf{I}_k$ and $\text{range}(\mathbf{E}) = \text{range}(\mathbf{B})^\perp$, the maps $x \mapsto \mathbf{E}x$ and $y \mapsto \mathbf{E}^\top y$ are inverse isometries between $\mathbb{S}^{k-1}$ and $\mathbb{S}_\mathbf{B}^\perp$. Hence, for every Borel set $A \subseteq \mathbb{S}_\mathbf{B}^\perp$,

$$\mathbb{1}\{\mathcal{E}_\sigma\} \Pr(\mathbf{d} \in A \mid \sigma(\mathcal{C}_n, \mathbf{Z})) = \mathbb{1}\{\mathcal{E}_\sigma\} \Pr(\mathbf{d}' \in \mathbf{E}^\top A \mid \sigma(\mathcal{C}_n, \mathbf{Z})).$$

The right-hand side is the uniform spherical probability of $\mathbf{E}^\top A$, which equals that of A because $\mathbf{E}^\top$ is an isometry. Thus $\mathbf{d}$ is conditionally uniform on $\mathbb{S}_\mathbf{B}^\perp$ on $\mathcal{E}_\sigma$. On $\neg \mathcal{E}_\sigma$, (121) gives, outside a null event,

$$\mathbf{d} = \frac{\mathbf{P}^\perp \boldsymbol{\xi}_n}{\|\mathbf{P}^\perp \boldsymbol{\xi}_n\|_2} = \mathbf{E} \frac{\mathbf{E}^\top \boldsymbol{\xi}_n}{\|\mathbf{E}^\top \boldsymbol{\xi}_n\|_2}.$$

The vector $\boldsymbol{\xi}_n$ is independent of $\sigma(\mathcal{C}_n, \mathbf{Z})$, and $\mathbf{E}$ is $\mathcal{C}_n$ -measurable. Hence, conditionally on $\sigma(\mathcal{C}_n, \mathbf{Z})$,

$$\mathbf{E}^\top \boldsymbol{\xi}_n \sim \mathcal{N}(\mathbf{0}_k, \mathbf{I}_k).$$

The Gaussian polar decomposition (Vershynin [2018], Exercise 3.3.7) therefore shows that $\mathbf{d}$ is uniform on $\mathbb{S}_\mathbf{B}^\perp$ and independent of $\mathbf{Z}$ on $\neg \mathcal{E}_\sigma$. Since $\mathcal{E}_\sigma \in \sigma(\mathbf{Z})$, the two cases give

$$\mathbf{d} \mid \mathcal{C}_n, \mathbf{Z} \stackrel{d}{=} \mathbf{d} \mid \mathcal{C}_n \stackrel{d}{=} \text{Unif}(\mathbb{S}_\mathbf{B}^\perp),$$

almost surely as required. $\square$

F.8 Minor auxiliary lemmas

F.8.1 Concentration results

Lemma 36 (Gaussian design spectral-norm bound). *Let $\widetilde{\mathbf{H}} = [\mathbf{1}, \mathbf{G}] \in \mathbb{R}^{n \times (p+1)}$, where $\mathbf{G}$ has i.i.d. $\mathcal{N}(0, 1)$ entries. There exists a universal constant $c_{\text{sv}} > 0$ such that, for every $t > 0$,*

$$\Pr\left(\|\widetilde{\mathbf{H}}\|_2 > 2\sqrt{n} + \sqrt{p} + t\right) \leq 2\exp\{-c_{\text{sv}}t^2\}.$$

Proof. The triangle inequality gives

$$\|\widetilde{\mathbf{H}}\|_2 \leq \sqrt{n} + \|\mathbf{G}\|_2,$$

and the claim follows from Vershynin [2018], Corollary 7.3.3. $\square$

Lemma 37 (Uniform row-projection bound). *Let $\widetilde{\mathbf{H}} = [\mathbf{1}, \mathbf{G}] \in \mathbb{R}^{n \times (p+1)}$. For $\epsilon \in (0, 1)$ and $\mathbf{v} \in \mathbb{R}^{p+1}$, define*

$$S_\epsilon(\mathbf{v}) = \left\{ i \in [n] : \left| \mathbf{v}^\top \widetilde{\mathbf{h}}_i \right| \leq \frac{3}{\sqrt{\epsilon}} \|\mathbf{v}\|_2 \right\},$$

where $\tilde{\mathbf{h}}_i^\top$ is the i th row of $\widetilde{\mathbf{H}}$. If $\|\widetilde{\mathbf{H}}\|_2 \leq 3\sqrt{n}$, then

$$\inf_{\mathbf{v} \in \mathbb{R}^{p+1}} |S_\epsilon(\mathbf{v})| \geq (1 - \epsilon)n.$$

If, moreover, $\mathbf{G}$ has i.i.d. $\mathcal{N}(0, 1)$ entries and $n > p$, then

$$\Pr \left(\inf_{\mathbf{v} \in \mathbb{R}^{p+1}} |S_\epsilon(\mathbf{v})| < (1 - \epsilon)n \right) \leq 2 \exp \left\{ -c_{\text{sv}} n \left(1 - \sqrt{\frac{p}{n}} \right)^2 \right\}.$$

Proof. The deterministic argument is similar to [Sur, 2019, Lemma 7]. On the event $\{\|\widetilde{\mathbf{H}}\|_2 \leq 3\sqrt{n}\}$, for every $\mathbf{v} \neq \mathbf{0}_{p+1}$,

$$\begin{aligned} 9n \|\mathbf{v}\|_2^2 &\geq \|\widetilde{\mathbf{H}}\mathbf{v}\|_2^2 \\ &\geq \sum_{i \notin S_\epsilon(\mathbf{v})} (\mathbf{v}^\top \tilde{\mathbf{h}}_i)^2 \\ &\geq \{n - |S_\epsilon(\mathbf{v})|\} \frac{9 \|\mathbf{v}\|_2^2}{\epsilon}. \end{aligned}$$

Hence $|S_\epsilon(\mathbf{v})| \geq (1 - \epsilon)n$. For $\mathbf{v} = \mathbf{0}_{p+1}$, $S_\epsilon(\mathbf{v}) = [n]$, so the deterministic claim follows. The probability bound follows from Lemma [36] with $t = \sqrt{n} - \sqrt{p}$. $\square$

F.8.2 Moreau-envelope analytic facts

Lemma 38 (Lipschitz continuity of the Moreau envelope). *Let $\rho : \mathbb{R} \rightarrow \mathbb{R}$ be convex and L_ρ -Lipschitz. For $\lambda > 0$, define*

$$m_\rho(x, \lambda) = \min_{\eta \in \mathbb{R}} \left\{ \rho(\eta) + \frac{(\eta - x)^2}{2\lambda} \right\}, \quad m_\rho(x, 0) = \rho(x).$$

Then:

- (i) *for every $\lambda \geq 0$, the map $x \mapsto m_\rho(x, \lambda)$ is L_ρ -Lipschitz on $\mathbb{R}$;*
- (ii) *for every $x \in \mathbb{R}$, the map $\lambda \mapsto m_\rho(x, \lambda)$ is $L_\rho^2/2$ -Lipschitz on $[0, \infty)$.*

In particular, for $\rho(x) = \log(1 + \exp\{x\})$, define, for $\lambda > 0$,

$$M_\rho(\mathbf{x}, \lambda) = \min_{\boldsymbol{\eta} \in \mathbb{R}^n} \left\{ \sum_{i=1}^n \rho(\eta_i) + \frac{\|\boldsymbol{\eta} - \mathbf{x}\|_2^2}{2\lambda} \right\}, \quad M_\rho(\mathbf{x}, 0) = \mathbf{1}^\top \boldsymbol{\rho}(\mathbf{x}).$$

Then M_ρ is $\sqrt{n}$ -Lipschitz in $\mathbf{x}$, $n/2$ -Lipschitz in λ , and $\sqrt{n + n^2/4}$ -Lipschitz jointly in $(\mathbf{x}, \lambda)$ on $\mathbb{R}^n \times [0, \infty)$.

Proof. We prove each claim in turn.

- (i) Fix $\lambda \geq 0$. For $\lambda = 0$, the claim follows from $m_\rho(\cdot, 0) = \rho$. Suppose that $\lambda > 0$, and set $\eta = \text{prox}_{\lambda\rho}(x)$. The standard Moreau-envelope derivative formula ([Salehi et al., 2019, Lemma 2, equation (29)], applied with $d = 1$, $\Phi = \rho$, and $t = \lambda$, gives

$$\partial_x m_\rho(x, \lambda) = \frac{x - \eta}{\lambda}.$$

The proximal optimality condition (cf. [Beck, 2017, Theorem 6.39] gives

$$\frac{x - \eta}{\lambda} \in \partial\rho\{\eta\}.$$

Since every subgradient of the L_ρ -Lipschitz function ρ has absolute value at most L_ρ (cf. [Beck, 2017, Theorem 3.61],

$$|\partial_x m_\rho(x, \lambda)| \leq L_\rho.$$

The mean-value theorem proves that $x \mapsto m_\rho(x, \lambda)$ is L_ρ -Lipschitz.

(ii) Fix $x \in \mathfrak{R}$. For $\lambda > 0$, write again $\eta = \text{prox}_{\lambda\rho}(x)$. The second Moreau-envelope derivative formula in [Salehi *et al.* \(2019\)](#) Lemma 2, equation (29)) gives

$$\partial_\lambda m_\rho(x, \lambda) = -\frac{\{x - \eta\}^2}{2\lambda^2}.$$

By the same proximal-optimality and bounded-subgradient argument as above,

$$|x - \eta| \leq L_\rho \lambda,$$

and therefore

$$|\partial_\lambda m_\rho(x, \lambda)| \leq \frac{L_\rho^2}{2}.$$

The mean-value theorem proves the claimed bound when both smoothing parameters are positive.

It remains to include $\lambda = 0$. For $\lambda > 0$, testing the defining minimum at $\eta = x$ gives $m_\rho(x, \lambda) \leq \rho(x)$, while $\rho(\eta) \geq \rho(x) - L_\rho|\eta - x|$ gives

$$m_\rho(x, \lambda) \geq \rho(x) + \min_{r \geq 0} \left\{ \frac{r^2}{2\lambda} - L_\rho r \right\} = \rho(x) - \frac{L_\rho^2}{2} \lambda.$$

Since $m_\rho(x, 0) = \rho(x)$,

$$|m_\rho(x, \lambda) - m_\rho(x, 0)| \leq \frac{L_\rho^2}{2} \lambda.$$

Hence $\lambda \mapsto m_\rho(x, \lambda)$ is $L_\rho^2/2$ -Lipschitz on $[0, \infty)$.

For logistic ρ , one has $L_\rho = 1$, and separability ([Salehi *et al.* \(2019\)](#) Lemma 6, equation (35)) gives $M_\rho(\mathbf{x}, \lambda) = \sum_{i=1}^n m_\rho(x_i, \lambda)$. Consequently, for $\mathbf{x}, \mathbf{z} \in \mathfrak{R}^n$ and $\lambda, \mu \geq 0$,

$$\begin{aligned} |M_\rho(\mathbf{x}, \lambda) - M_\rho(\mathbf{z}, \mu)| &\leq \|\mathbf{x} - \mathbf{z}\|_1 + \frac{n}{2} |\lambda - \mu| \\ &\leq \sqrt{n} \|\mathbf{x} - \mathbf{z}\|_2 + \frac{n}{2} |\lambda - \mu| \\ &\leq \sqrt{n + \frac{n^2}{4}} \left\| \begin{bmatrix} \mathbf{x} \\ \lambda \end{bmatrix} - \begin{bmatrix} \mathbf{z} \\ \mu \end{bmatrix} \right\|_2. \end{aligned}$$

The separate vector bounds follow by taking $\lambda = \mu$ or $\mathbf{x} = \mathbf{z}$. □

References

- Aliprantis CD, Border KC (2006). *Infinite dimensional analysis: a hitchhiker's guide*. Springer, 3 edition. doi:10.1007/3-540-29587-9.
- Beck A (2017). *First-order methods in optimization*. SIAM. ISBN 978-1-61197-498-0. doi:10.1137/1.9781611974997.
- Bhatia R (1997). *Matrix Analysis*, volume 169 of *Graduate Texts in Mathematics*. Springer, New York. doi:10.1007/978-1-4612-0653-8.
- Donoho D, Montanari A (2016). “High dimensional robust M-estimation: Asymptotic variance via approximate message passing.” *Probability Theory and Related Fields*, **166**, 935–969. doi:10.1007/s00440-015-0675-z.
- Dudley RM (2018). *Real analysis and probability*. CRC Press. doi:10.1201/9781351076197.
- Eaton ML (1989). *Group Invariance Applications in Statistics*, volume 1 of *NSF-CBMS Regional Conference Series in Probability and Statistics*. Institute of Mathematical Statistics, Hayward, California.
- Magnus JR, Neudecker H (2019). *Matrix Differential Calculus with Applications in Statistics and Econometrics*. John Wiley & Sons. ISBN 9781119541202. doi:10.1002/9781119541219.
- Newey WK, McFadden D (1994). “Chapter 36 Large sample estimation and hypothesis testing.” volume 4 of *Handbook of Econometrics*, pp. 2111–2245. Elsevier. doi:https://doi.org/10.1016/S1573-4412(05)80005-4.
- Rigon T, Aliverti E (2023). “Conjugate priors and bias reduction for logistic regression models.” *Statistics & Probability Letters*, **202**, 109901. doi:10.1016/j.spl.2023.109901.
- Rudelson M, Vershynin R (2010). *Non-asymptotic theory of random matrices: Extreme singular values*, pp. 1576–1602. doi:10.1142/9789814324359_0111.
- Rudin W (1976). *Principles of Mathematical Analysis*. McGraw-hill New York, 3rd edition. ISBN 978-0070542358.
- Salehi F, Abbasi E, Hassibi B (2019). “The impact of regularization on high-dimensional logistic regression.” In “Proceedings of the 33rd International Conference on Neural Information Processing Systems,” Curran Associates Inc., Red Hook, NY, USA. [1906.03761](#)
- Serfling RJ (1980). *Approximation Theorems of Mathematical Statistics*. Wiley. ISBN 9780470316481. doi:10.1002/9780470316481.
- Shiryaev AN (2016). *Probability-1*, volume 95. Springer. ISBN 978-1-4939-7905-9. doi:10.1007/978-0-387-72206-1.
- Simons S (1995). *Minimax Theorems and Their Proofs*, pp. 1–23. Springer US, Boston, MA. ISBN 978-1-4613-3557-3. doi:10.1007/978-1-4613-3557-3.1.
- Stein CM (1981). “Estimation of the mean of a multivariate normal distribution.” *The Annals of Statistics*, **9**(6), 1135–1151. doi:10.1214/aos/1176345632.
- Sterzinger P, Kosmidis I (2026). “Diaconis–Ylvisaker Prior Penalized Likelihood for $p/n \rightarrow \kappa \in (0, 1)$ Logistic Regression.” *Biometrika*, **113**(2), asag014. doi:10.1093/biomet/asag014. URL <https://doi.org/10.1093/biomet/asag014>.
- Sur P (2019). *A Modern Maximum Likelihood Theory for High-dimensional Logistic Regression*. Ph.D. thesis, Stanford University. URL <http://purl.stanford.edu/jw604jq1260>
- Thrampoulidis C, Abbasi E, Hassibi B (2018). “Precise error analysis of regularized M -estimators in high dimensions.” *IEEE Transactions on Information Theory*, **64**(8), 5592–5628. doi:10.1109/TIT.2018.2840720.

- Vershynin R (2018). *High-dimensional Probability: An Introduction with Applications in Data Science*, volume 47. Cambridge university press. doi:10.1017/9781108231596.
- Villani C (2009). *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer, Berlin. doi:10.1007/978-3-540-71050-9.
- Yadlowsky S, Yun T, McLean CY, D'Amour A (2021). “SLOE: A Faster Method for Statistical Inference in High-Dimensional Logistic Regression.” In M Ranzato, A Beygelzimer, Y Dauphin, P Liang, JW Vaughan (eds.), “Advances in Neural Information Processing Systems,” volume 34, pp. 29517–29528. Curran Associates, Inc. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/f6c2a0c4b566bc99d596e58638e342b0-Paper.pdf
- Zhao Q, Sur P, Candes EJ (2022). “The asymptotic distribution of the MLE in high-dimensional logistic models: Arbitrary covariance.” *Bernoulli*, **28**(3), 1835–1861. doi:10.3150/21-BEJ1401.